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Pacella

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(54) **BASSINET ACCESSORY FOR A PLAYARD**

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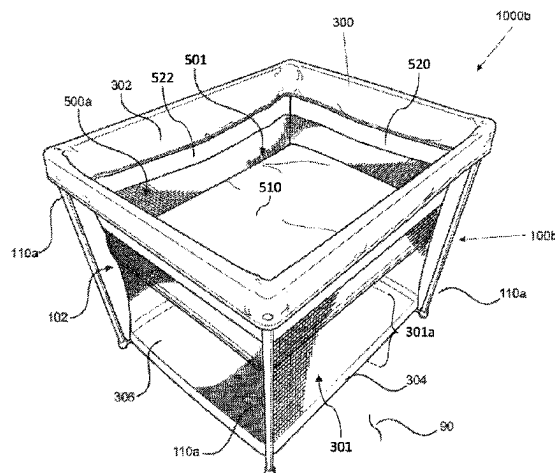
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ABSTRACT

A bassinet accessory for a foldable playard, wherein the bassinet and playard are folded and unfolded together. When unfolded, the bassinet is relatively shallow from a top of the playard so that a caregiver may more easily place a child into, and remove the child from, the bassinet. A bassinet mattress is supported by a central hub on the bottom of the bassinet and four support tubes coupled to the hub and extending to respective bottom corners of the bassinet. During folding, the bassinet hub travels in a vertical direction and the support tubes rotate from a horizontal to a vertical position, becoming parallel with respective legs of the playard. Thus, in the compact folded configuration, the bassinet hub is surrounded by and within the folded frame of the playard or flush with the top of the playard (so as not to appreciably extend beyond a top of the playard).

20 Claims, 134 Drawing Sheets



(58) **Field of Classification Search**

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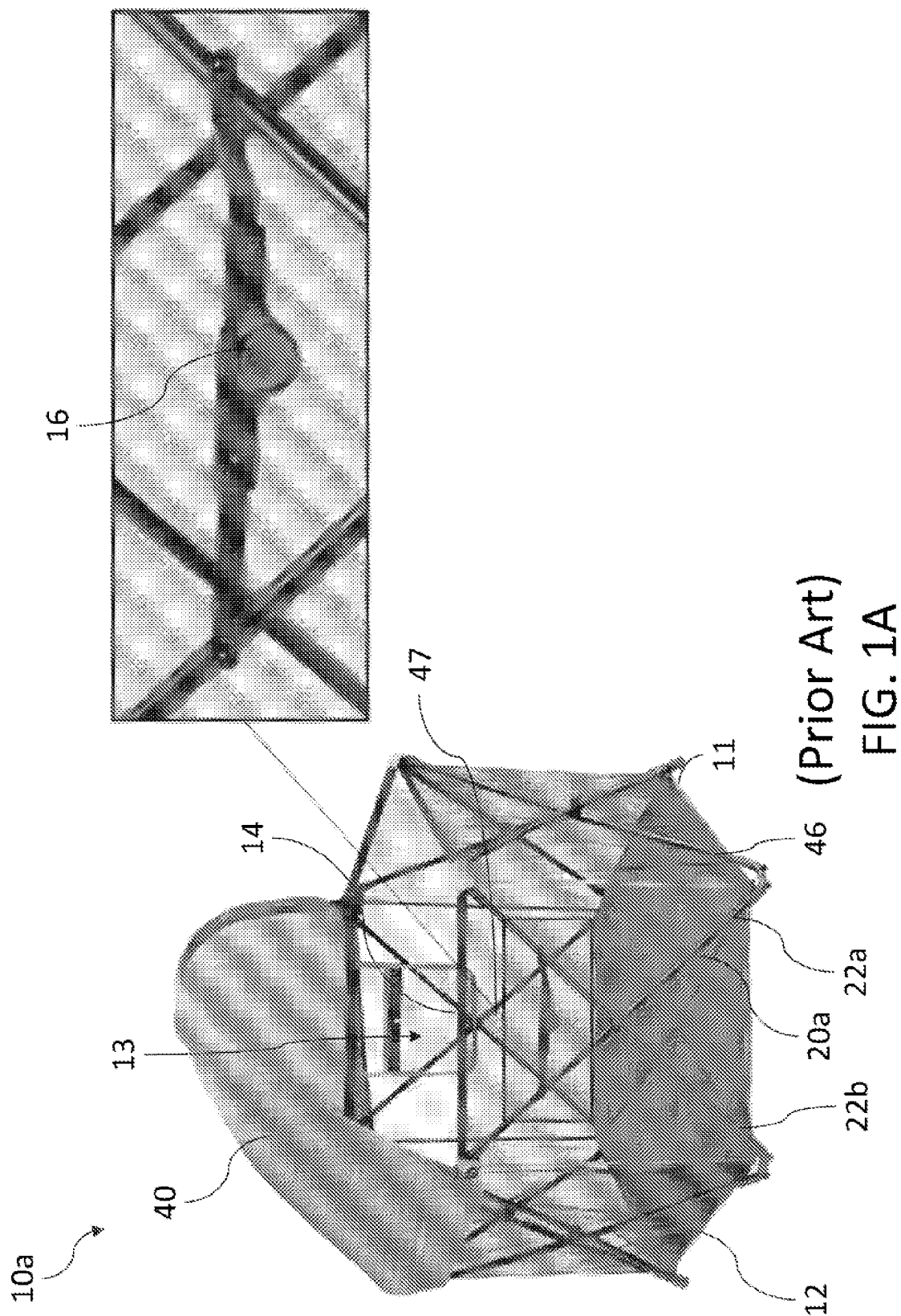
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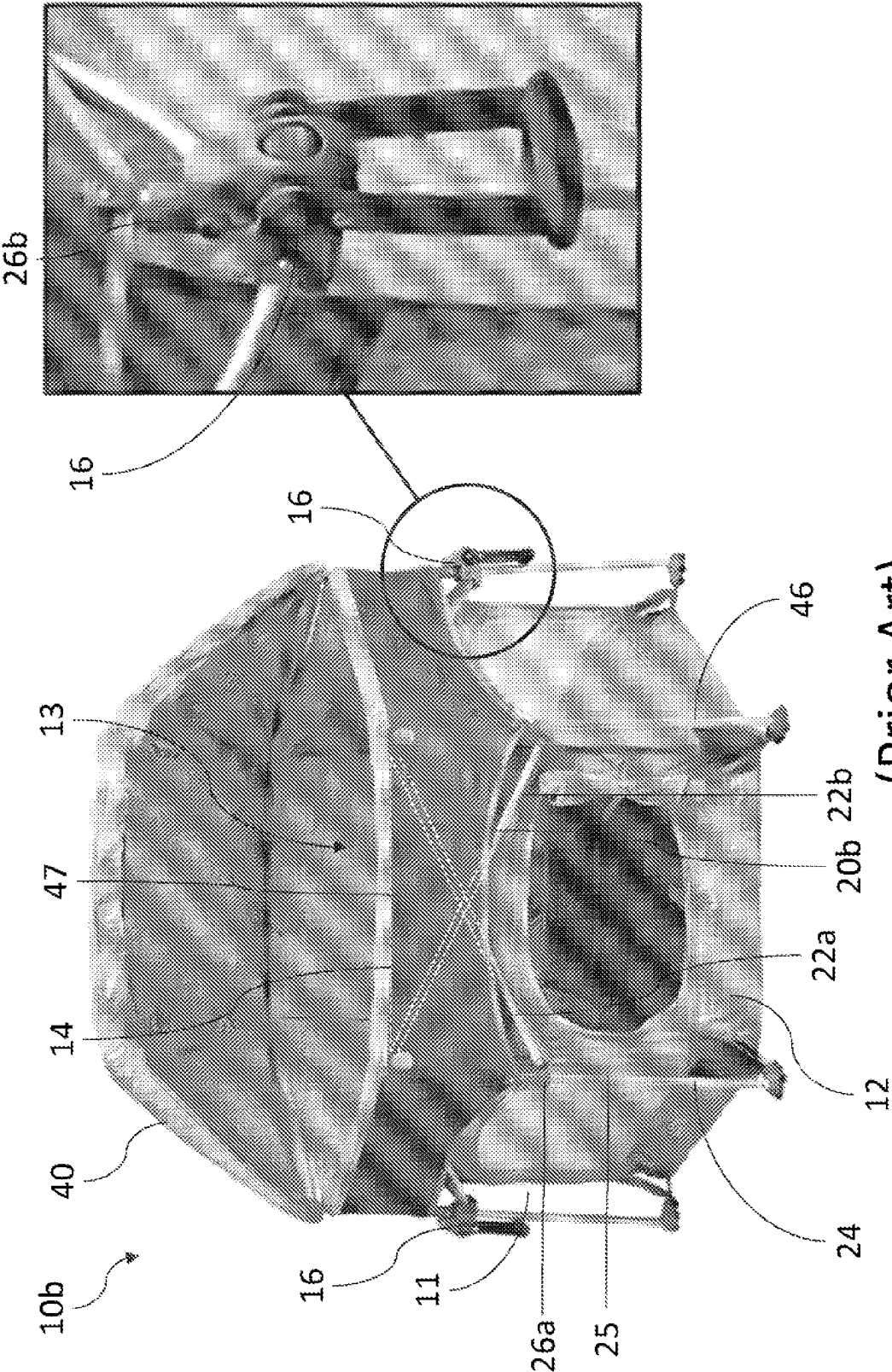
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"Final Office Action Issued in Corresponding U.S. Appl. No. 17/781,939", Mailed Date: Aug. 29, 2024, 12 pages.

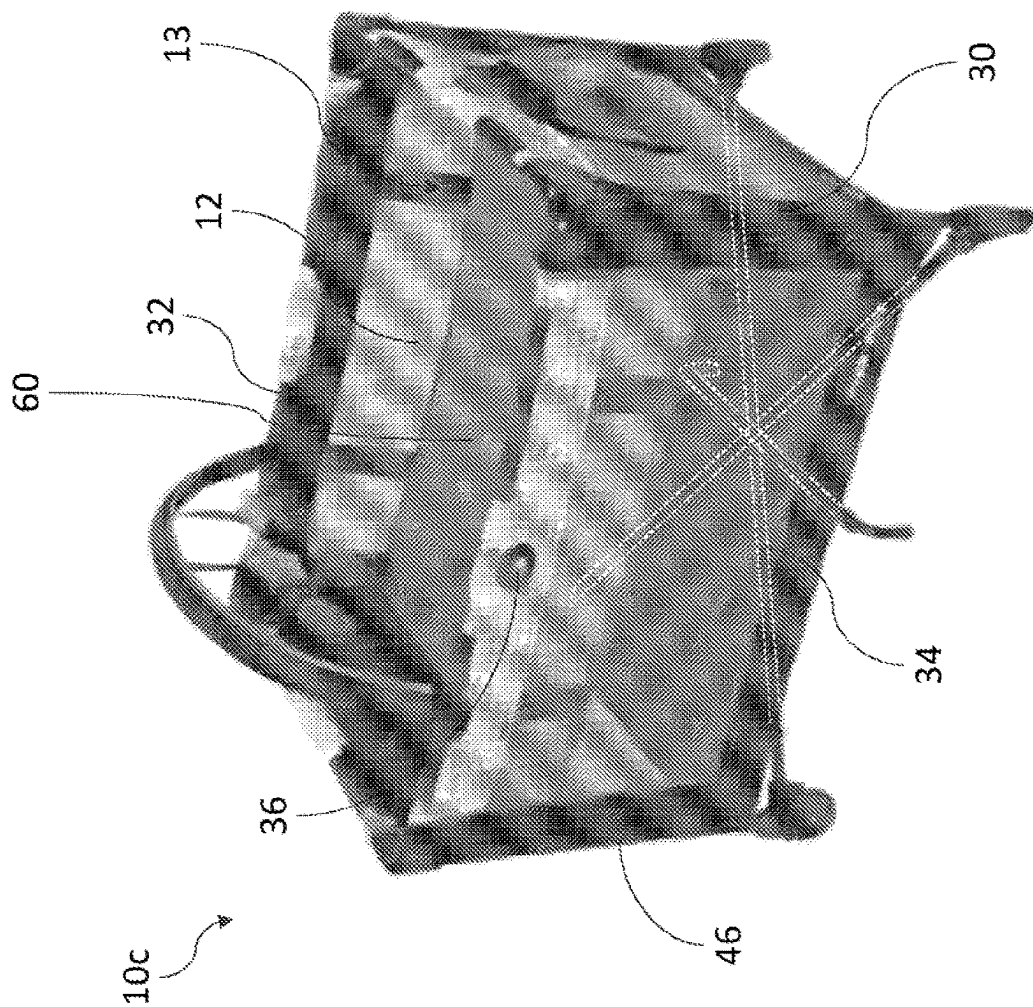
Second Office Action for corresponding Chinese Application No. 202080056114.1 filed Aug. 7, 2020 mailed Feb. 18, 2025, 19 pages.

* cited by examiner

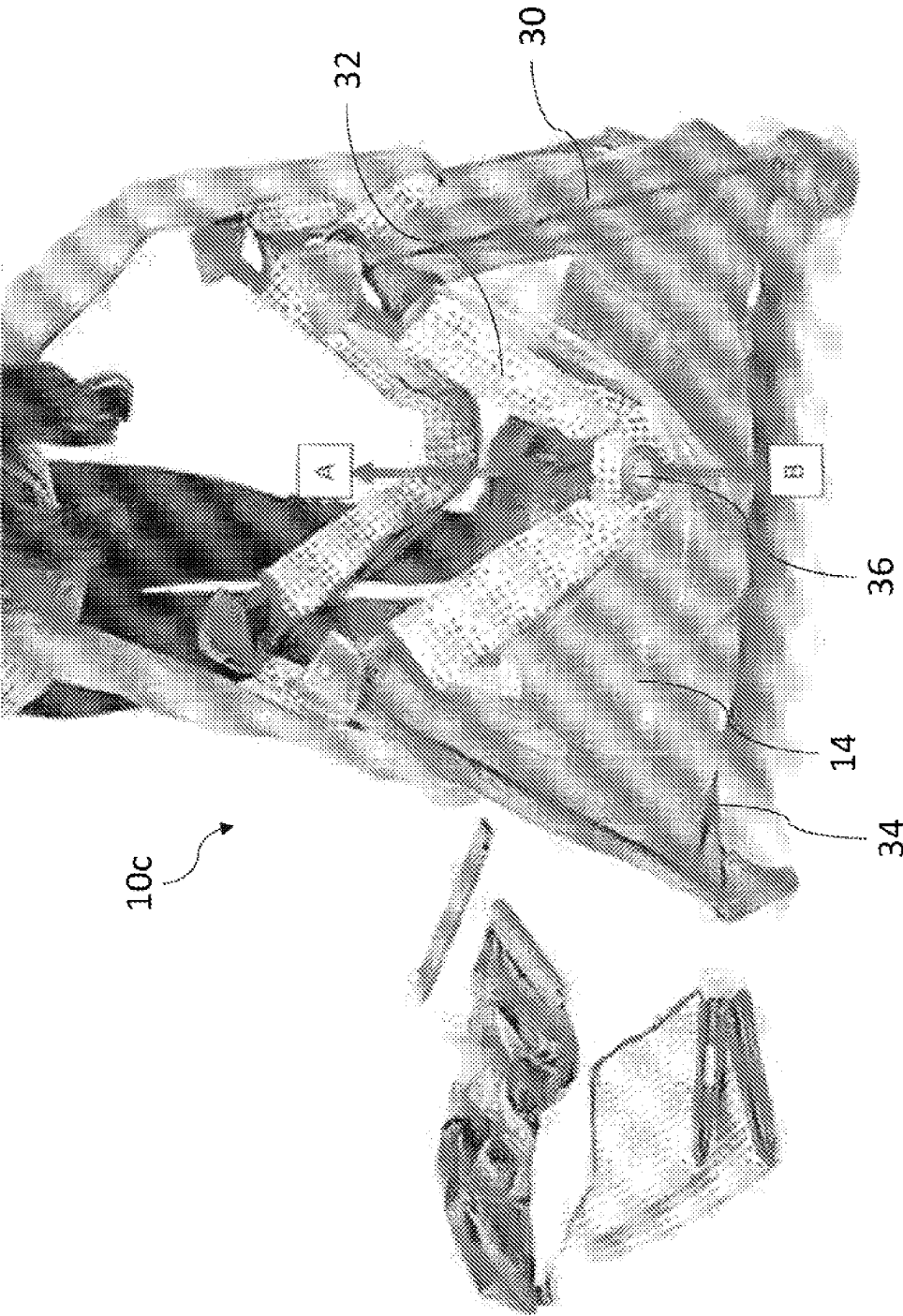




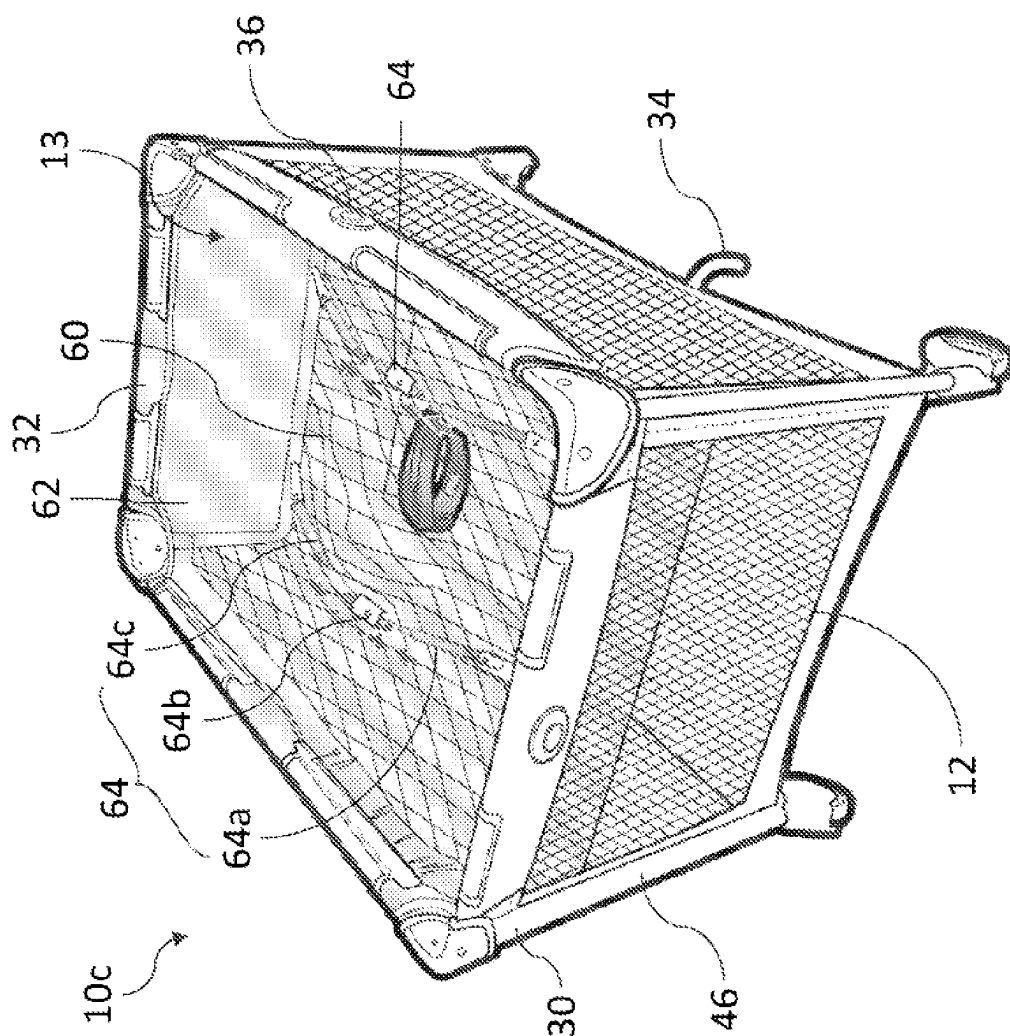
(Prior Art)
FIG. 1B



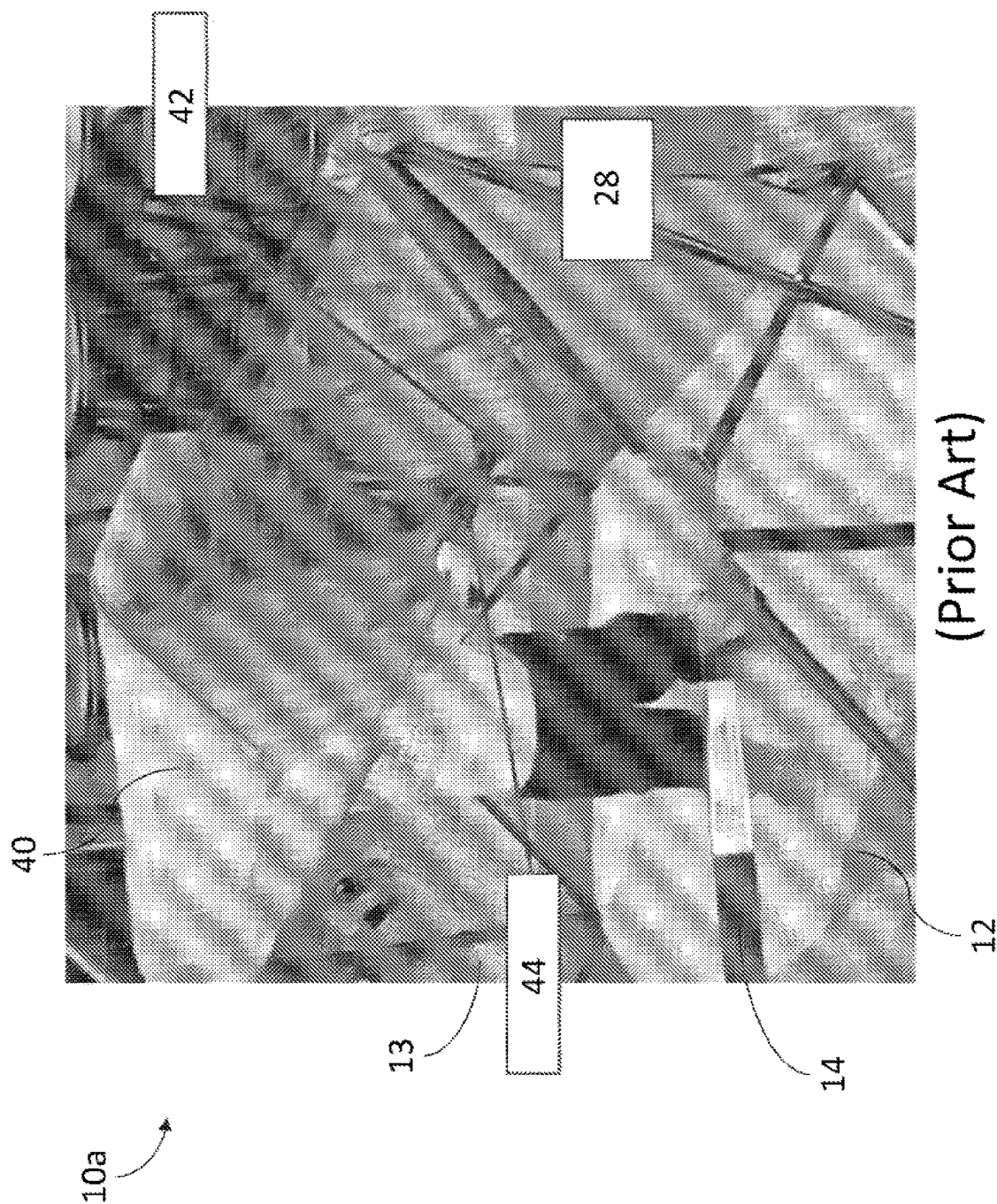
(Prior Art)
FIG. 1C



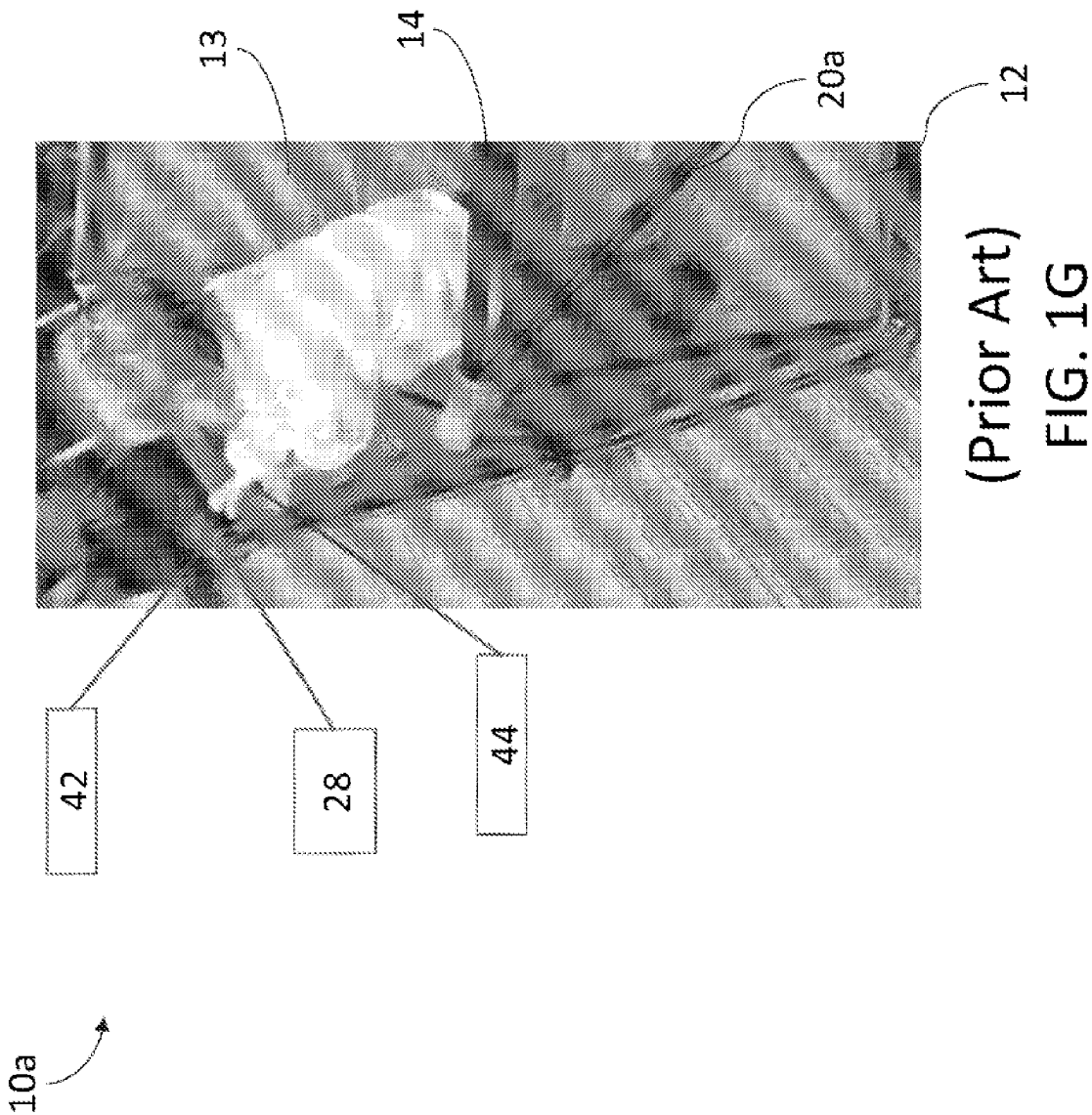
(Prior Art)
FIG. 1D



(Prior Art)
FIG. 1E



(Prior Art)
FIG. 1F



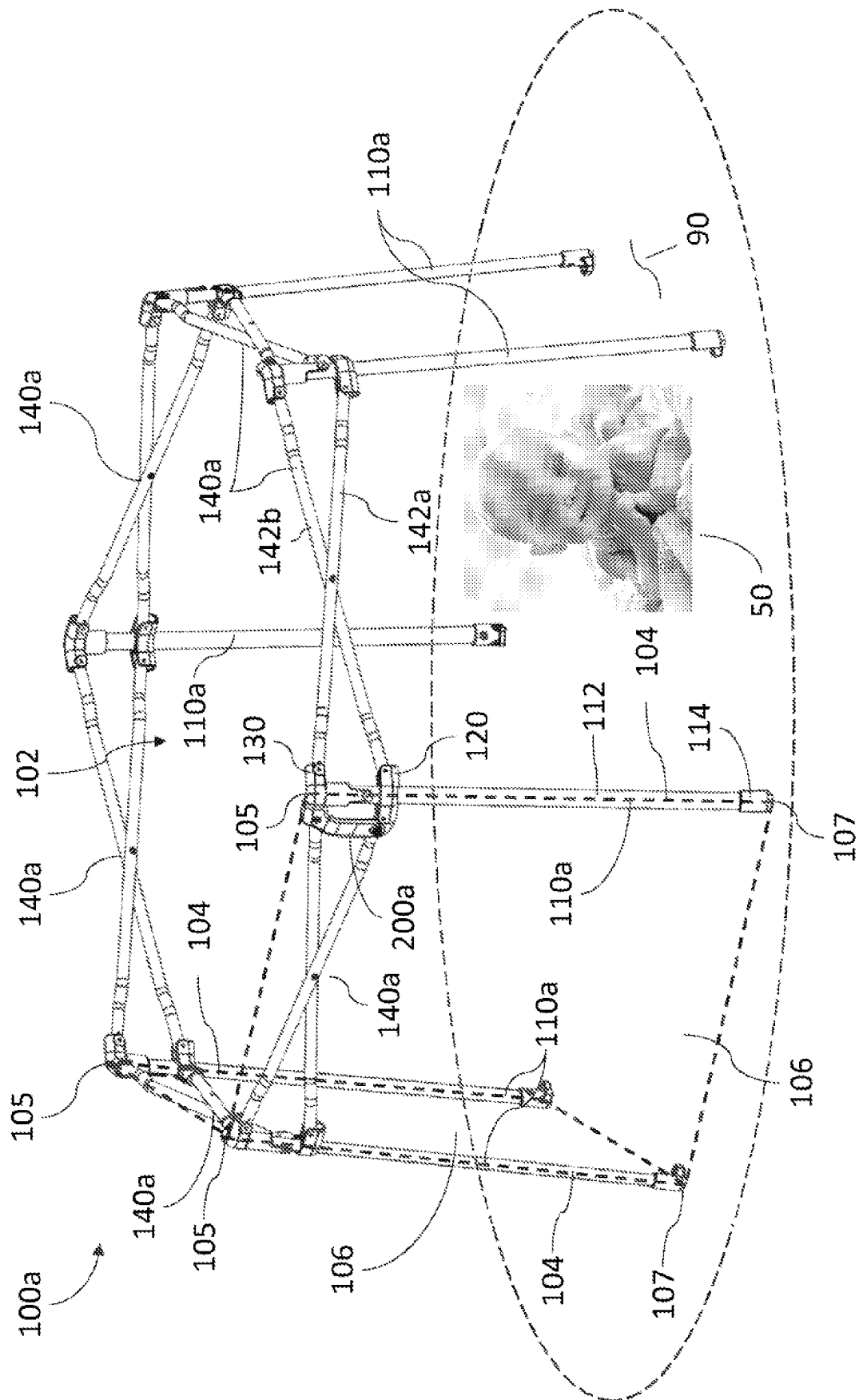
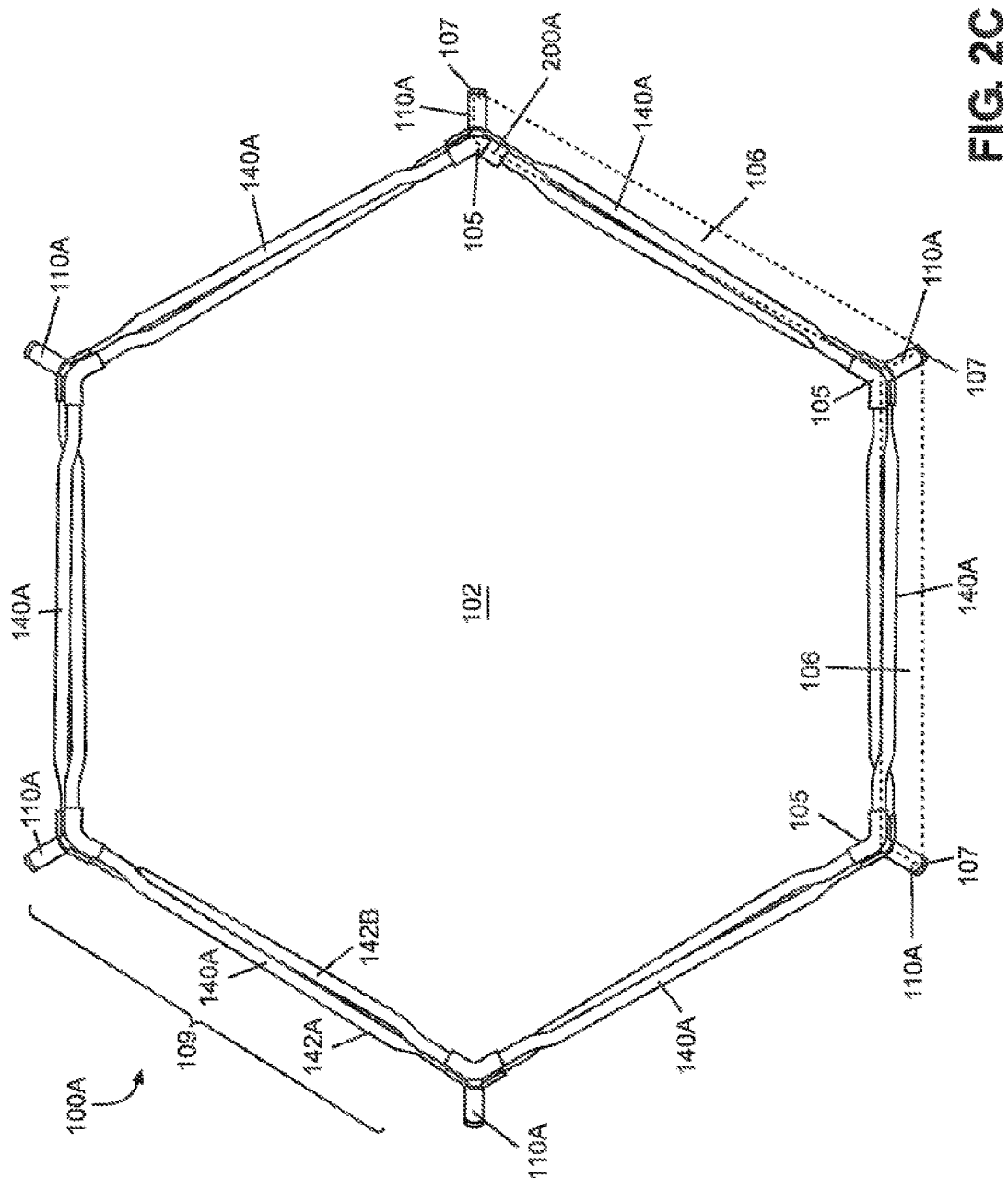
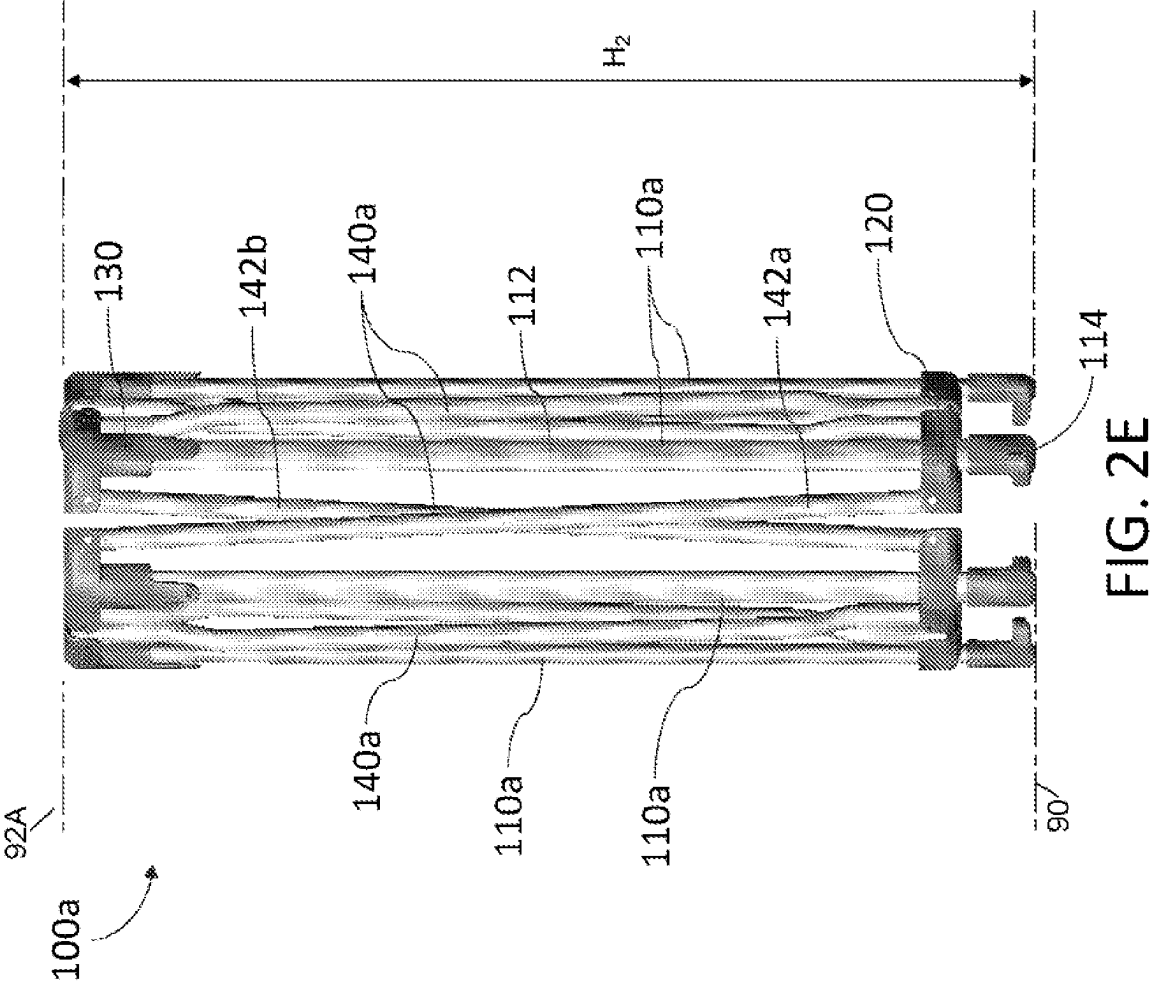


FIG. 2A





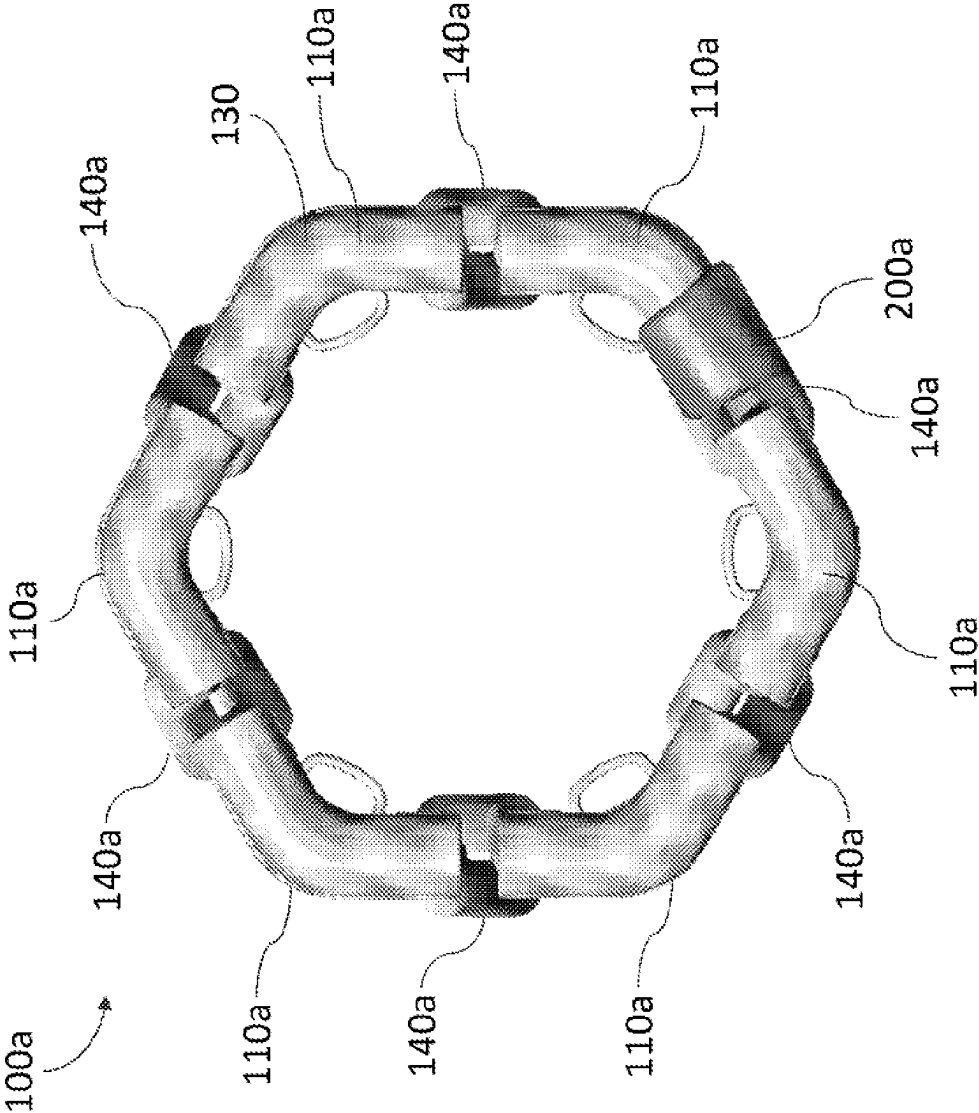


FIG. 2F

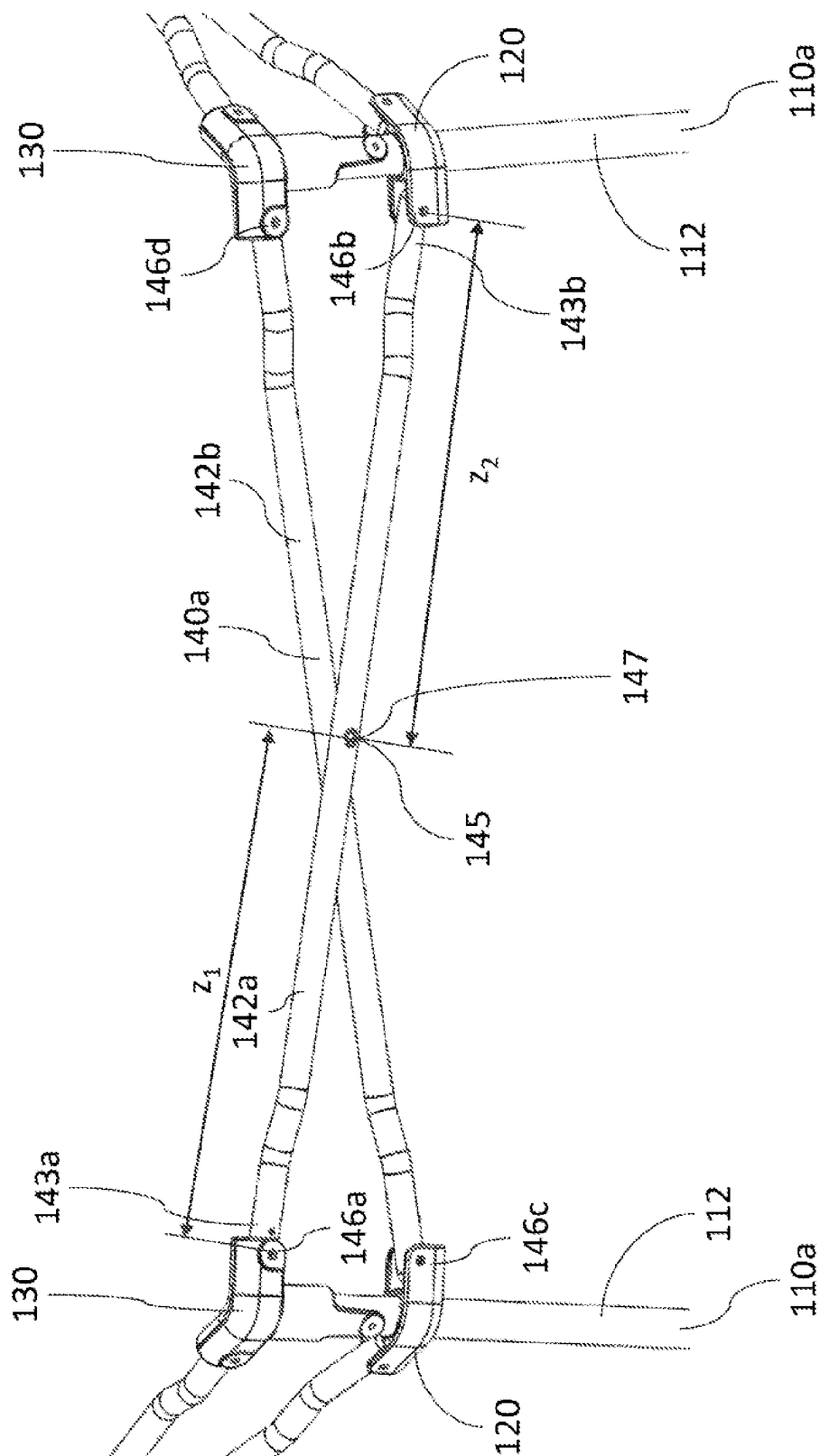


FIG. 3A

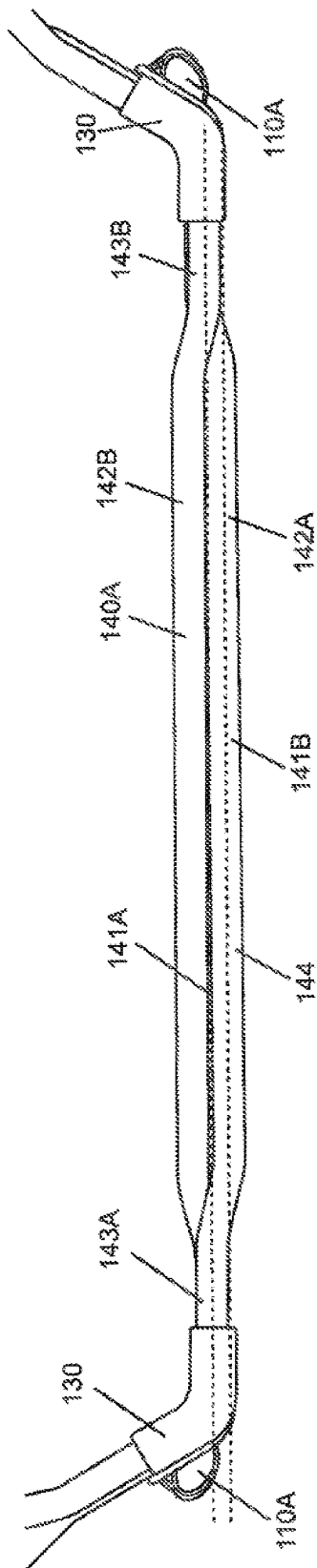


FIG. 3B

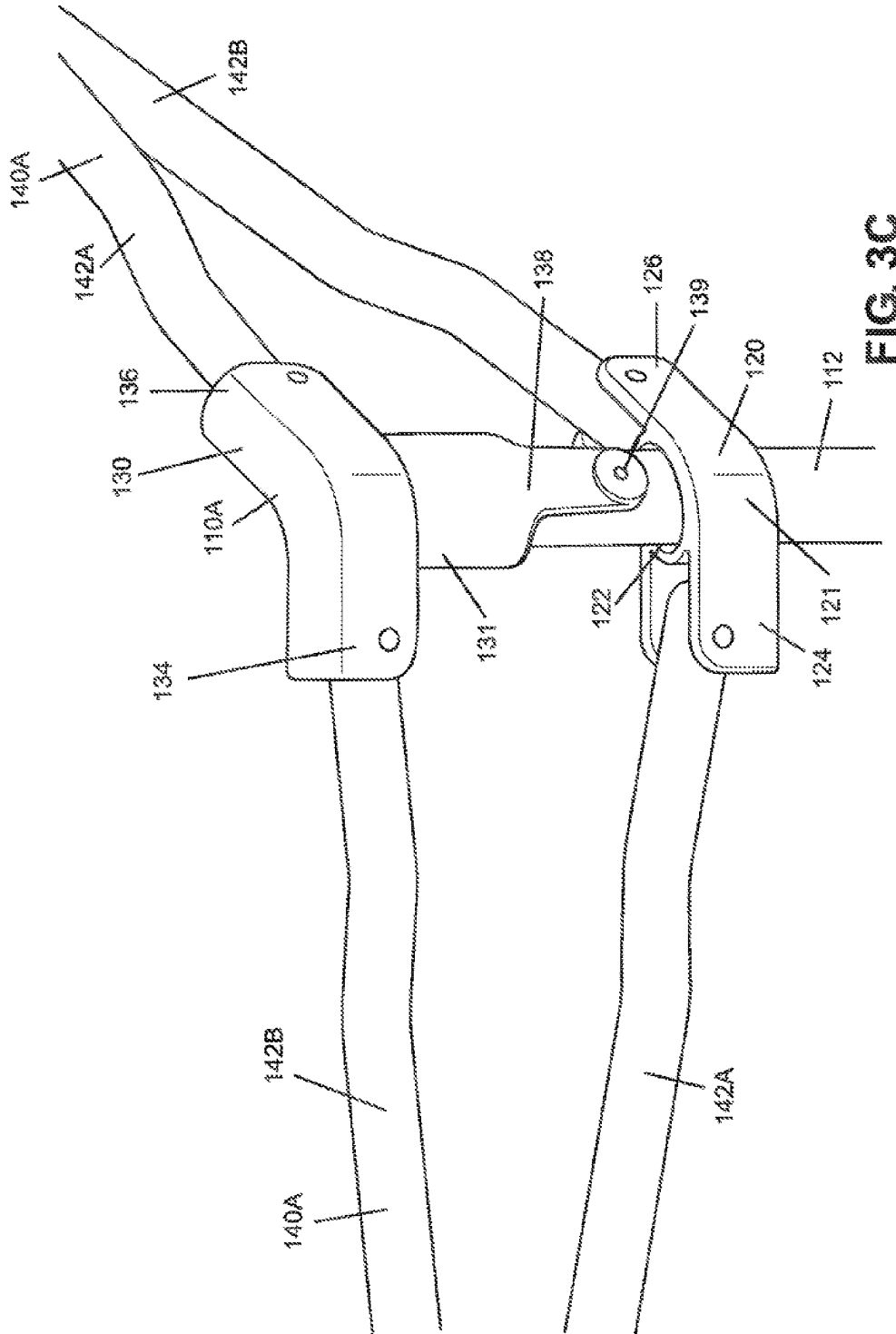


FIG. 3C

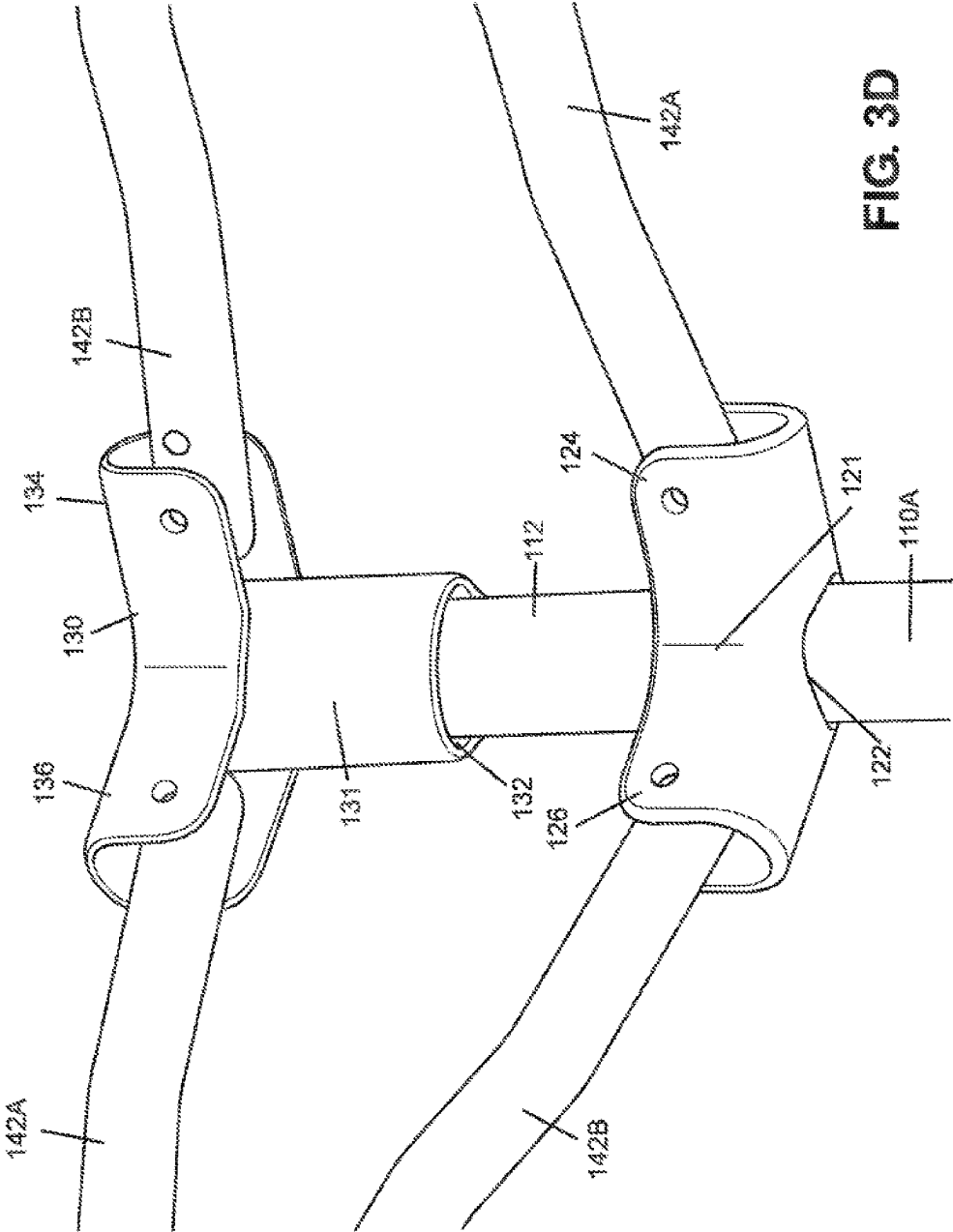


FIG. 3D

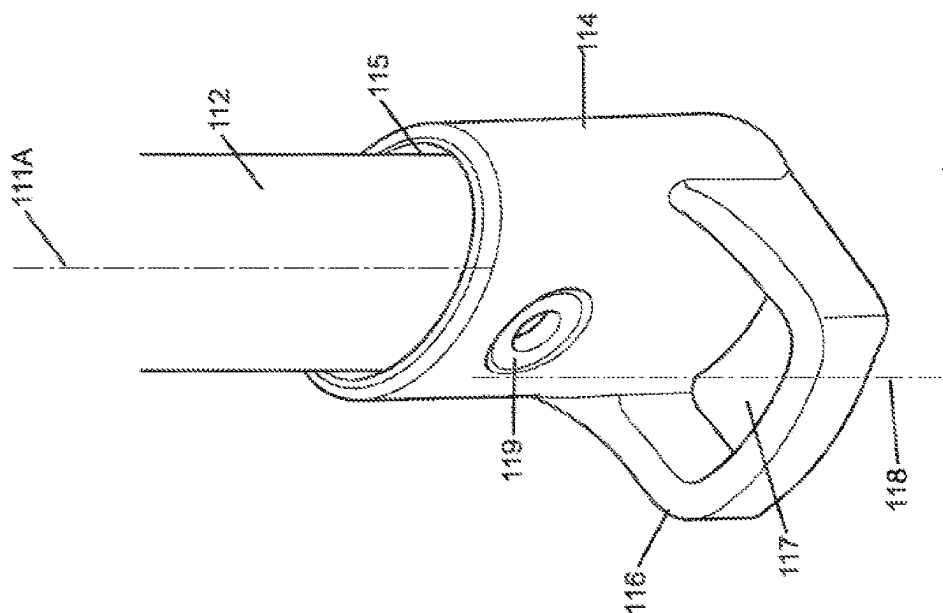


FIG. 3E

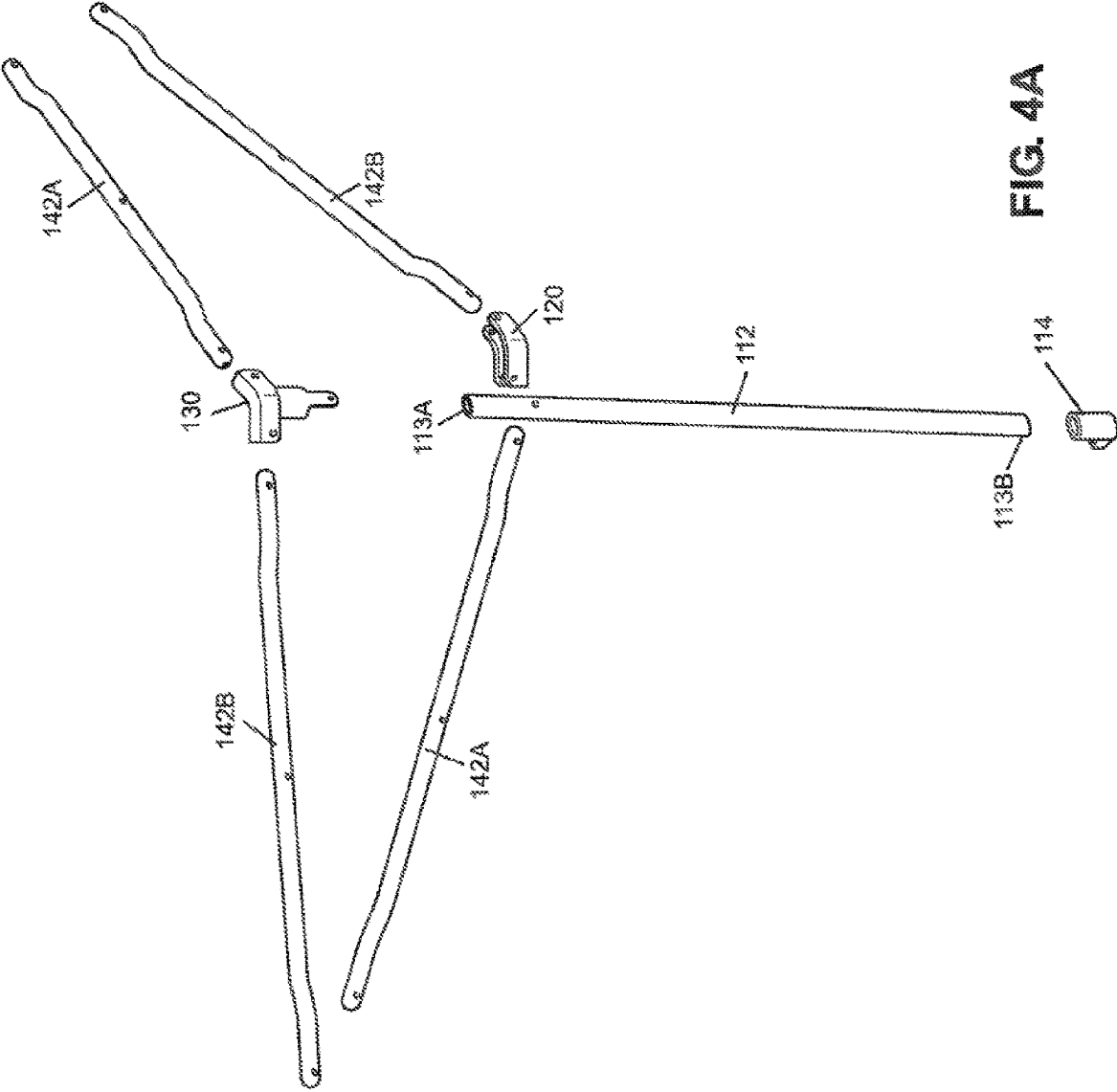


FIG. 4A

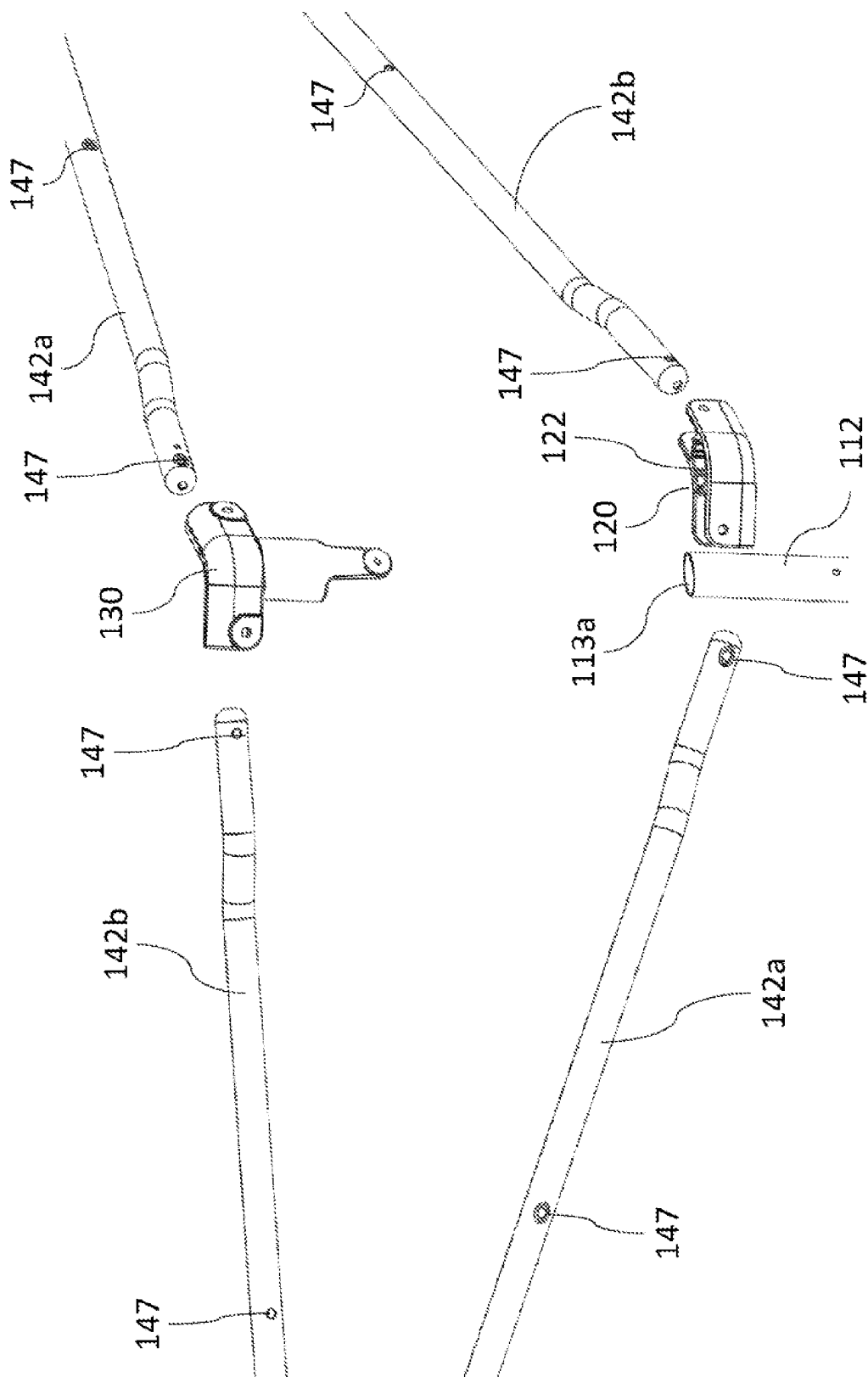


FIG. 4B

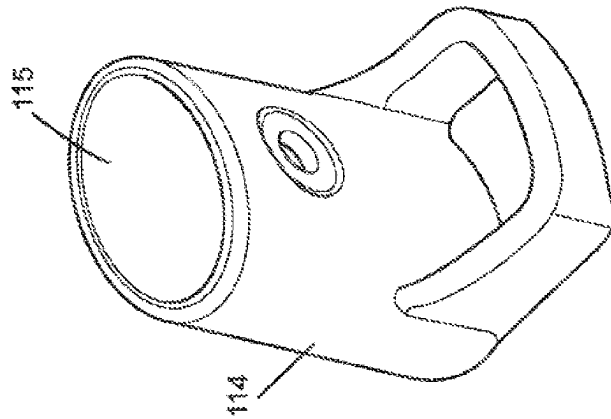
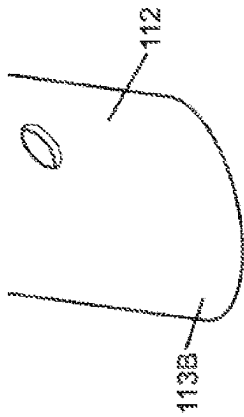
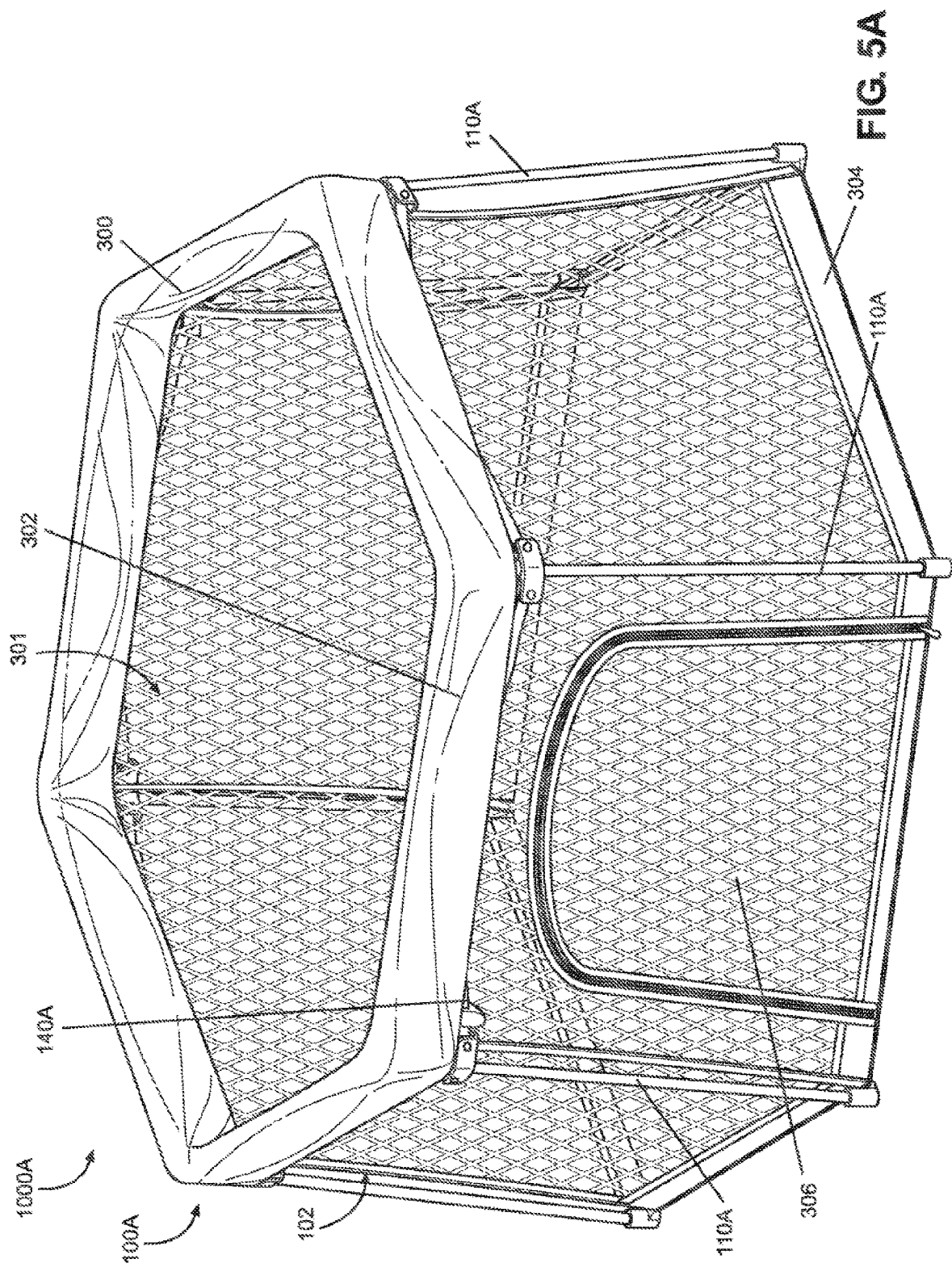


FIG. 4C



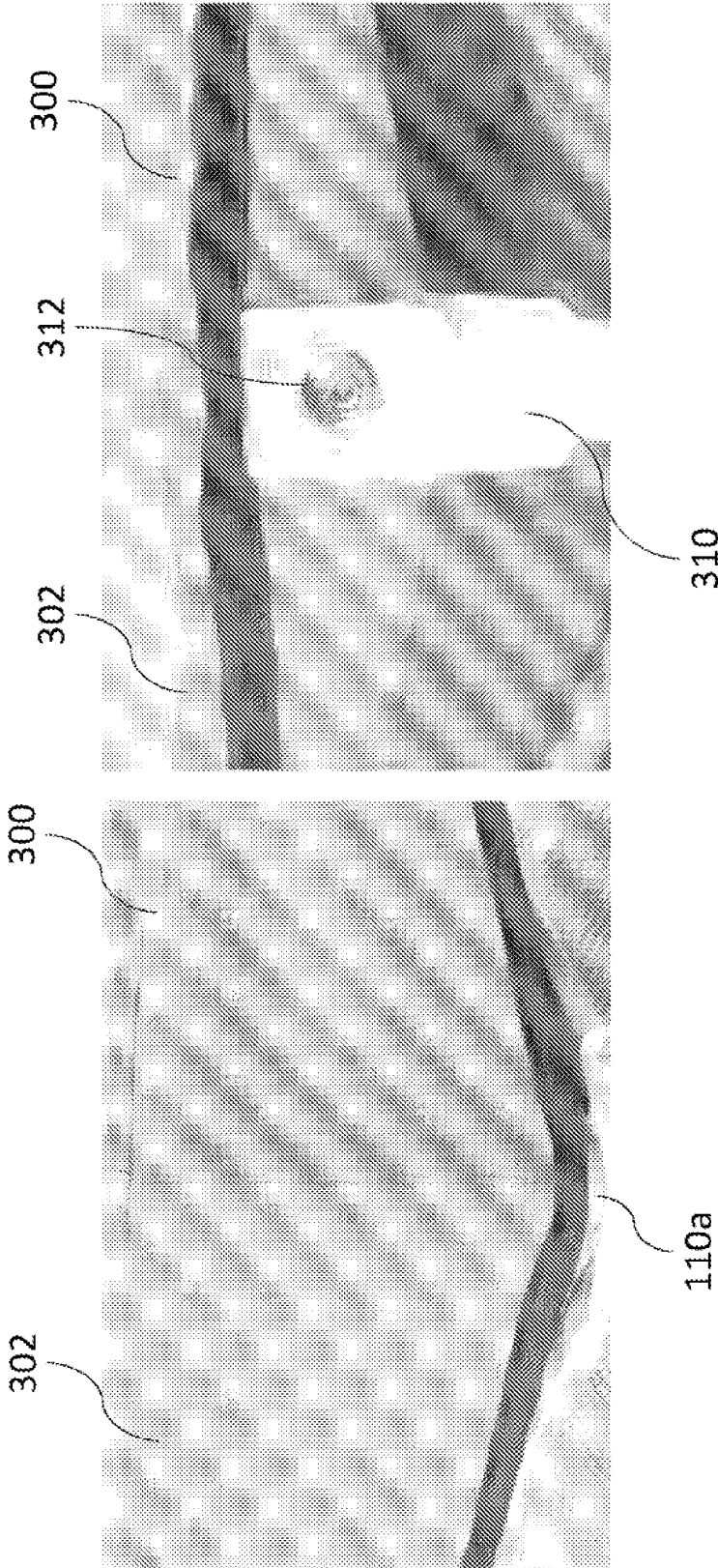


FIG. 5C

FIG. 5B

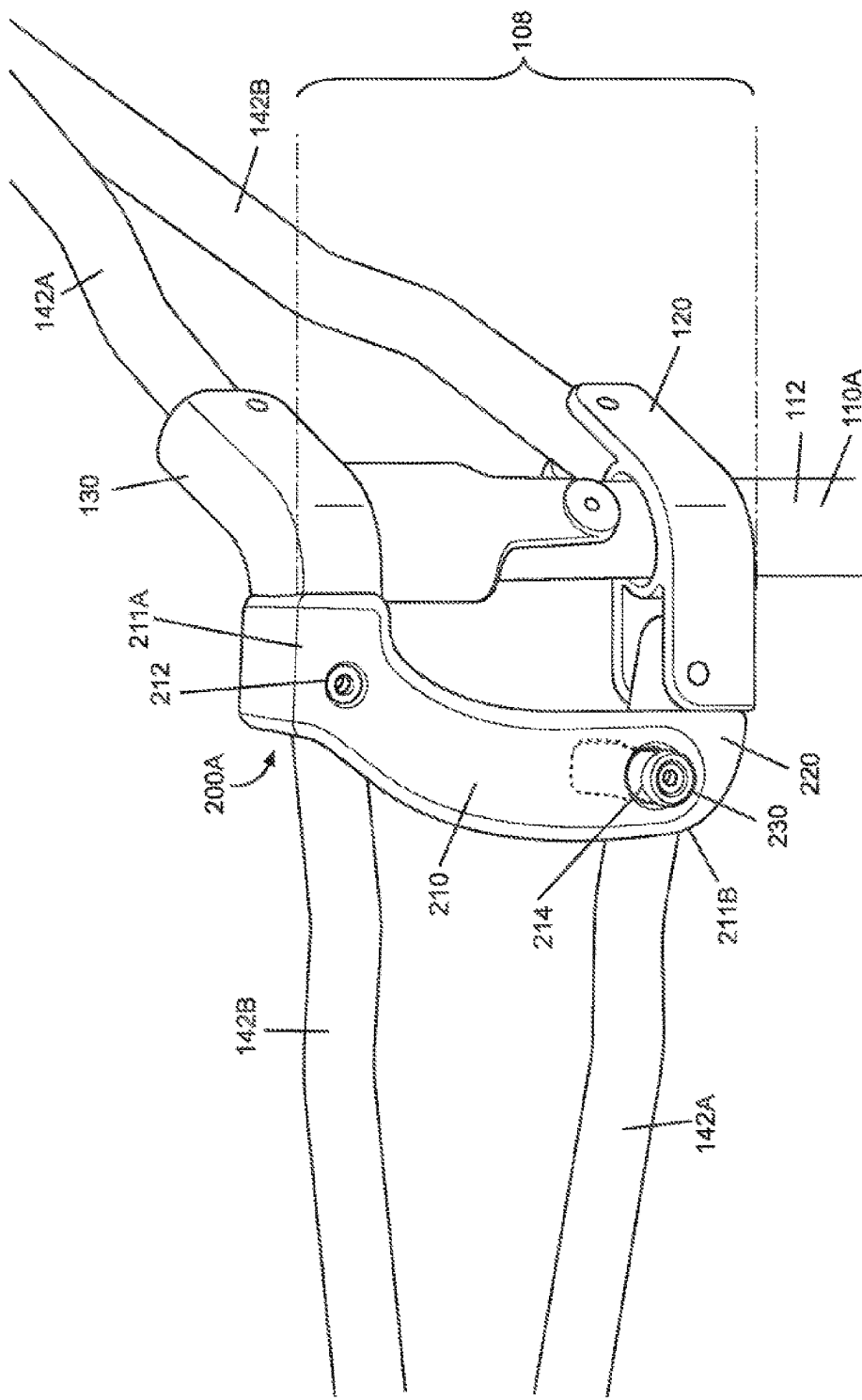


FIG. 6A

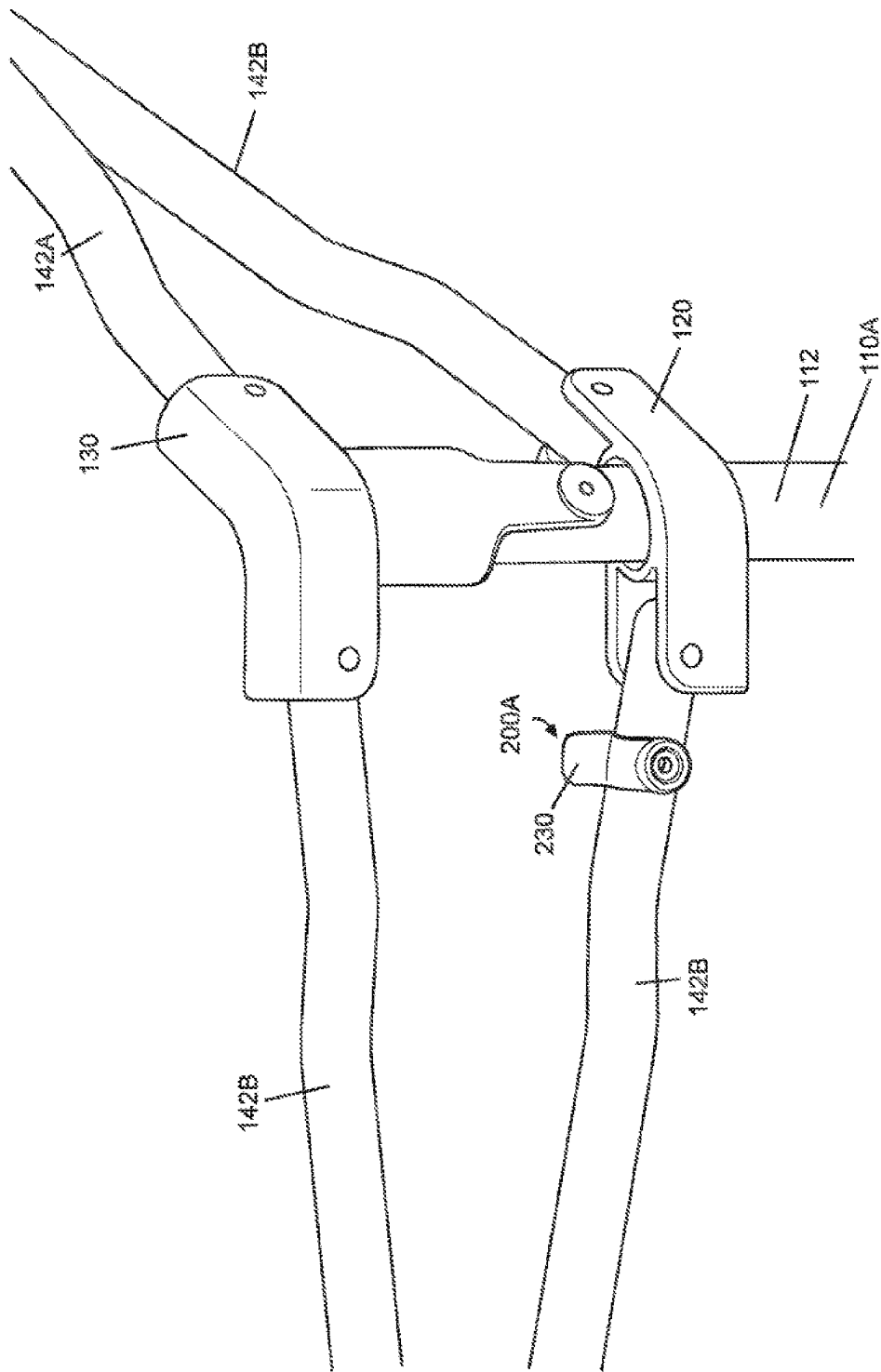


FIG. 6B

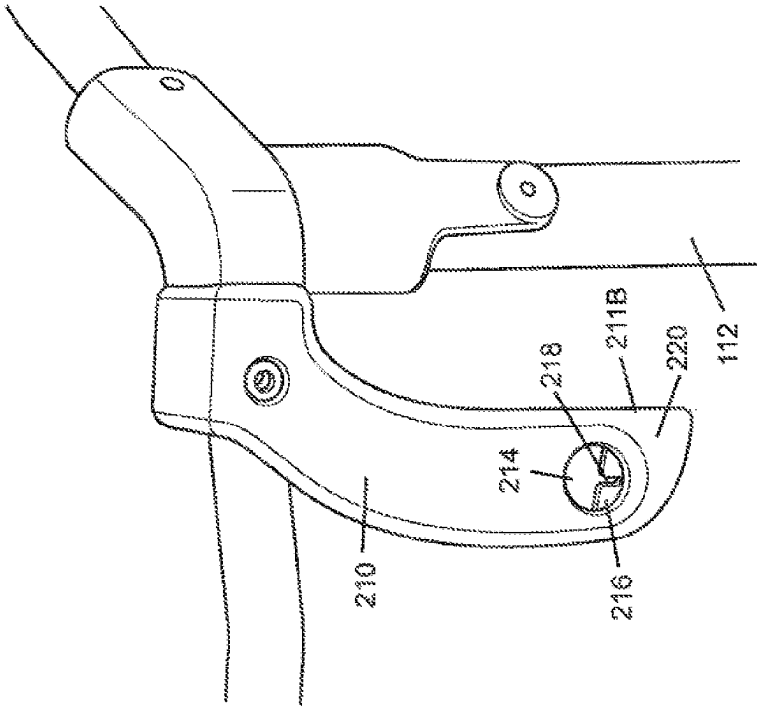


FIG. 6C

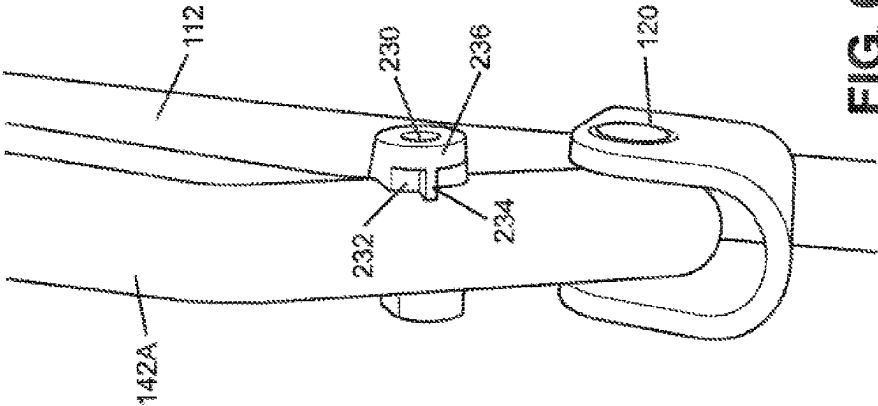


FIG. 6D

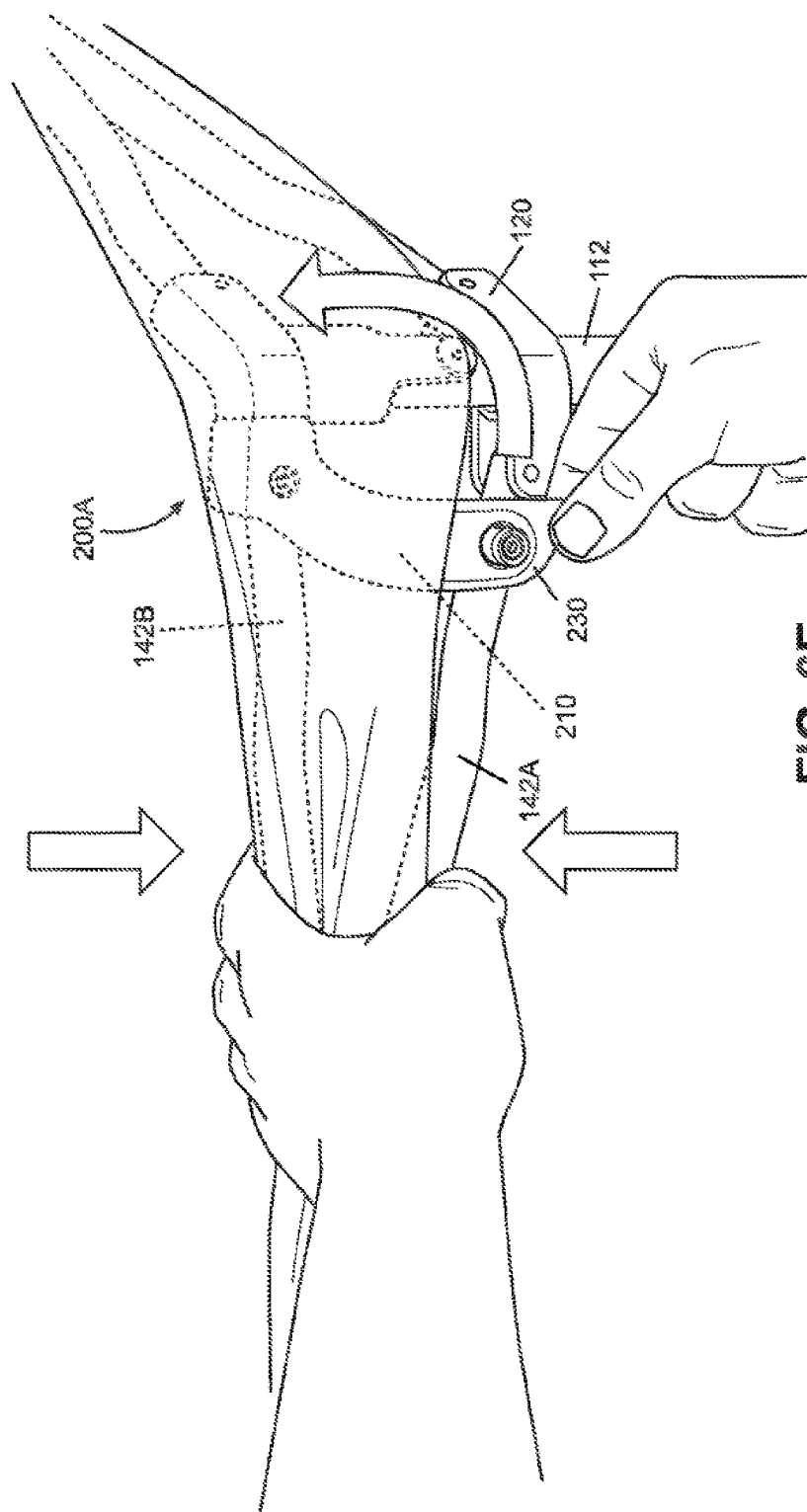


FIG. 6E

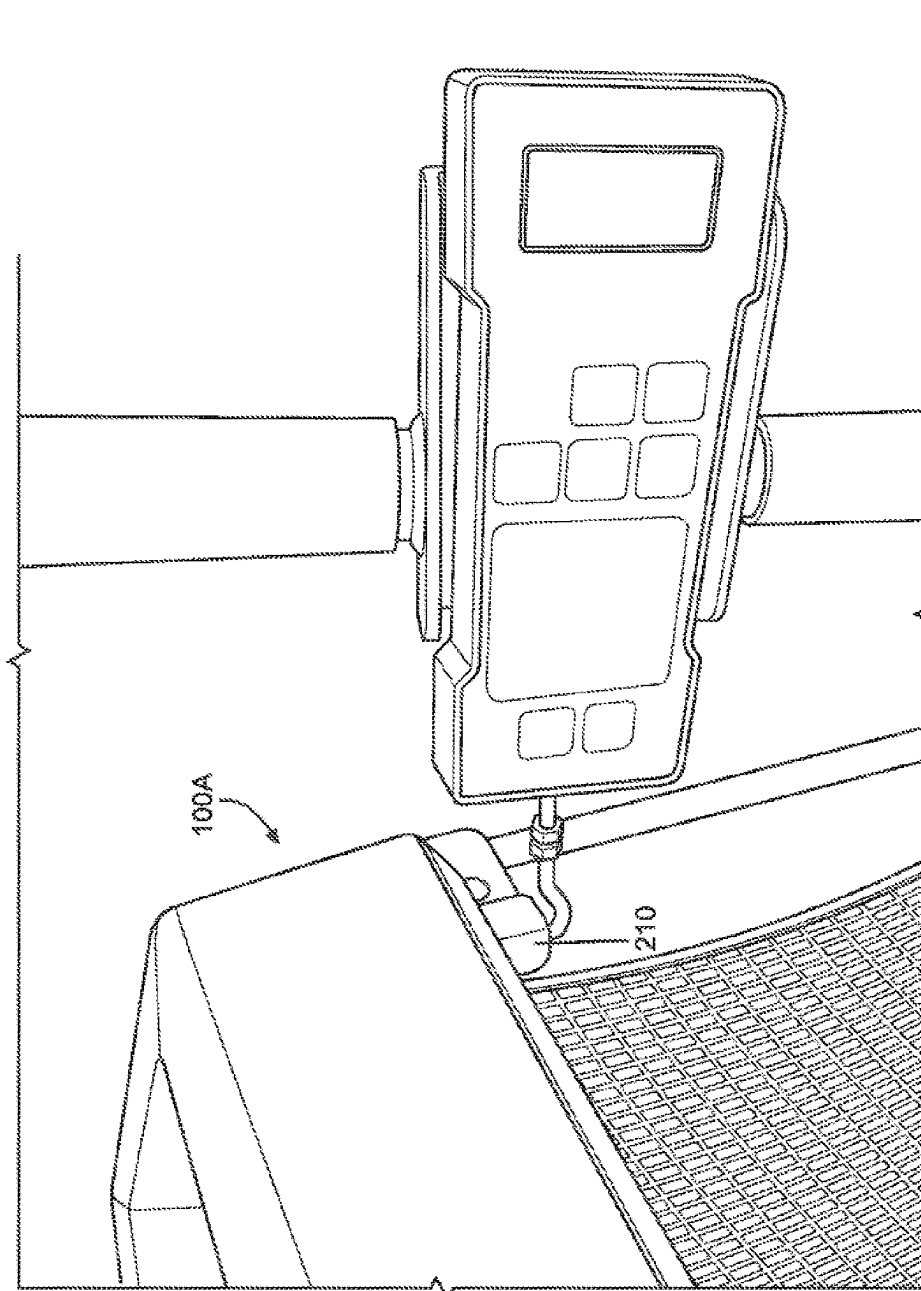
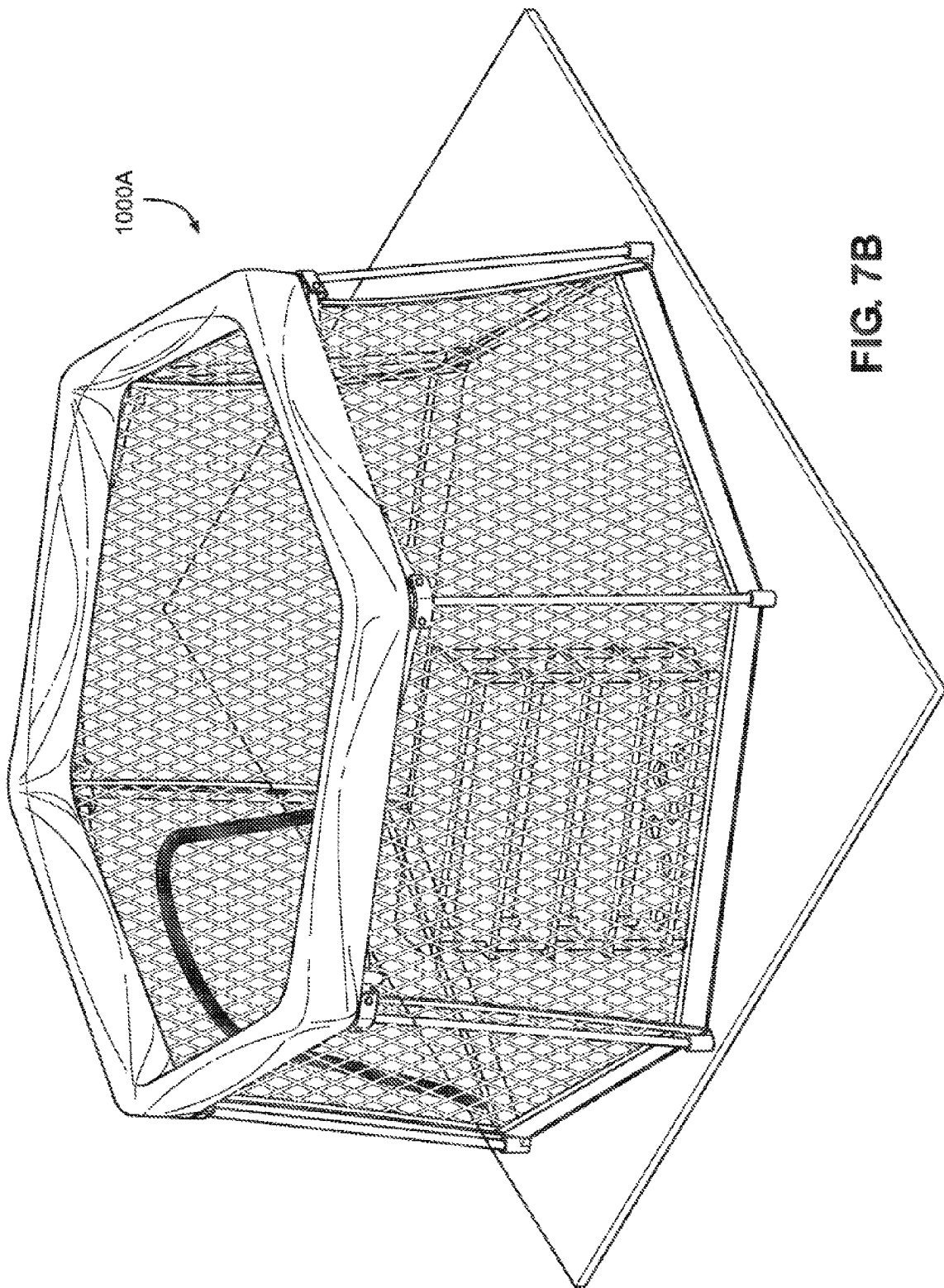
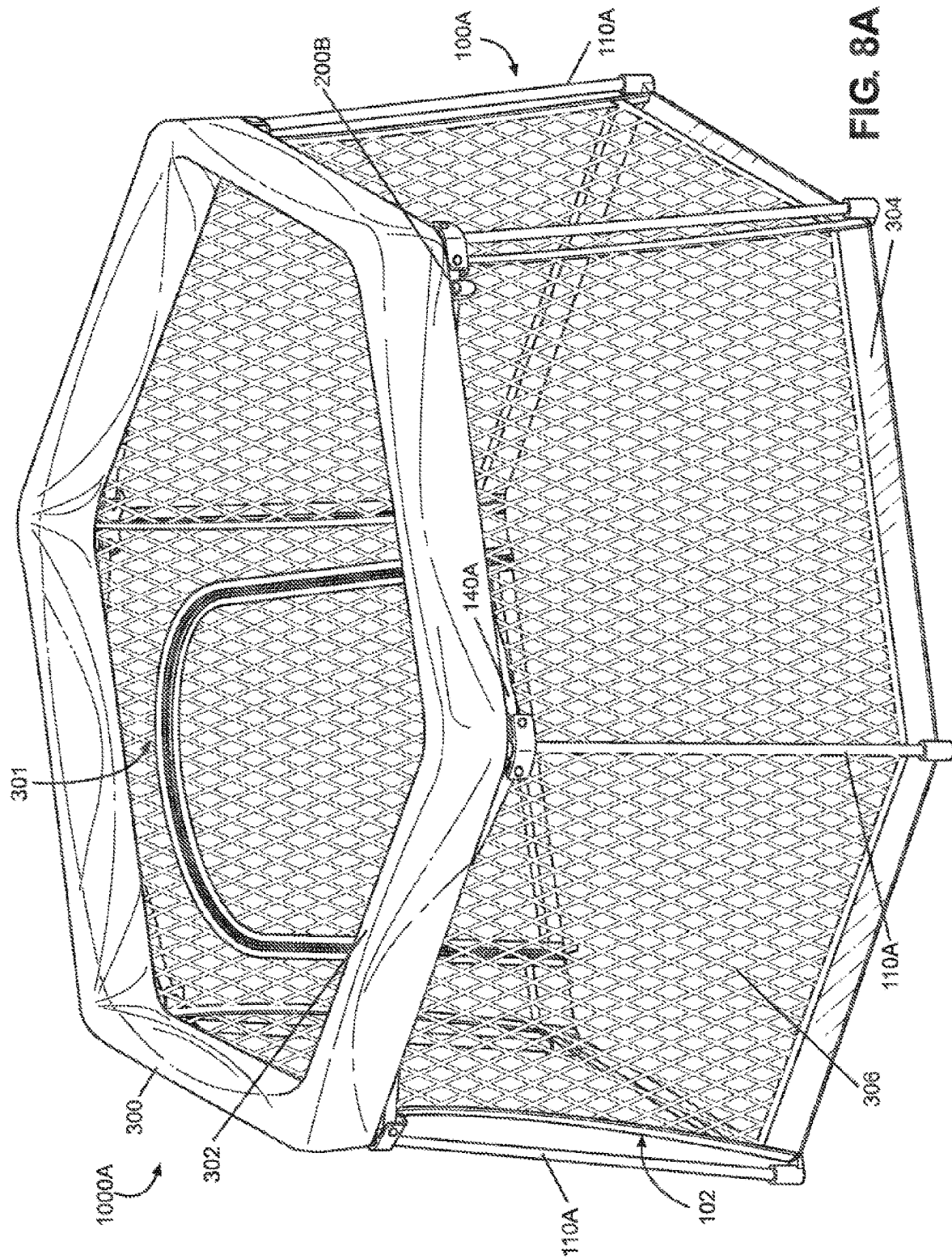


FIG. 7A





AGILE

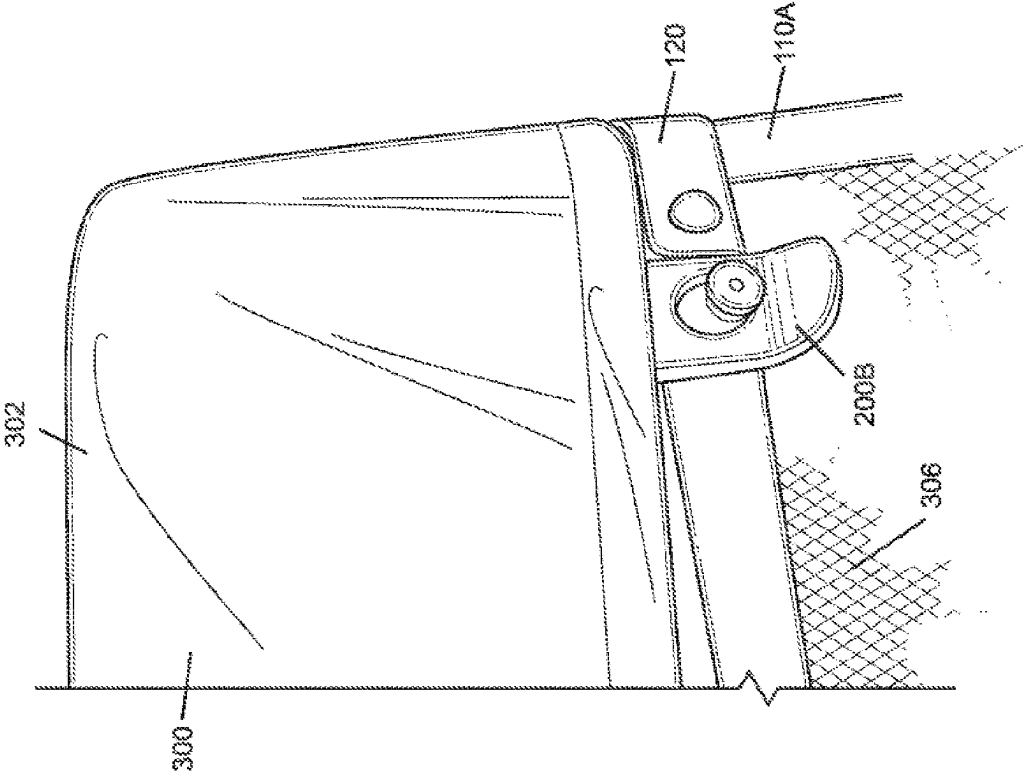


FIG. 8B

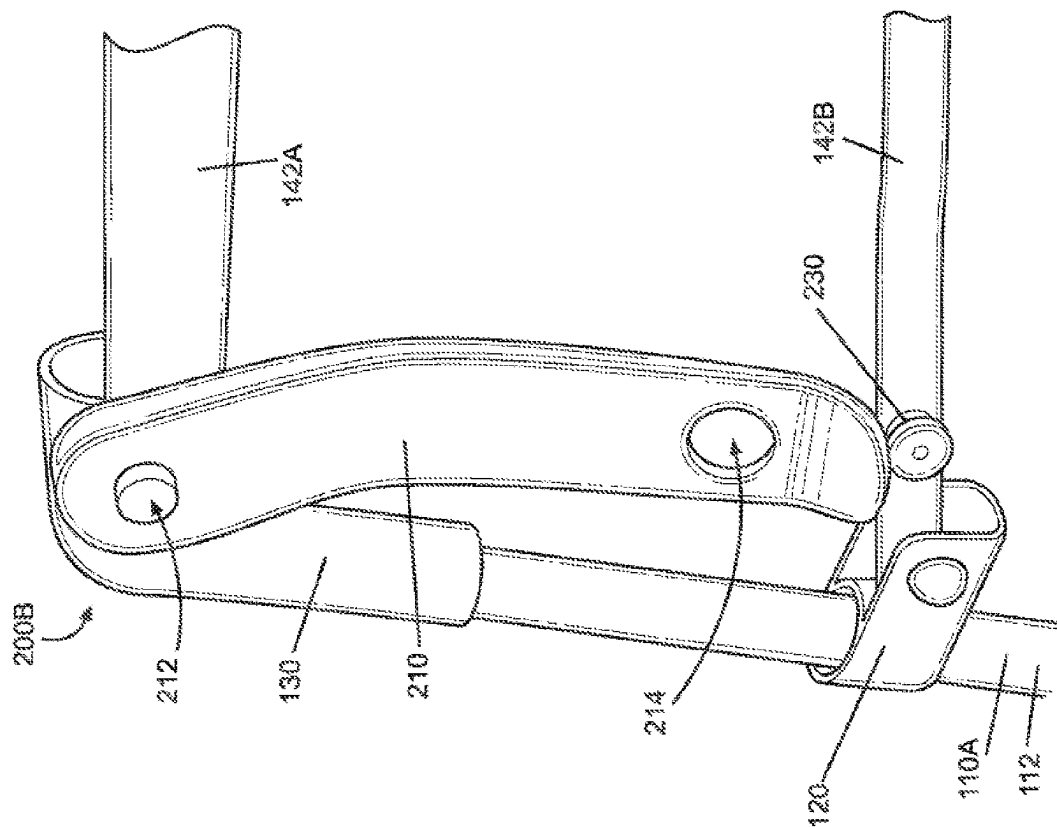


FIG. 8D

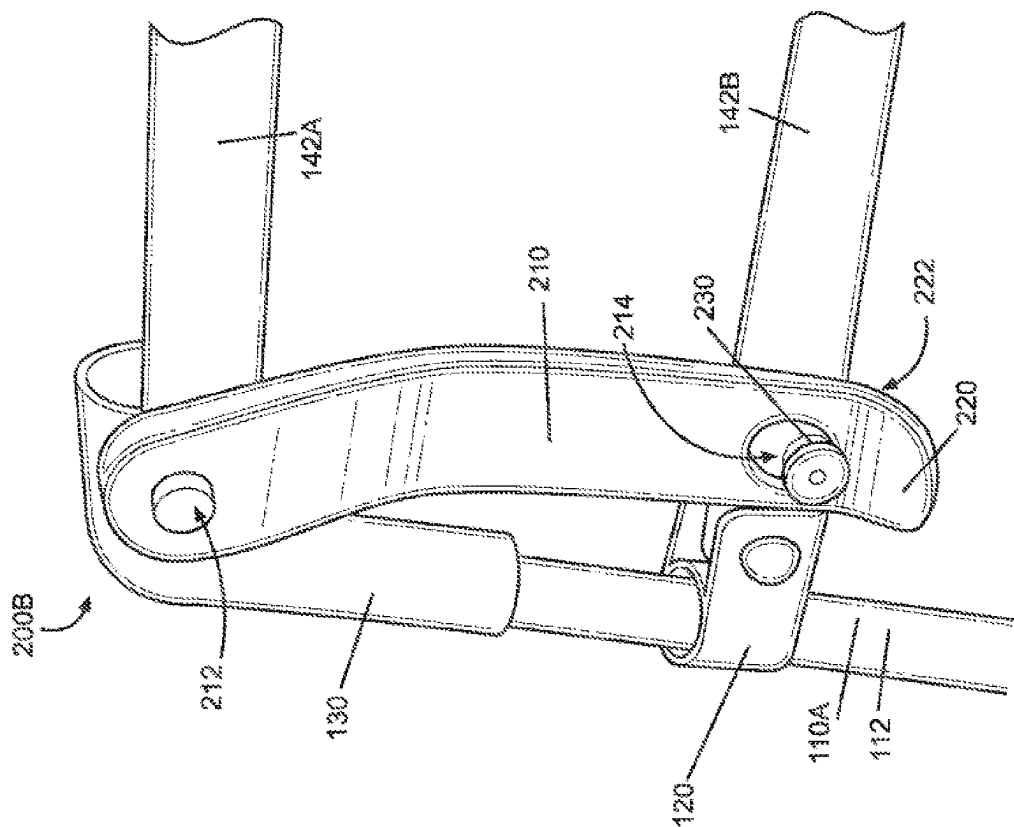


FIG. 8C

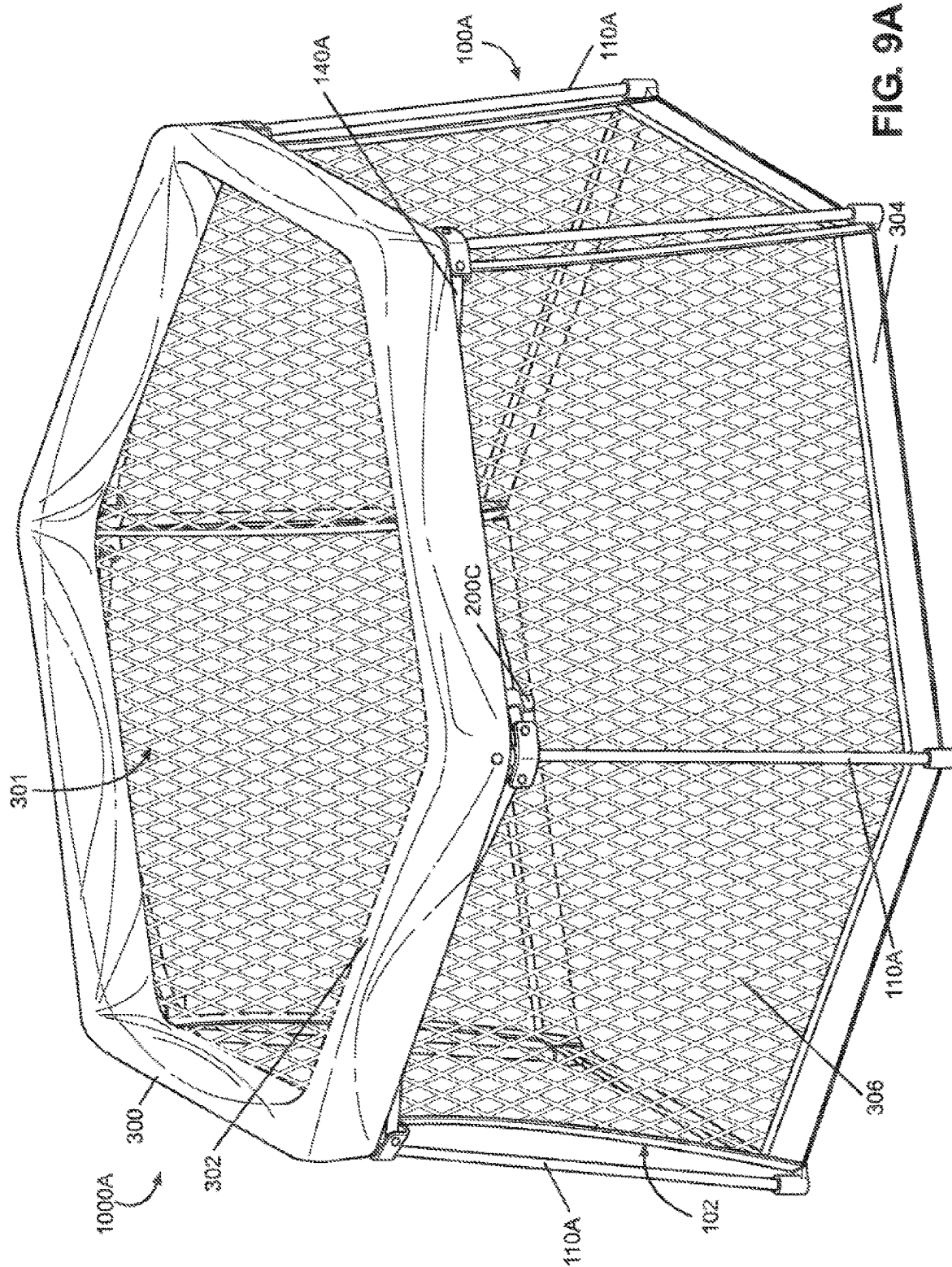


FIG. 9A

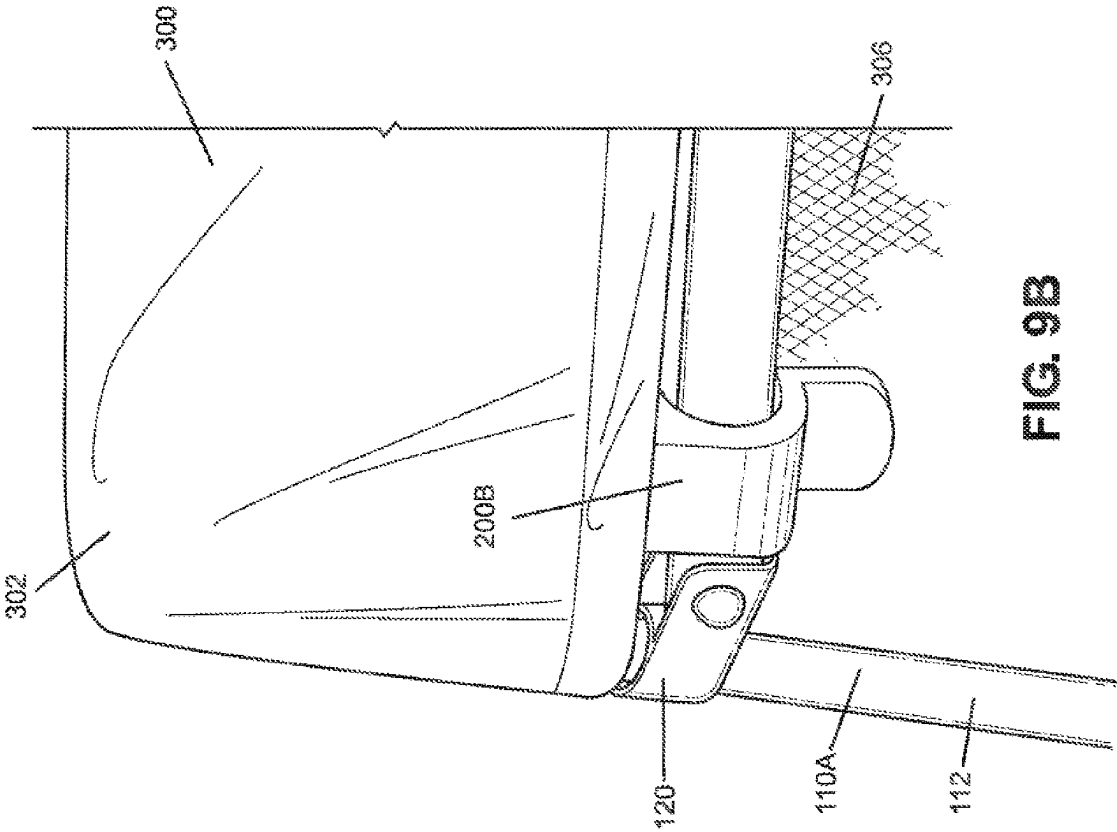


FIG. 9B

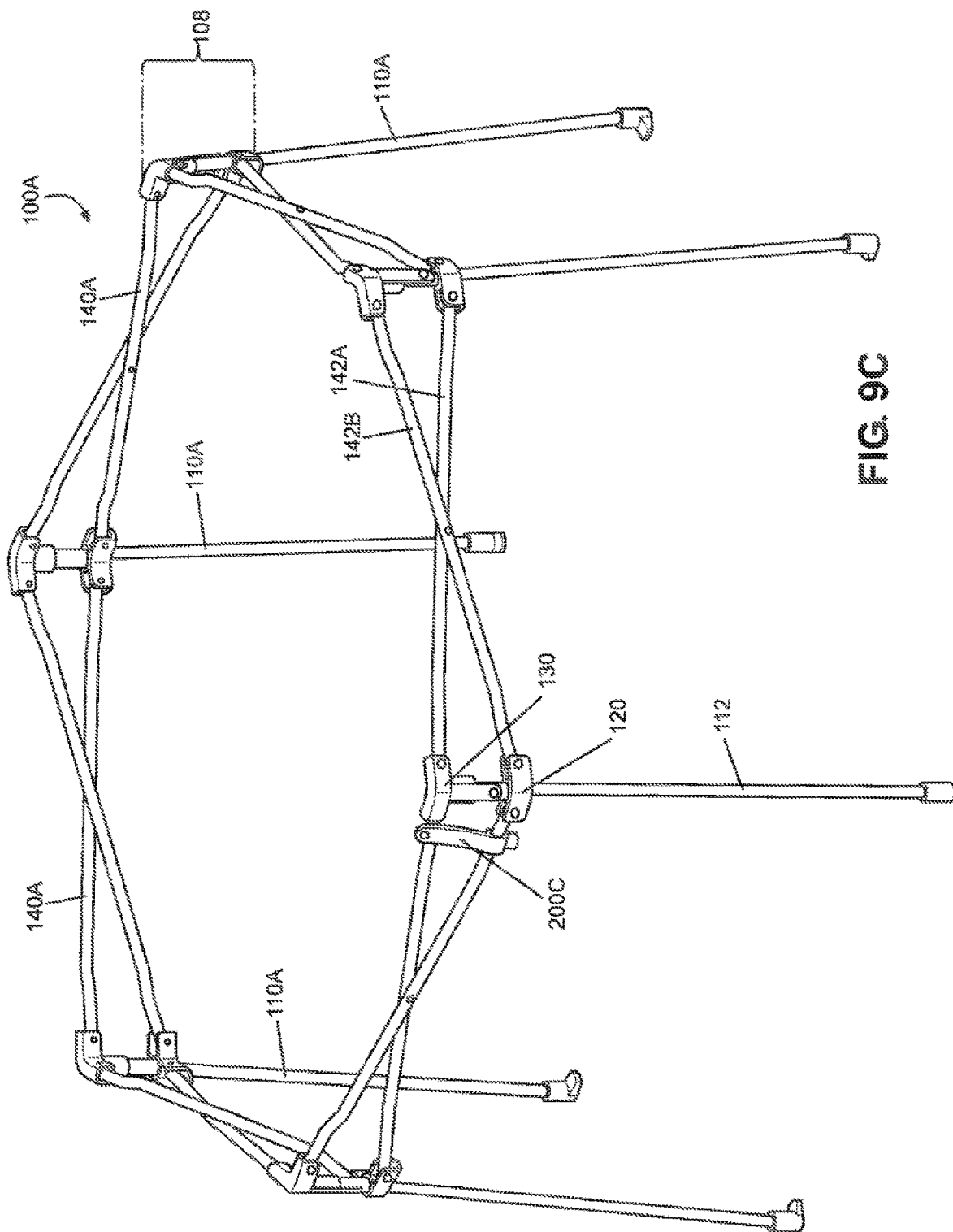


FIG. 9C

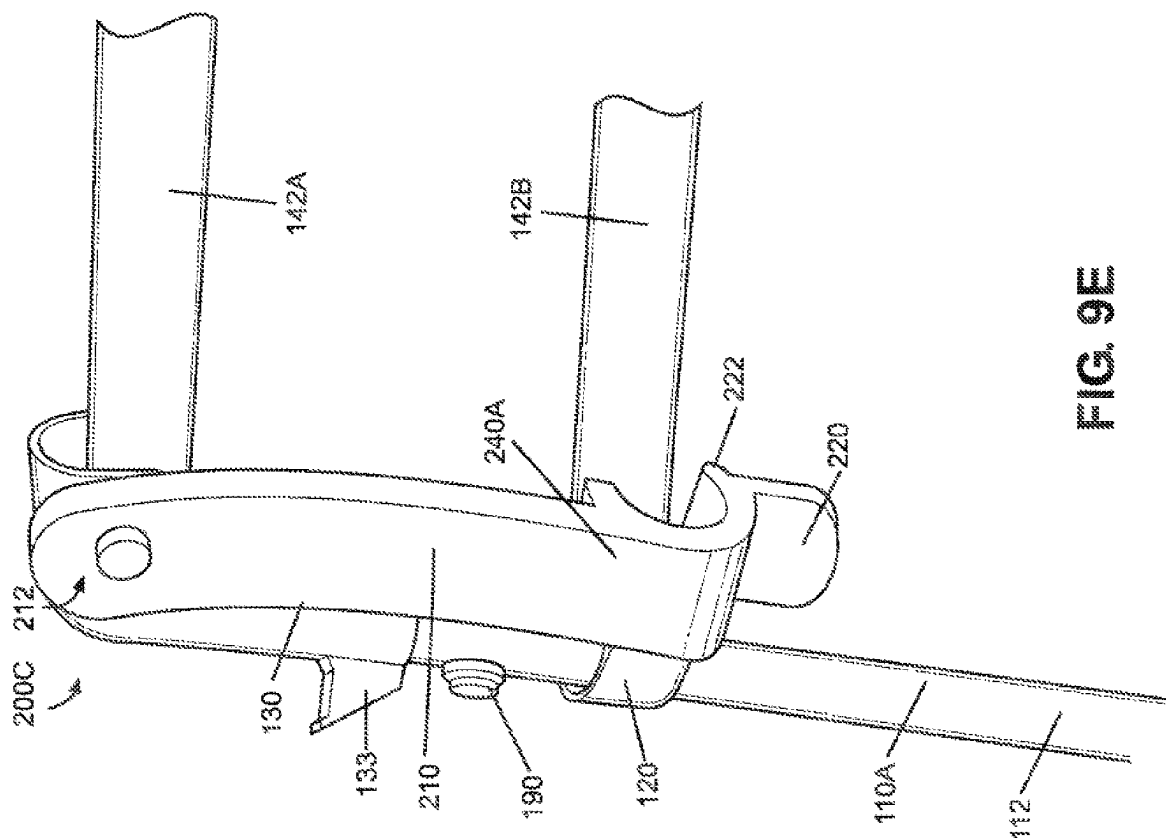


FIG. 9E

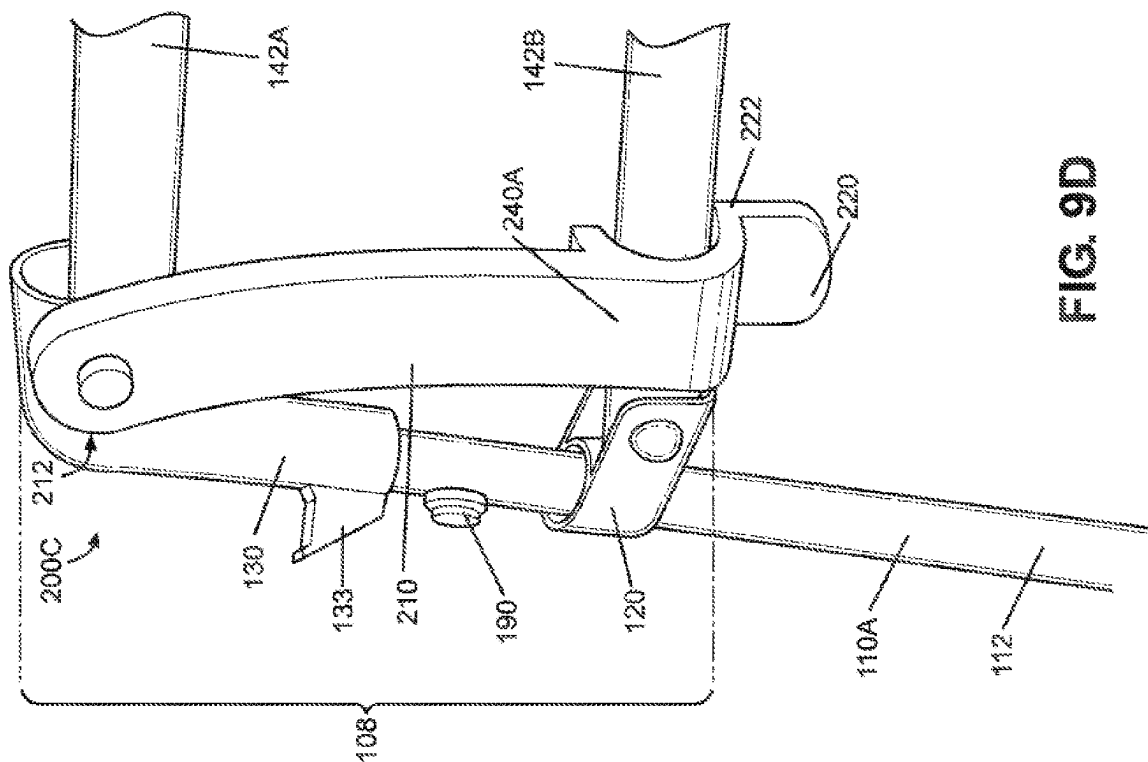


FIG. 9D

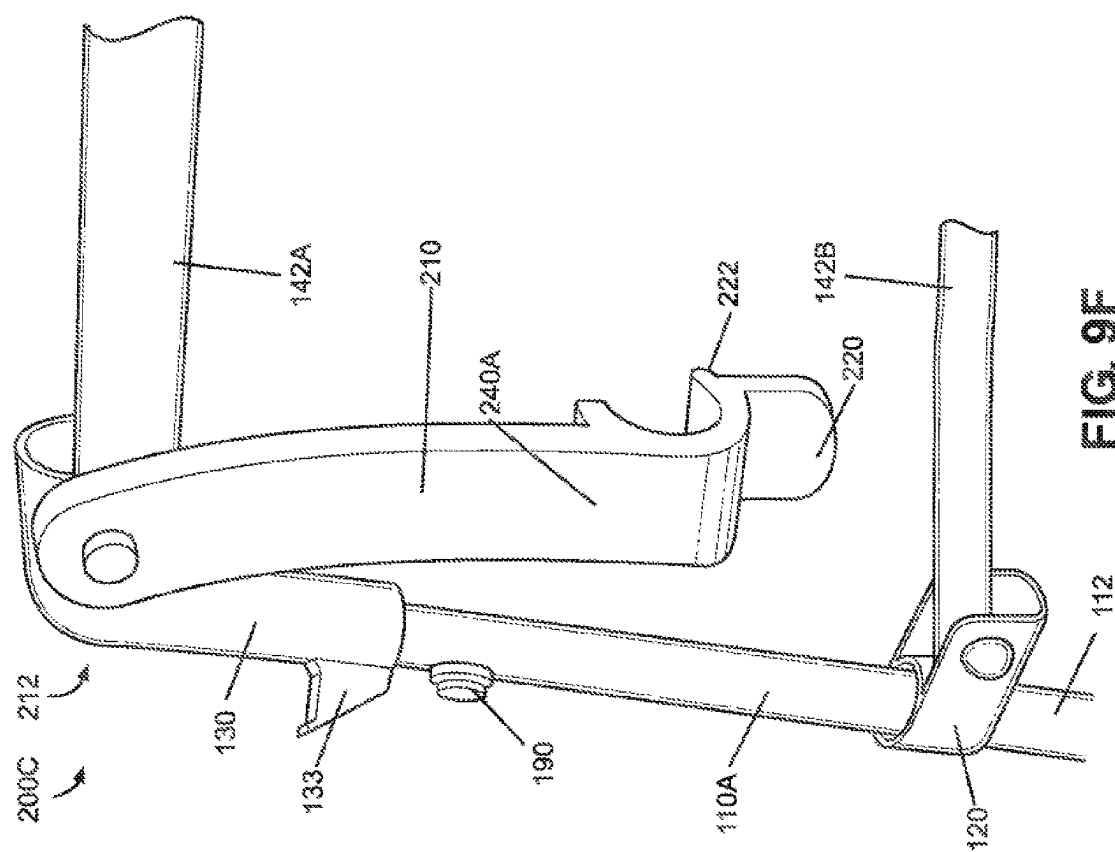


FIG. 9F

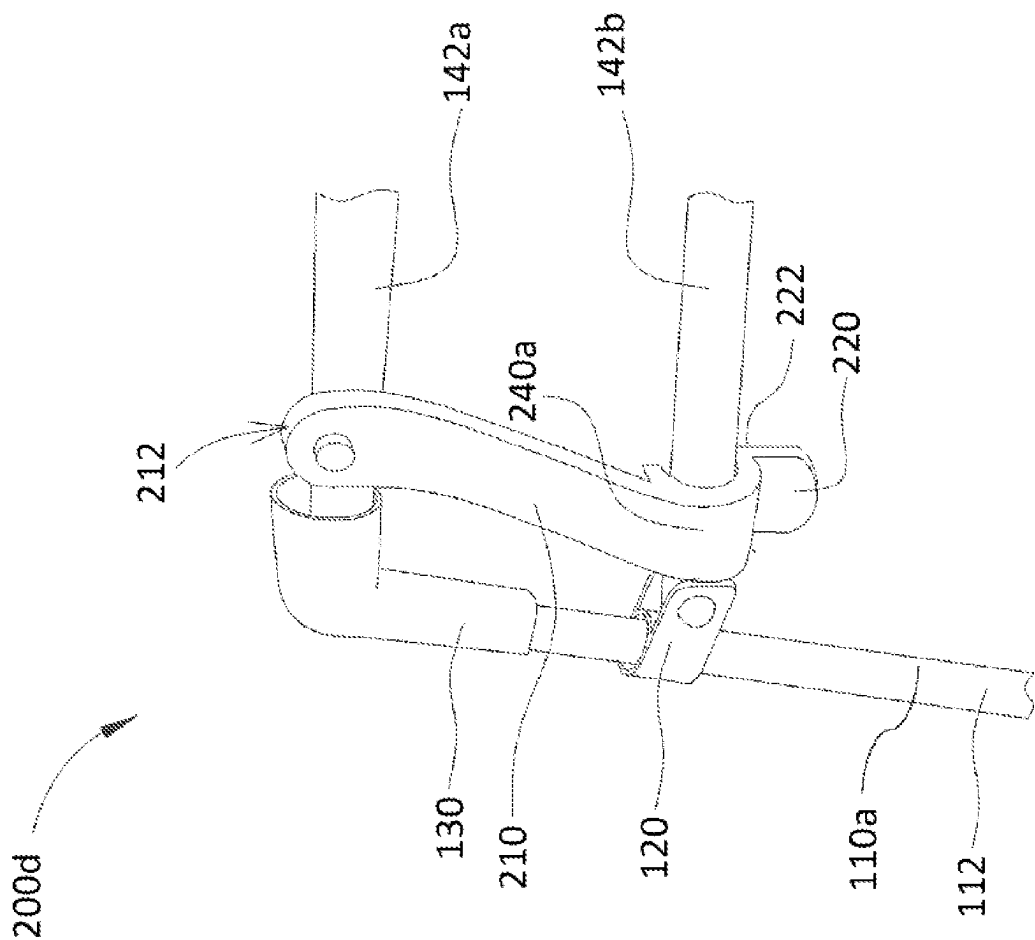
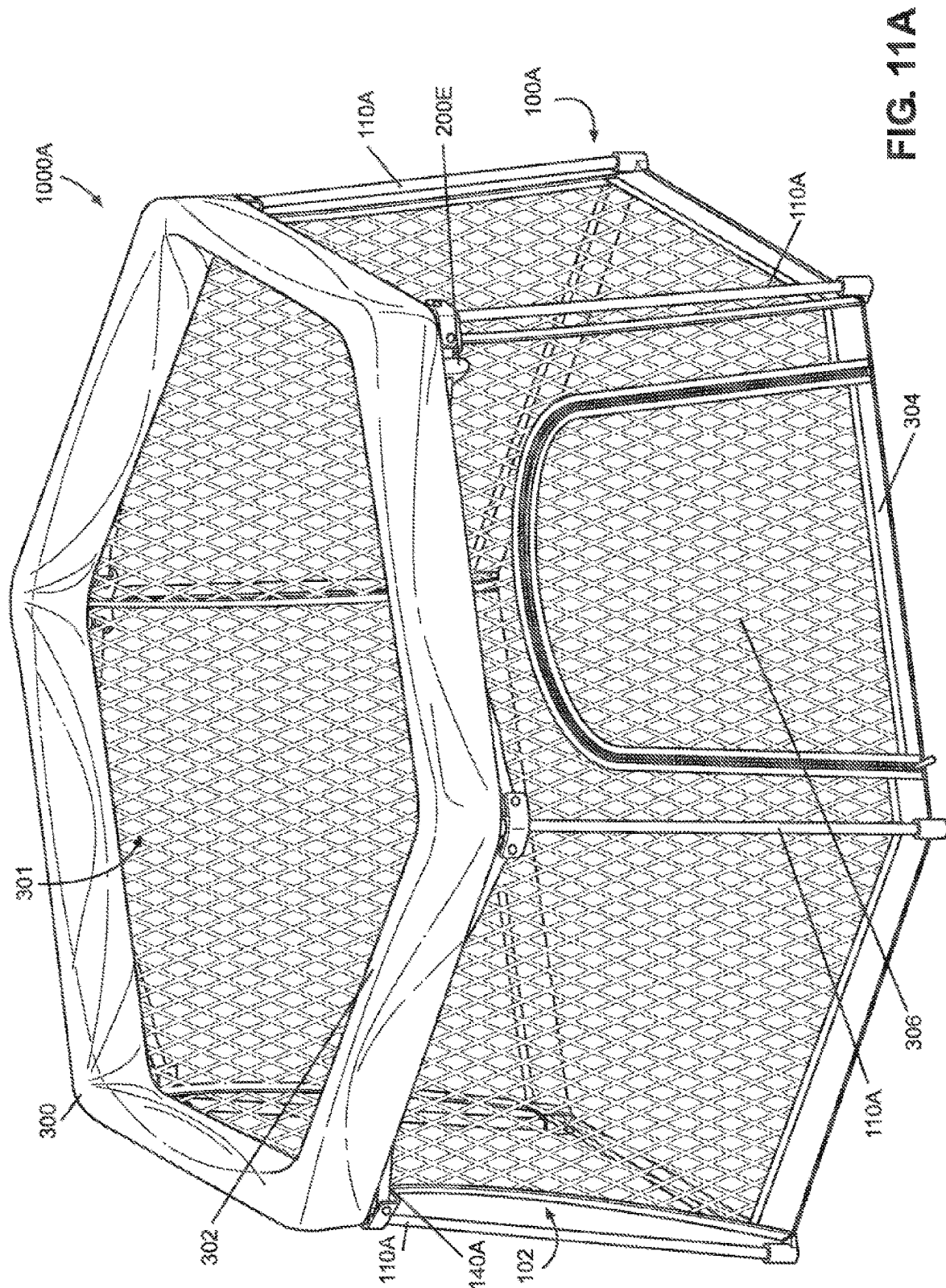


FIG. 10



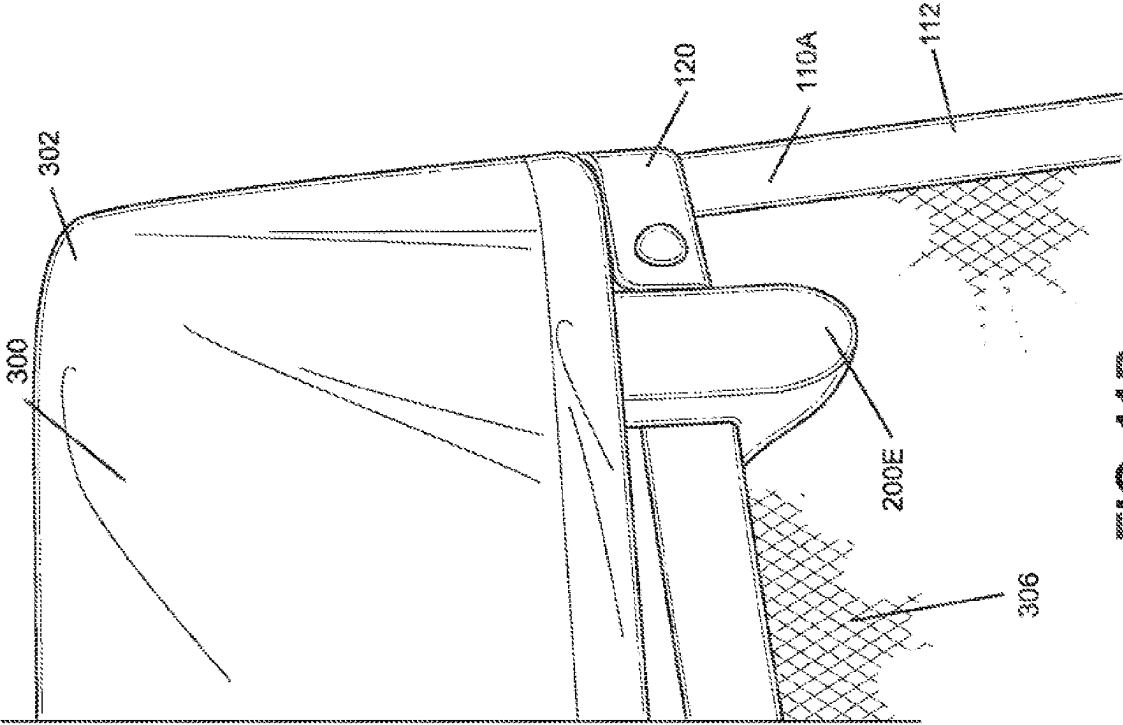


FIG. 11B

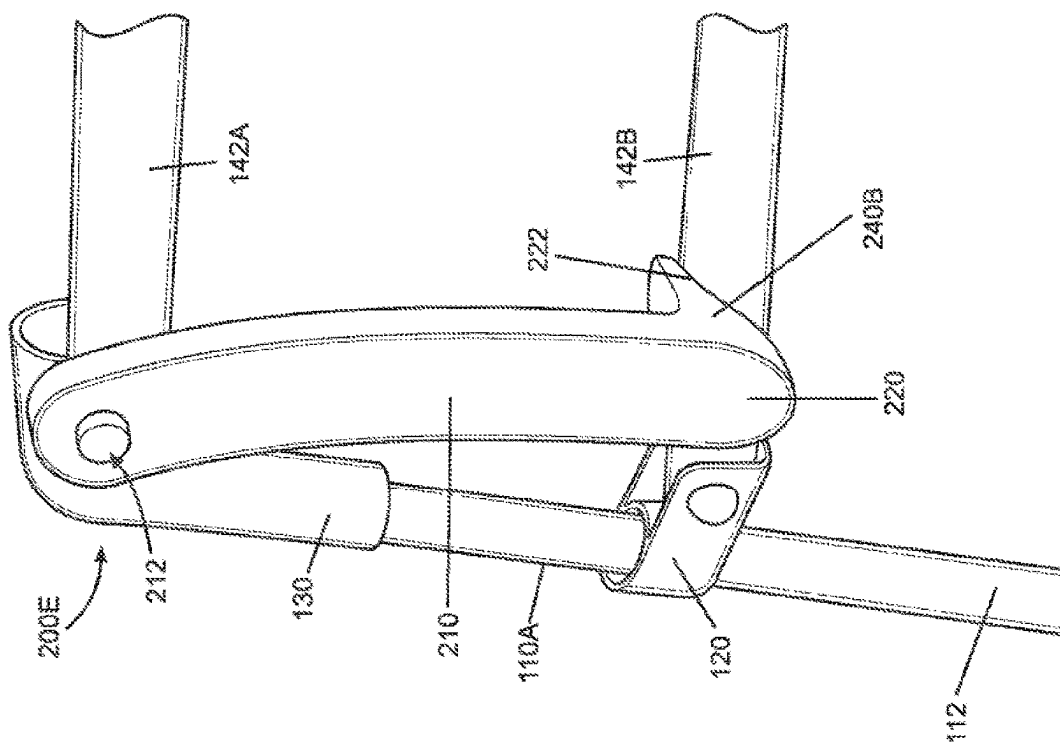


FIG. 11D

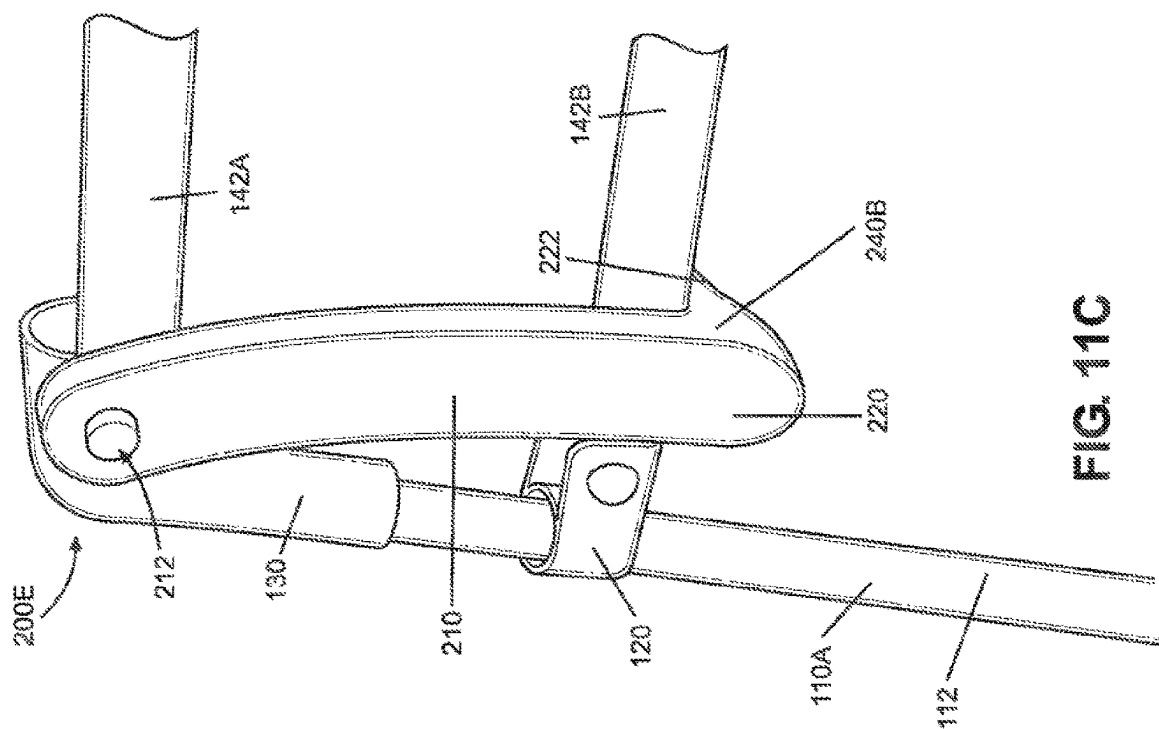


FIG. 11C

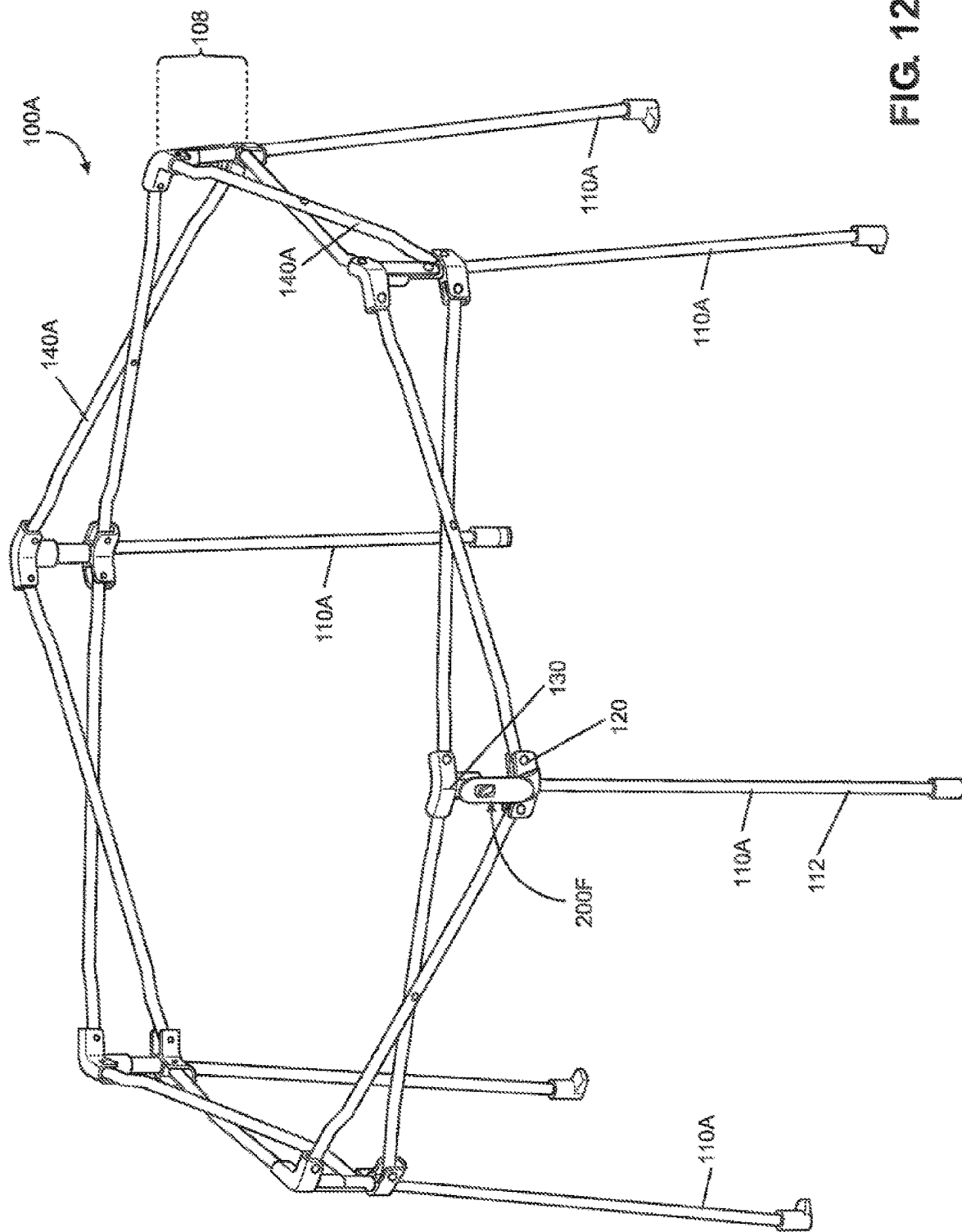


FIG. 12A

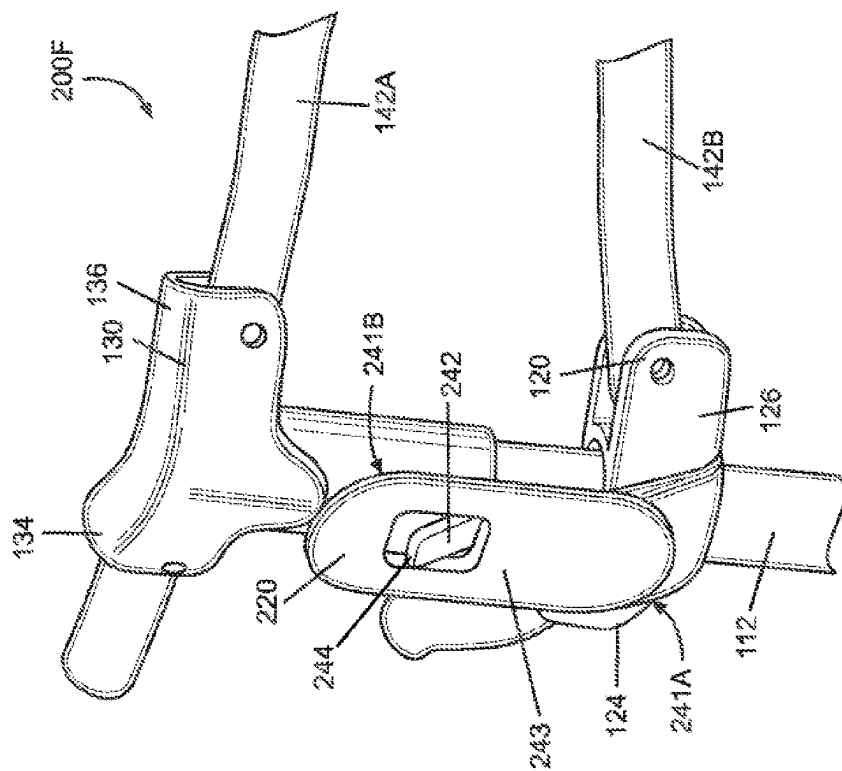


FIG. 12B

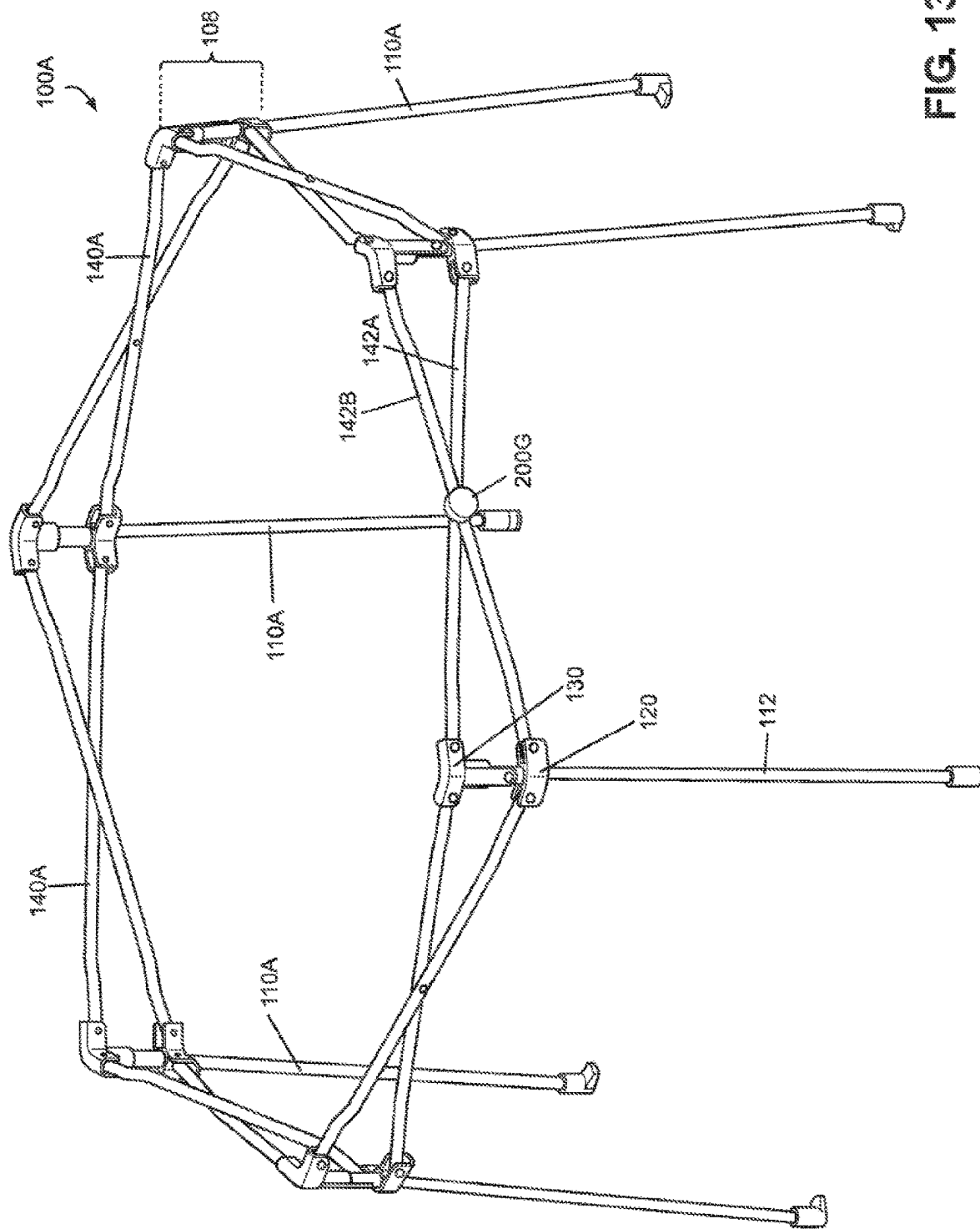


FIG. 13A

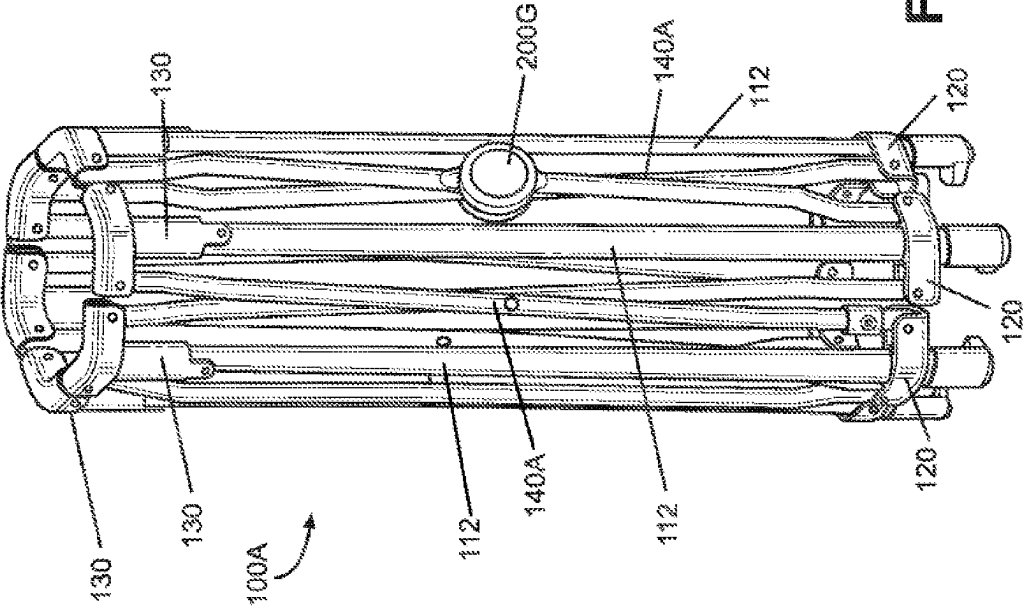


FIG. 13B

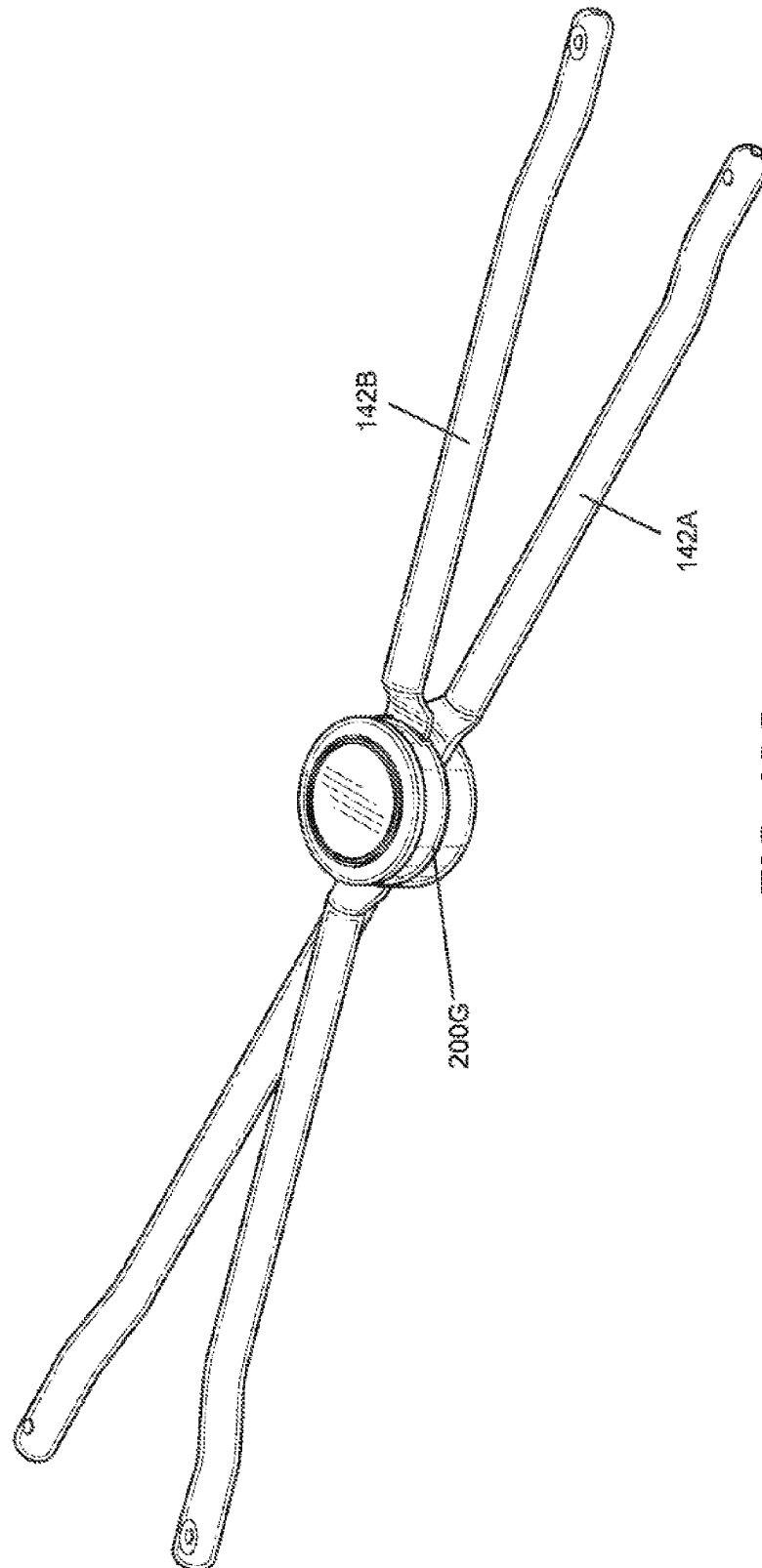
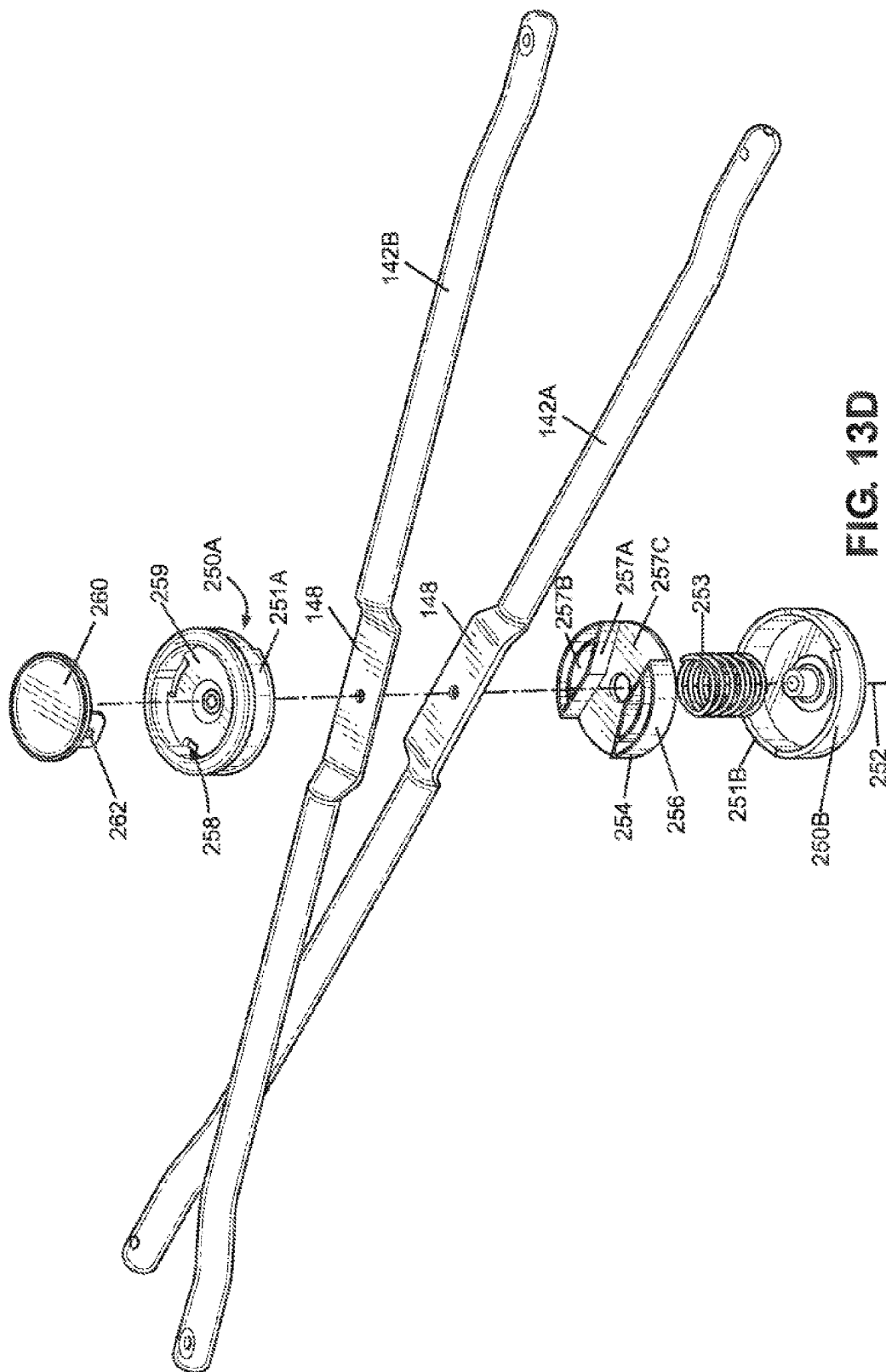


FIG. 13C



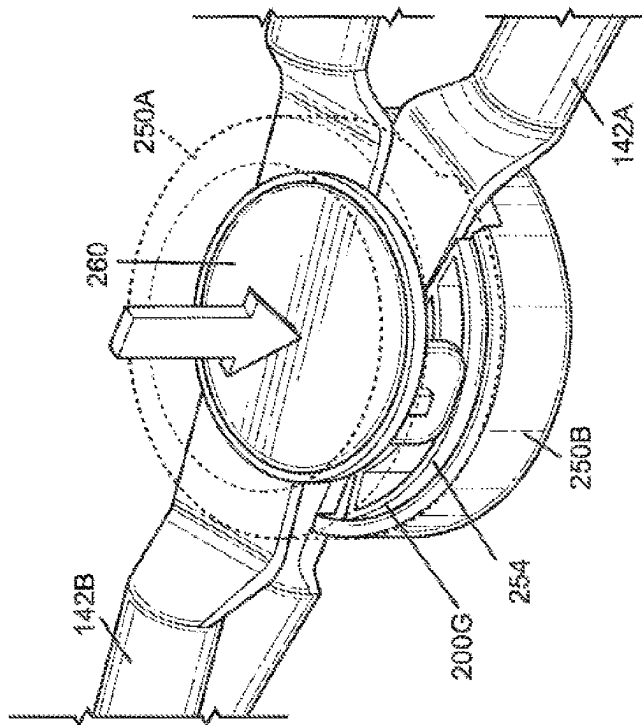


FIG. 13F

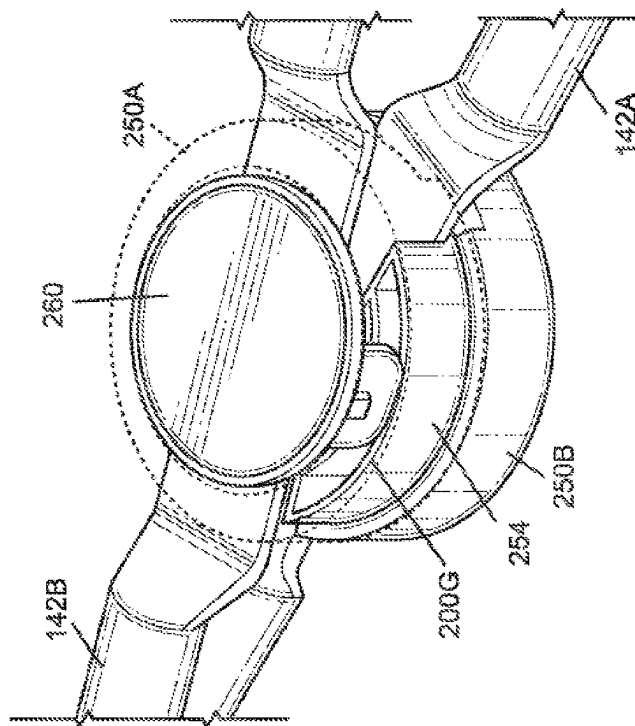


FIG. 13E

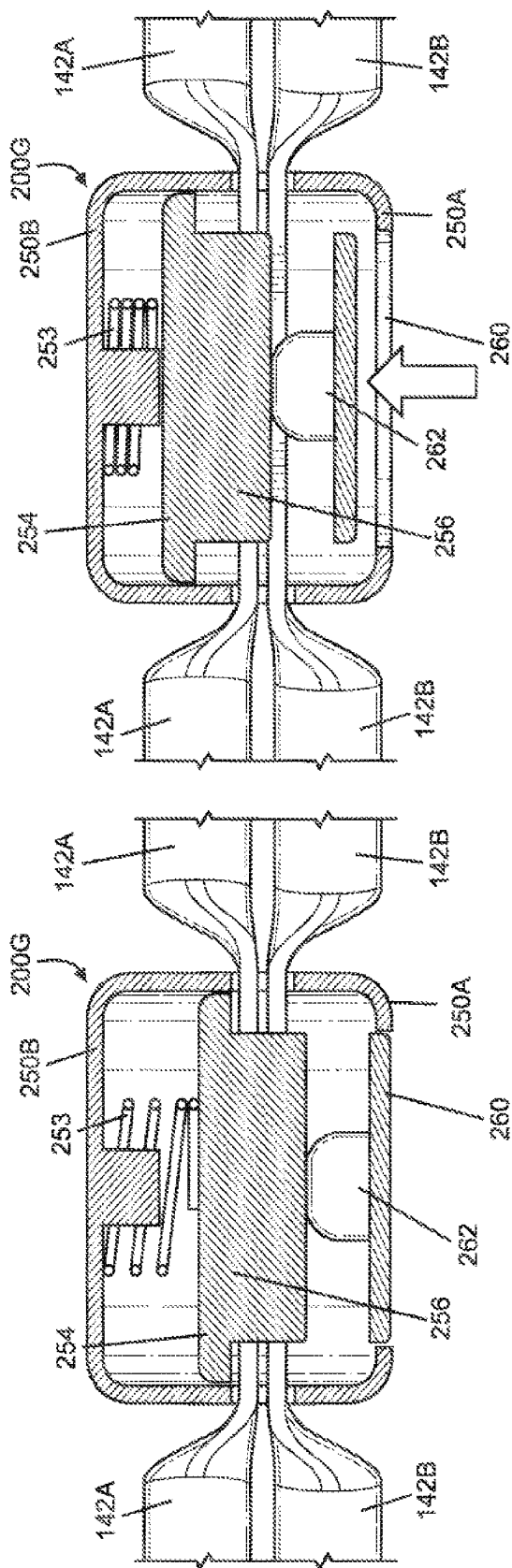
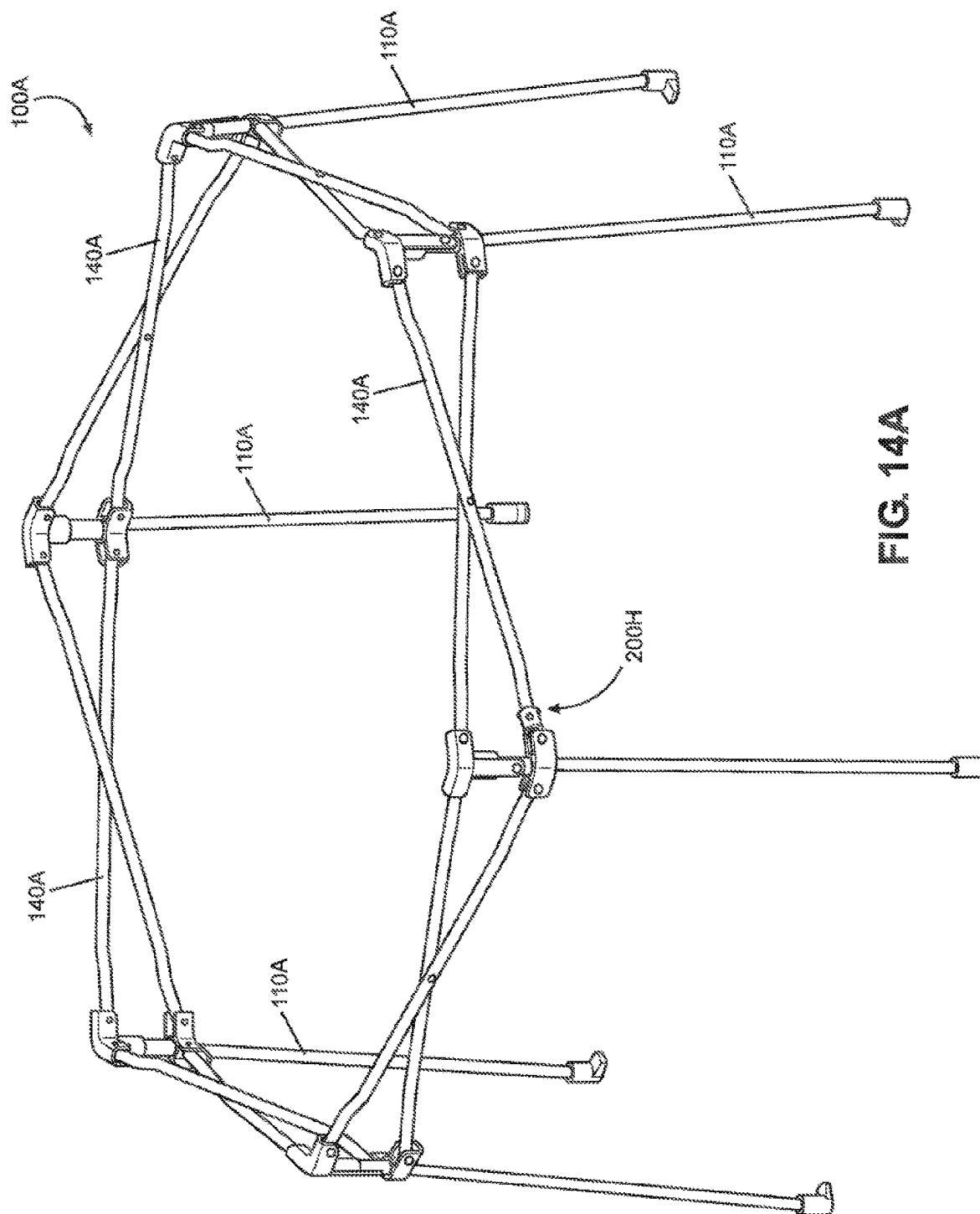
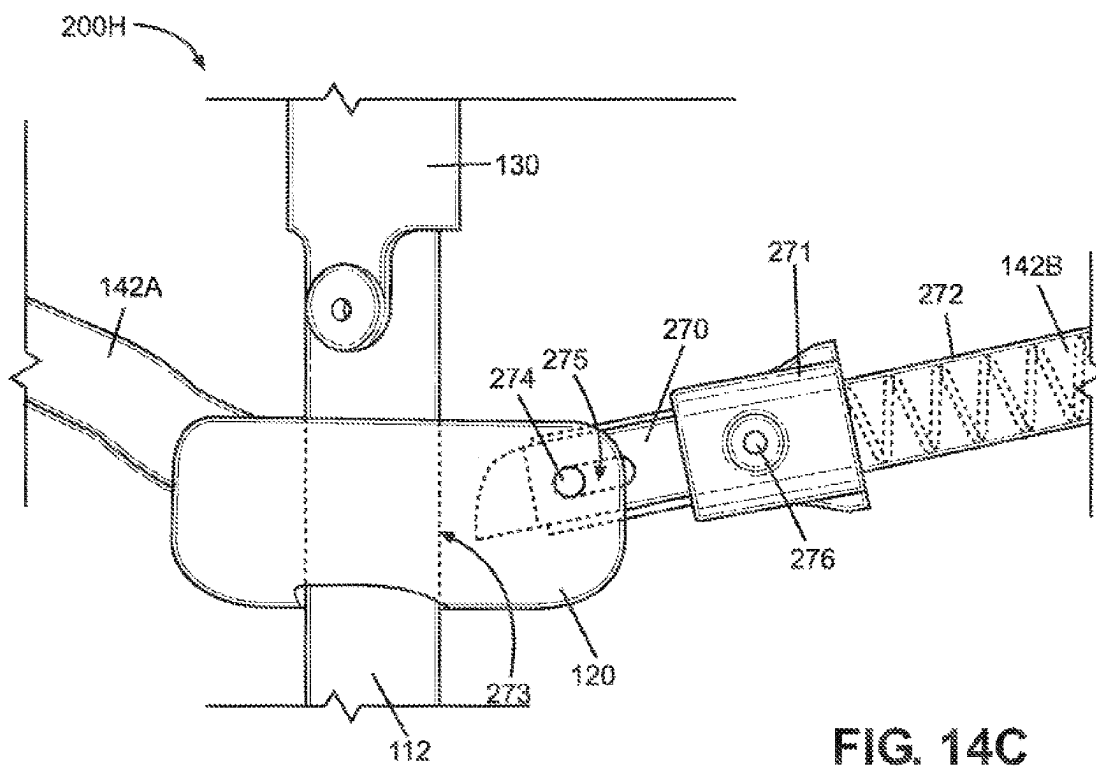
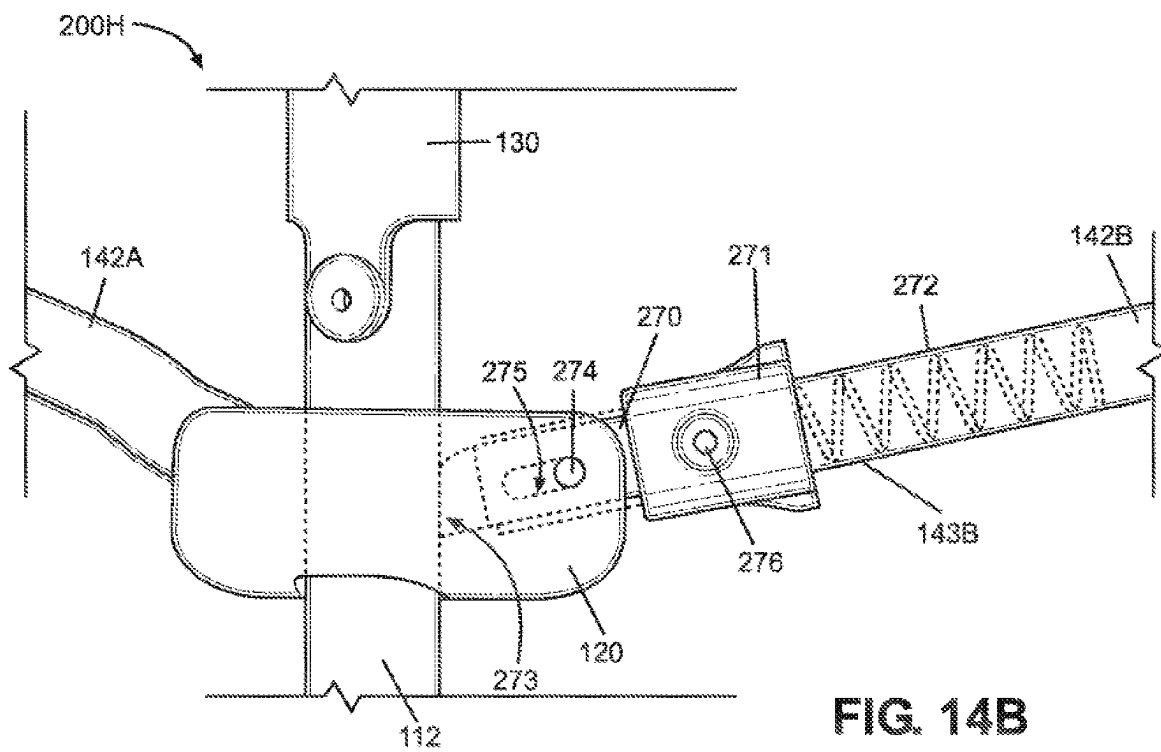


FIG. 13H

FIG. 13G





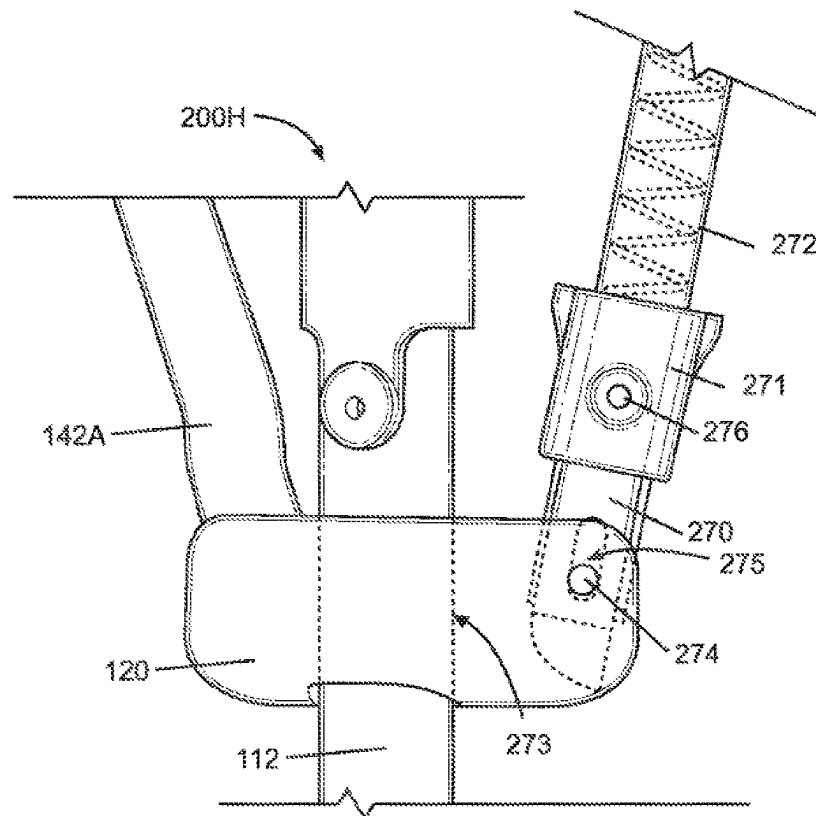
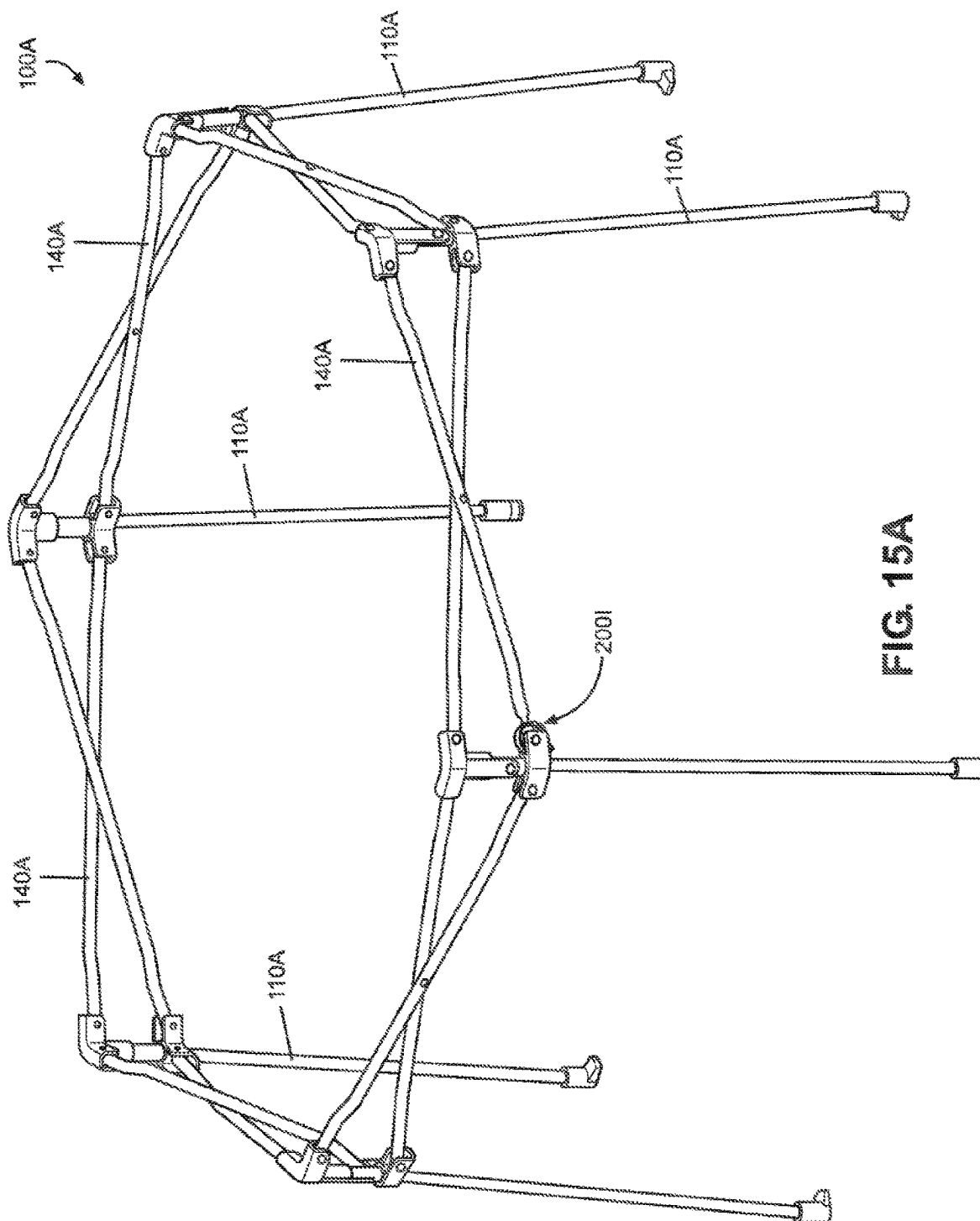


FIG. 14D



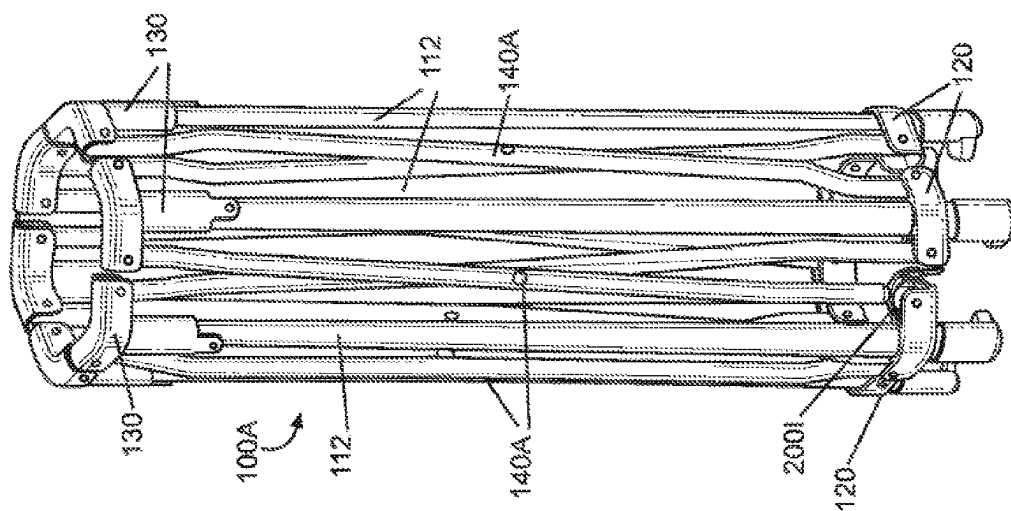


FIG. 15B

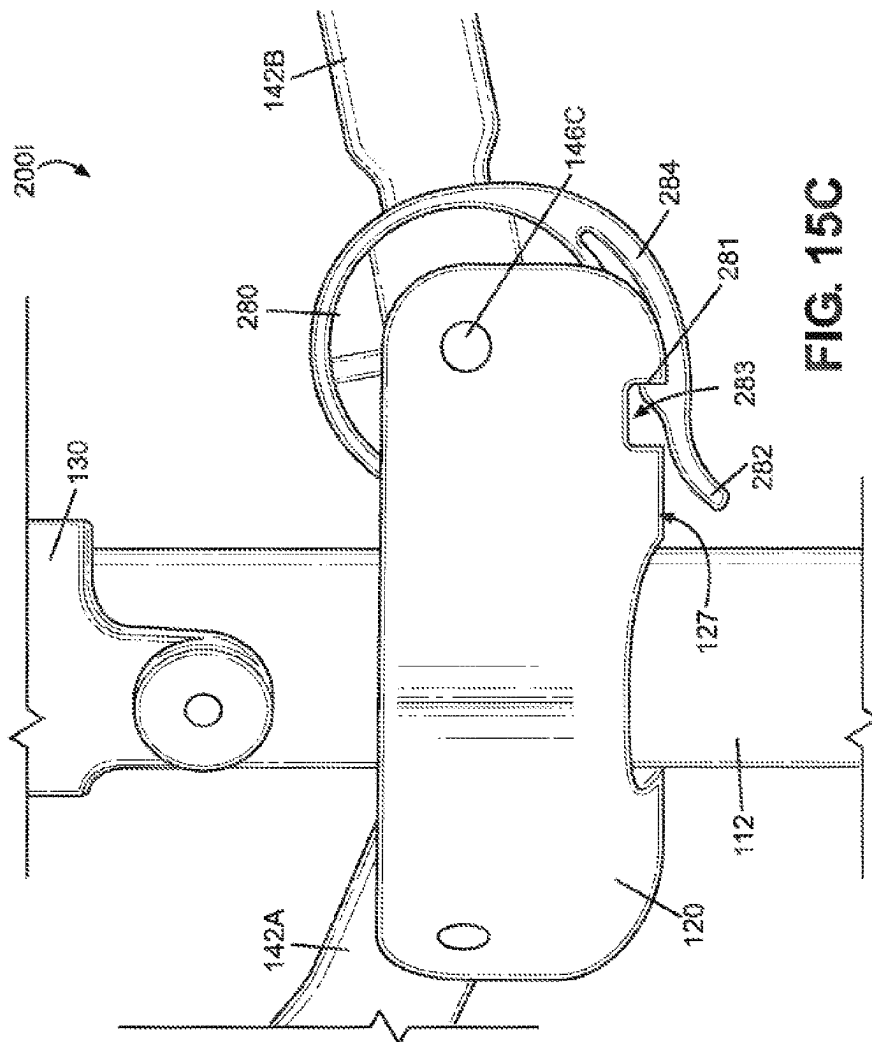


FIG. 15C

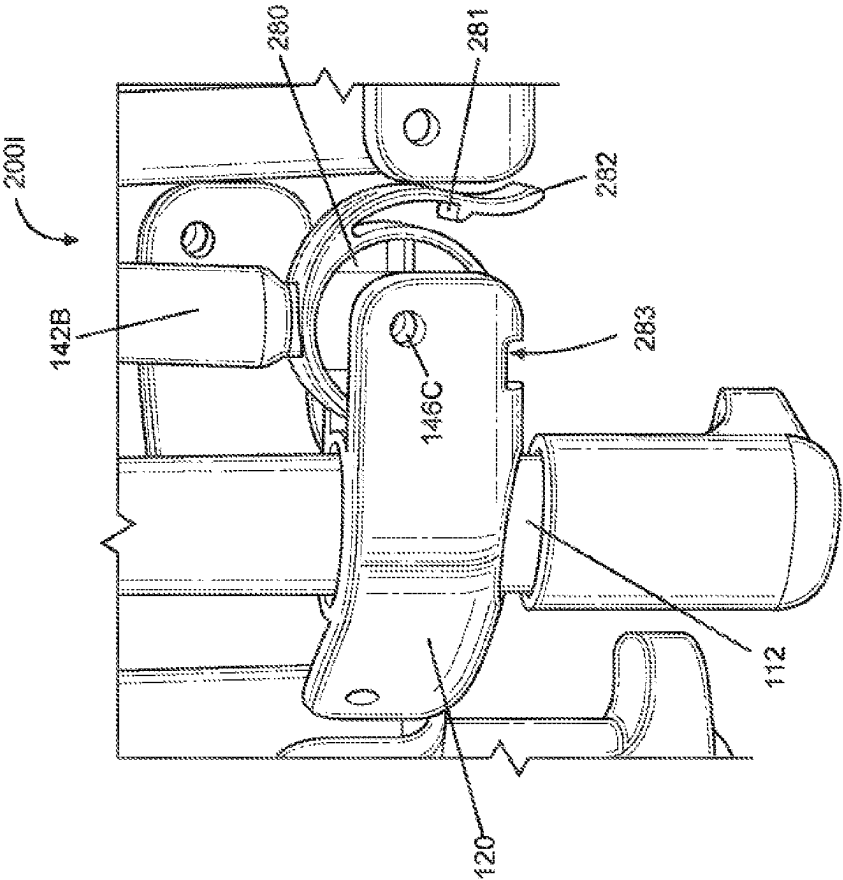


FIG. 15D

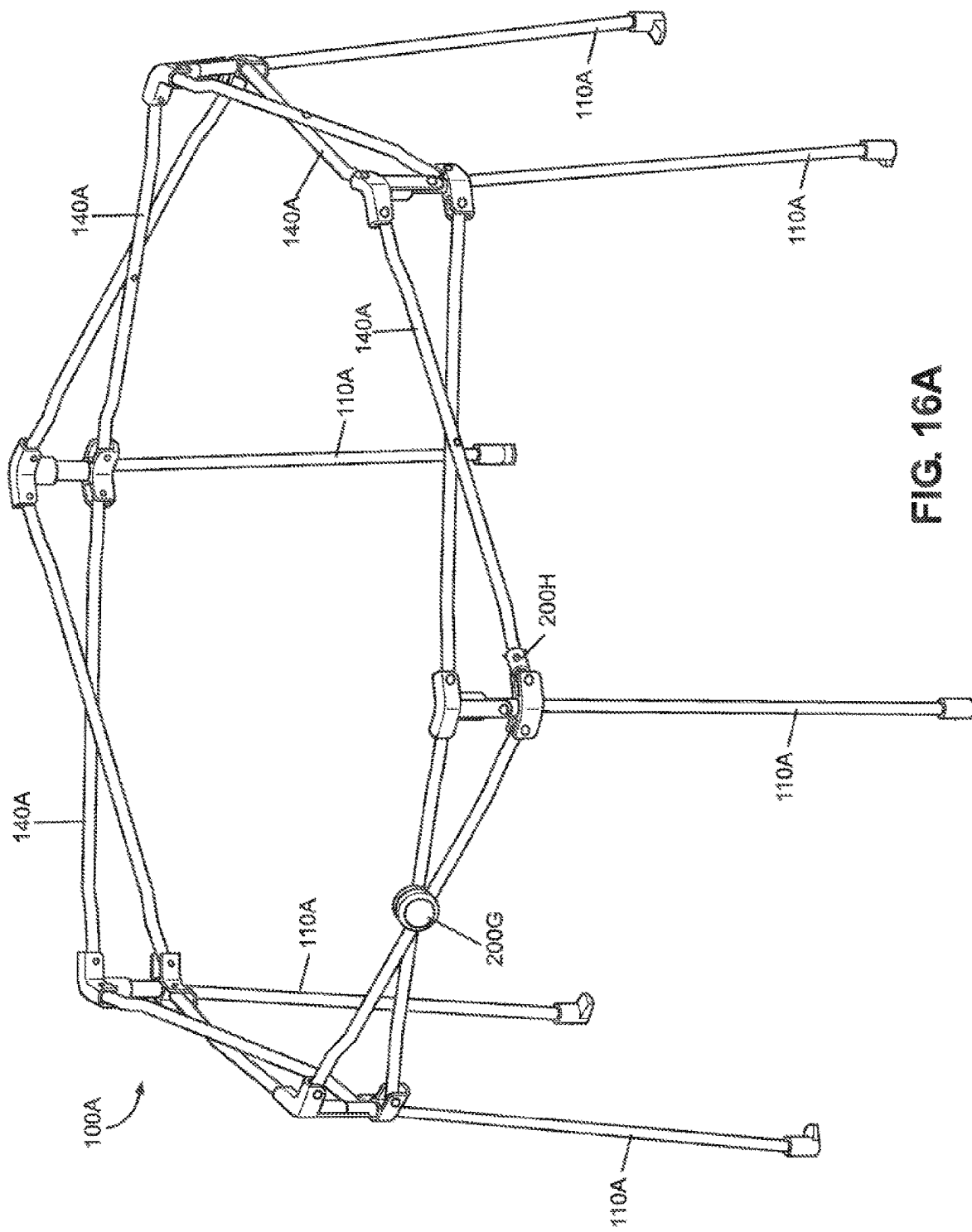
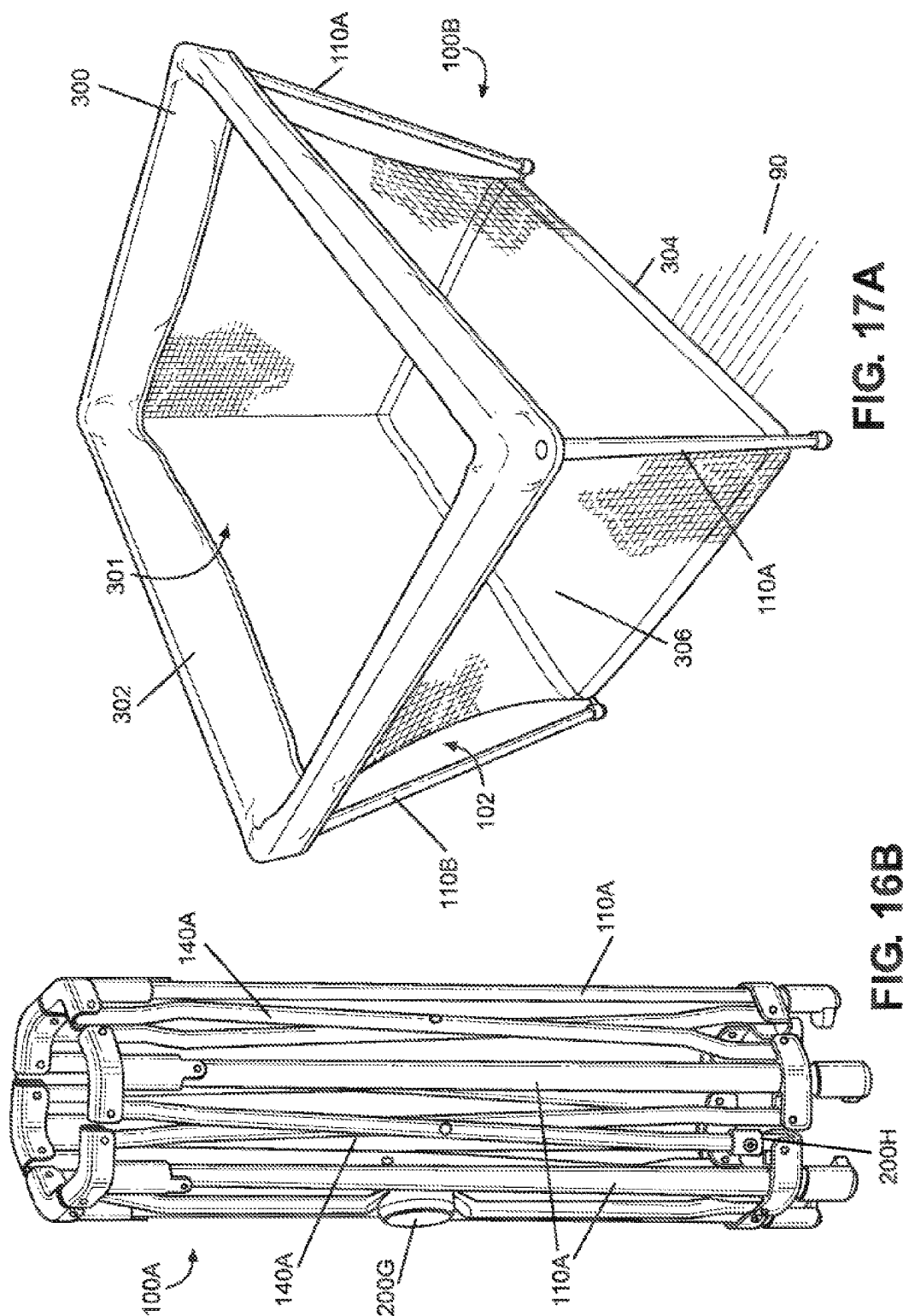


FIG. 16A



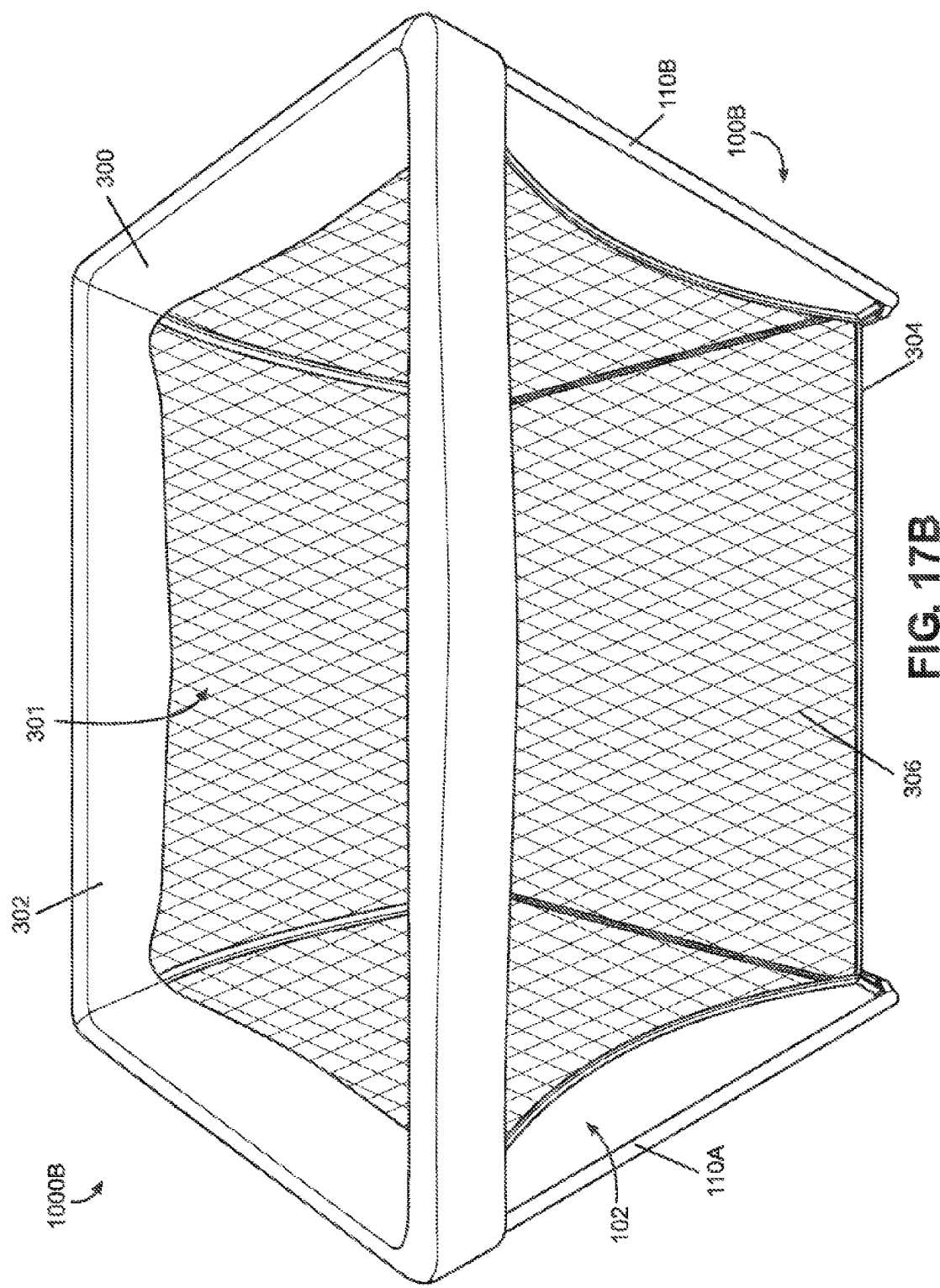
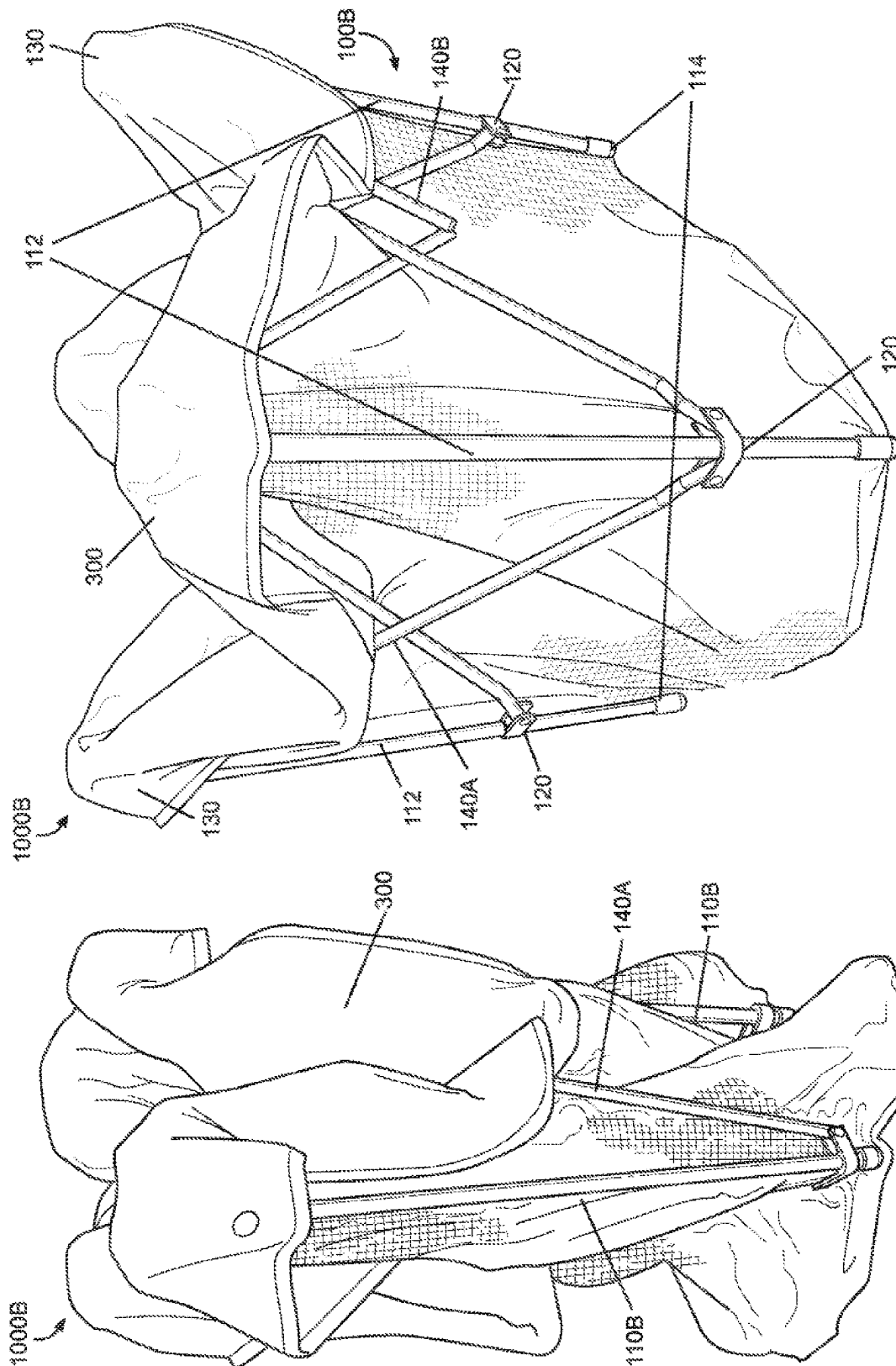
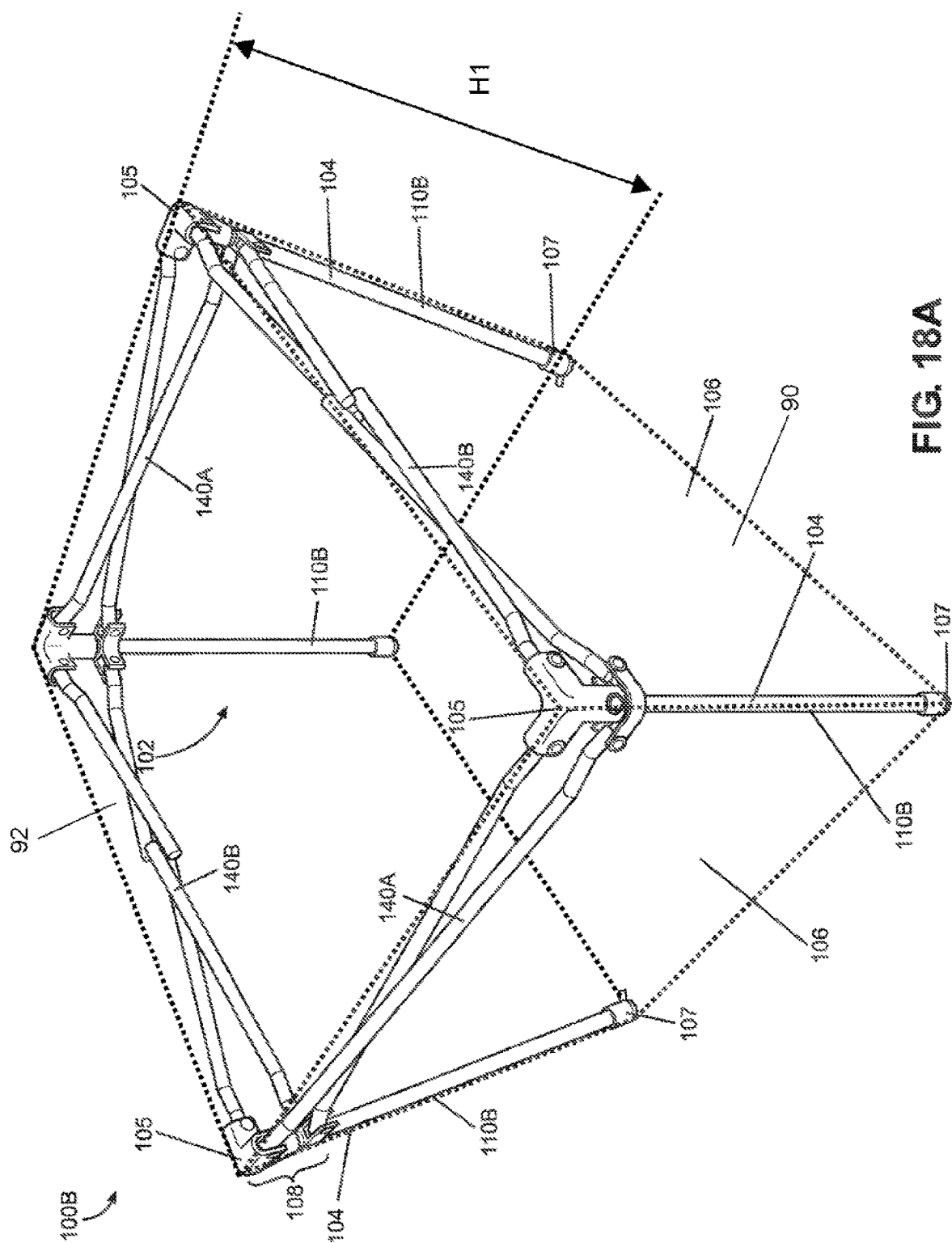


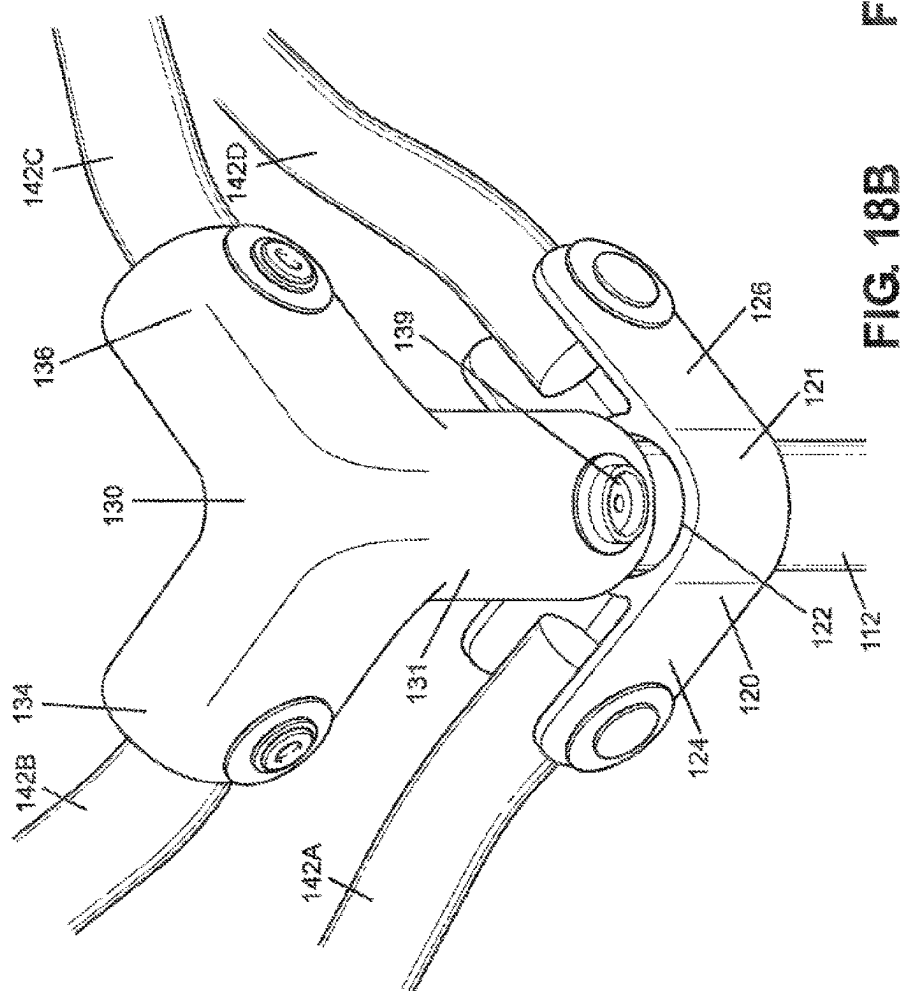
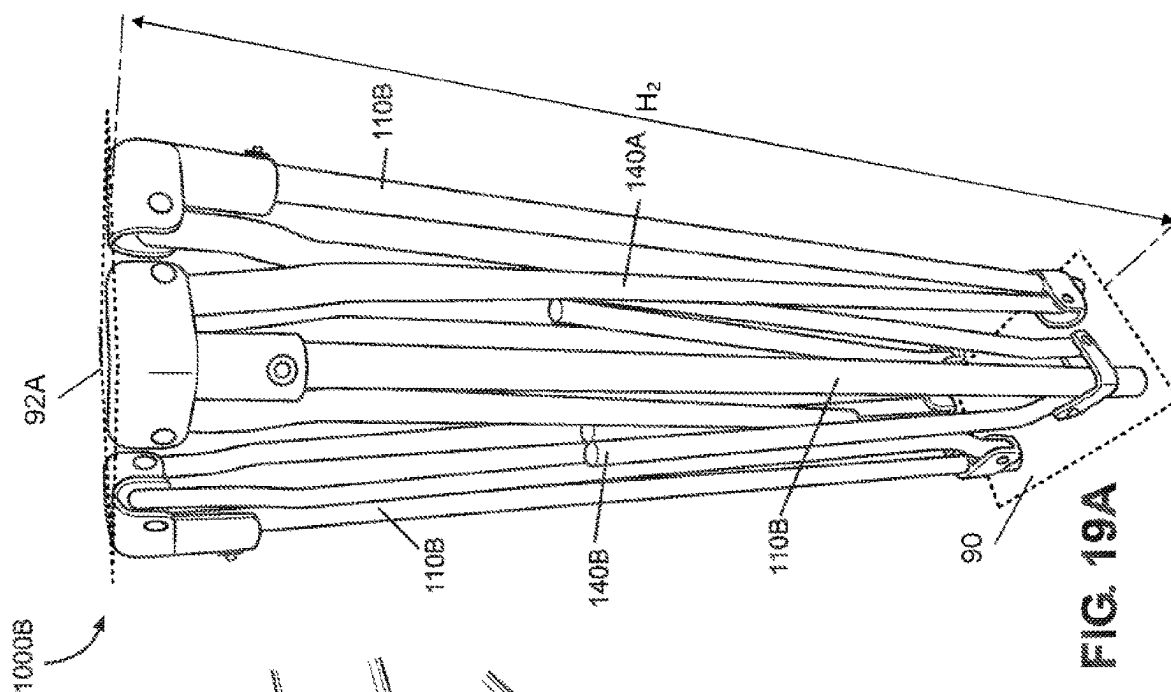
FIG. 17B



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FIG 17C





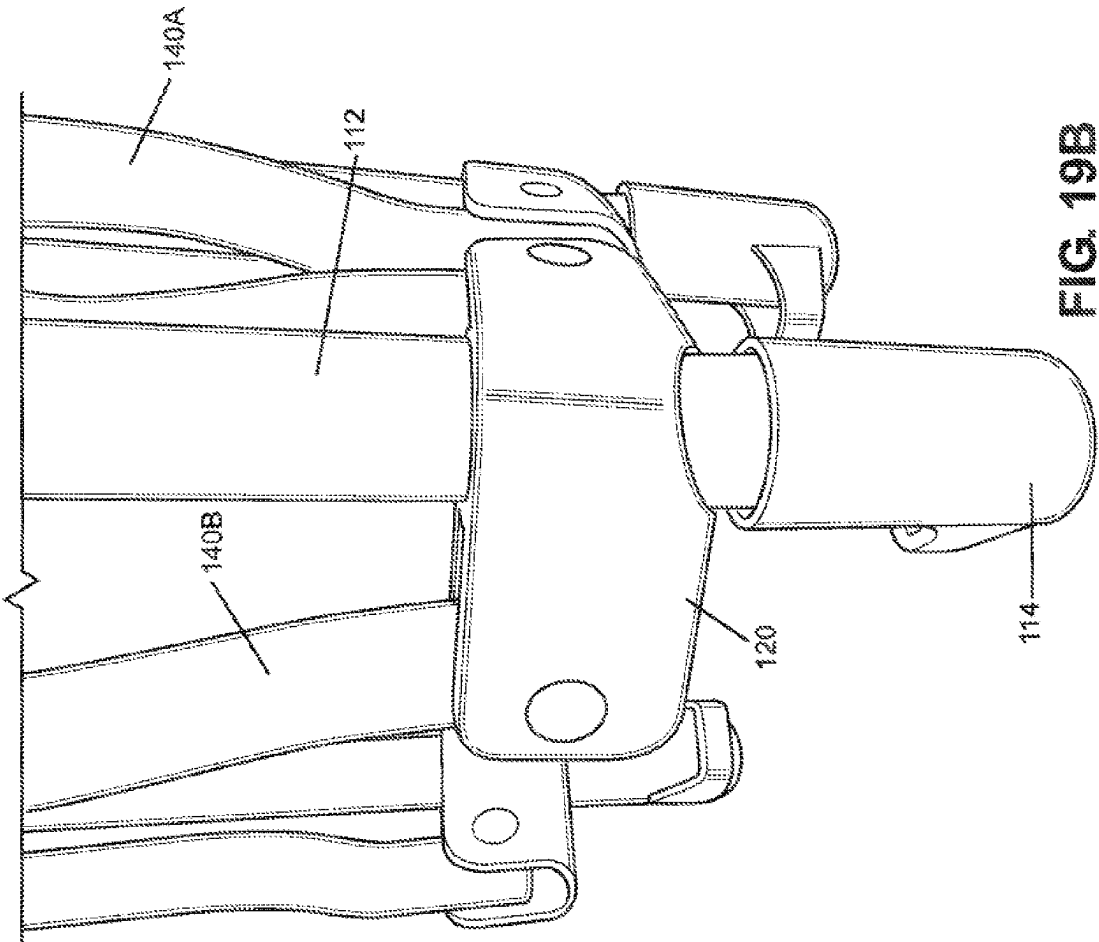
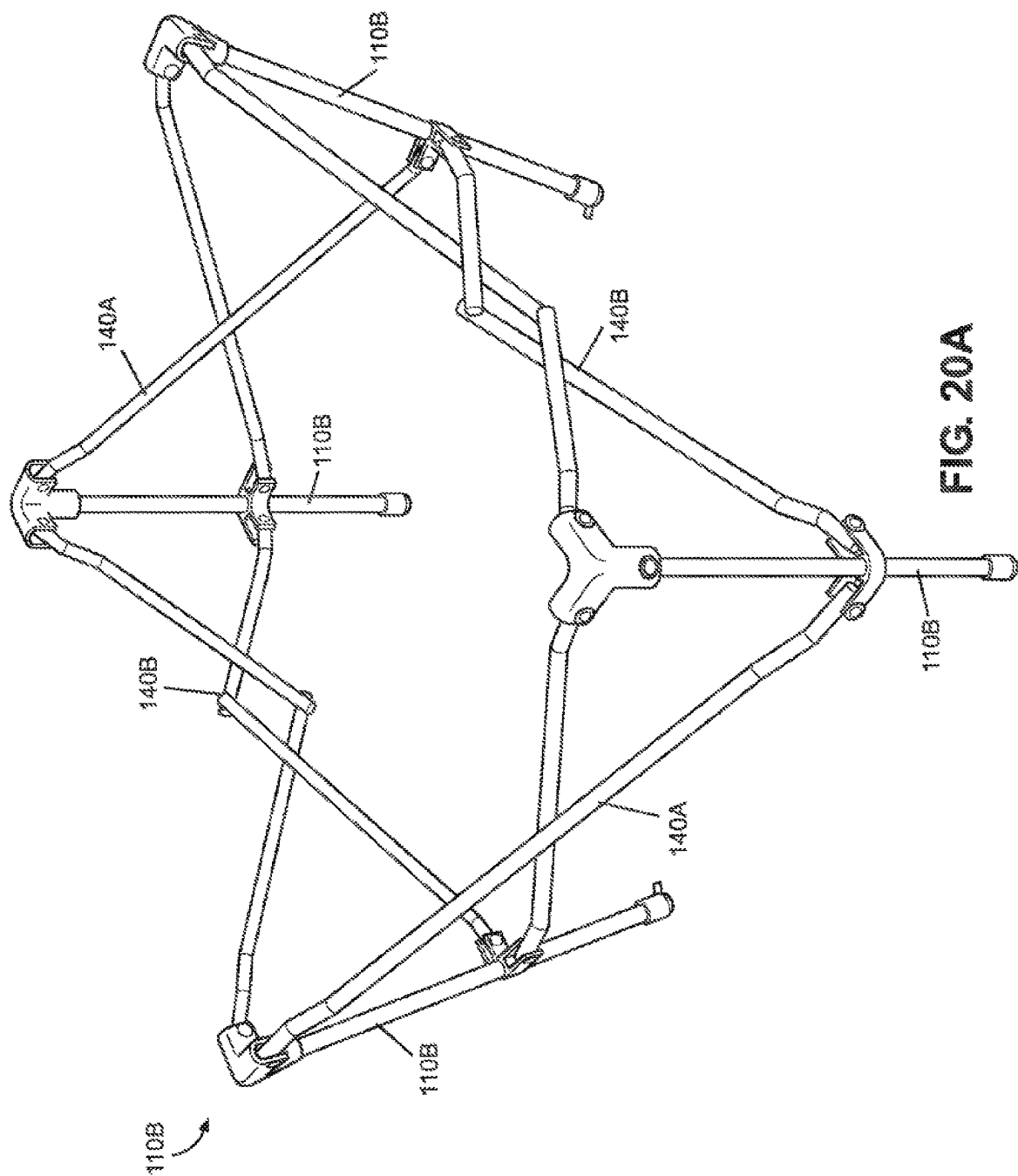


FIG. 19B



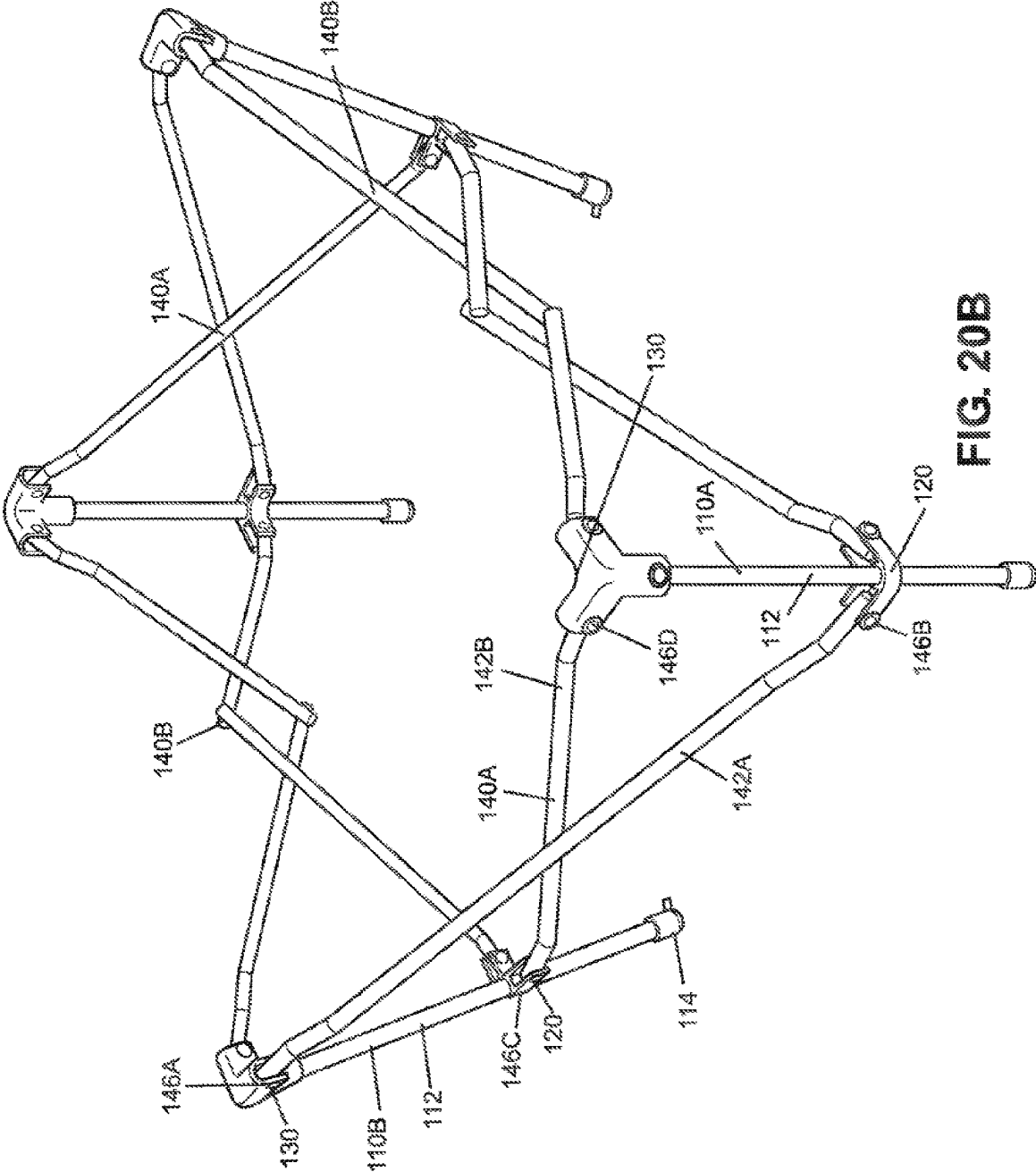


FIG. 20B

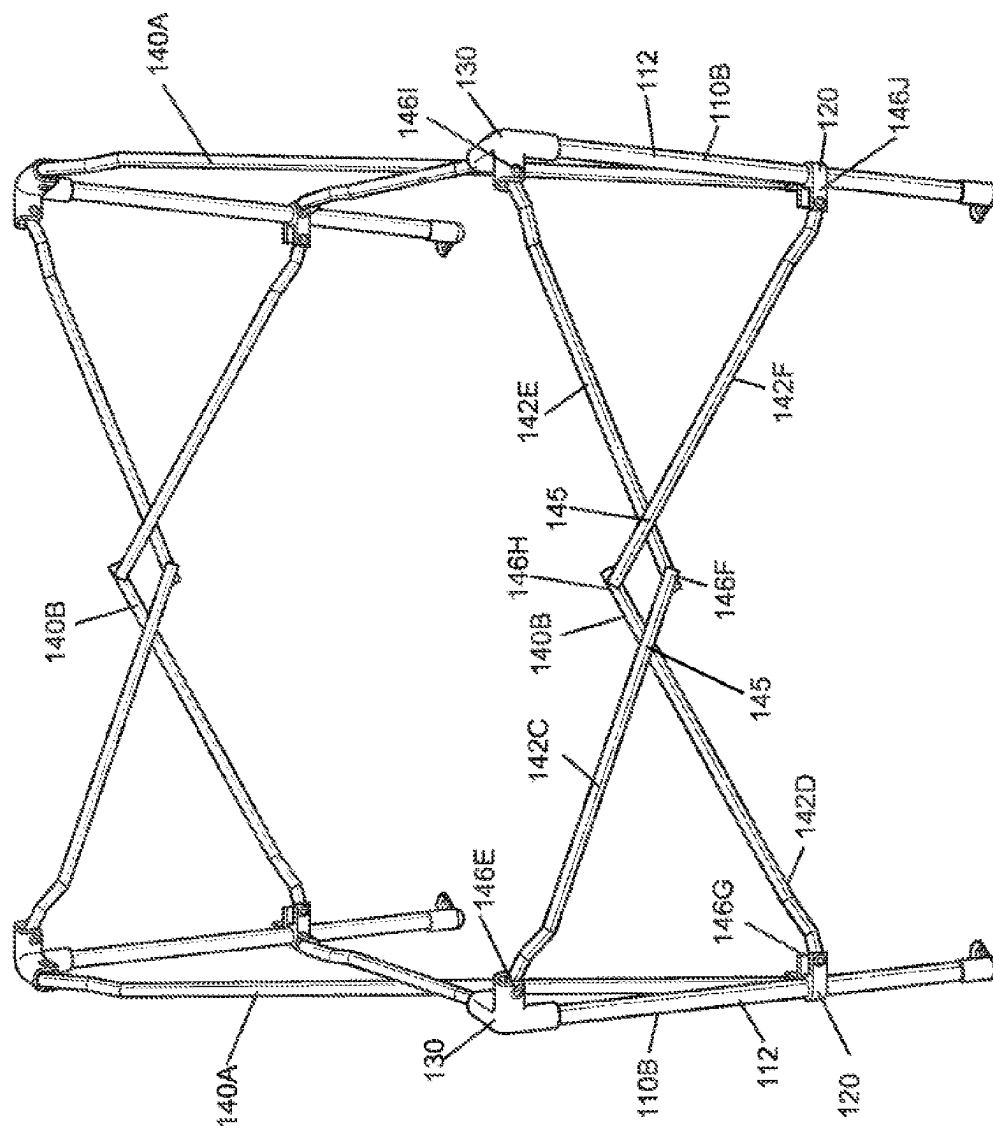


FIG. 20C

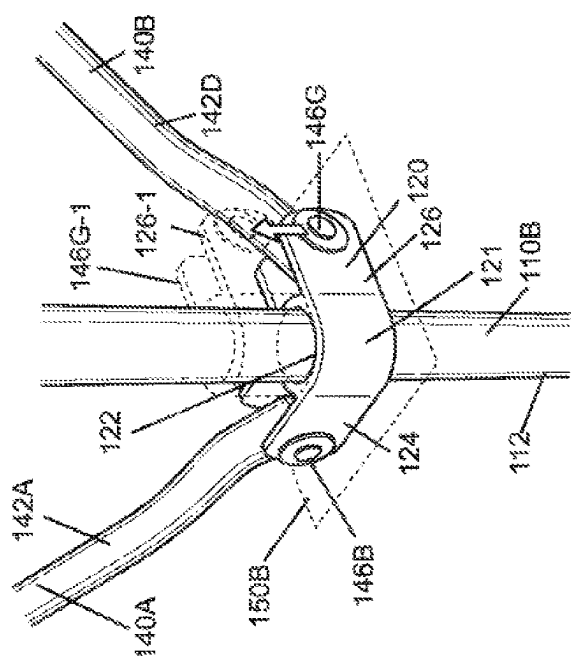


FIG. 20D

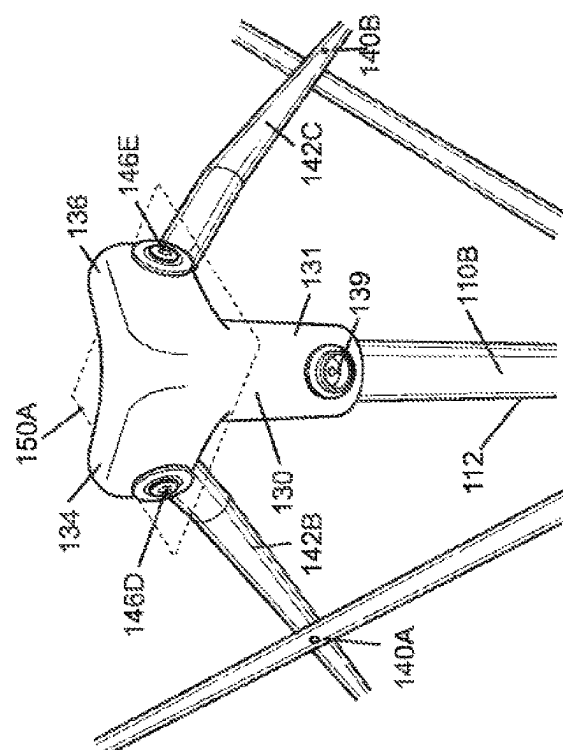
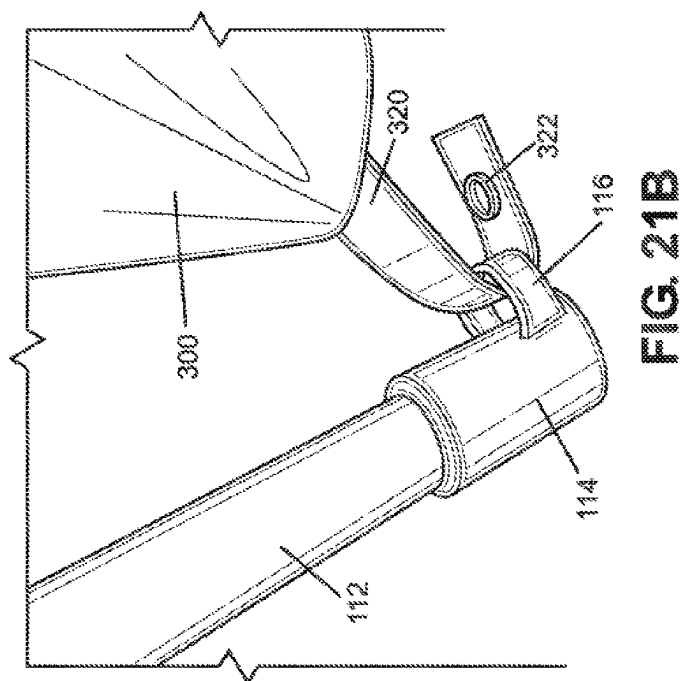
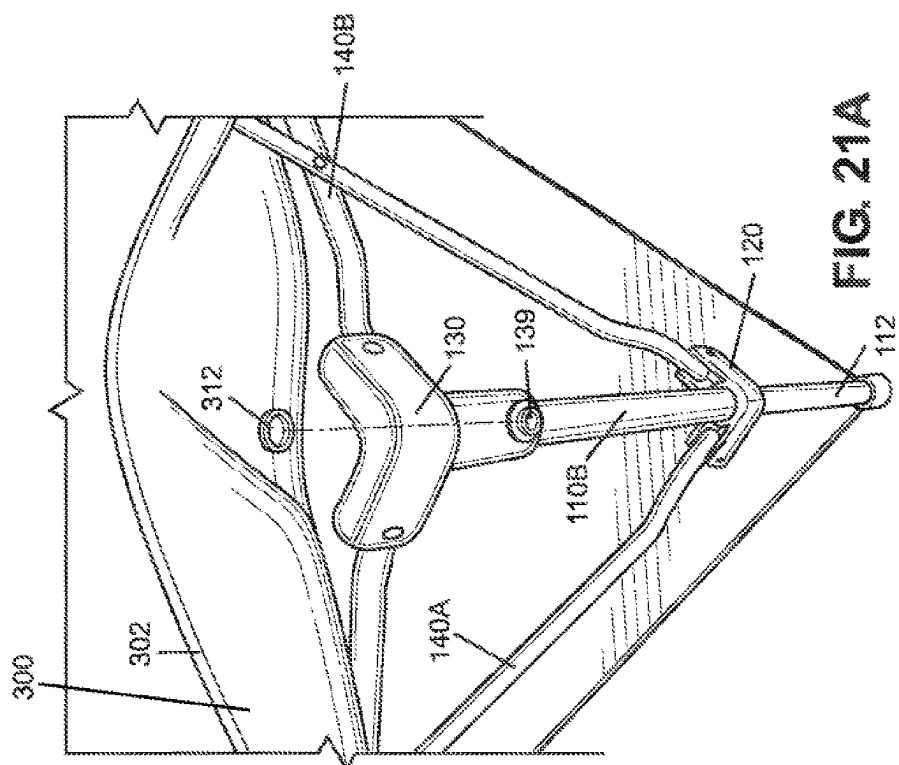


FIG. 20E



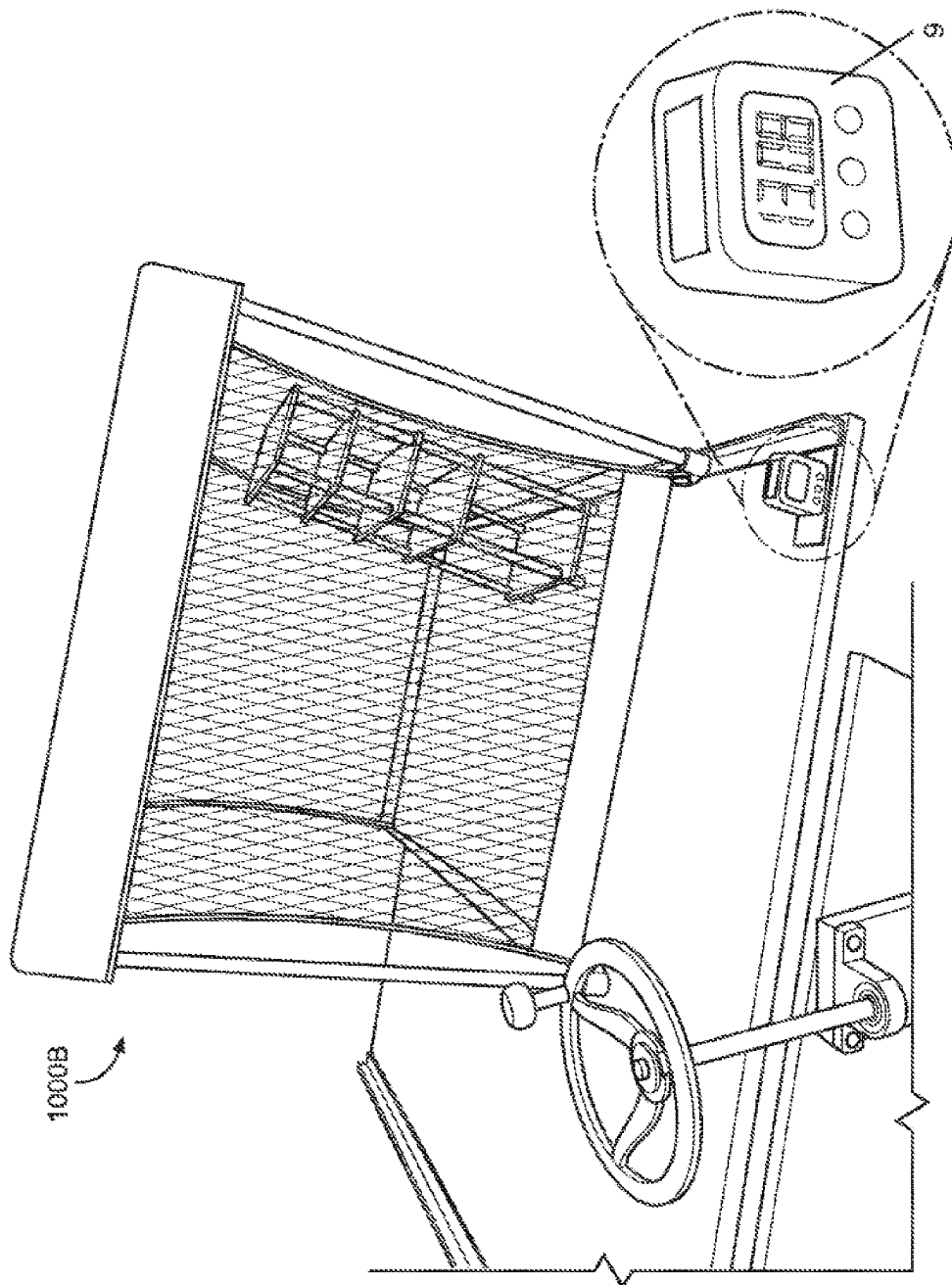


FIG. 22

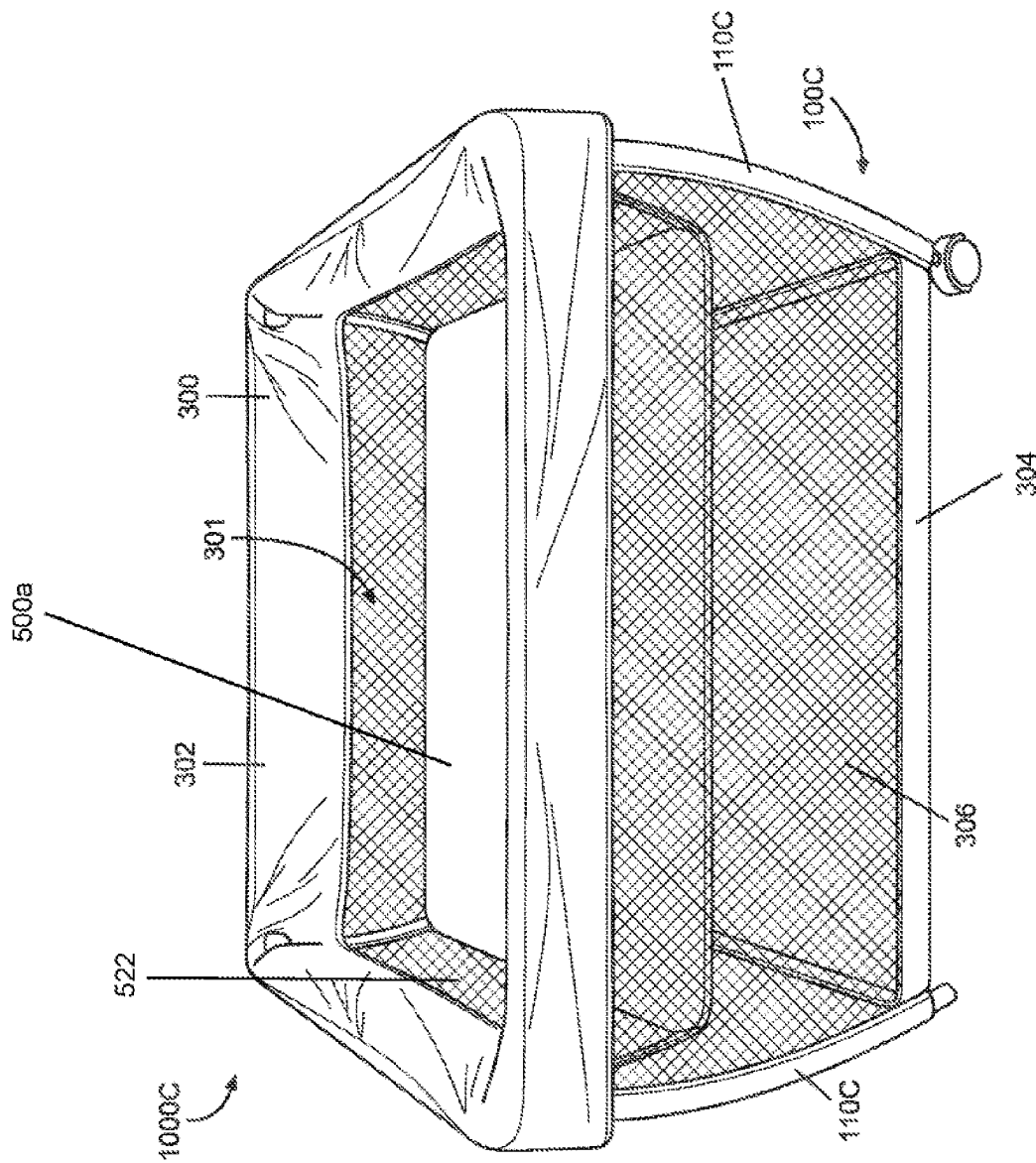
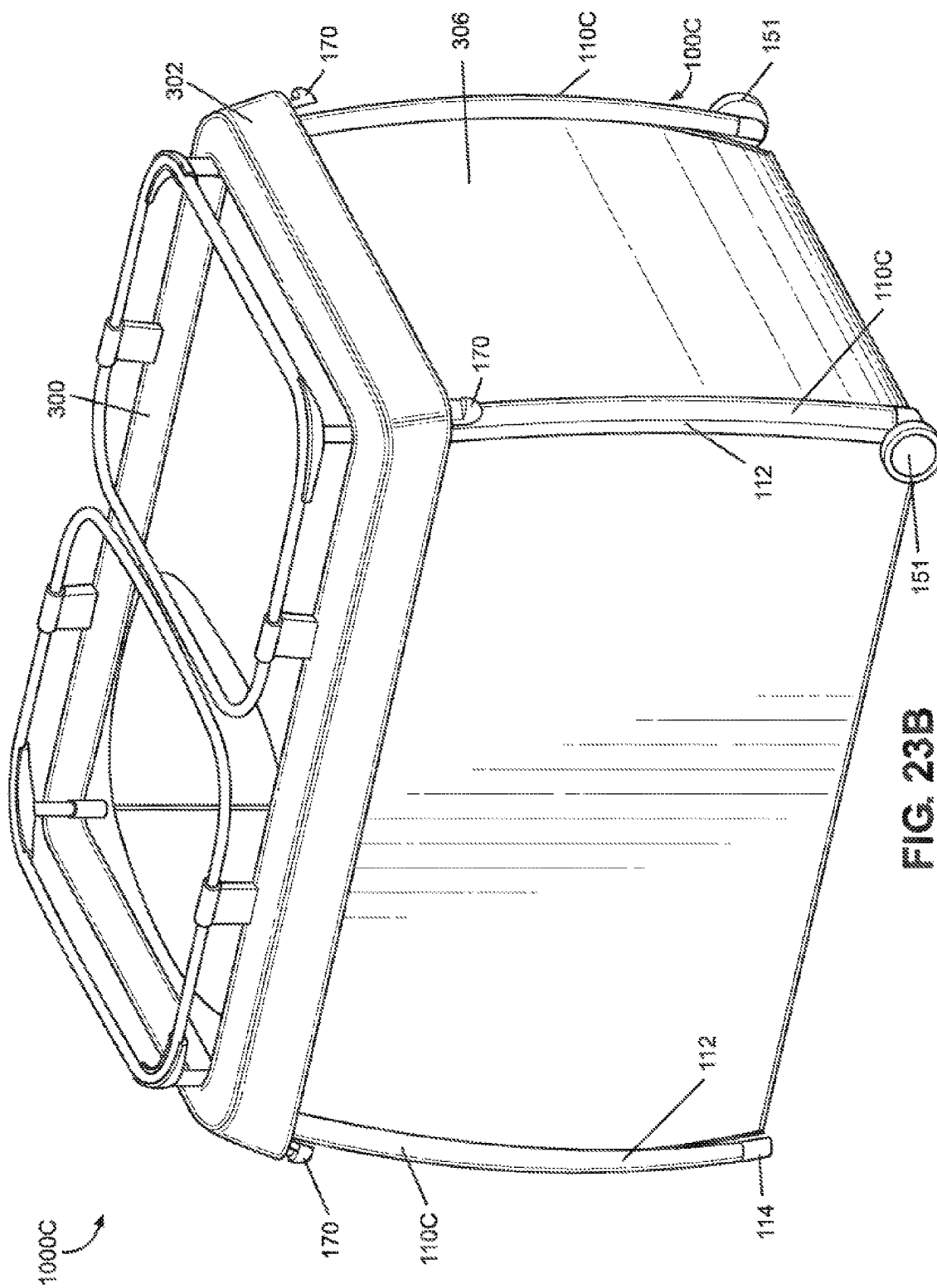


FIG 23A



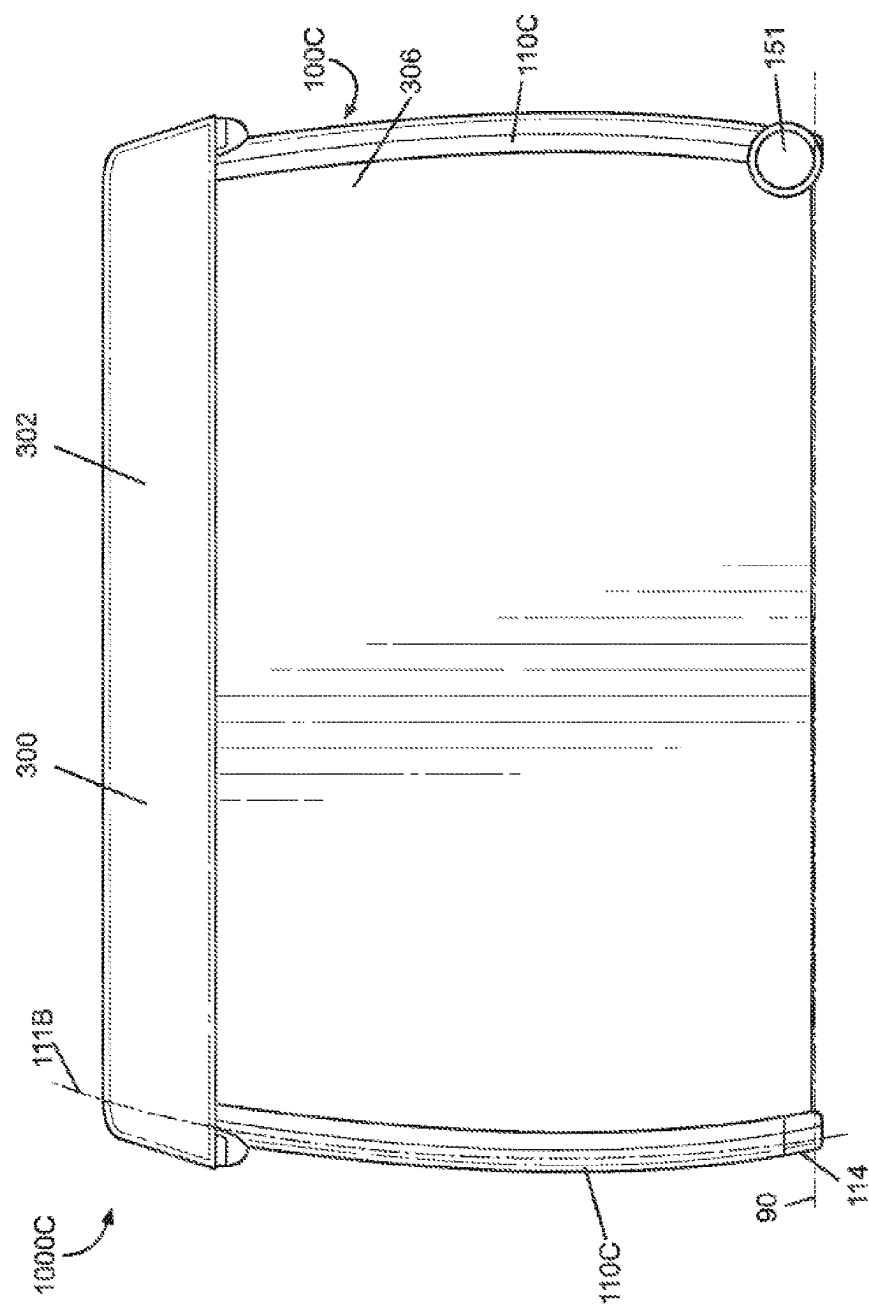


FIG. 23C

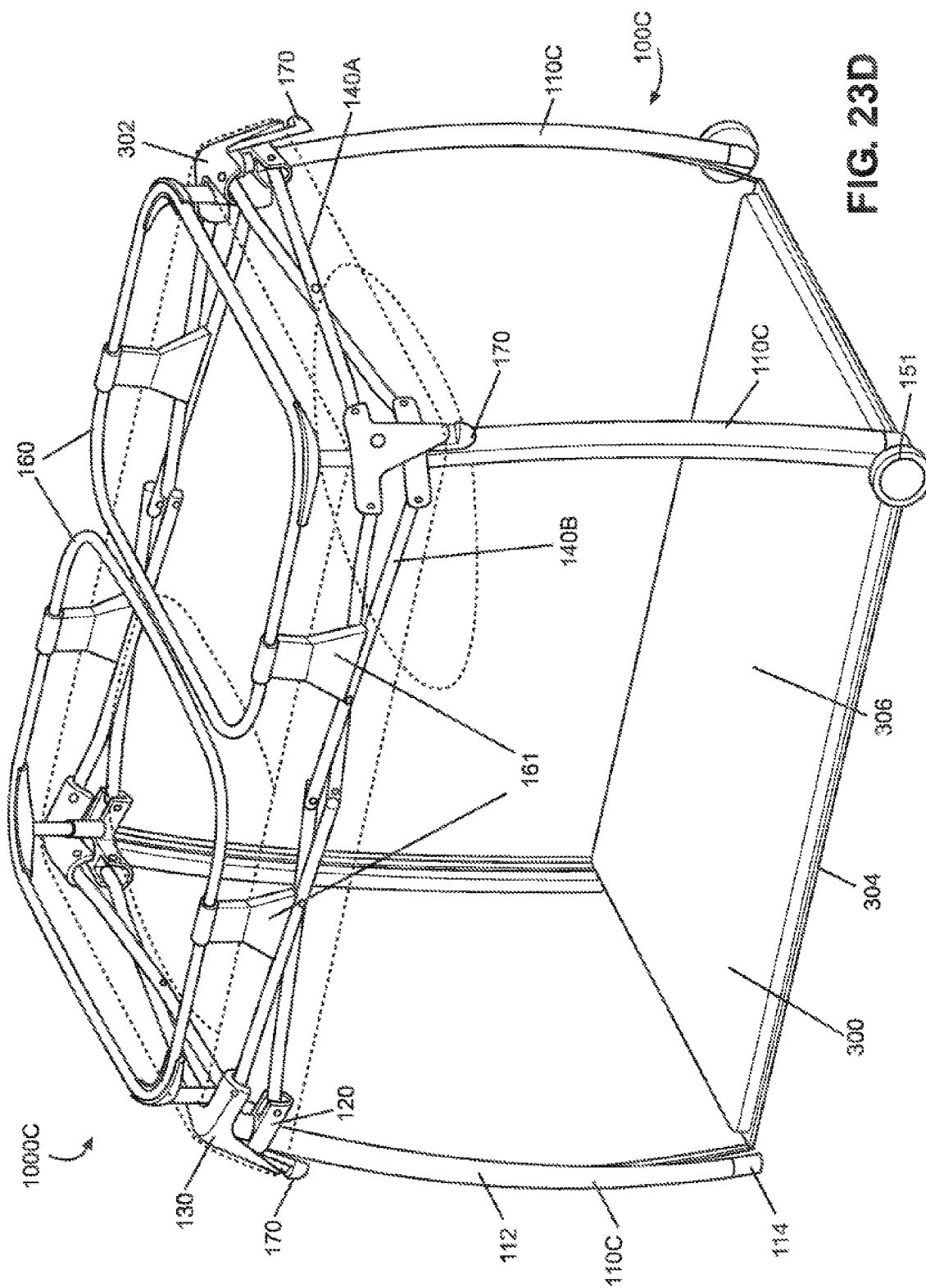


FIG. 23D

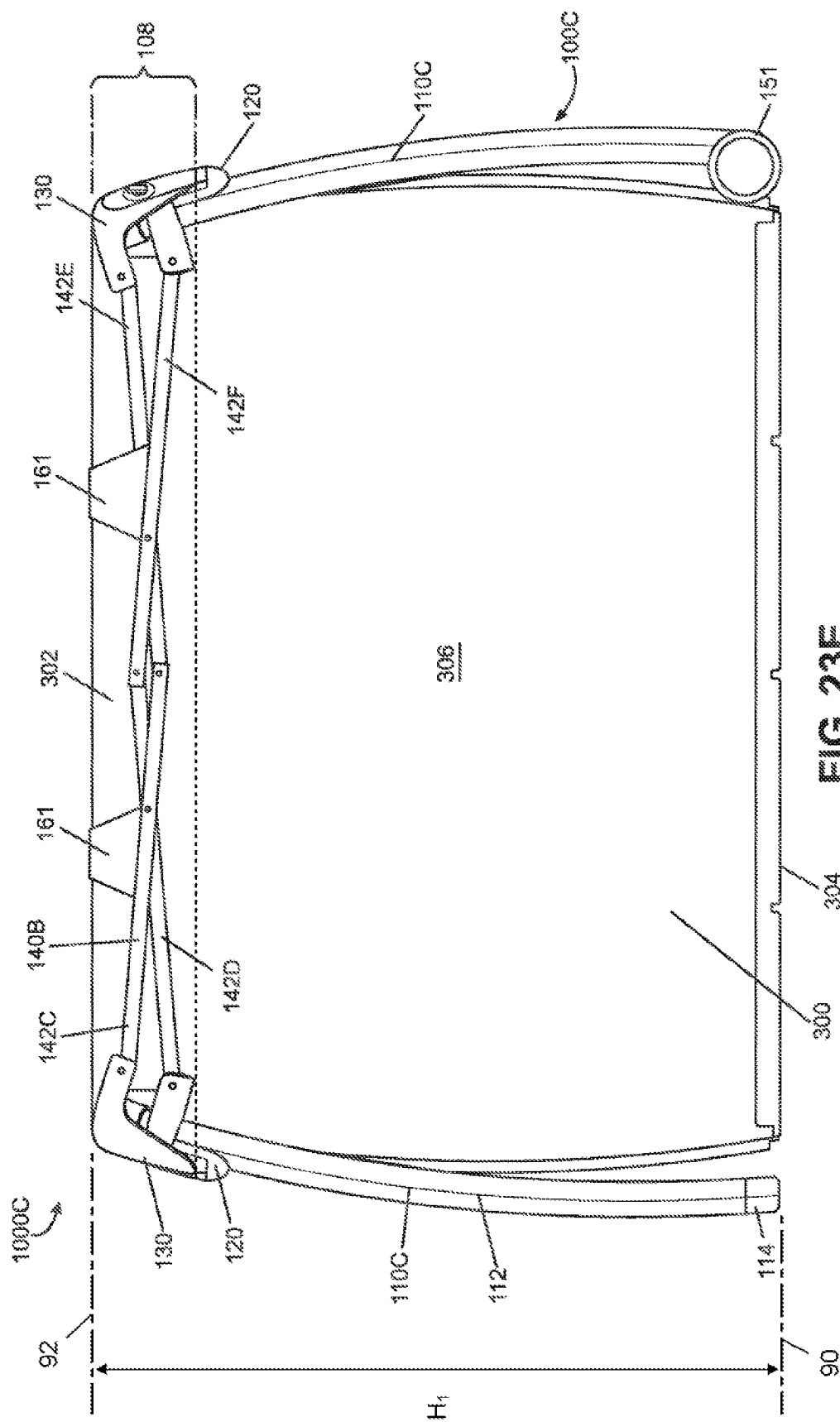
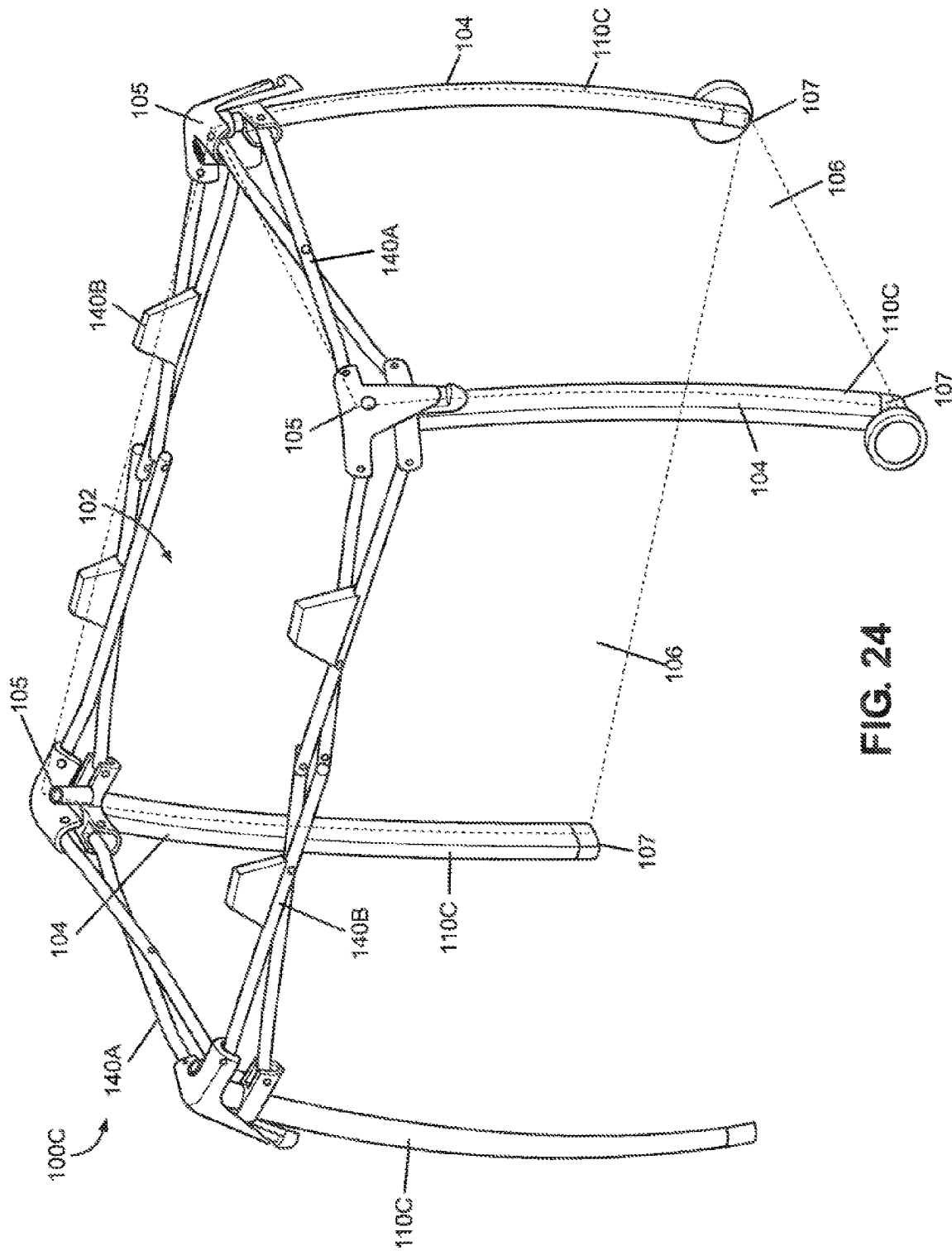


FIG. 23E



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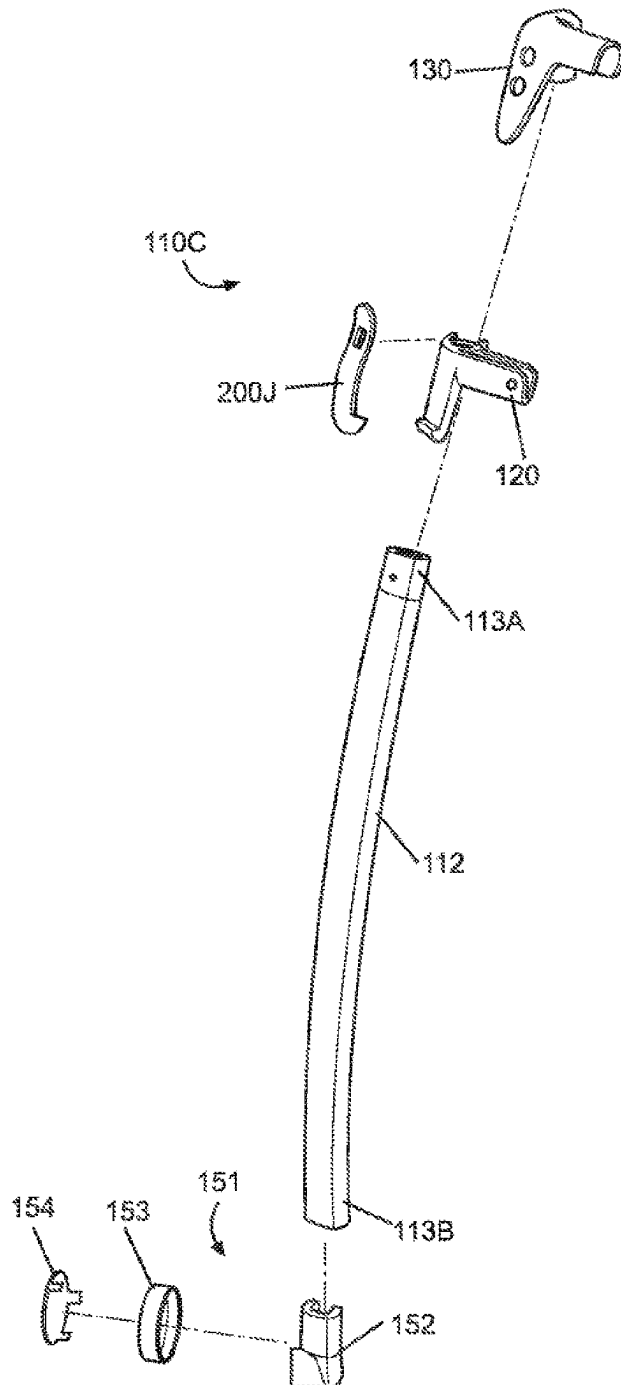


FIG. 25A

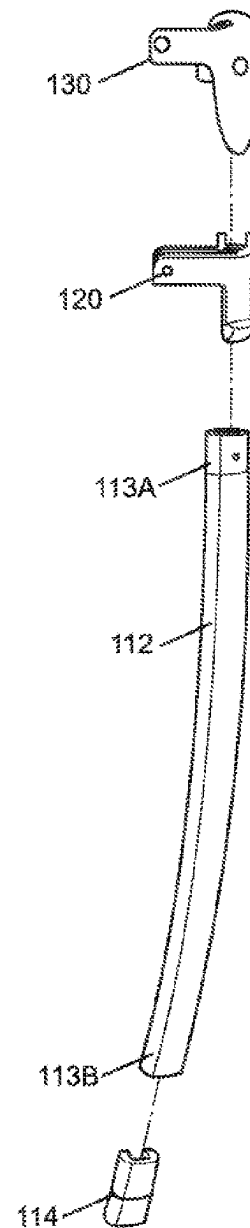
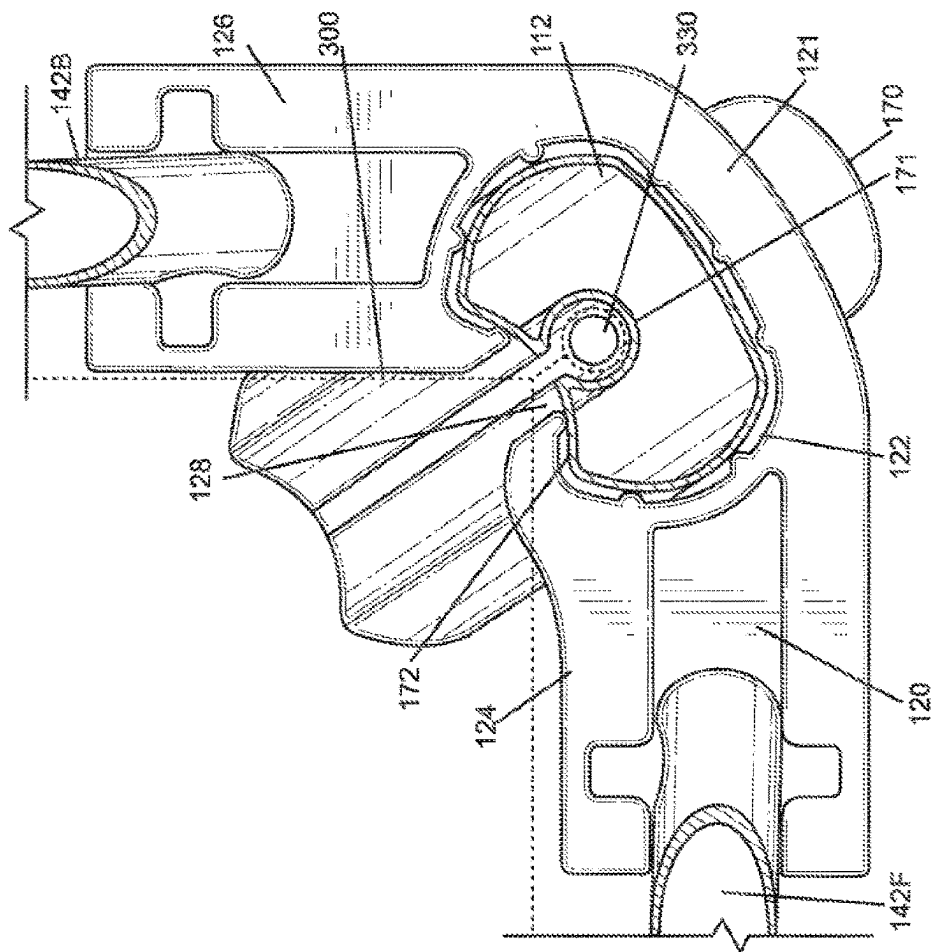
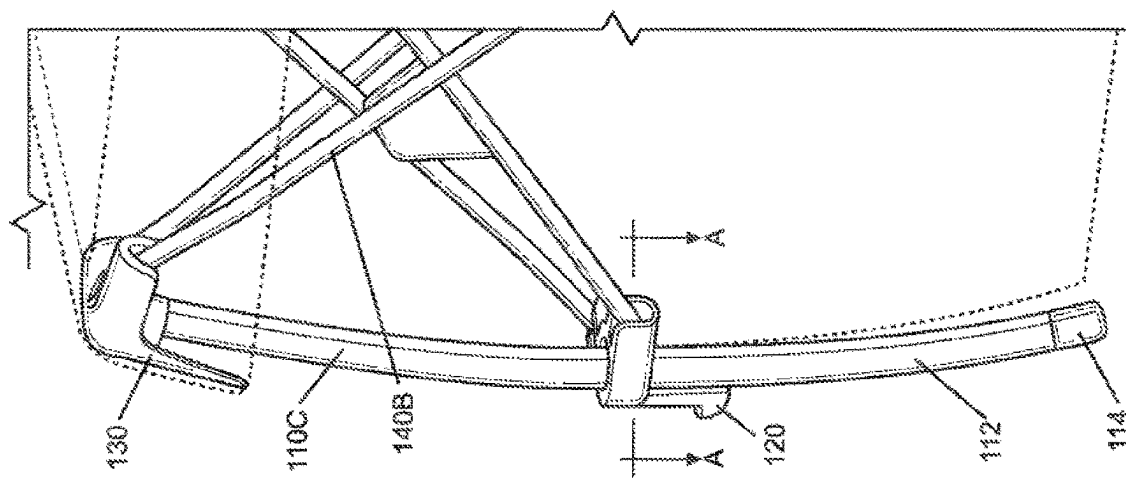
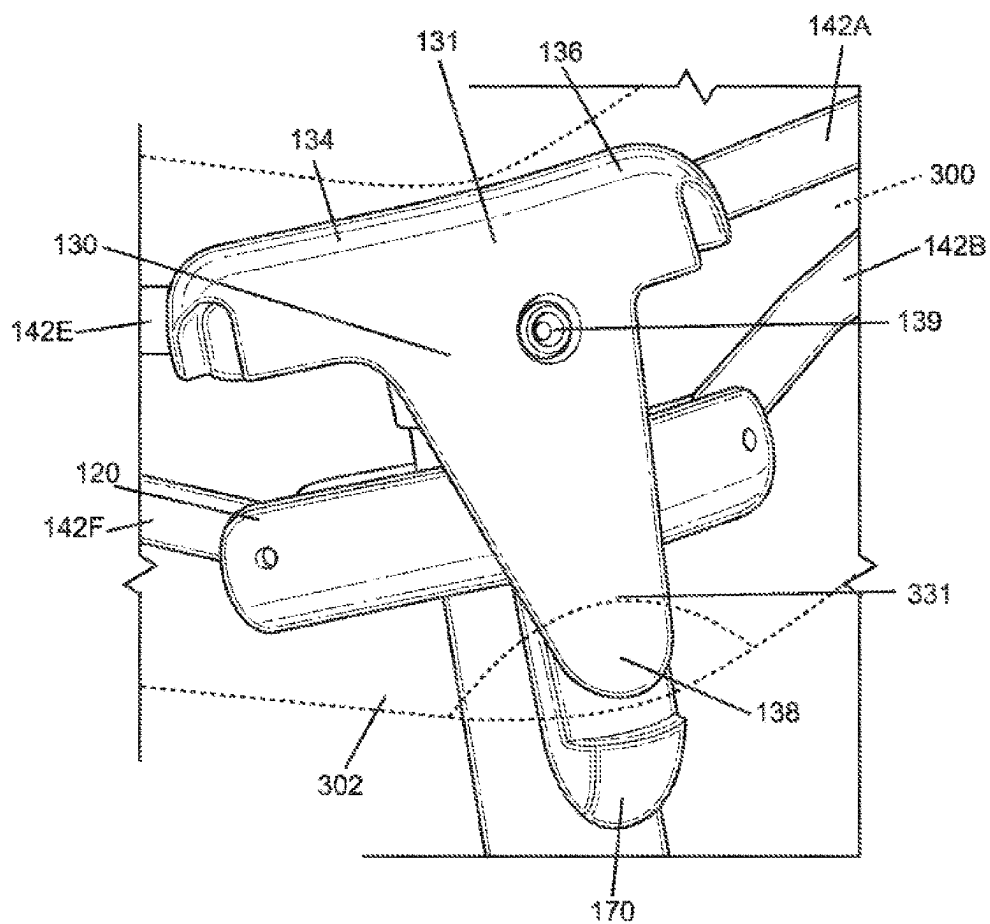


FIG. 25B



**FIG. 27A**

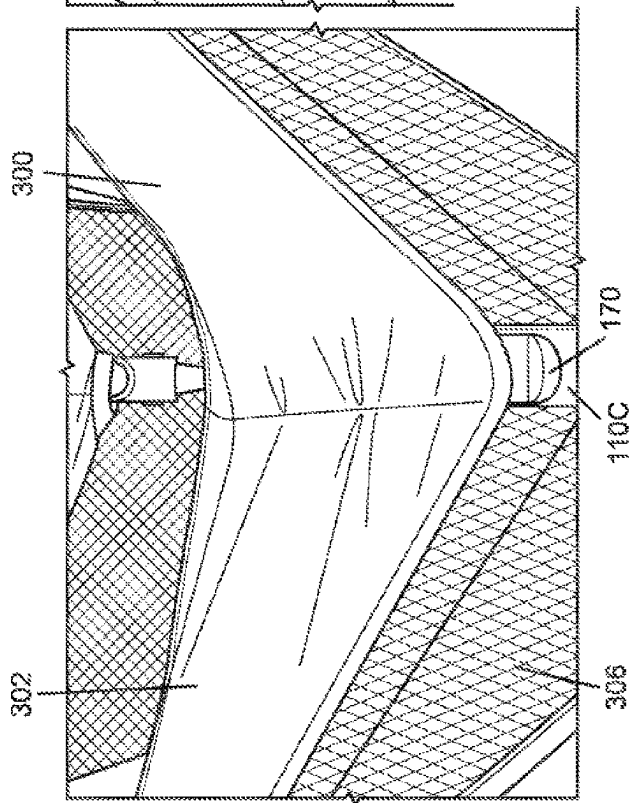


FIG. 27B

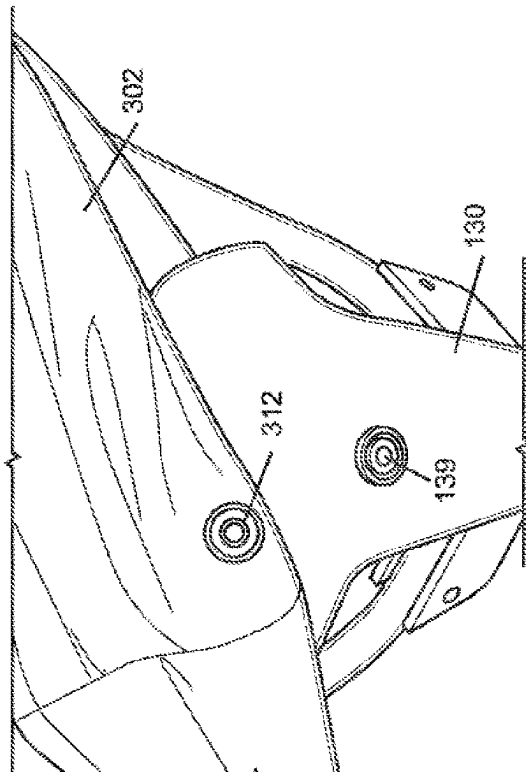


FIG. 27C

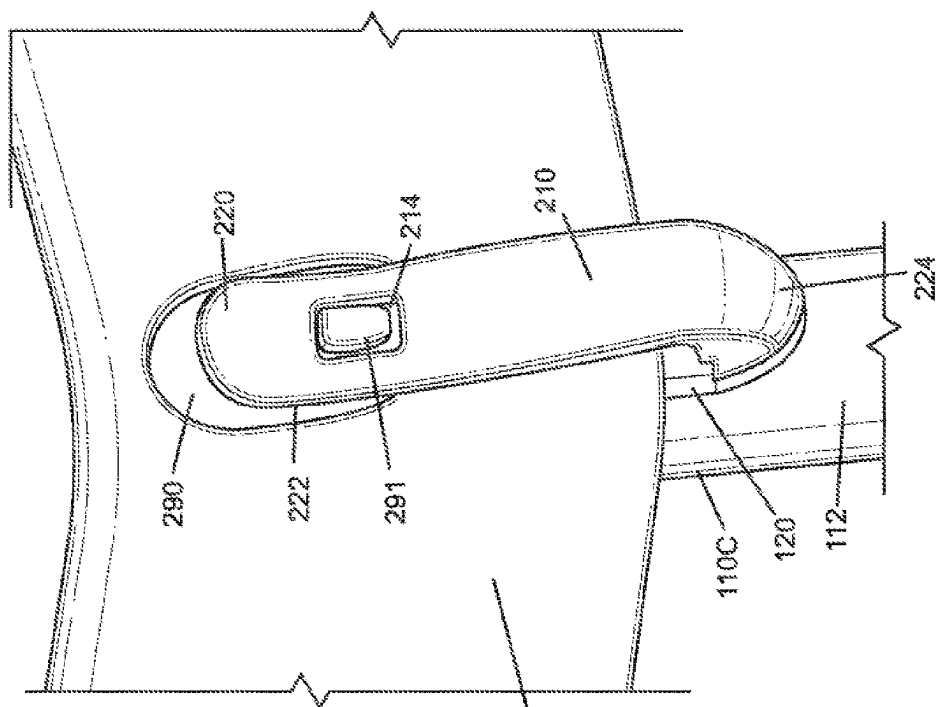


FIG. 28B

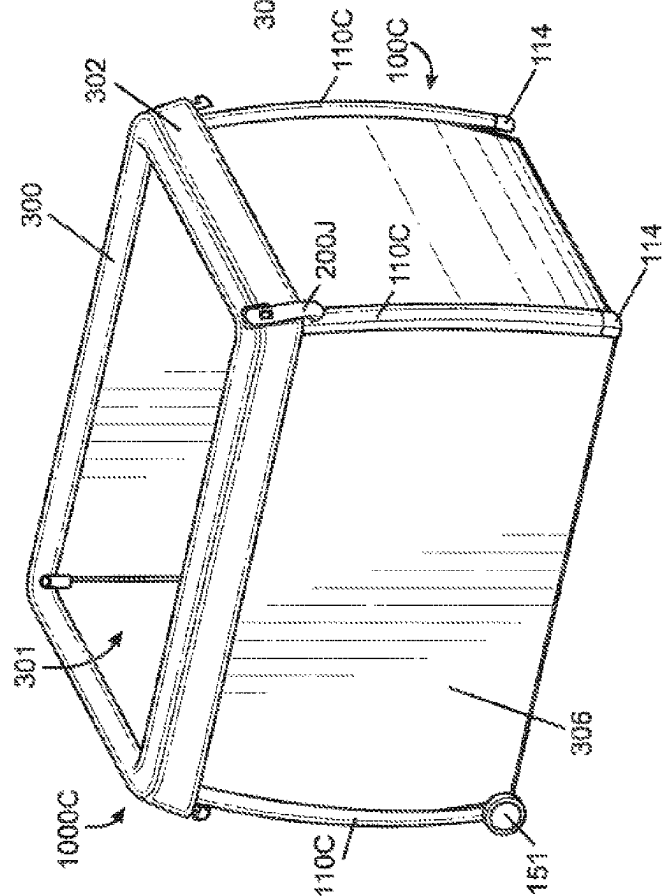
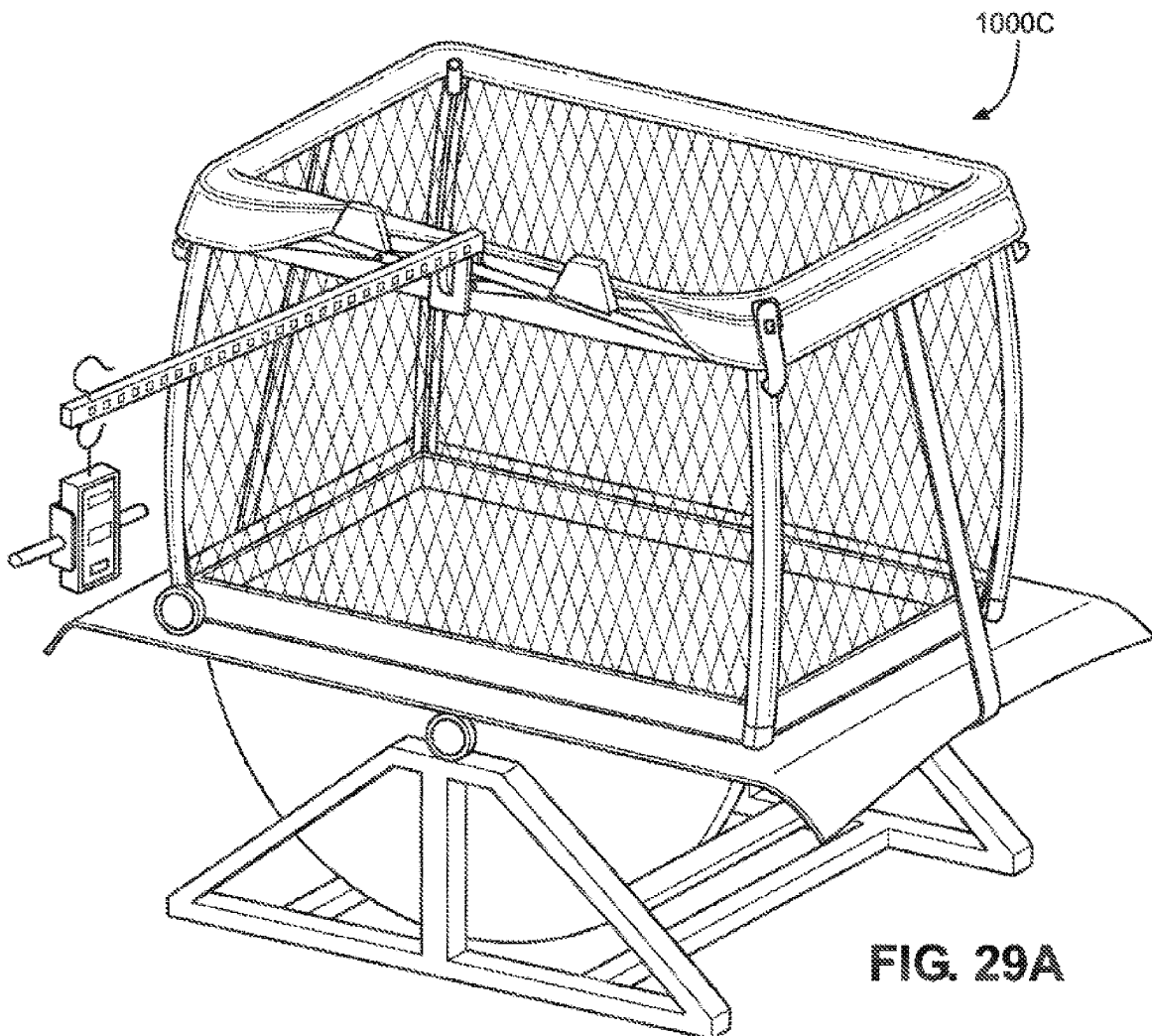
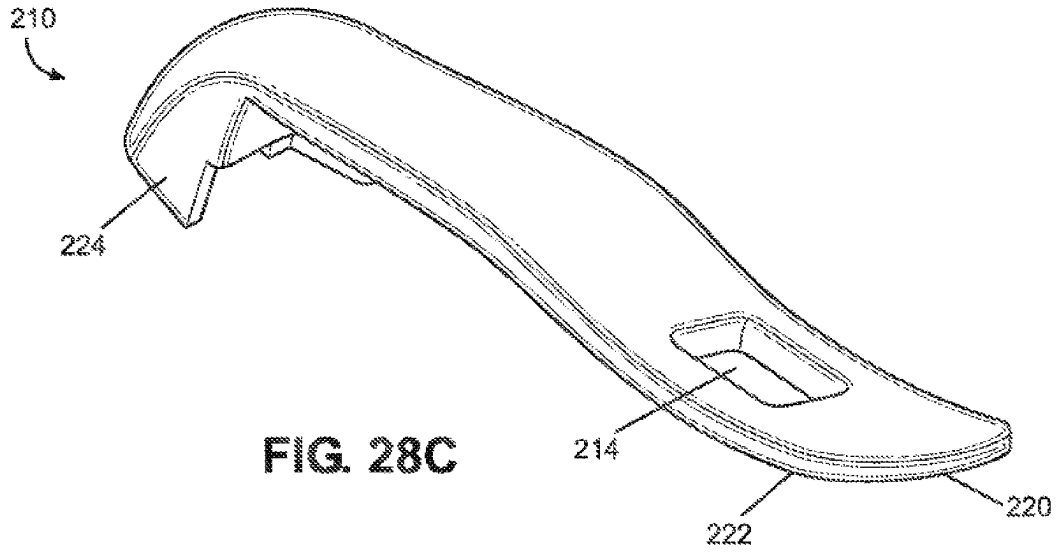


FIG. 28A



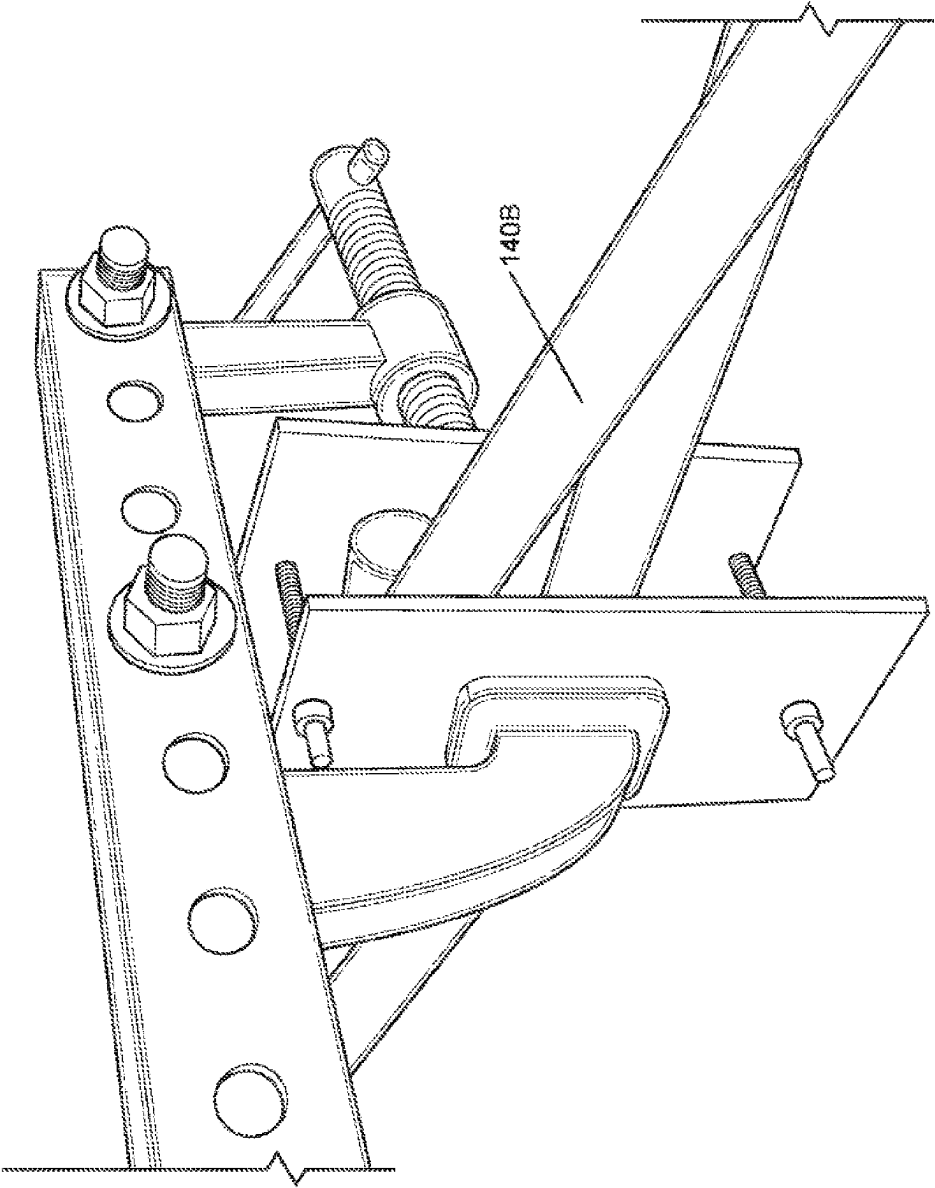
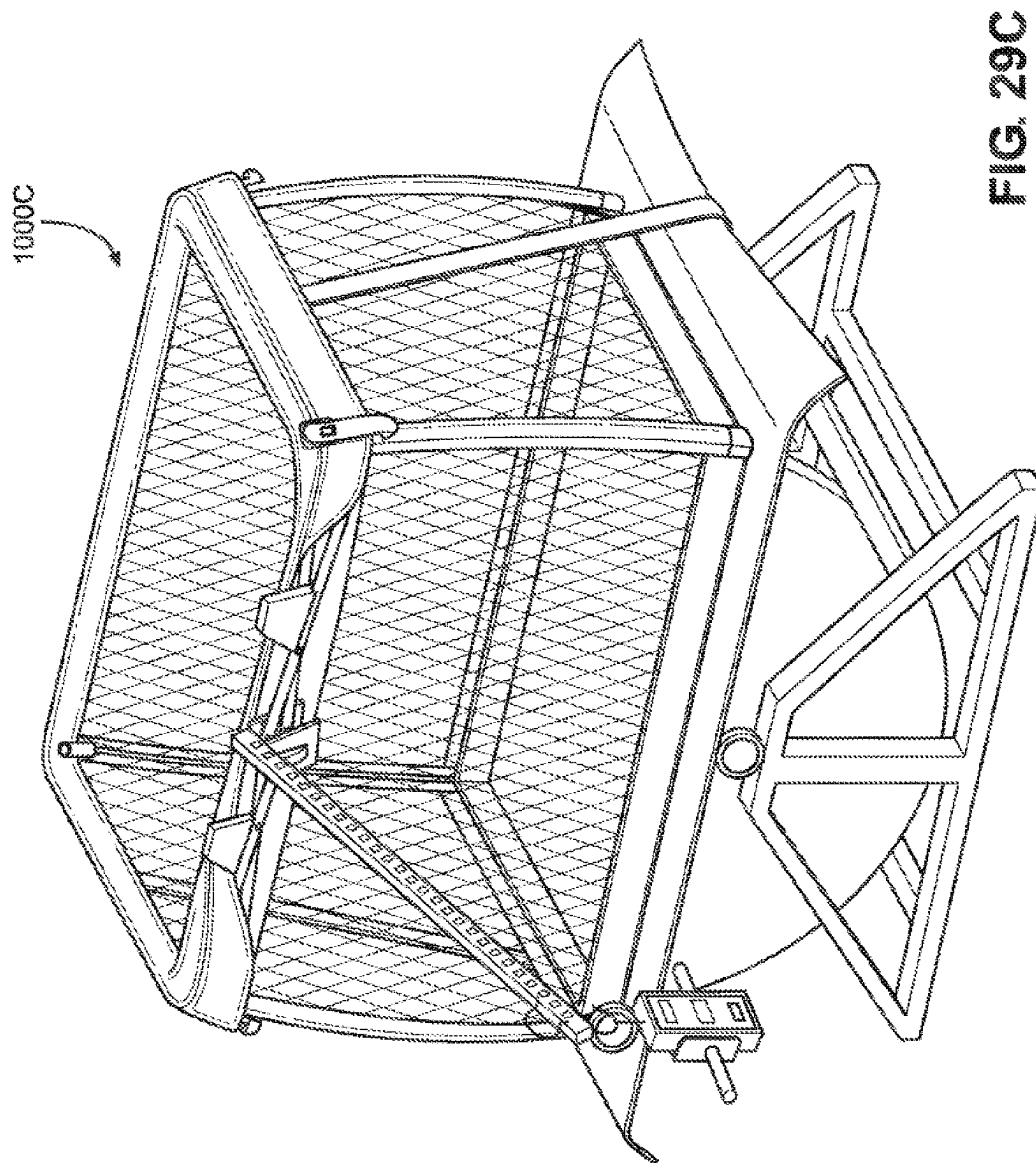
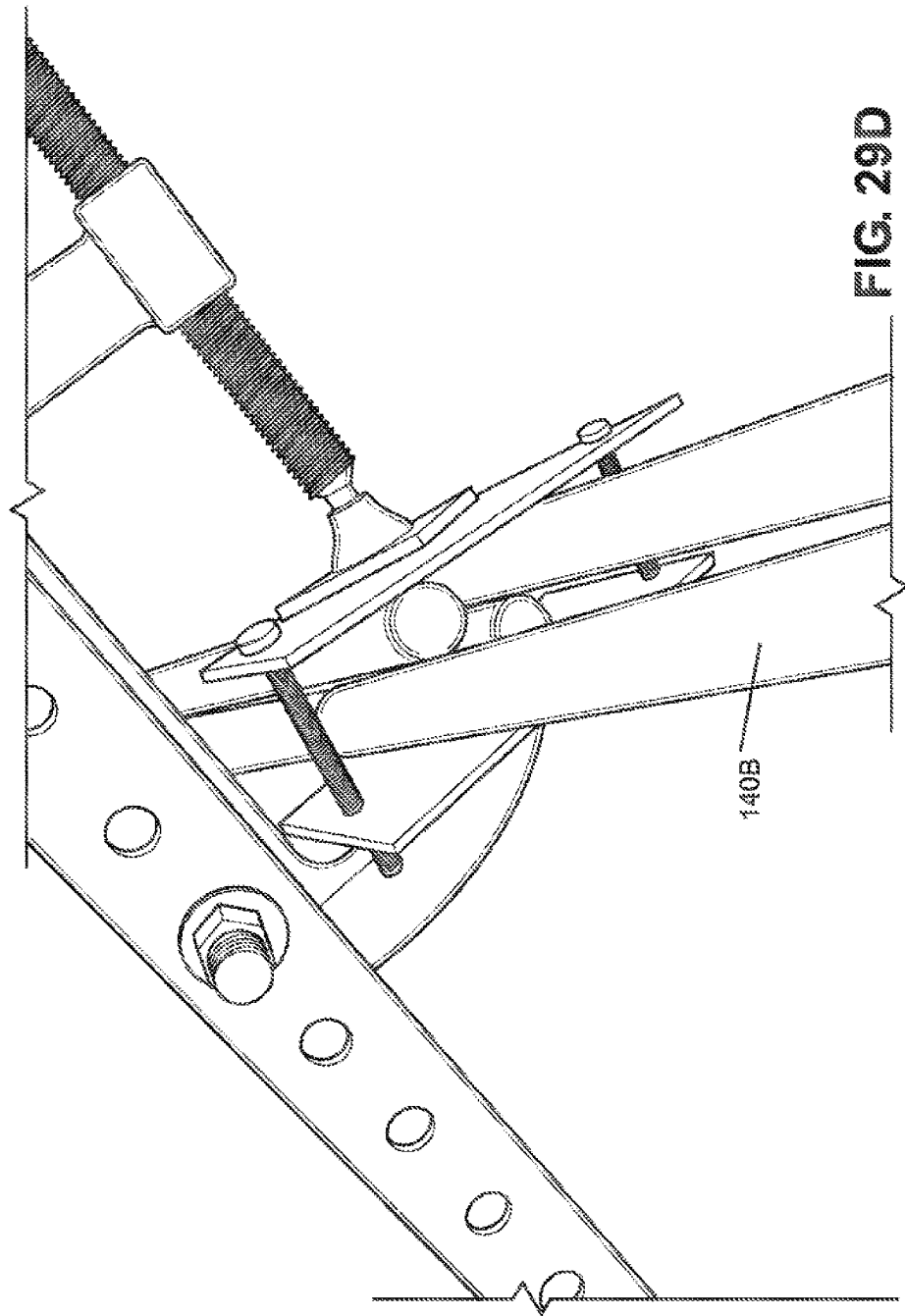


FIG. 29B





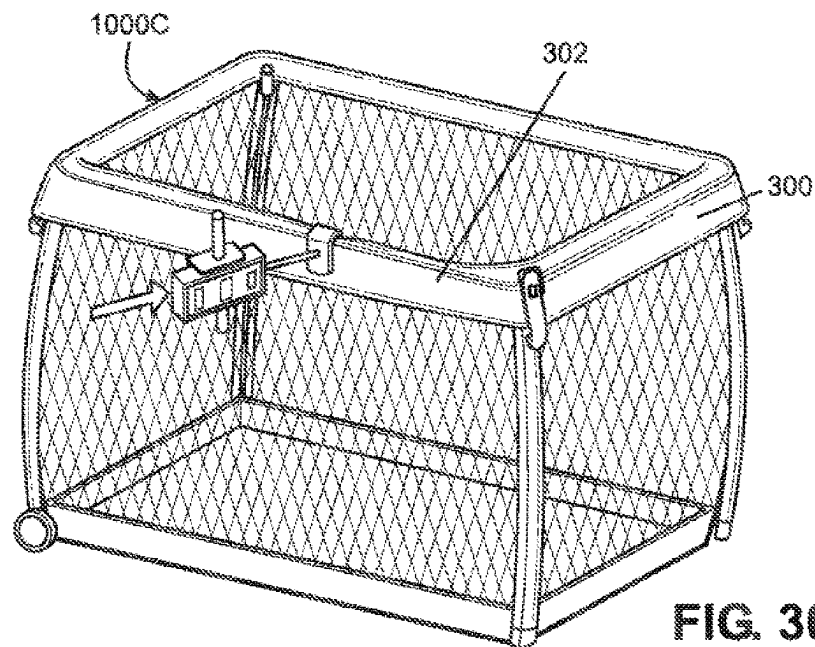


FIG. 30A

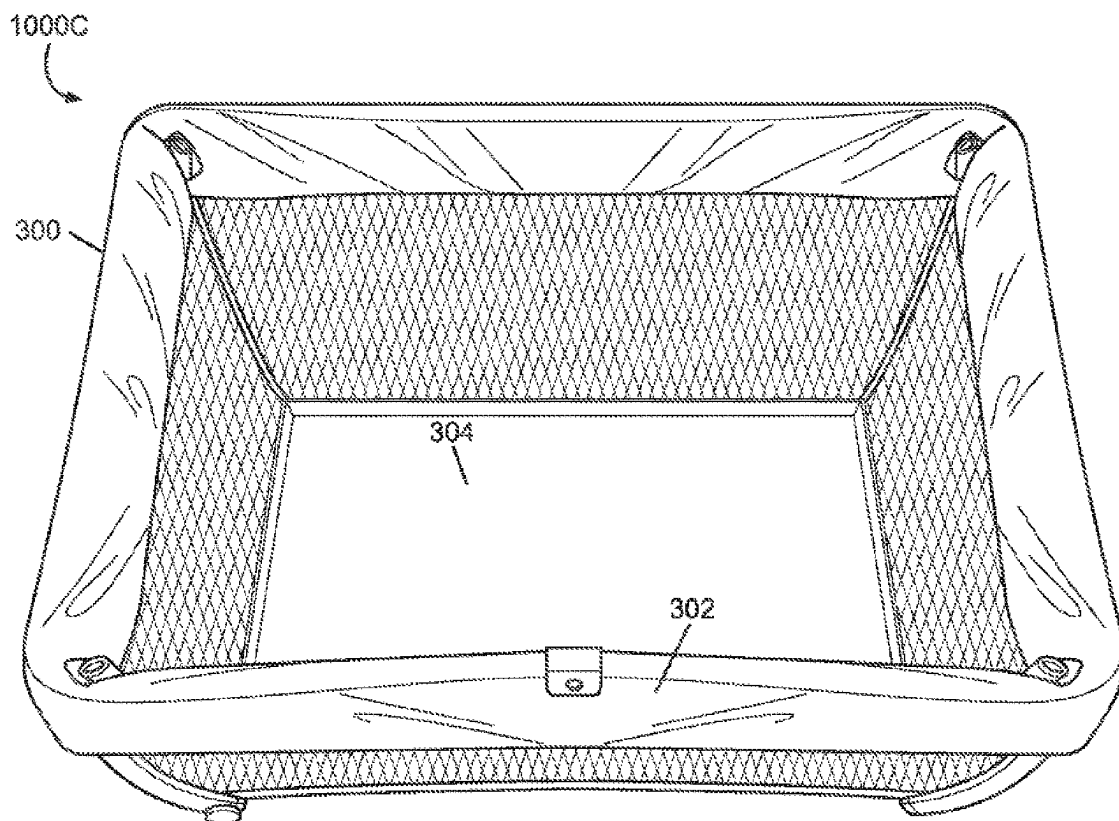
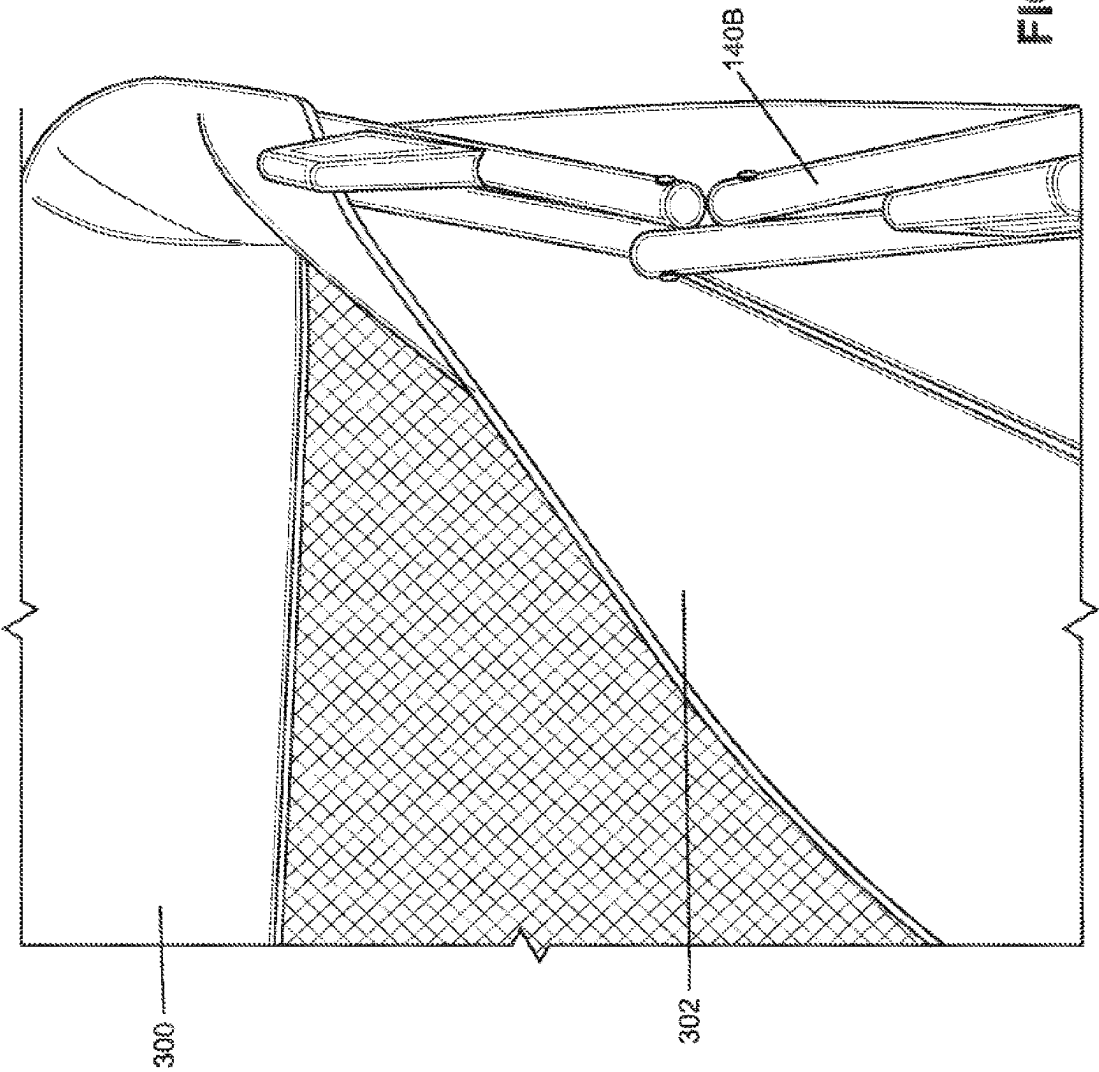


FIG. 30B



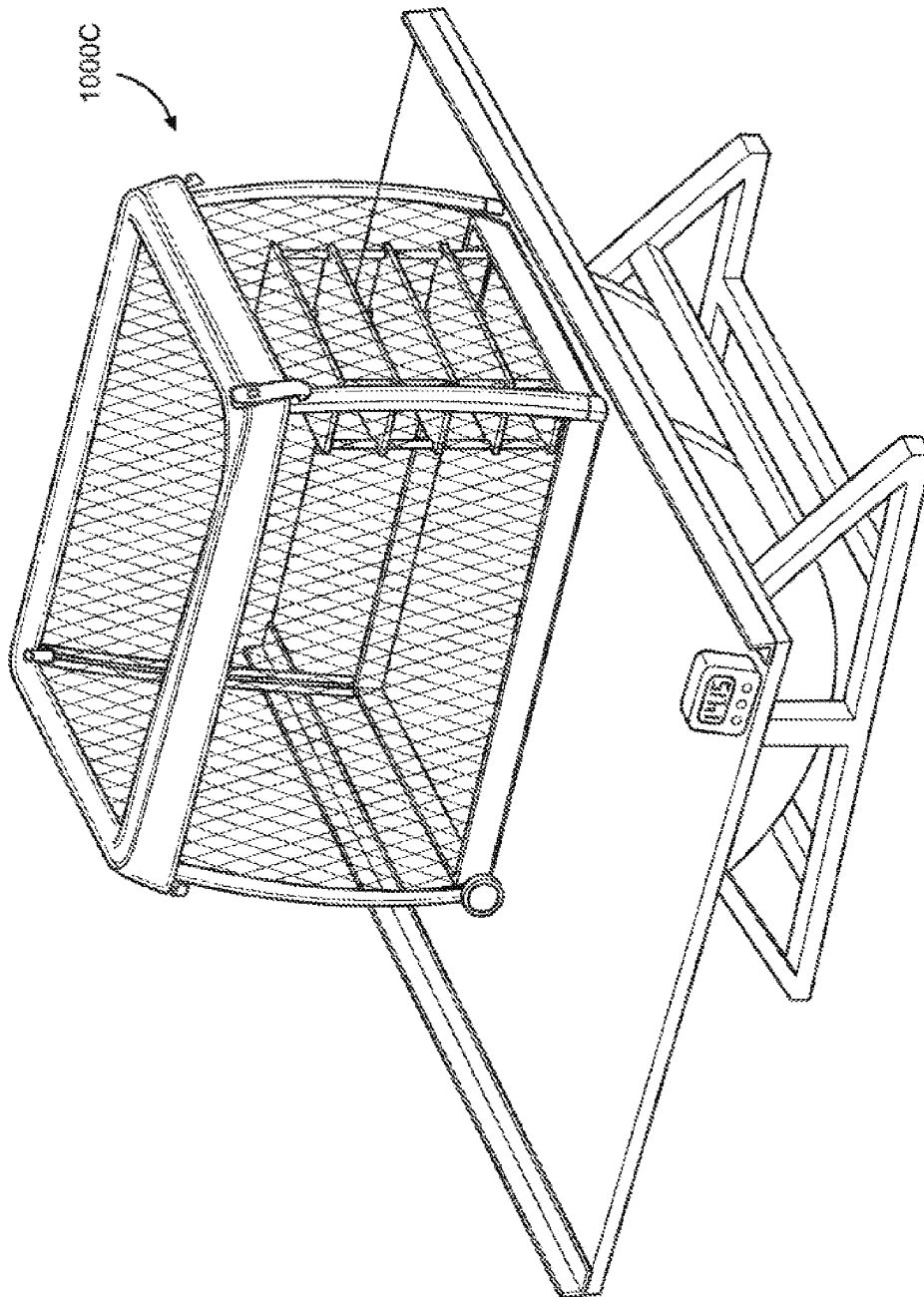


FIG. 31

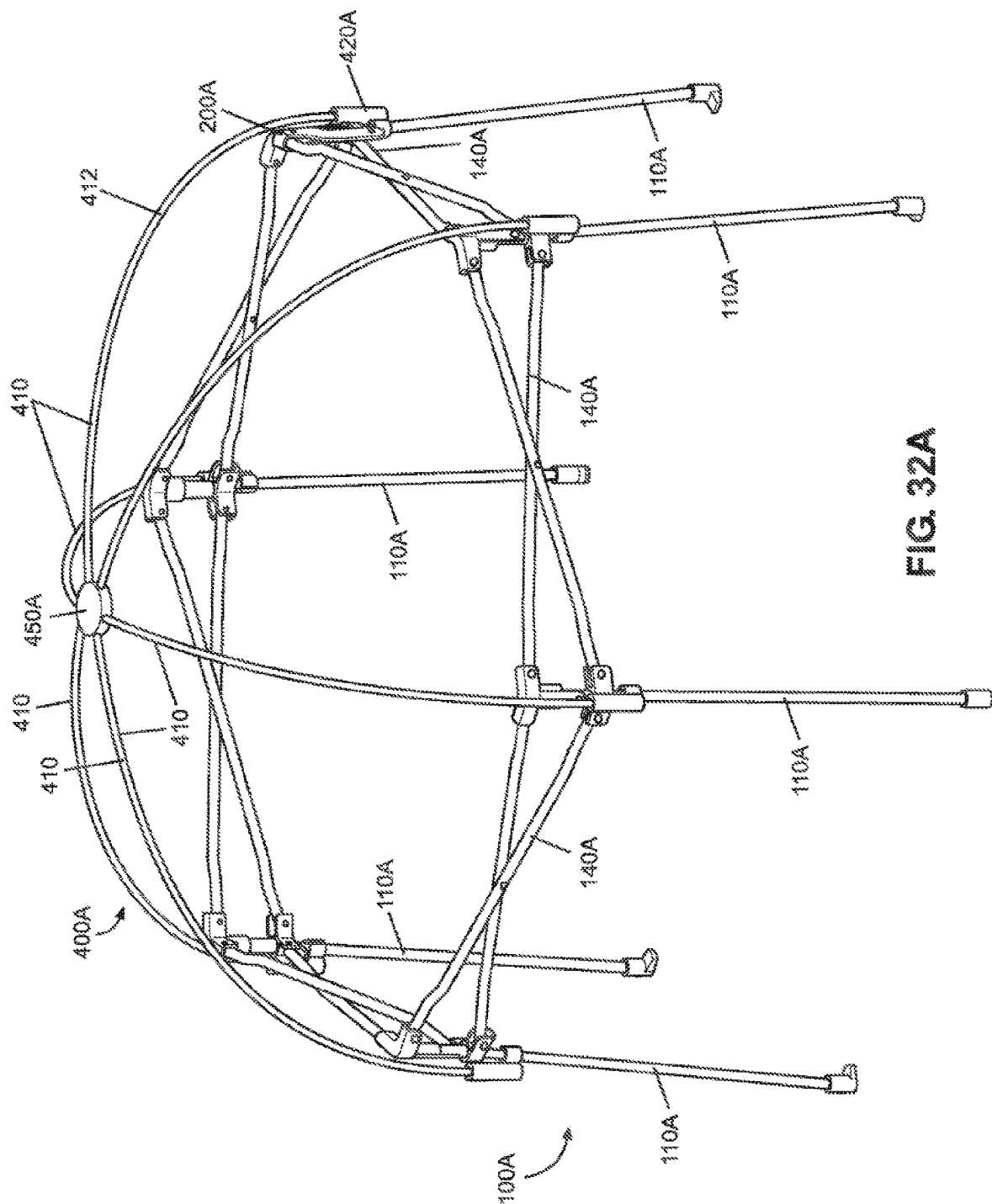


FIG. 32A

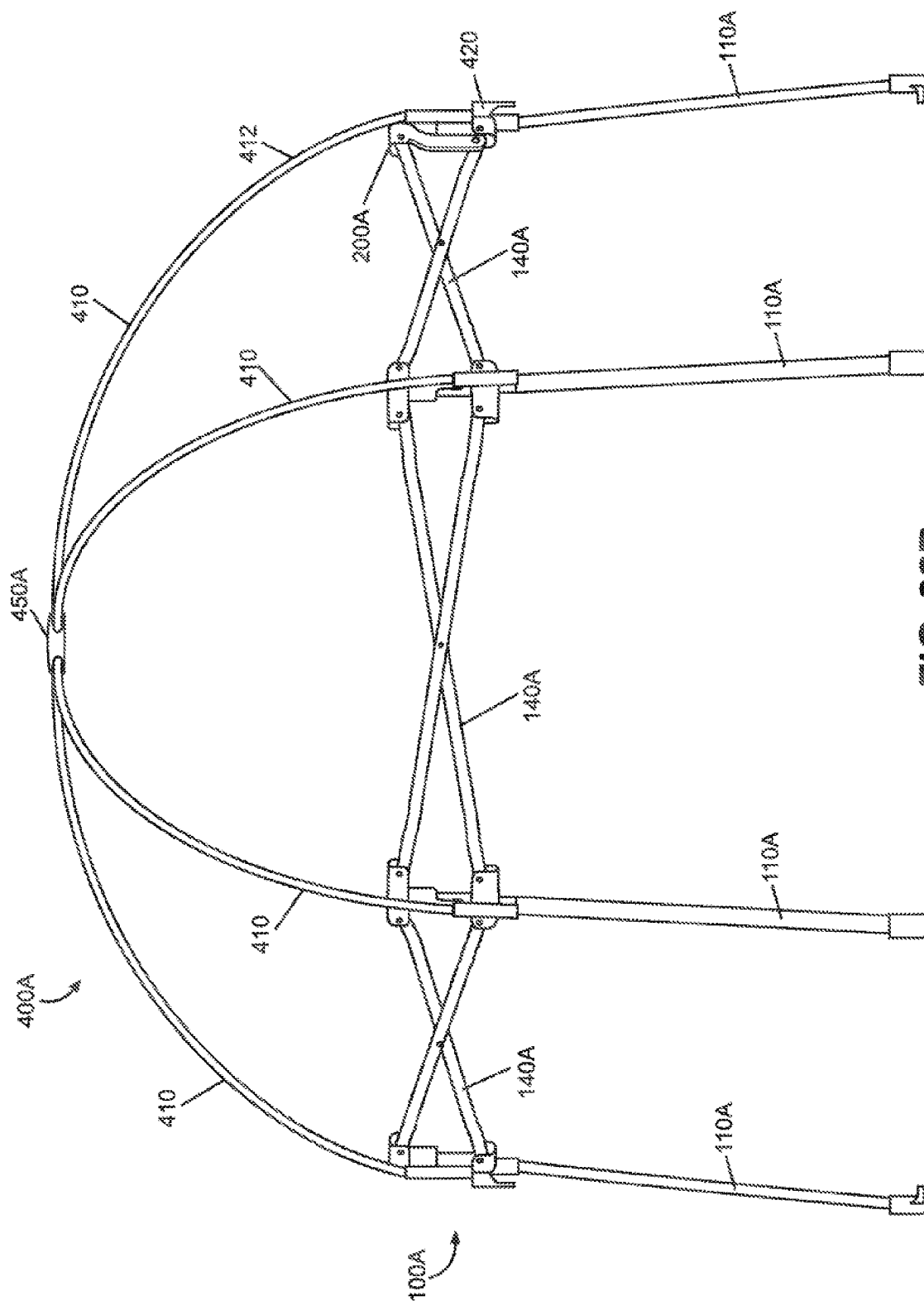


FIG. 32B

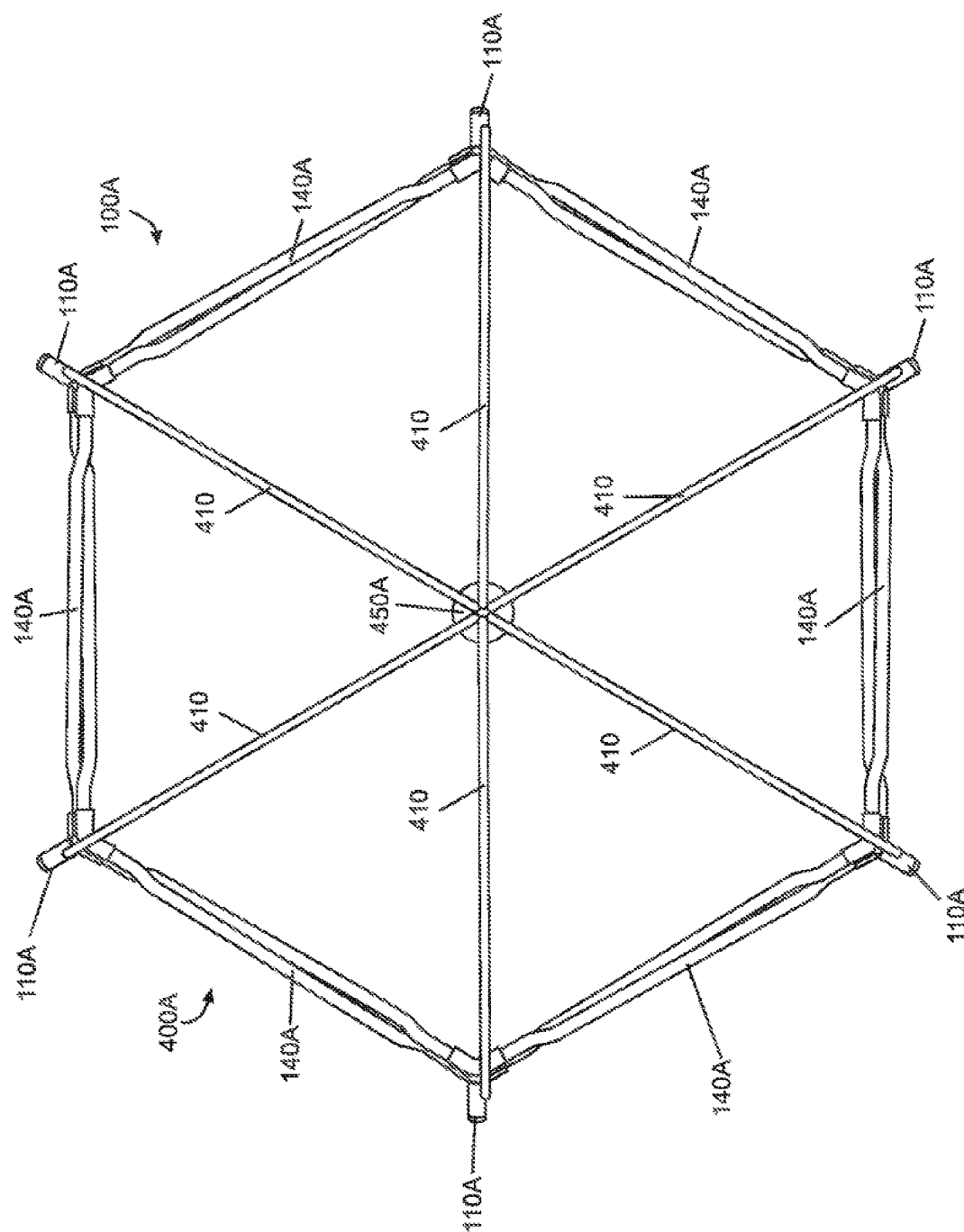
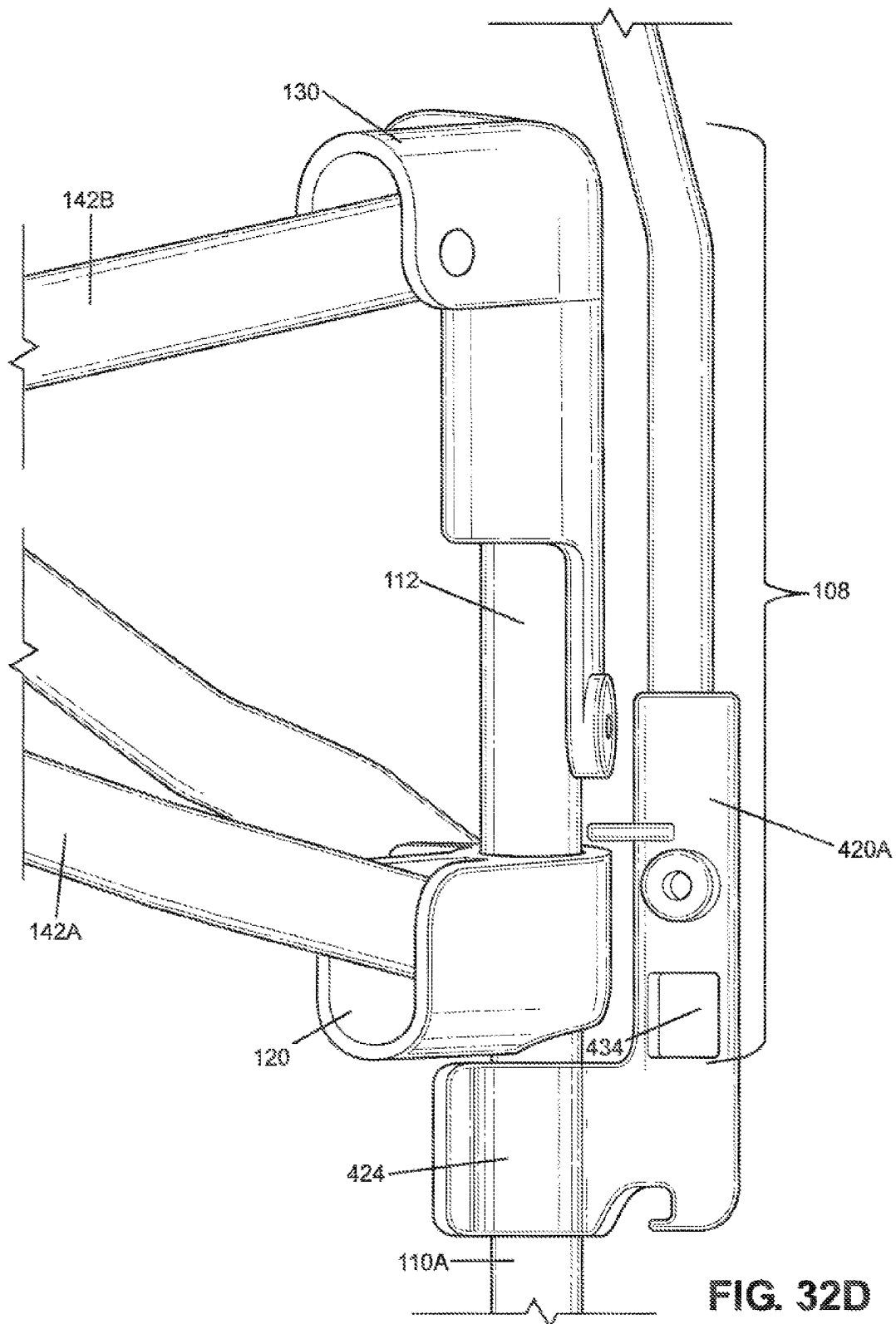


FIG. 32C



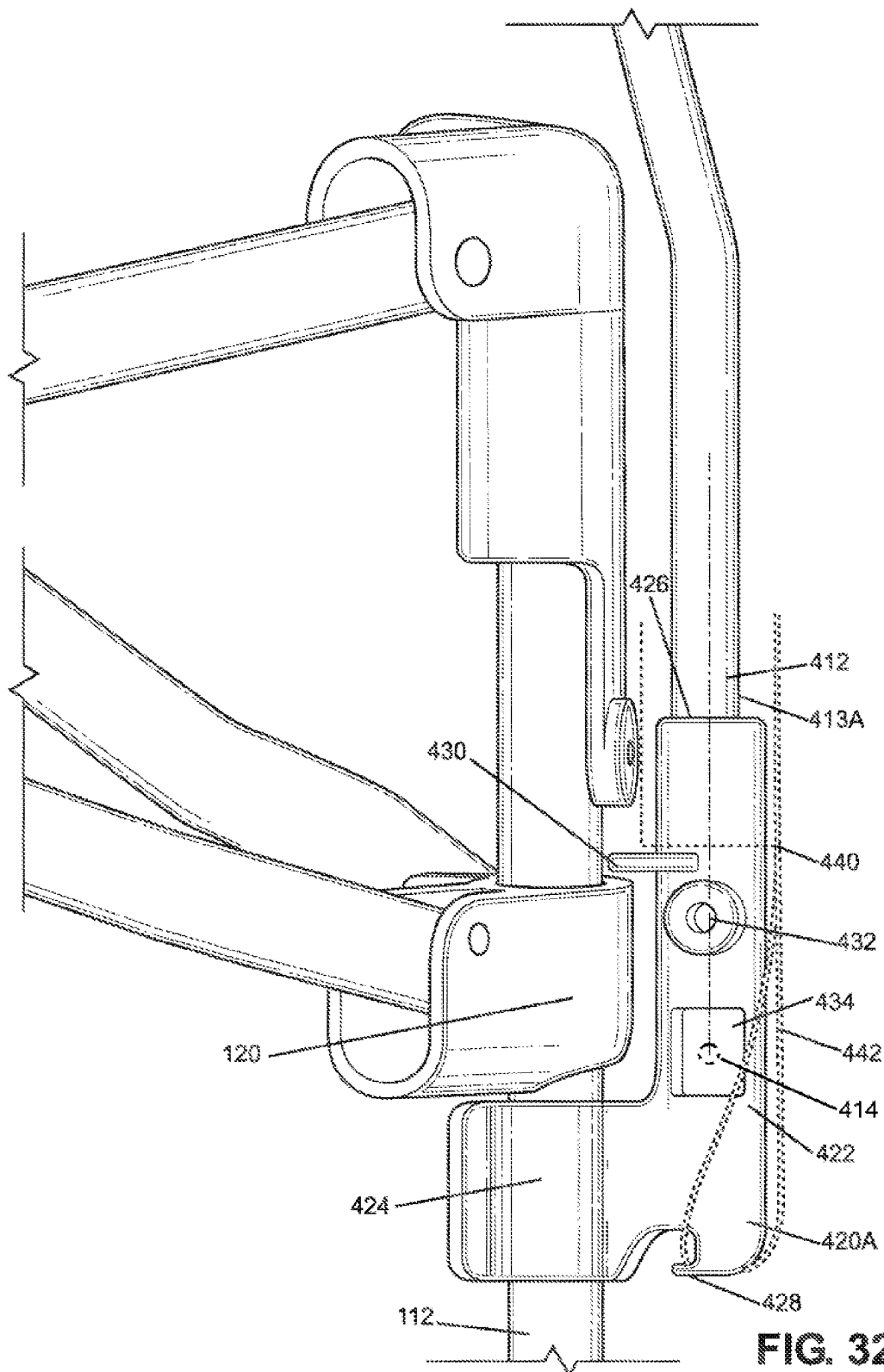


FIG. 32E

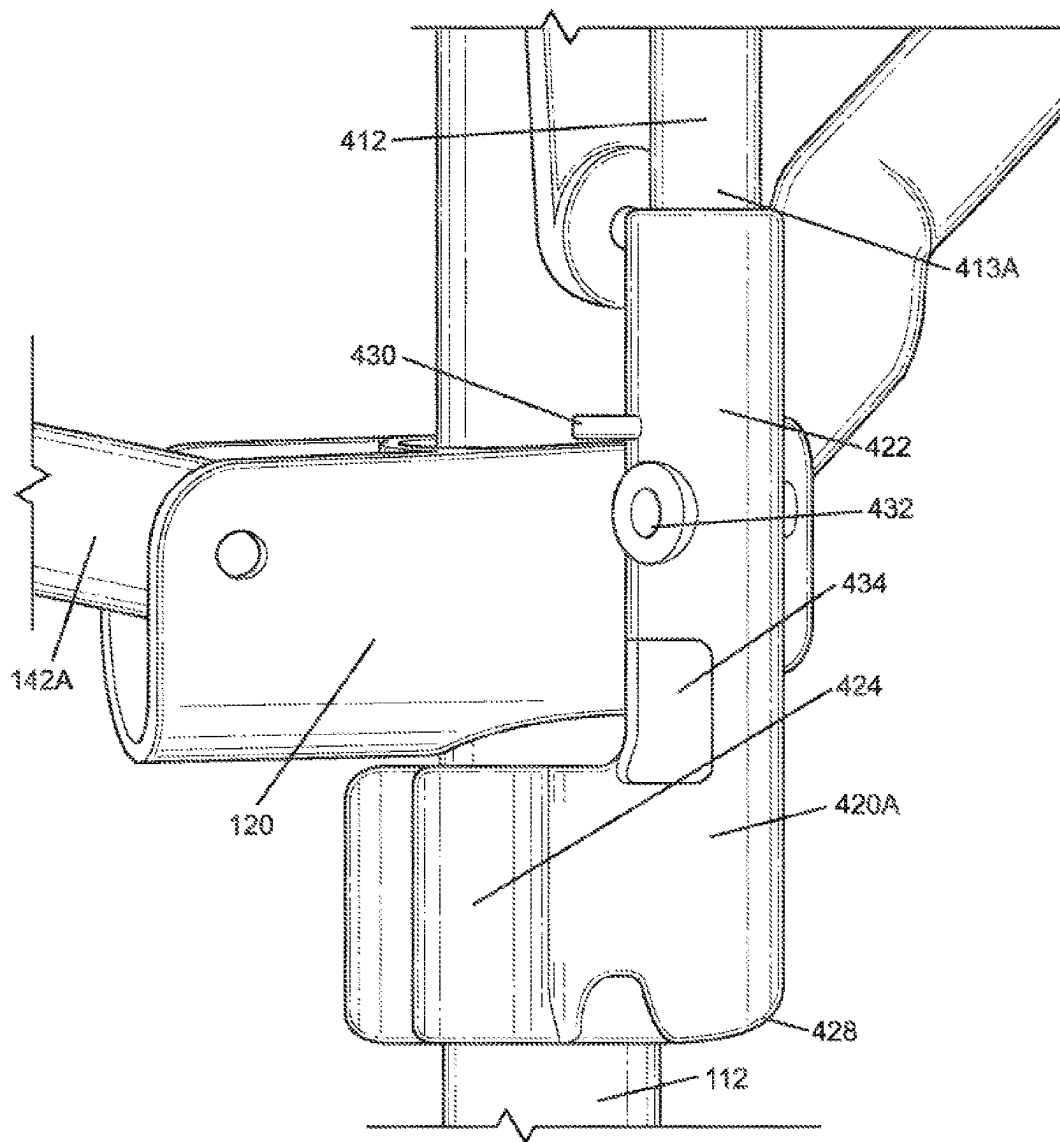
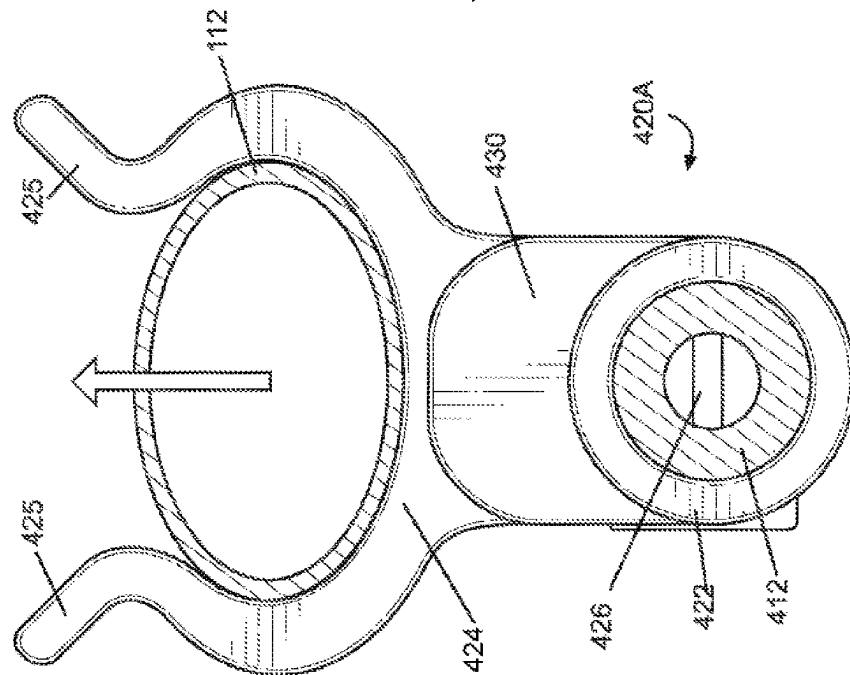
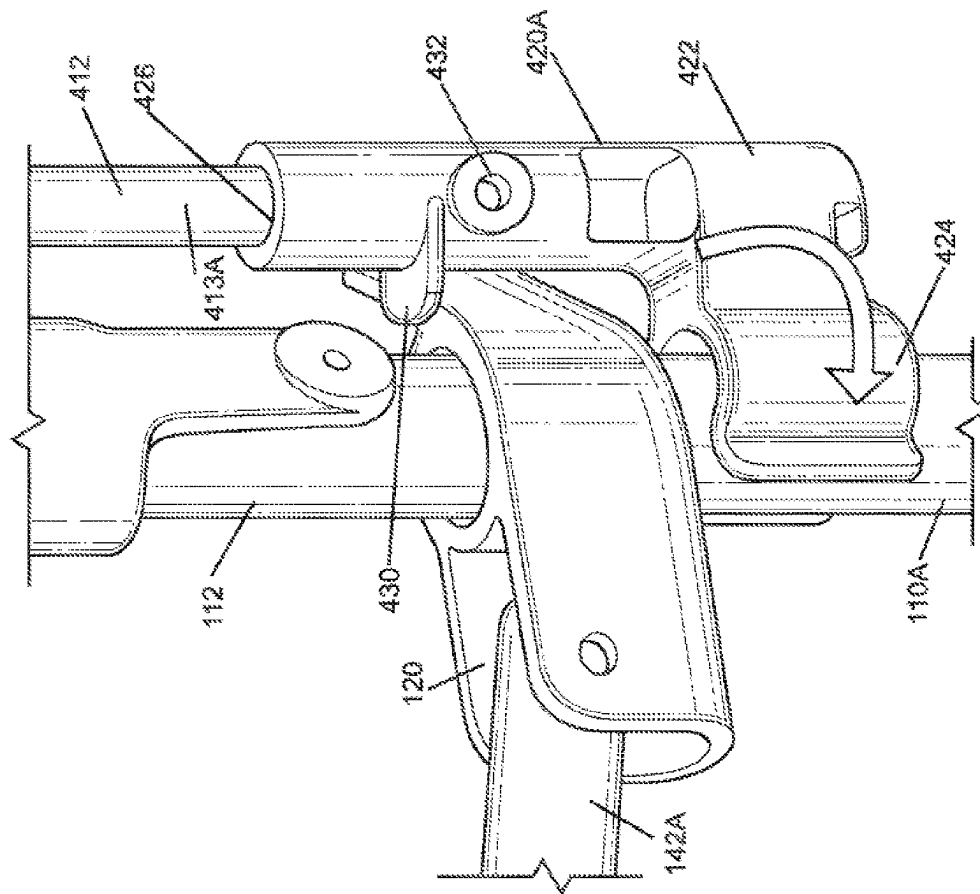


FIG. 32F



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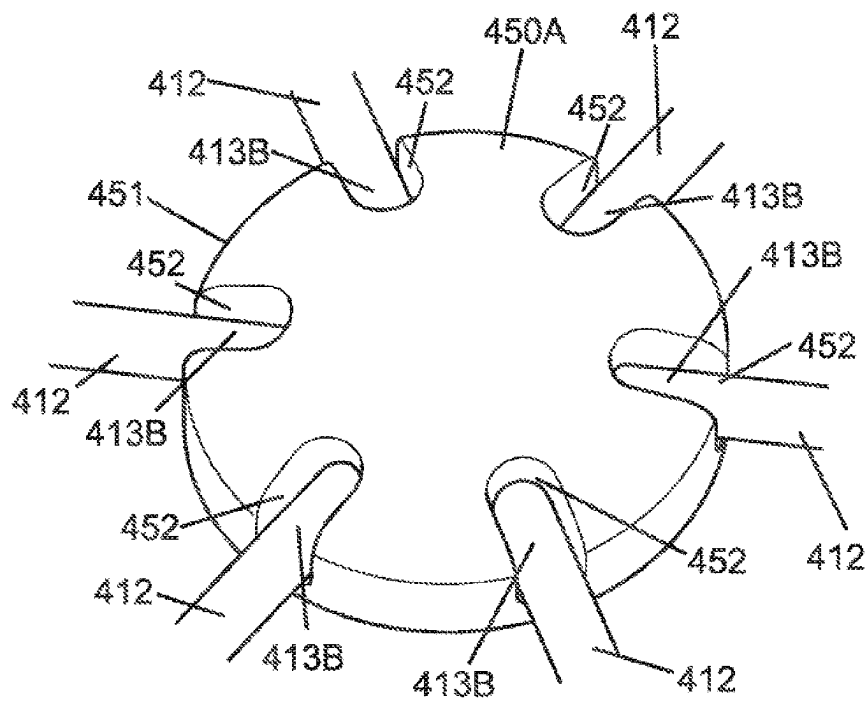


FIG. 34A

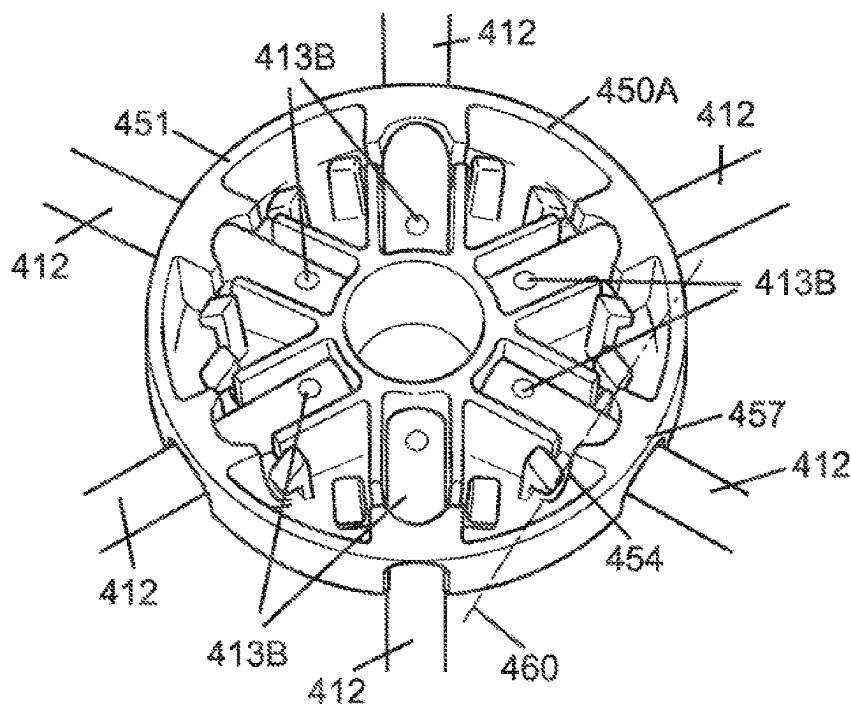


FIG. 34B

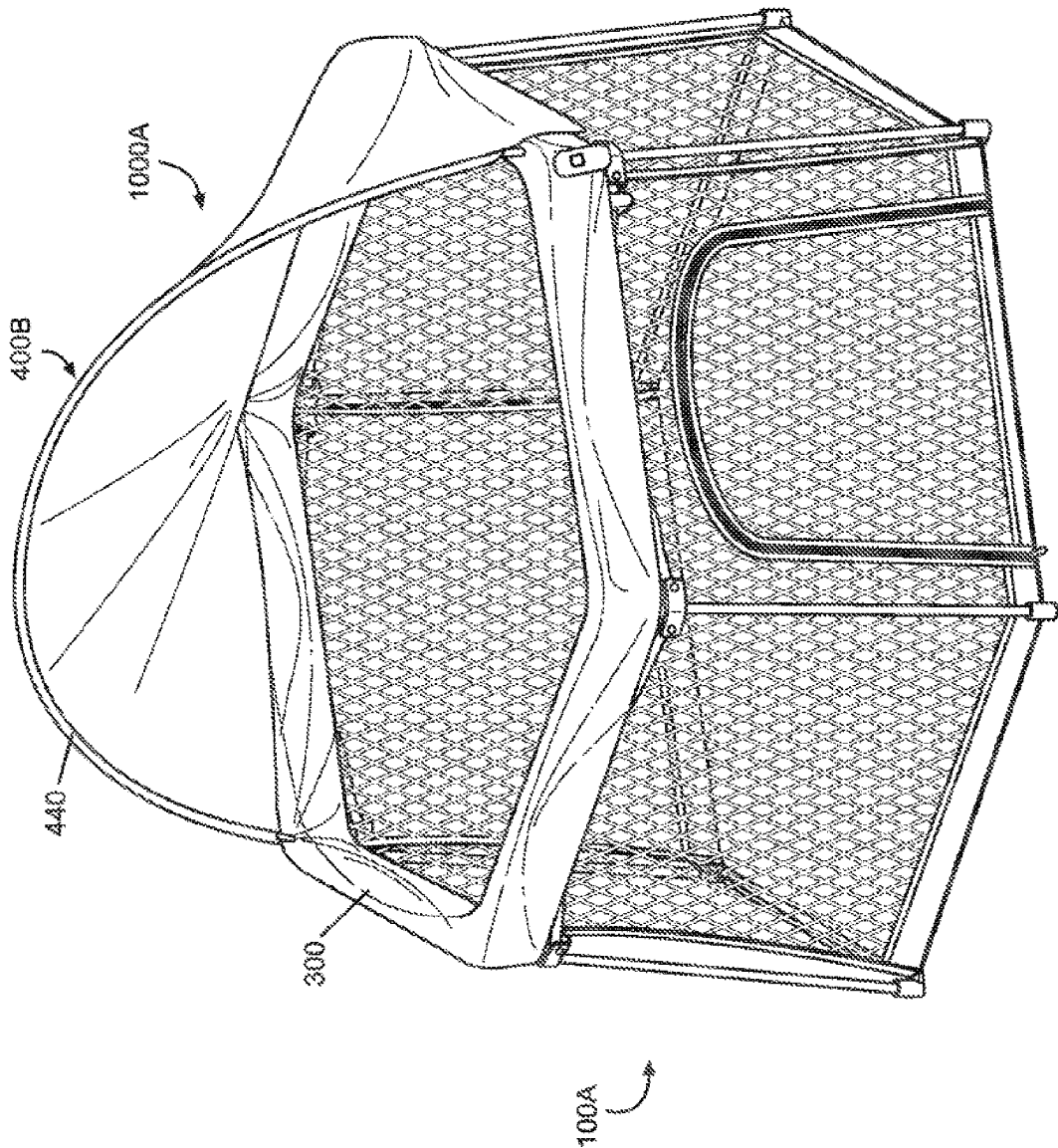


FIG. 35A

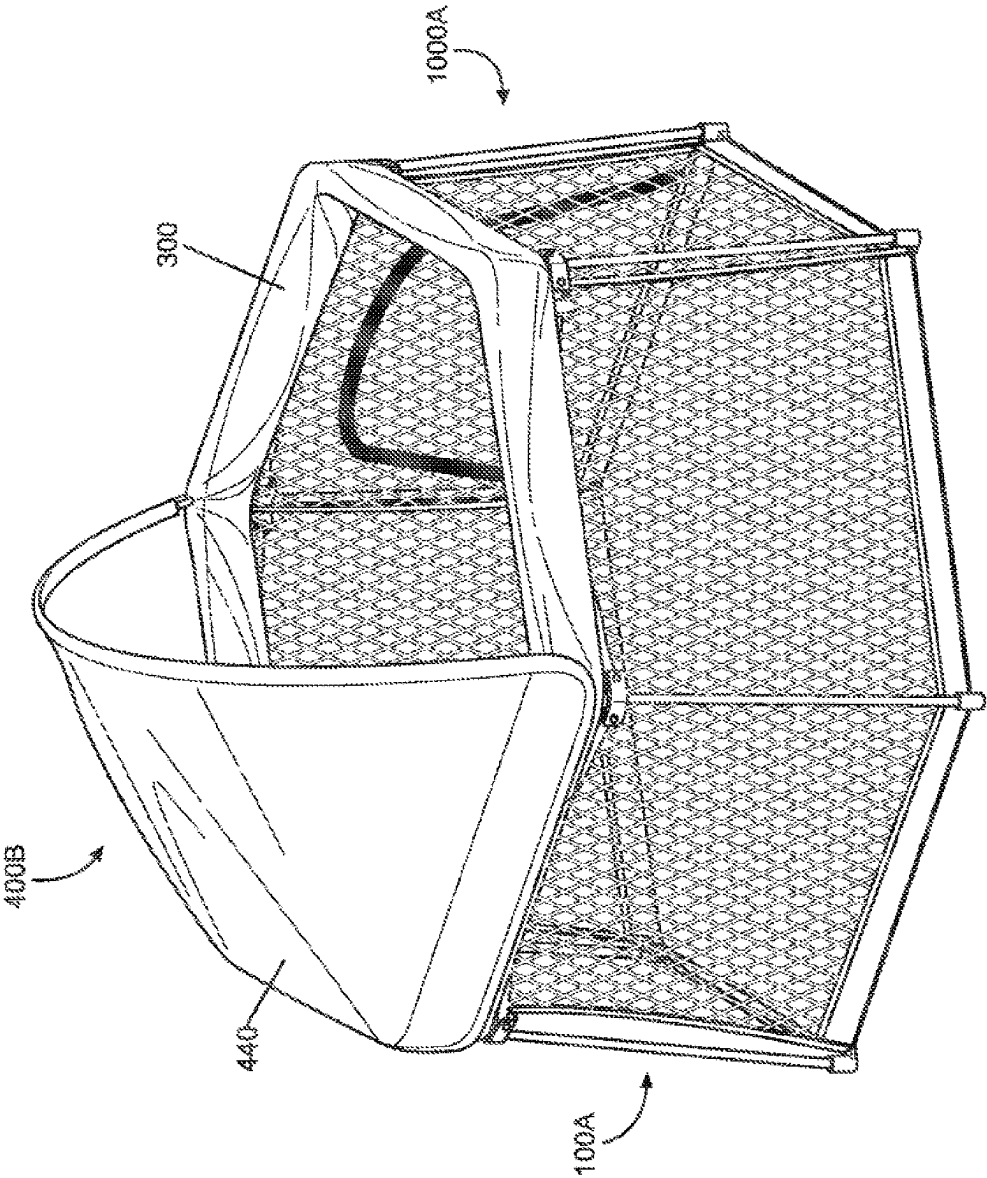
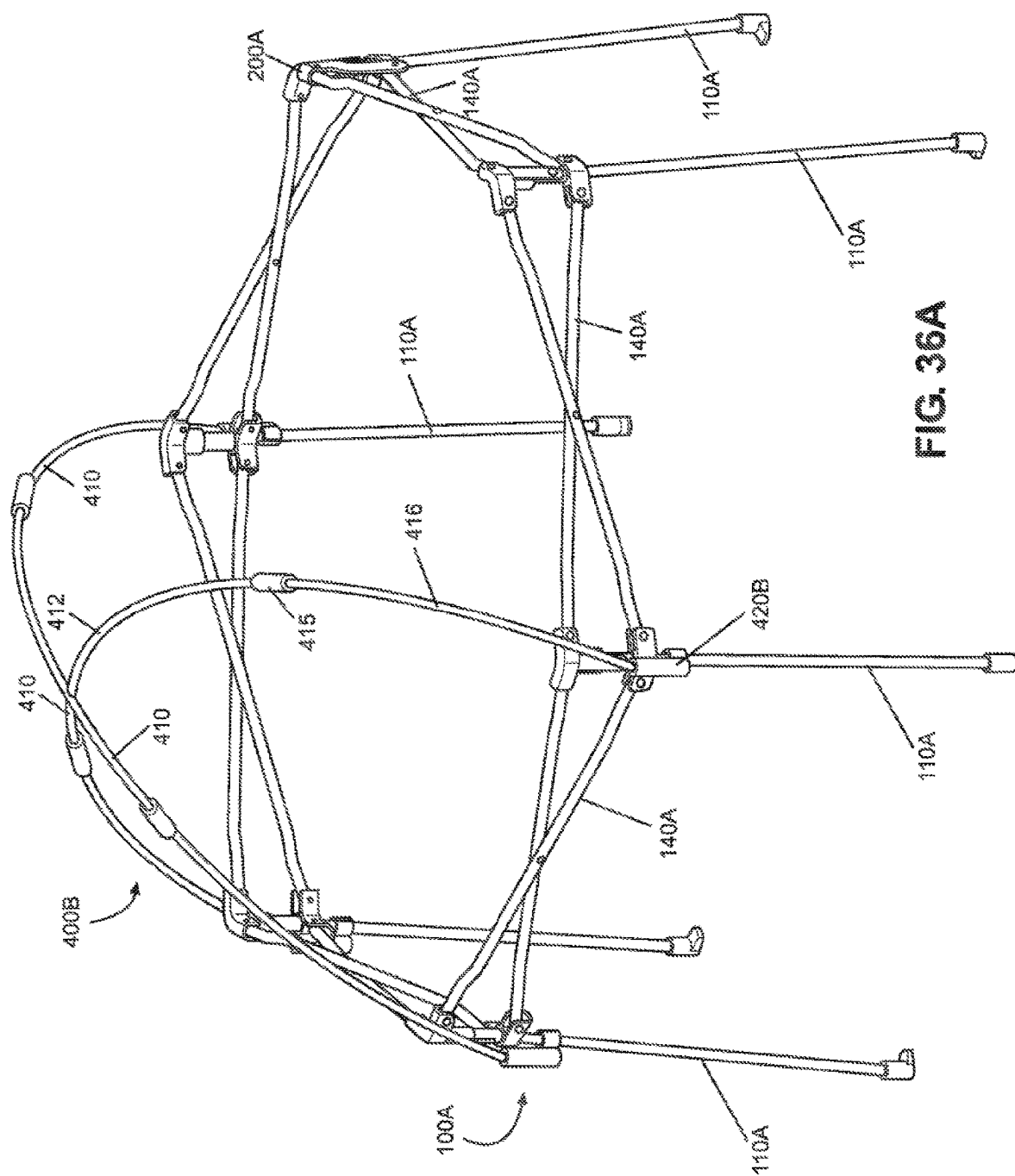


FIG. 35B



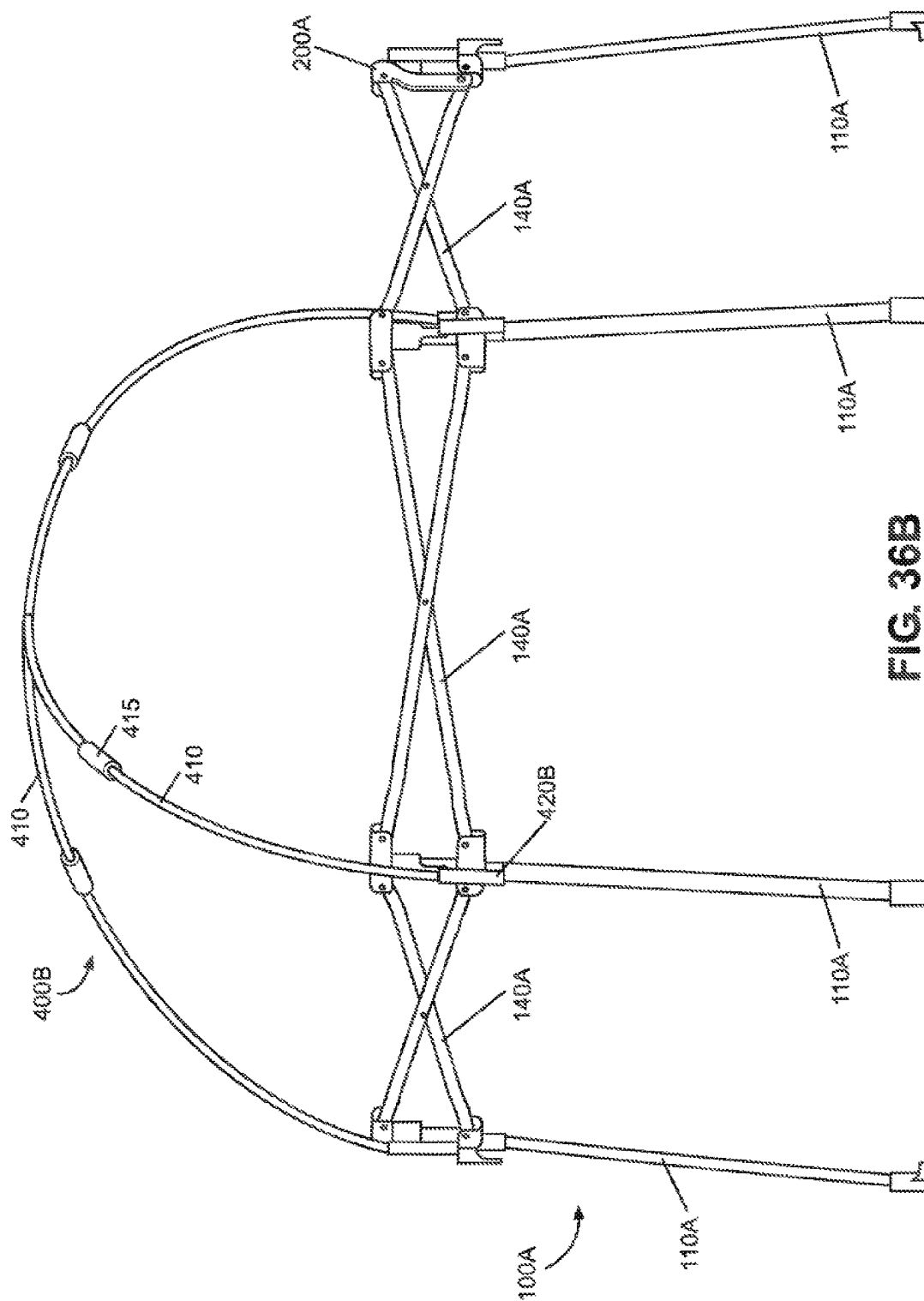


FIG. 36B

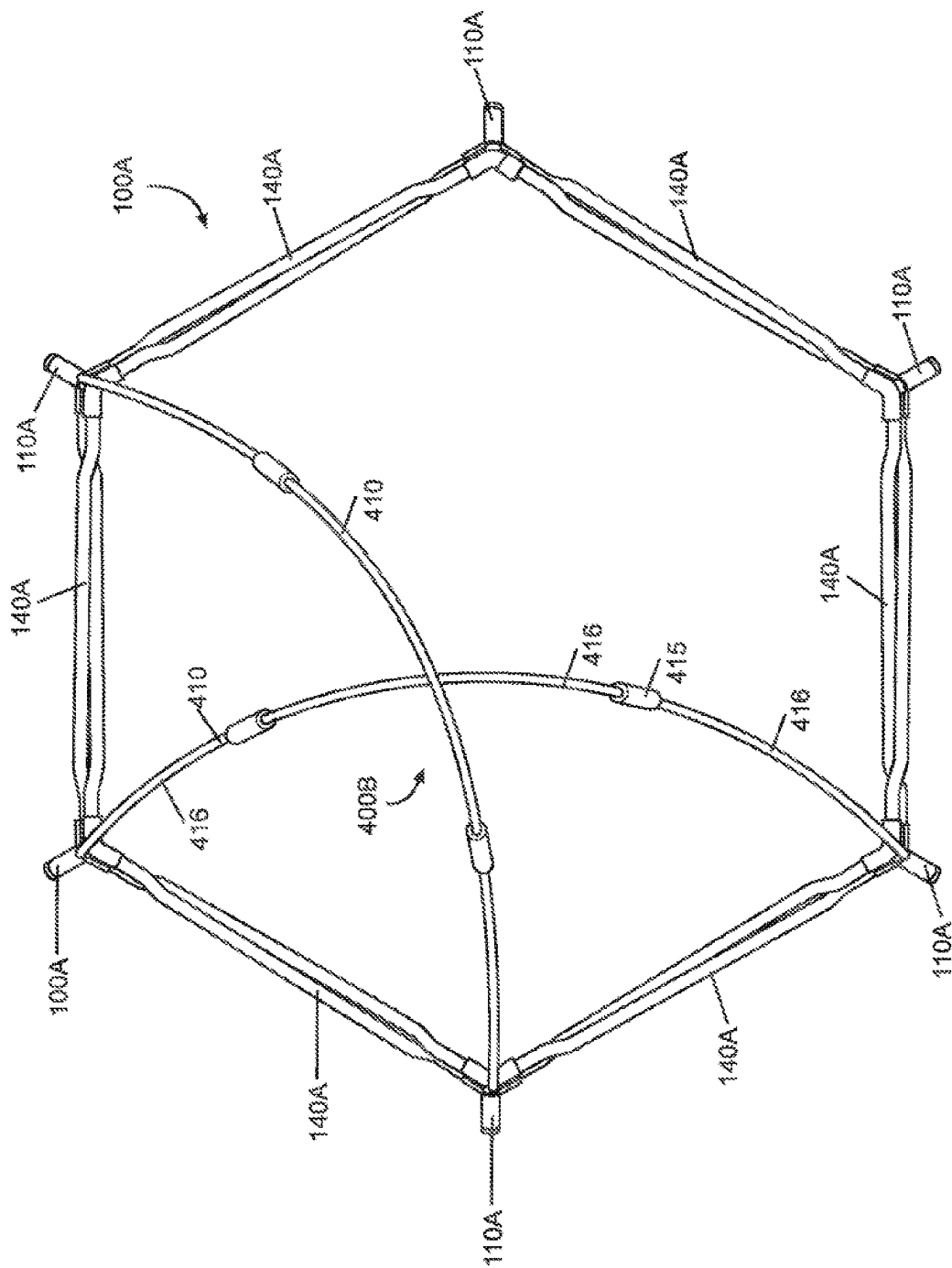


FIG. 36C

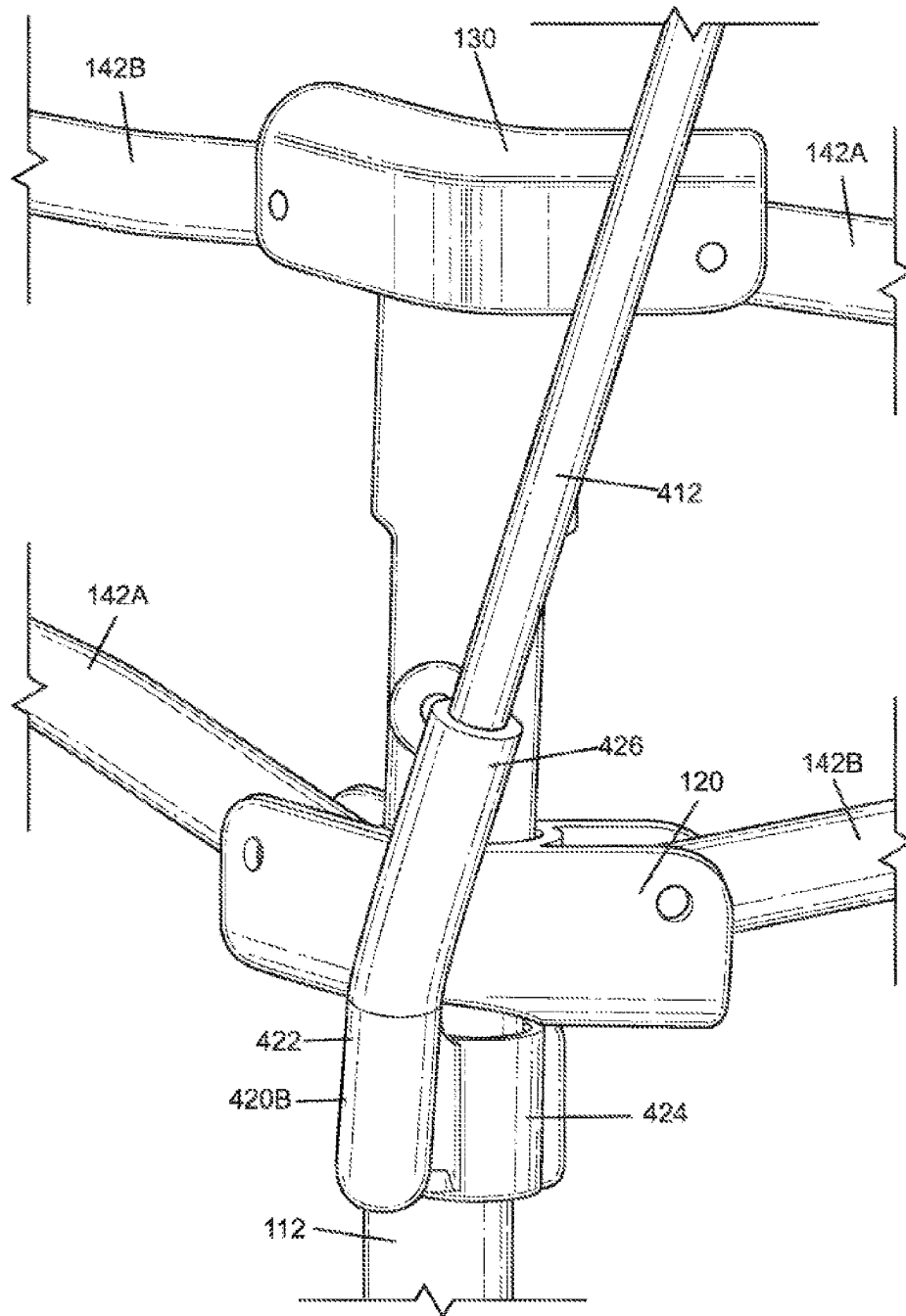


FIG. 36D

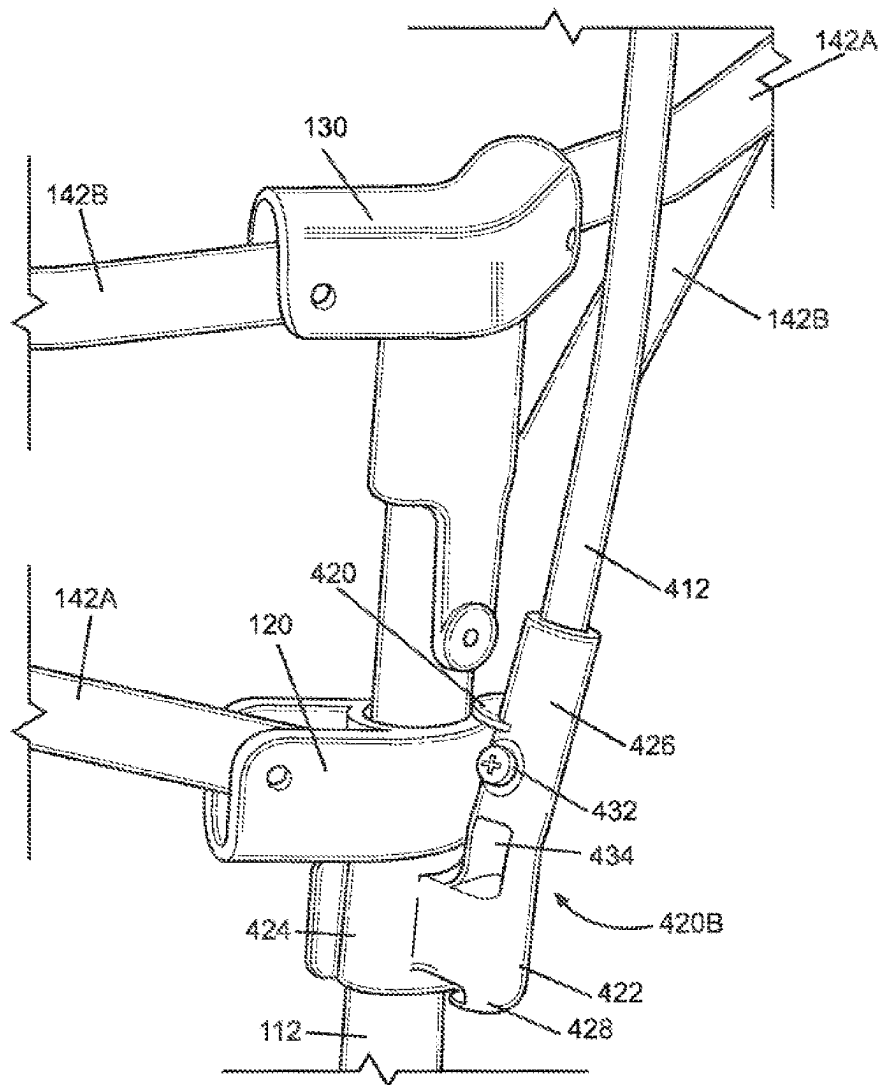
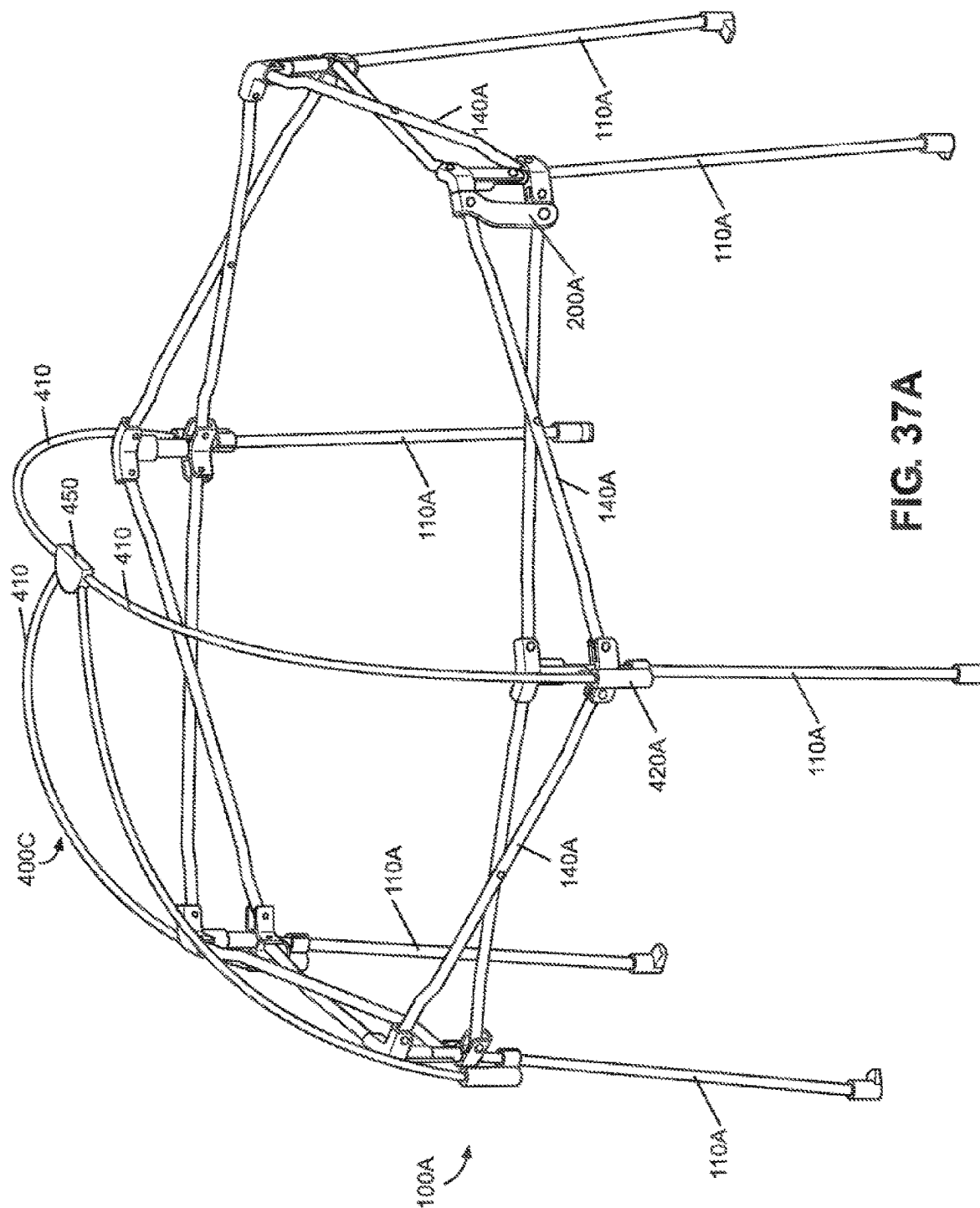
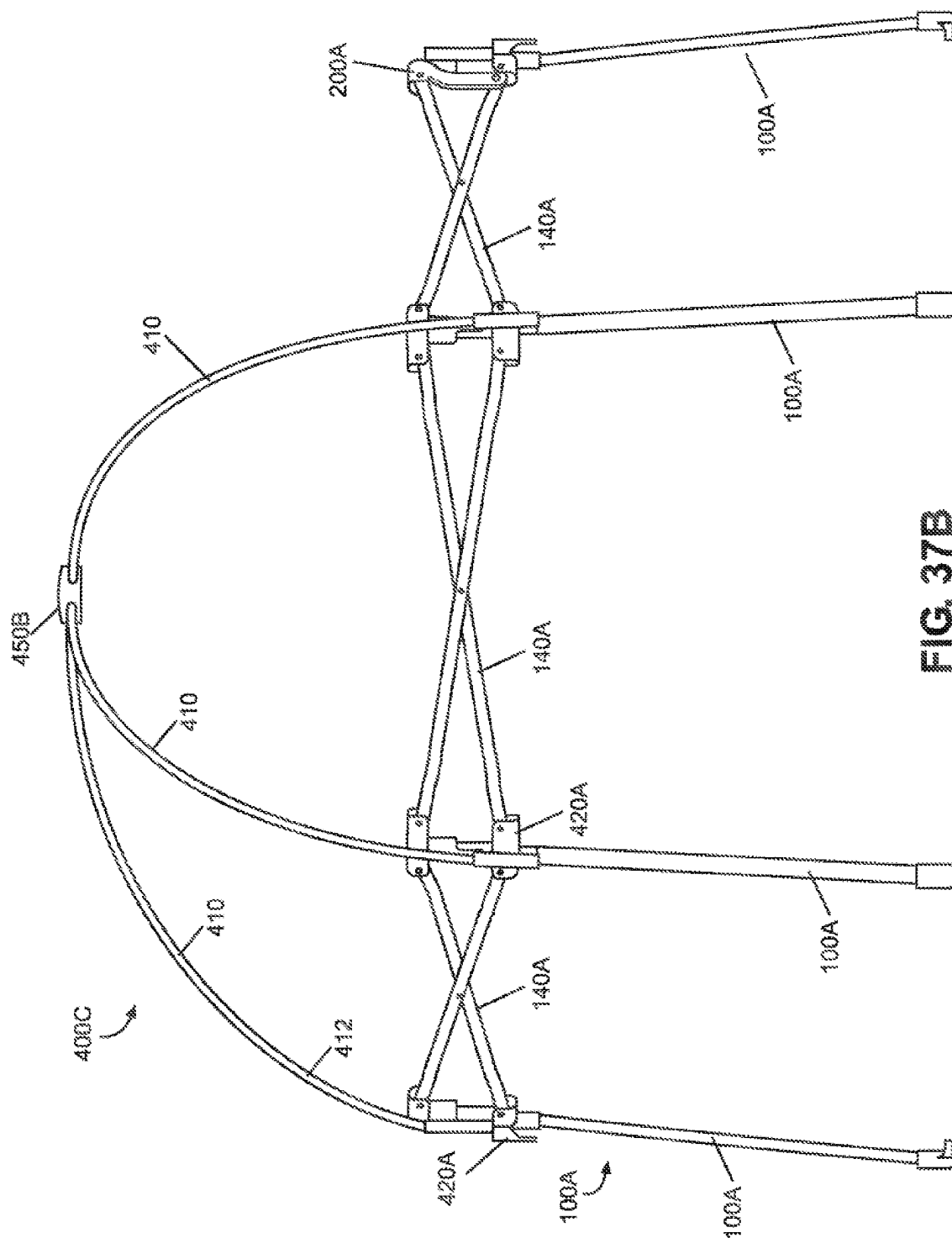
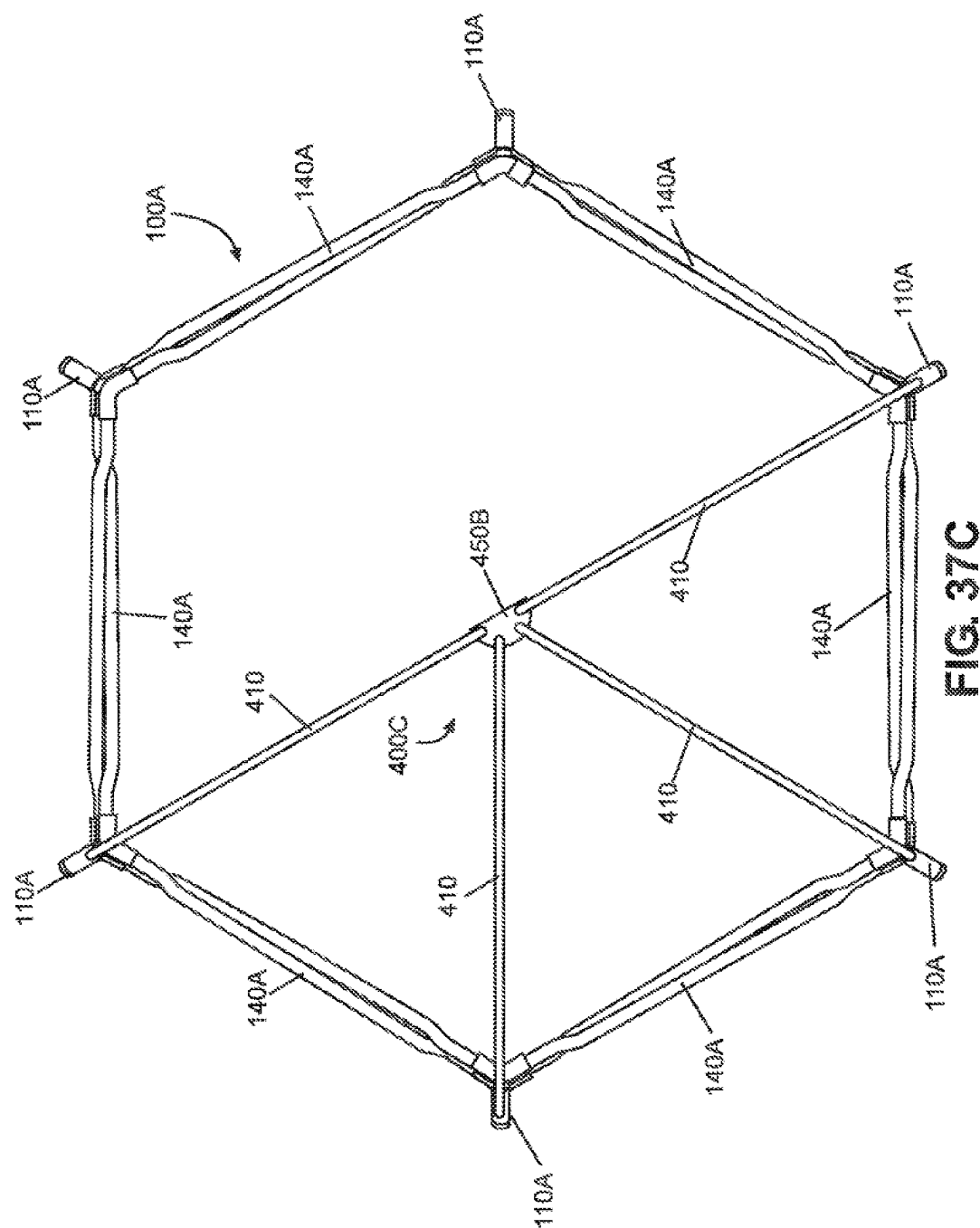


FIG. 36E







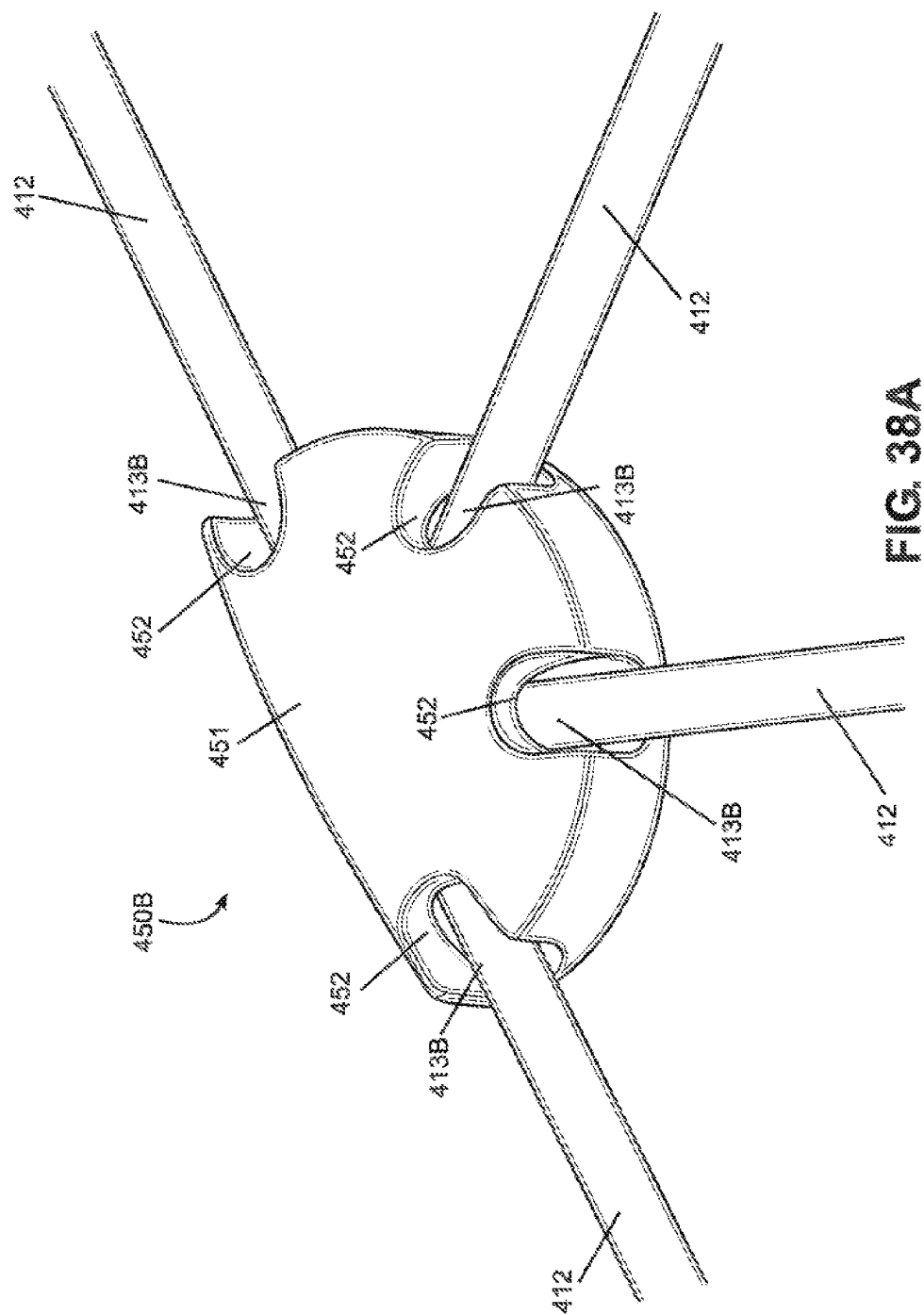


FIG. 38A

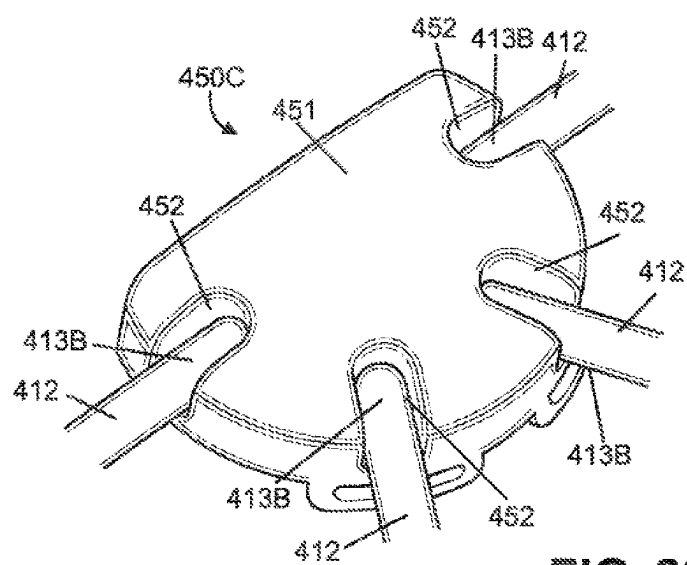
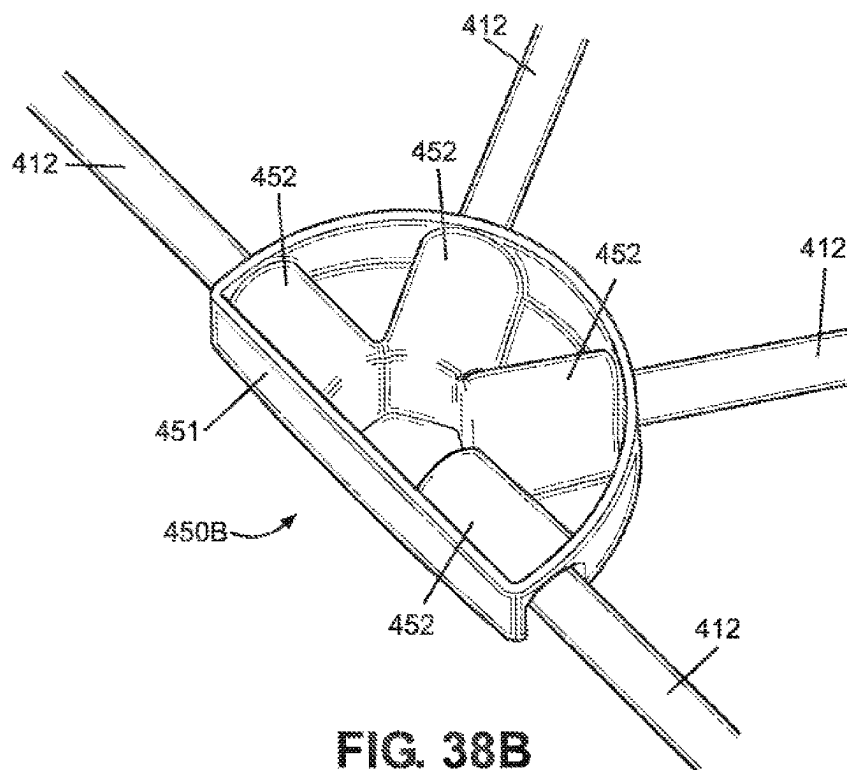


FIG. 39A

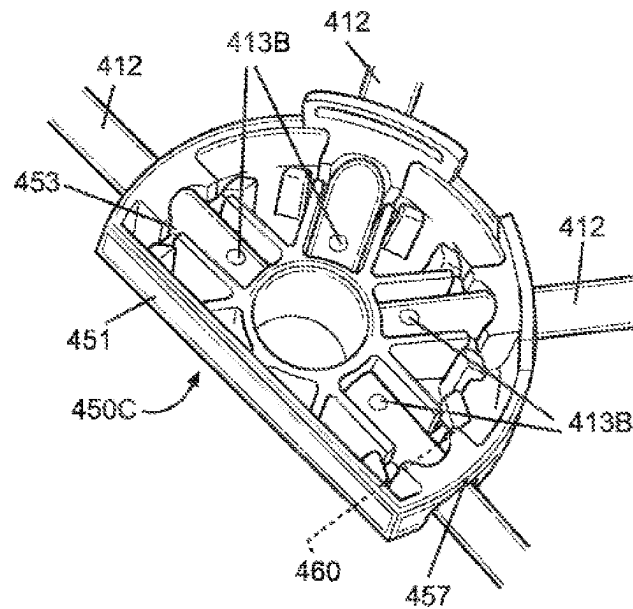


FIG. 39B

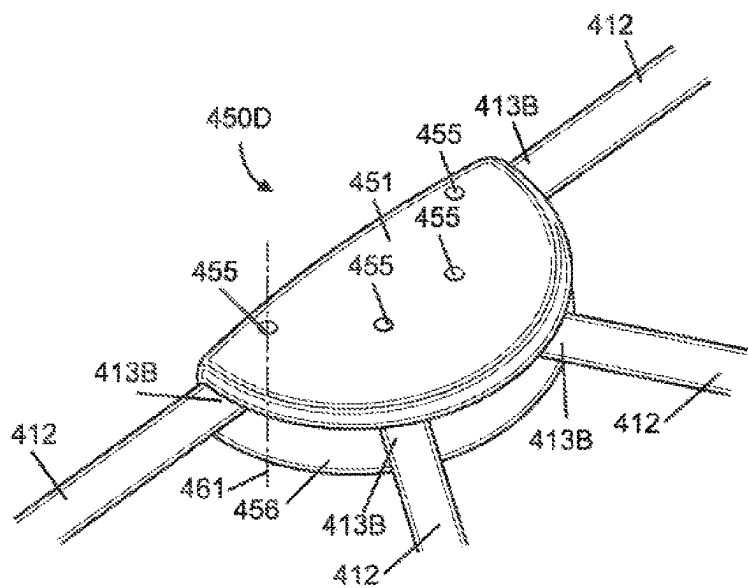


FIG. 40A

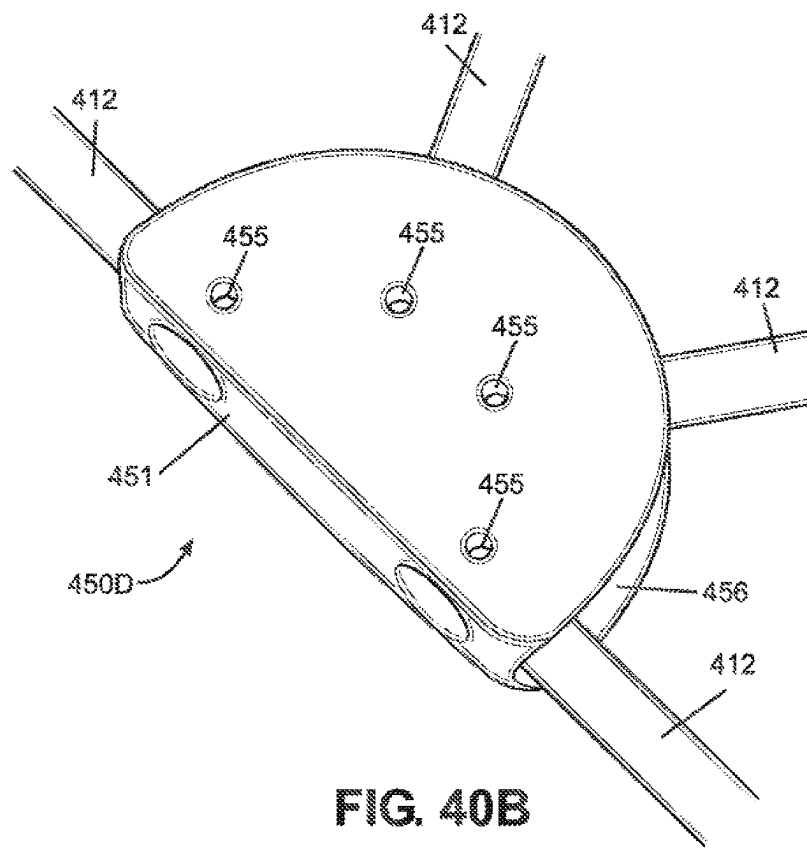
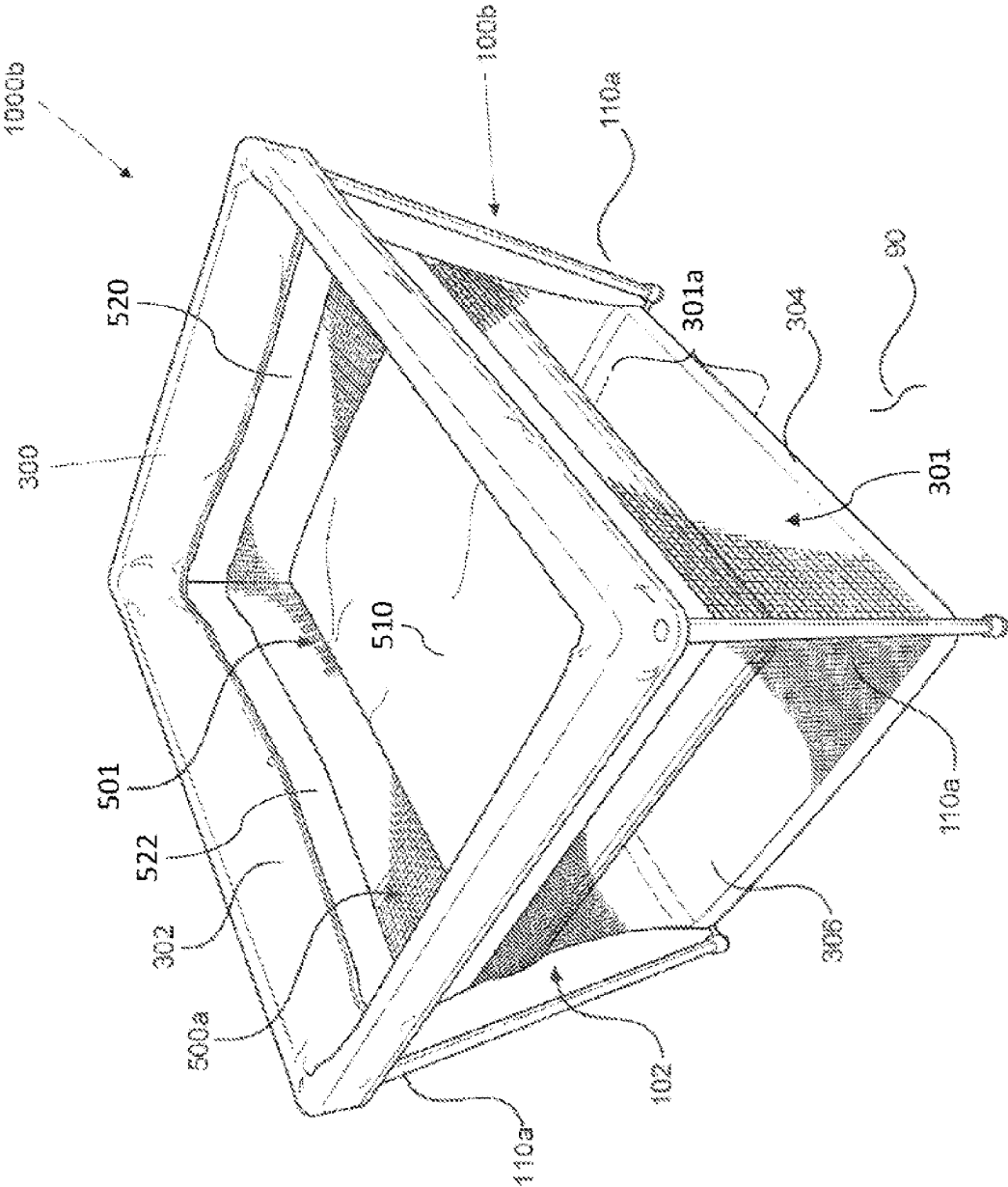


FIG. 40B



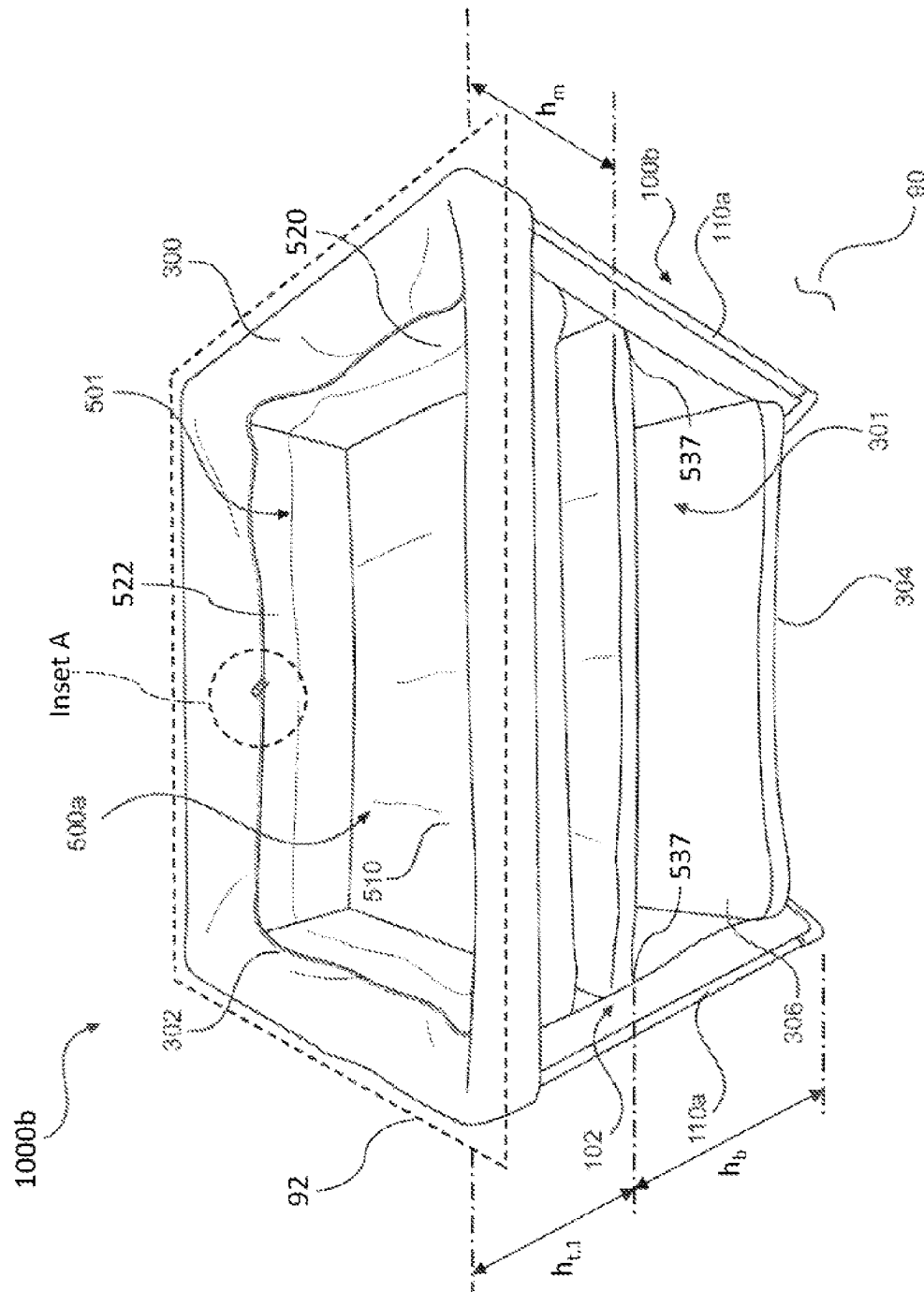


FIG. 41B

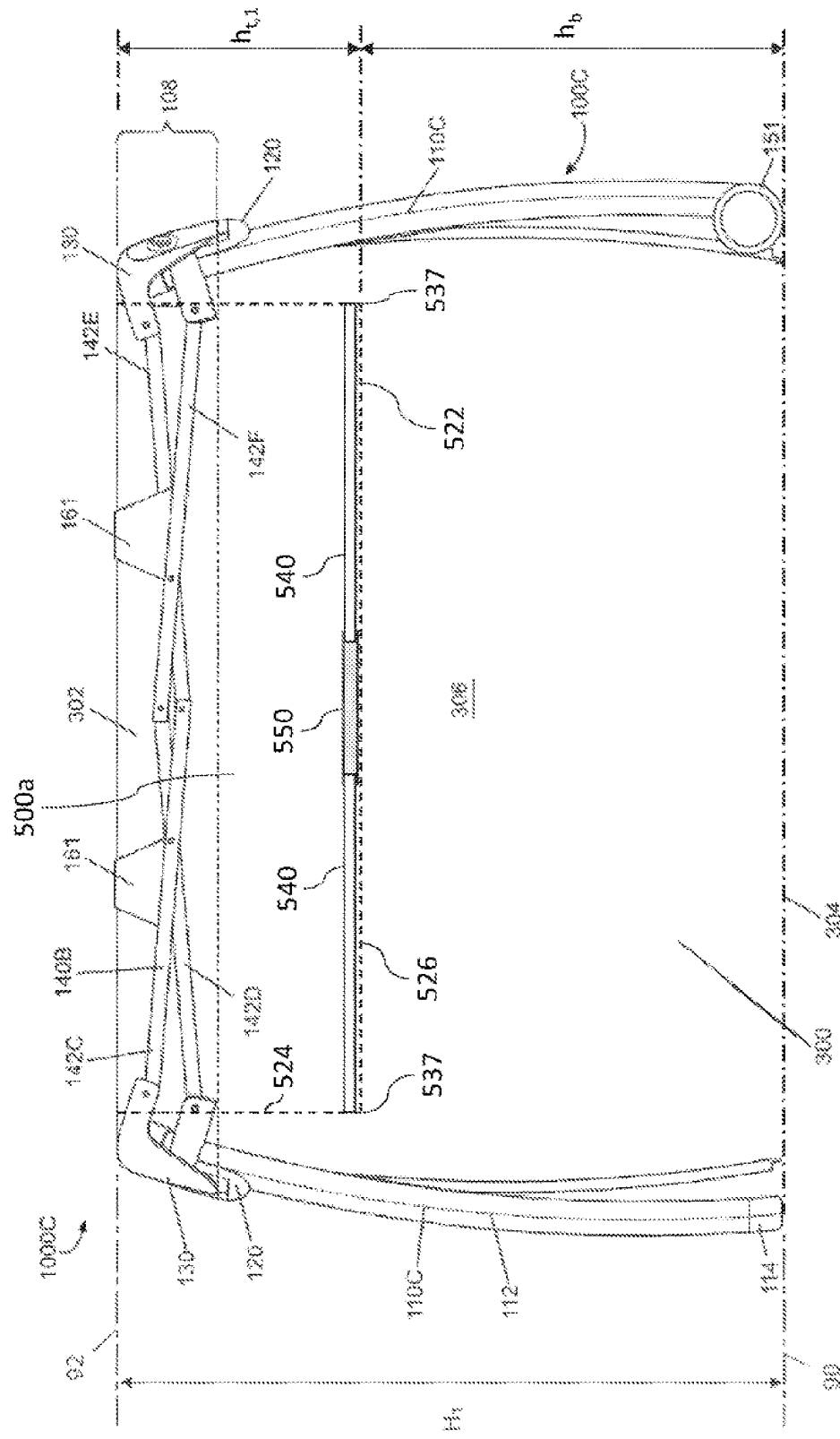


FIG. 41C

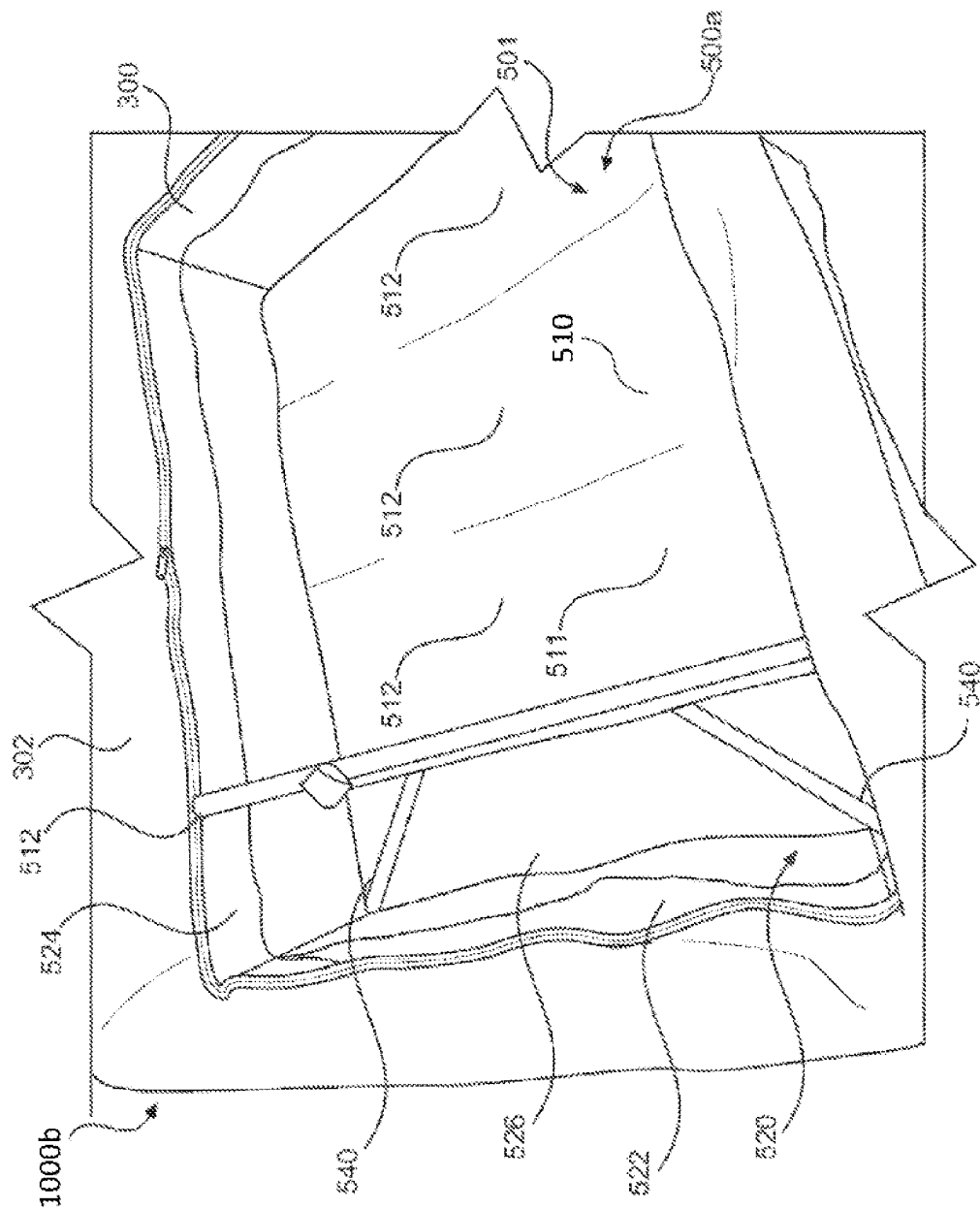


FIG. 42A

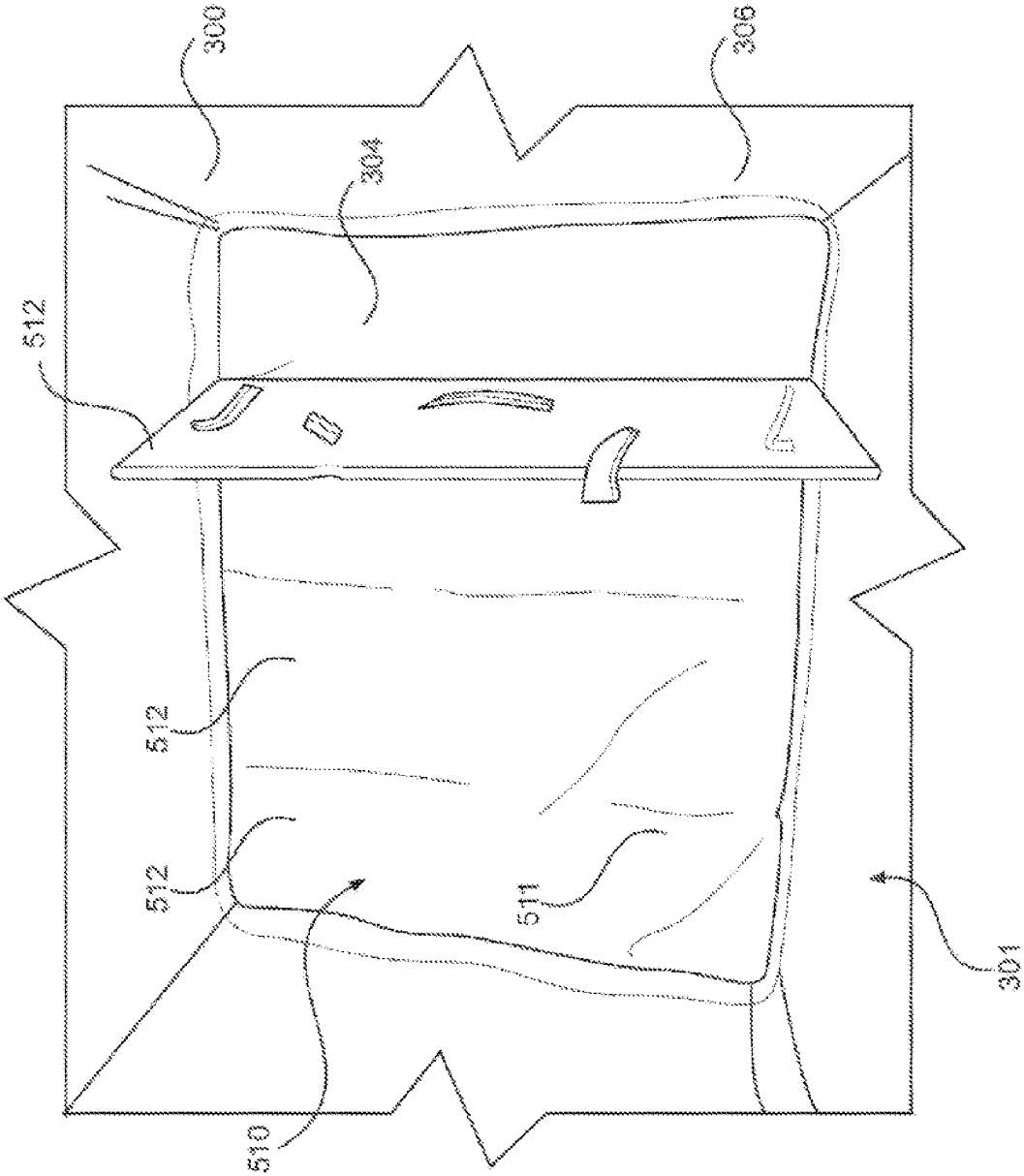


FIG. 42B

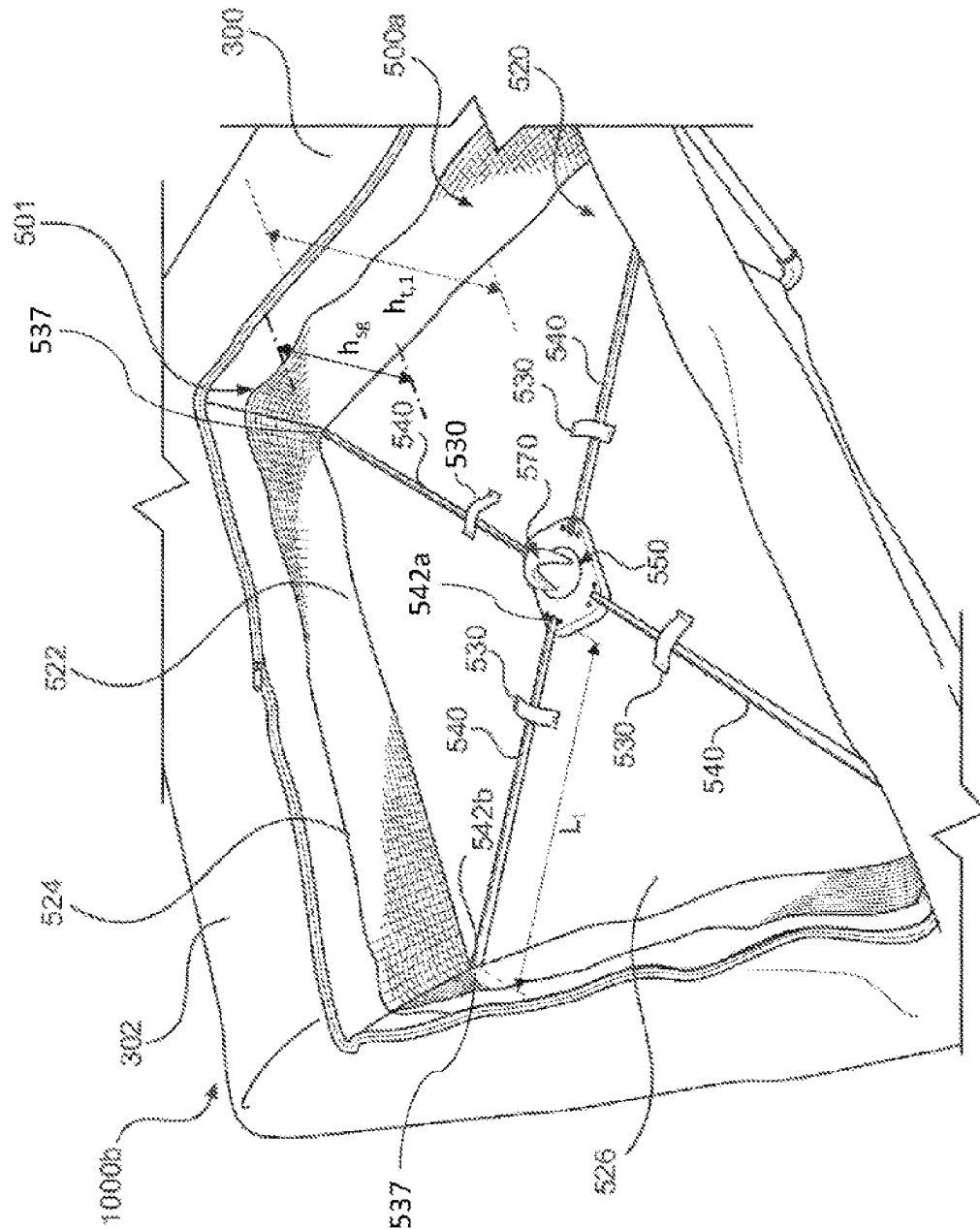


FIG. 43

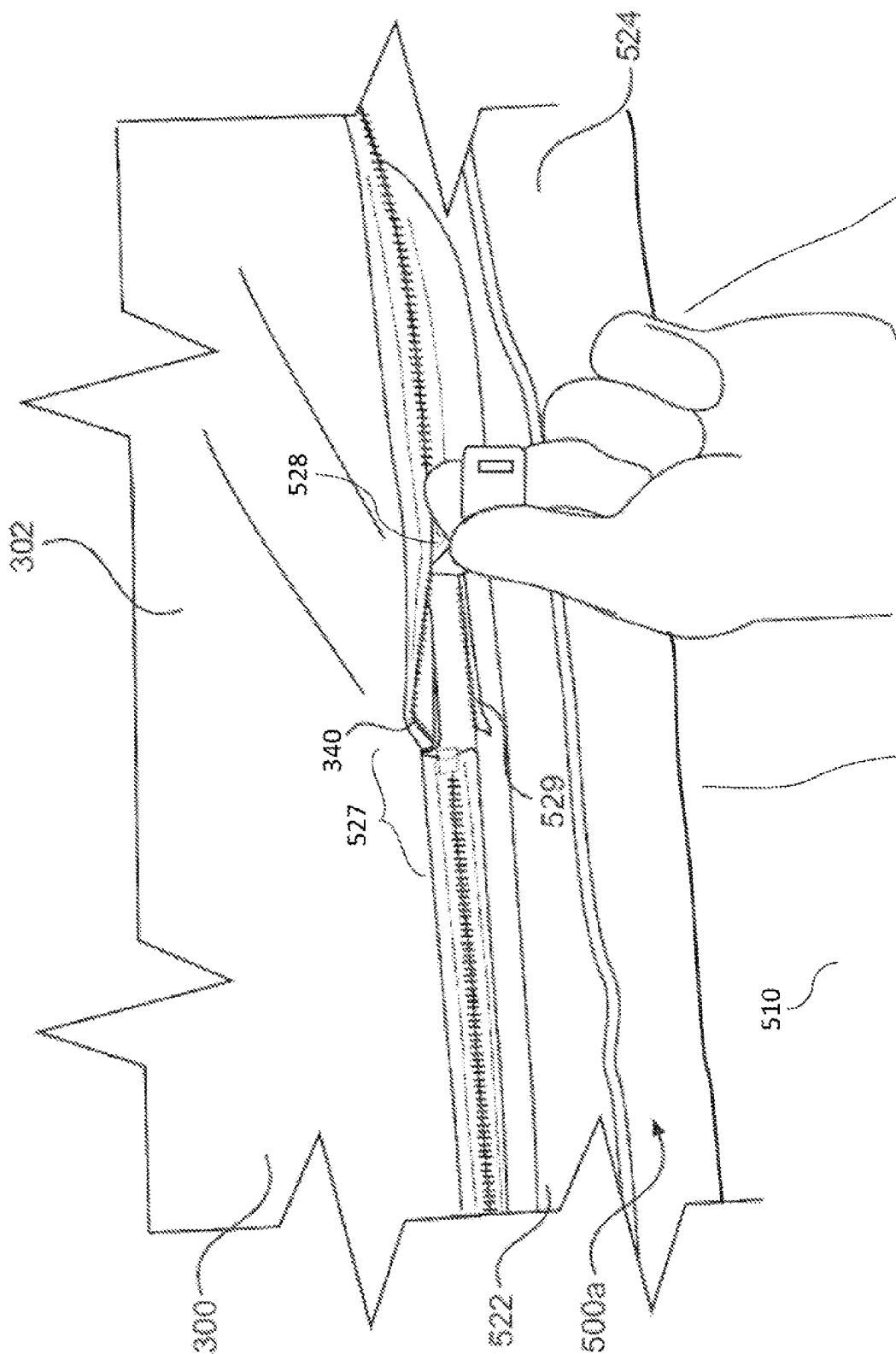


FIG. 44A

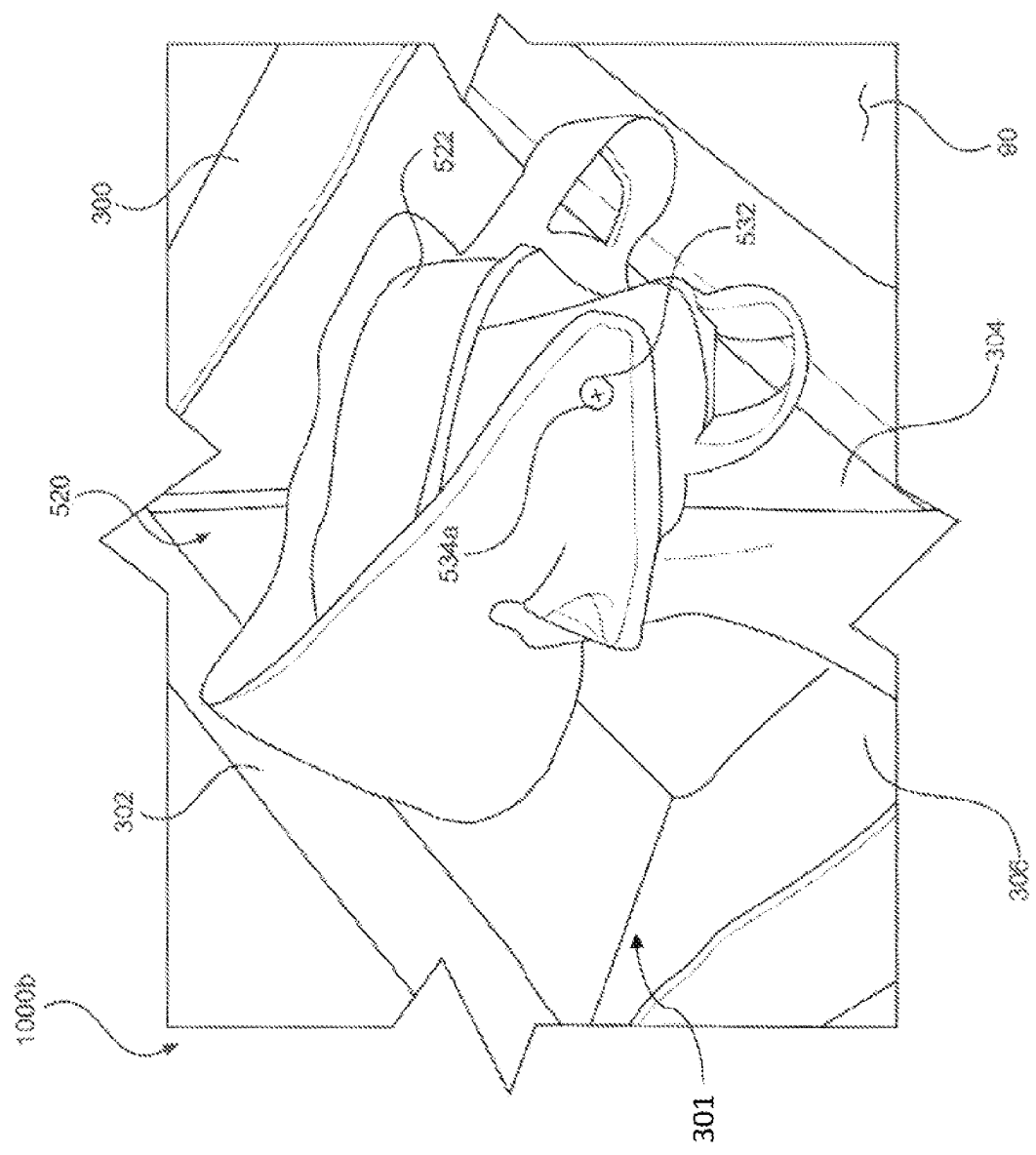


FIG. 44B

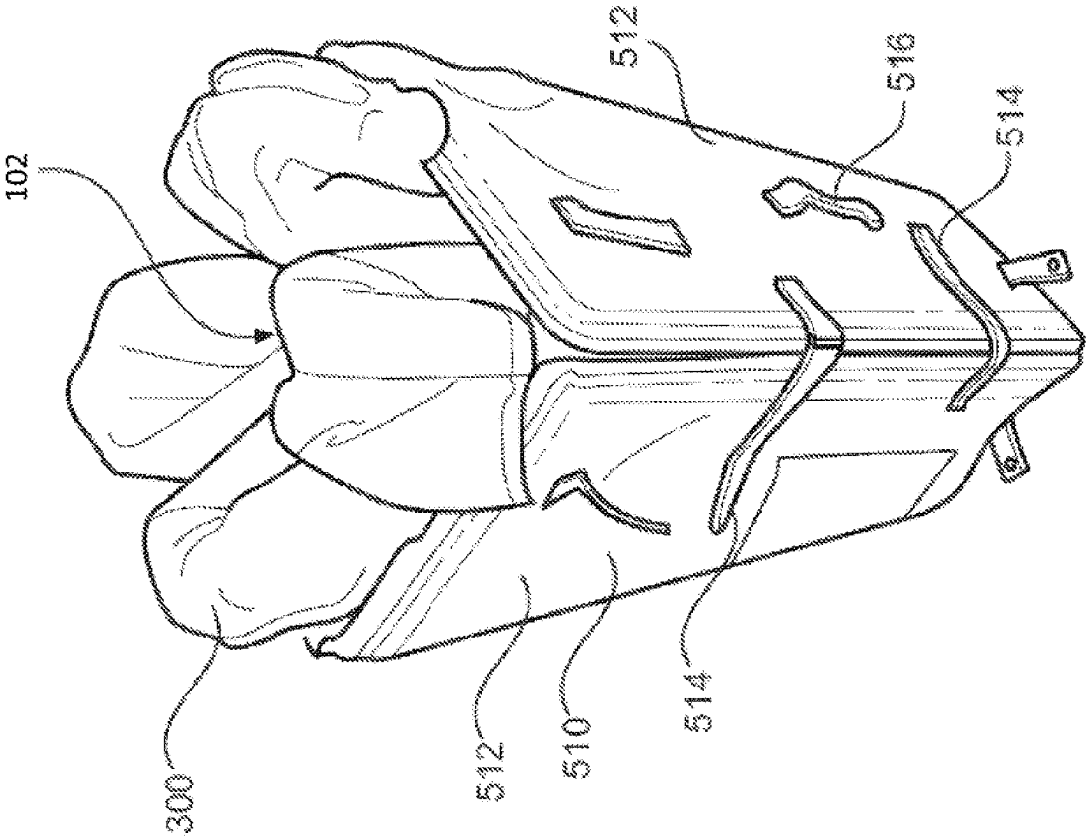


FIG. 45A

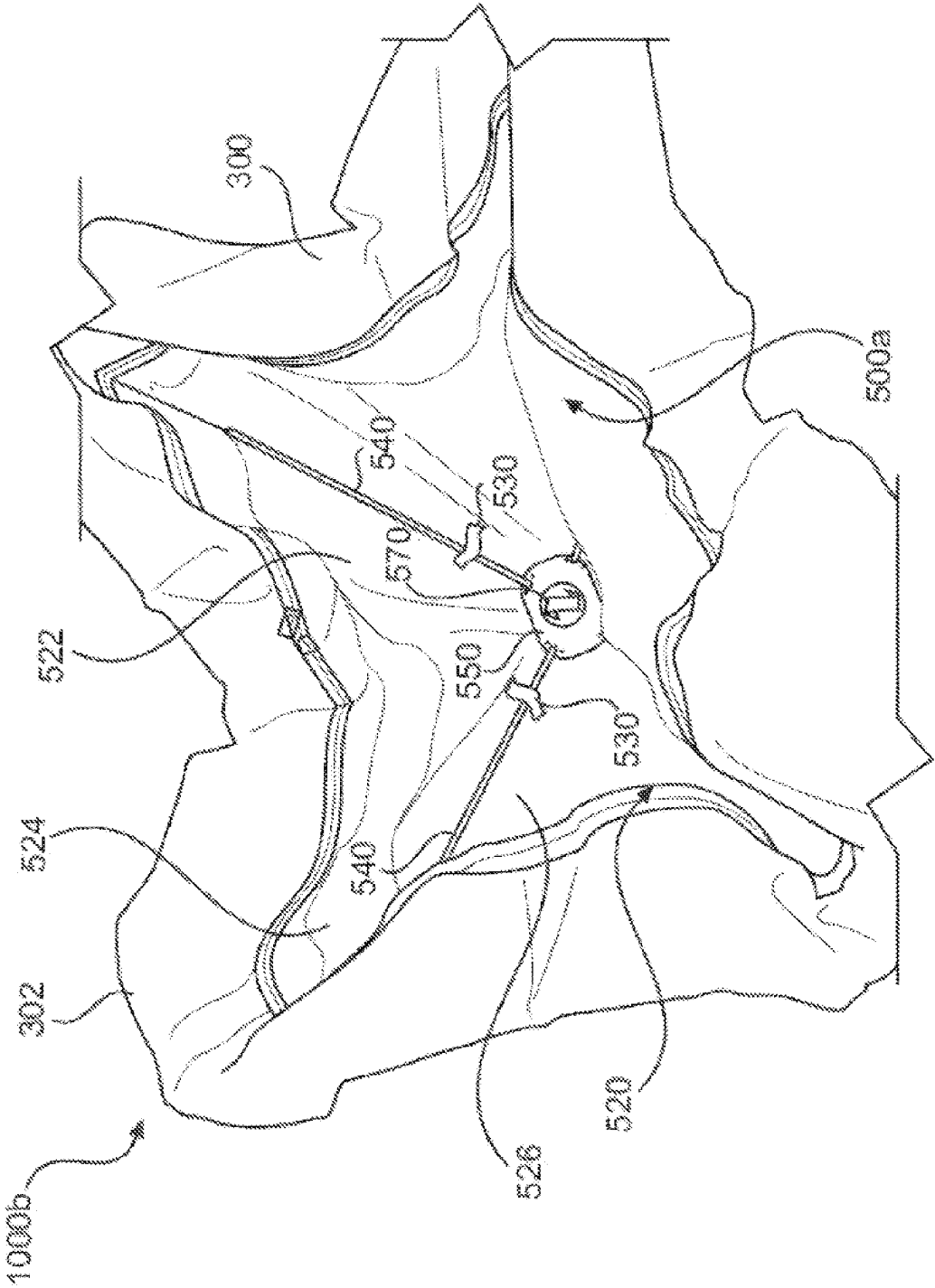


FIG. 45B

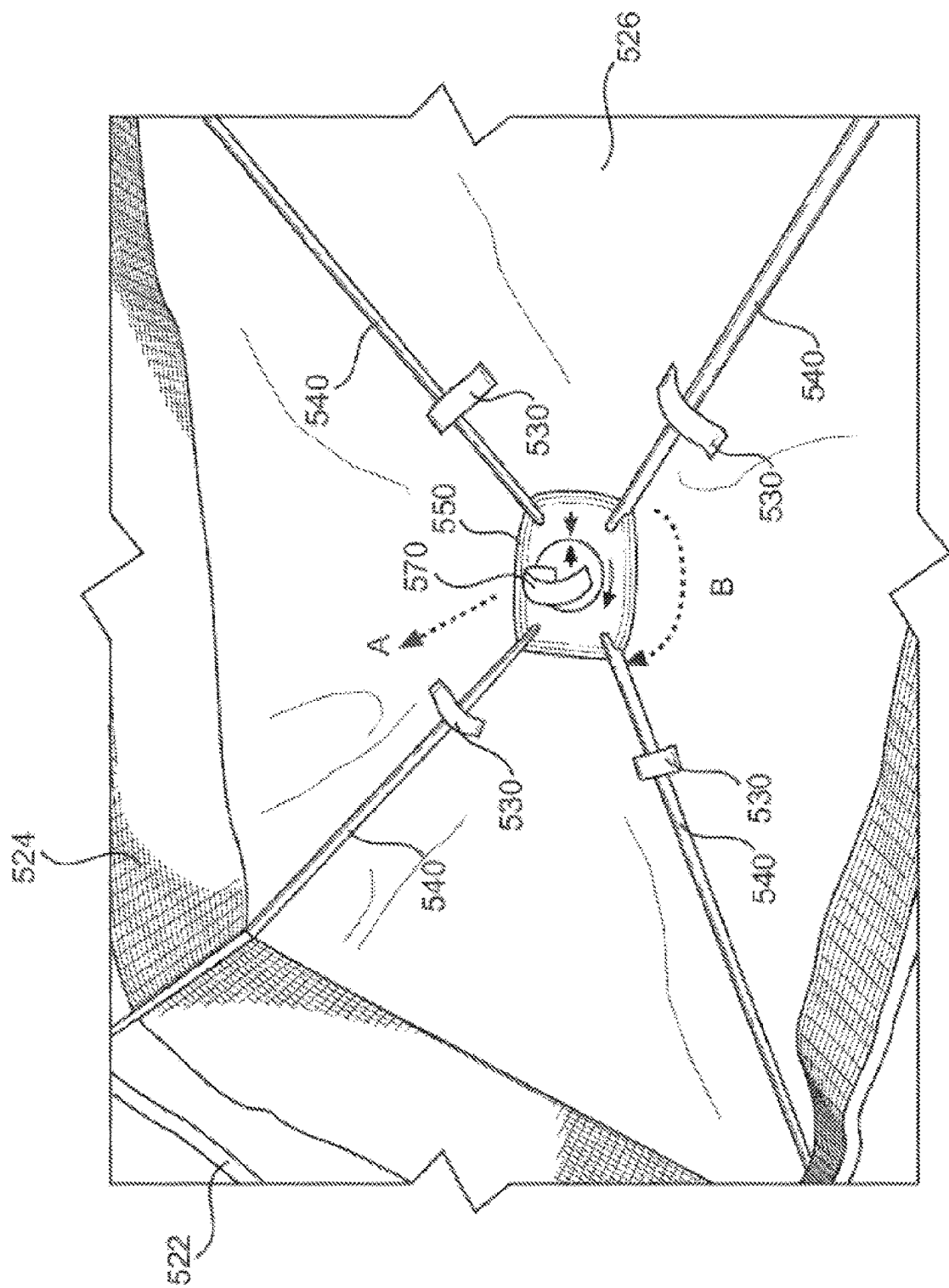


FIG. 45C

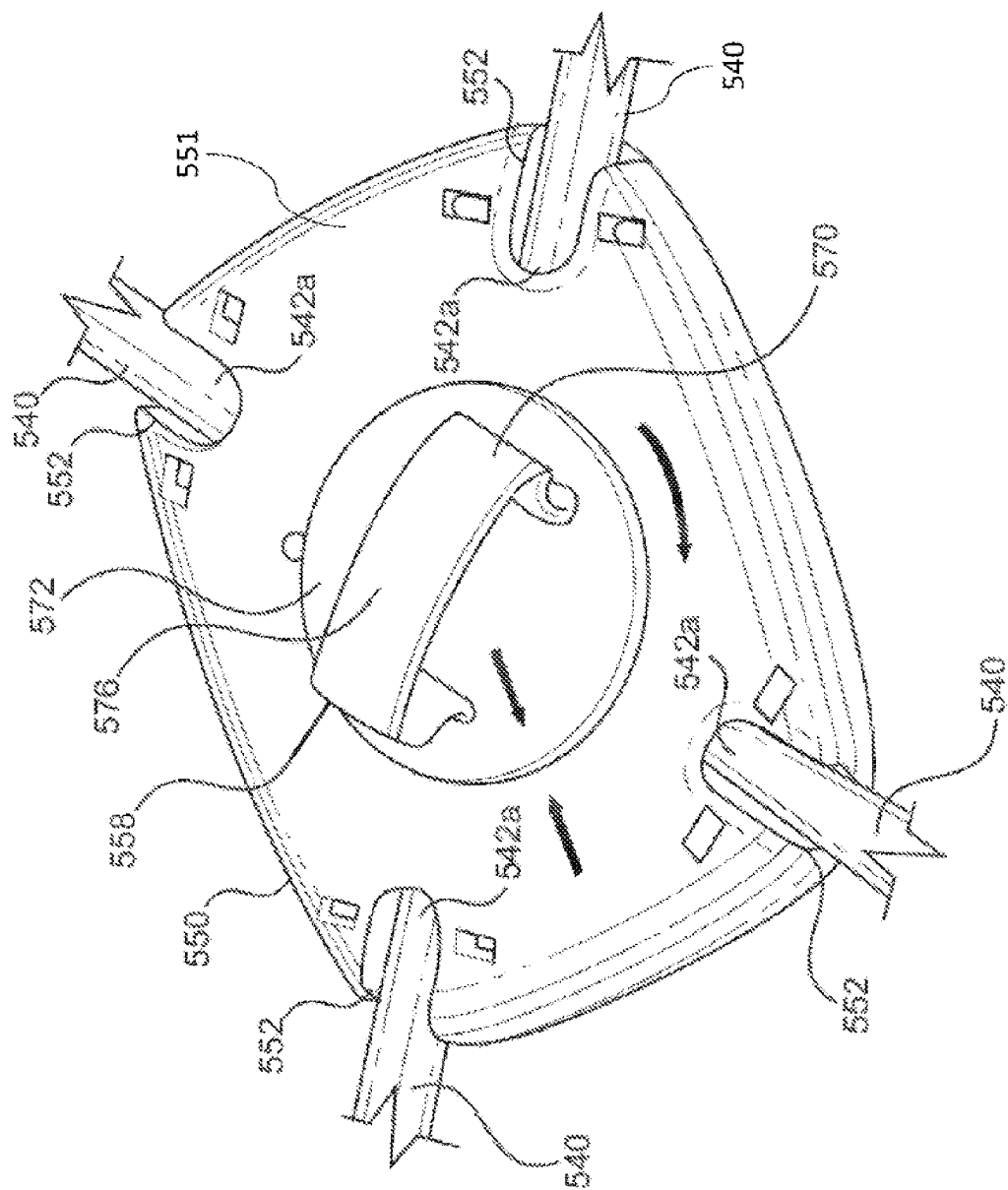


FIG. 46A

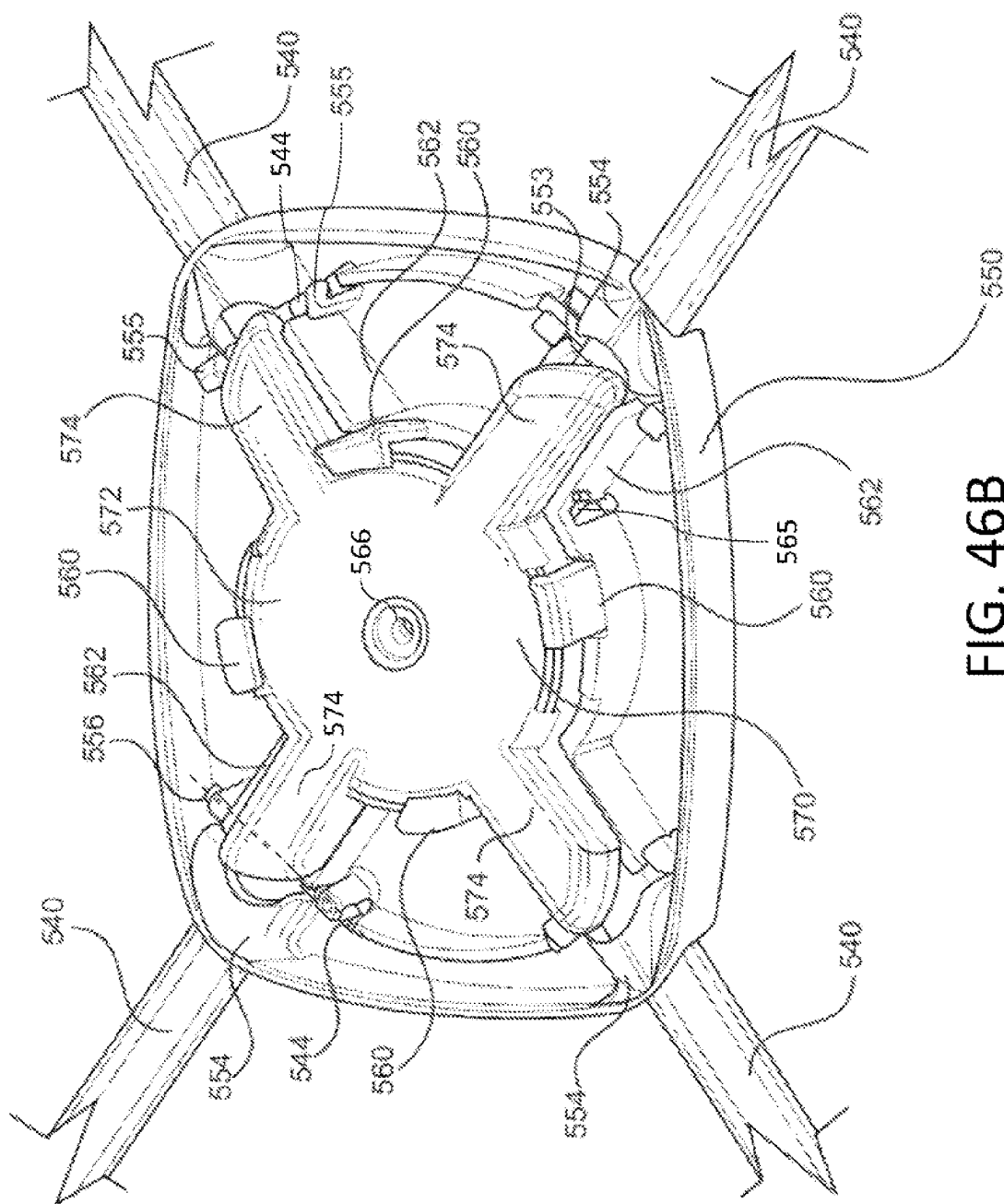
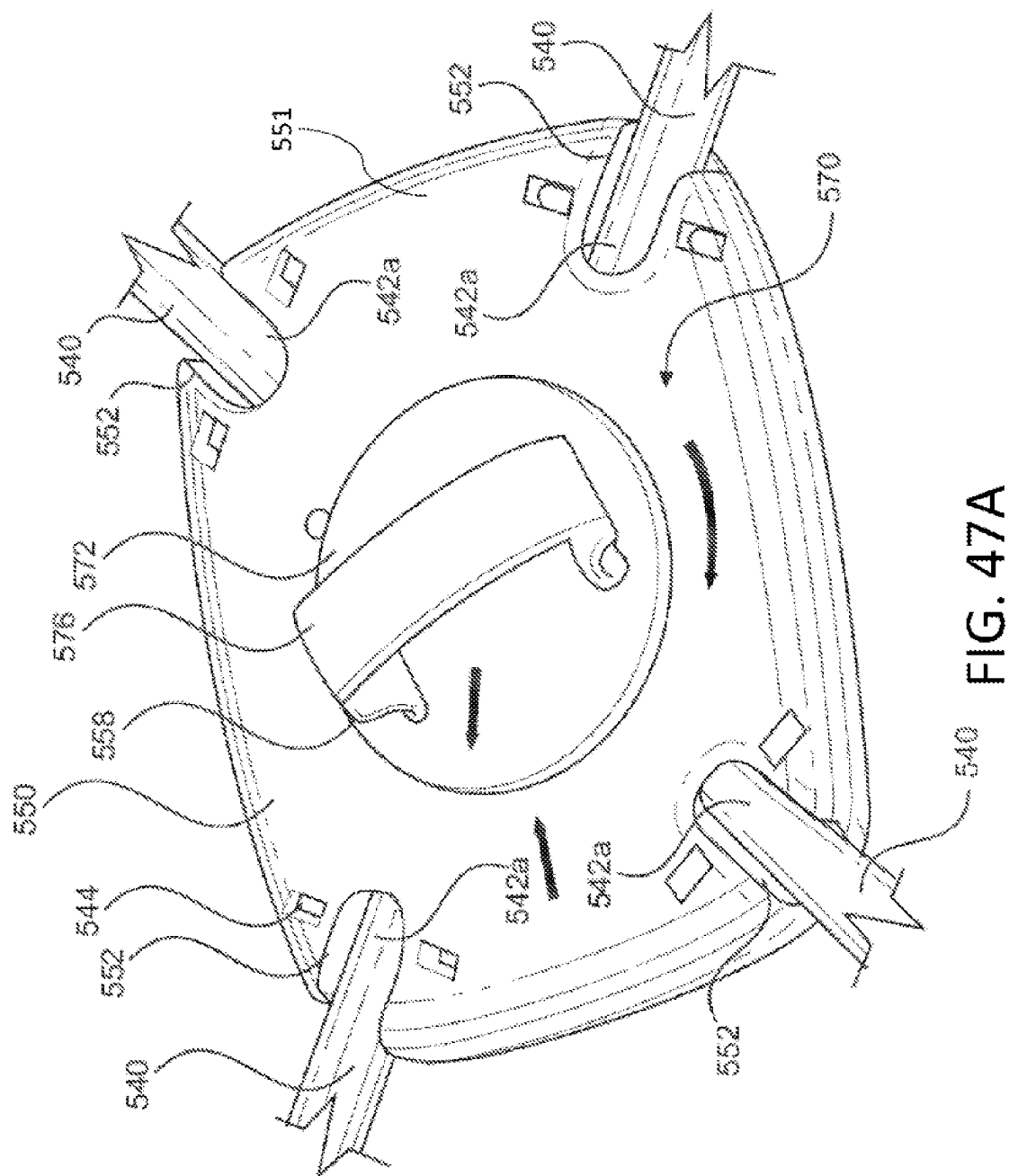


FIG. 46B



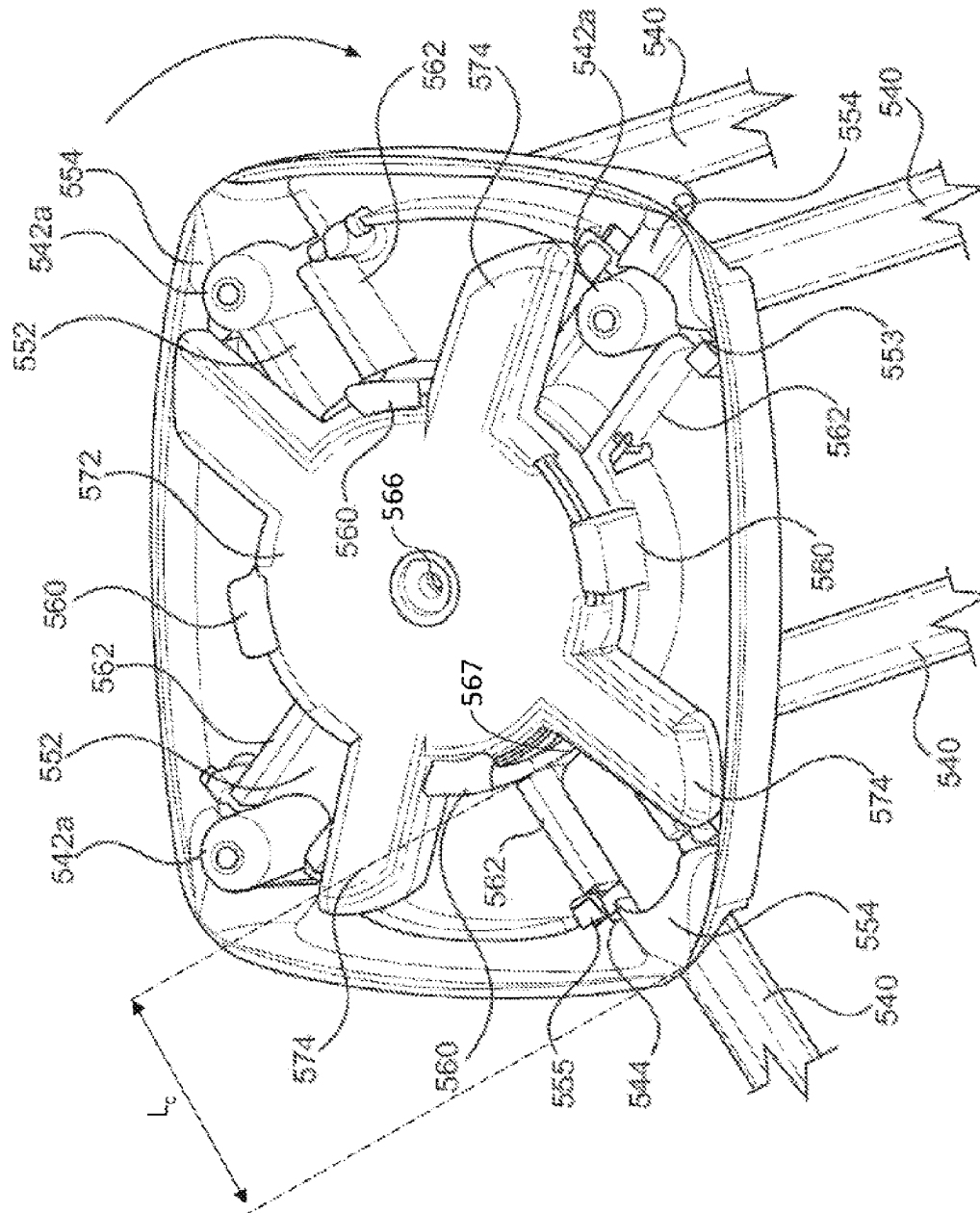


FIG. 47B

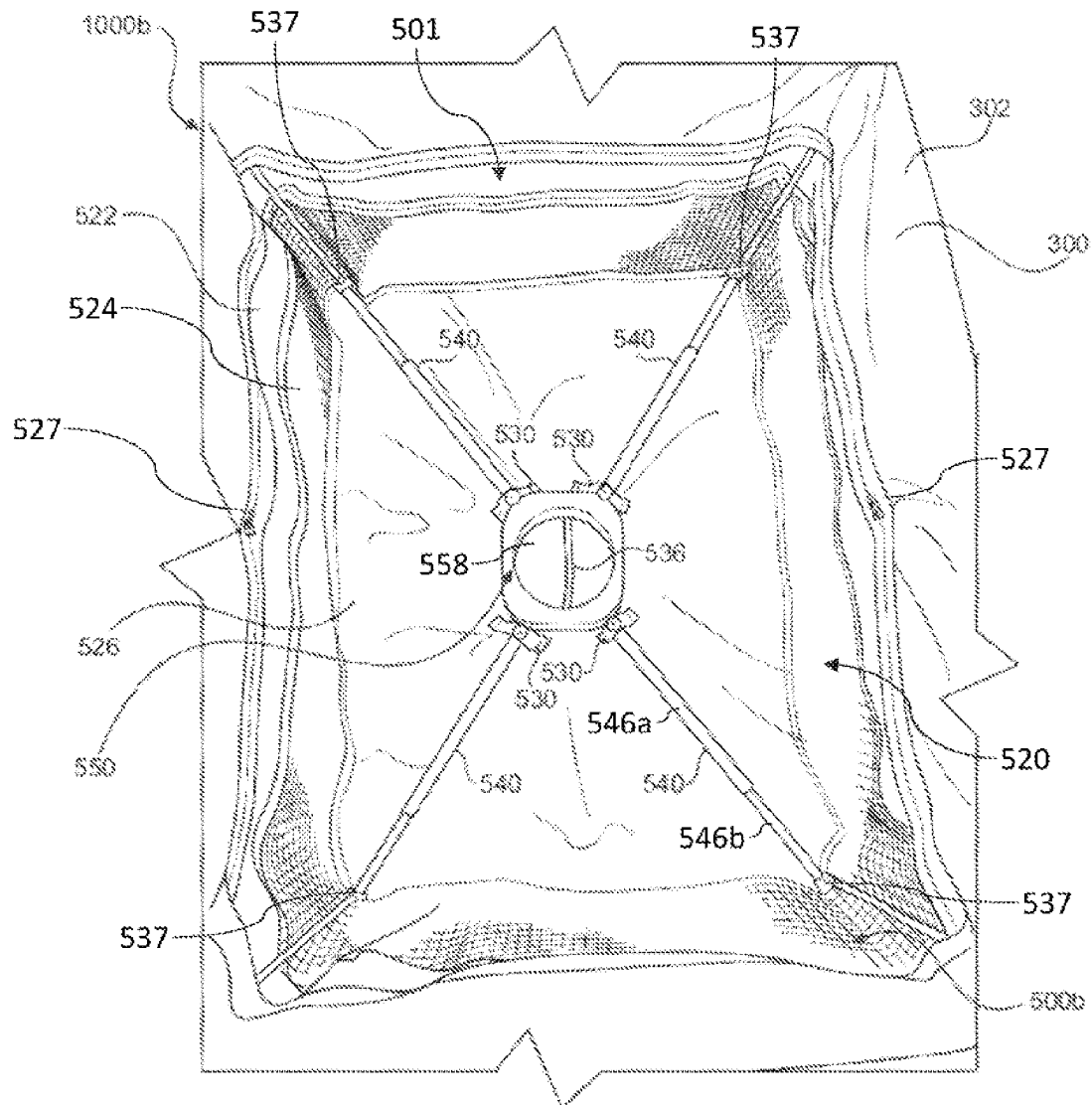


FIG. 48

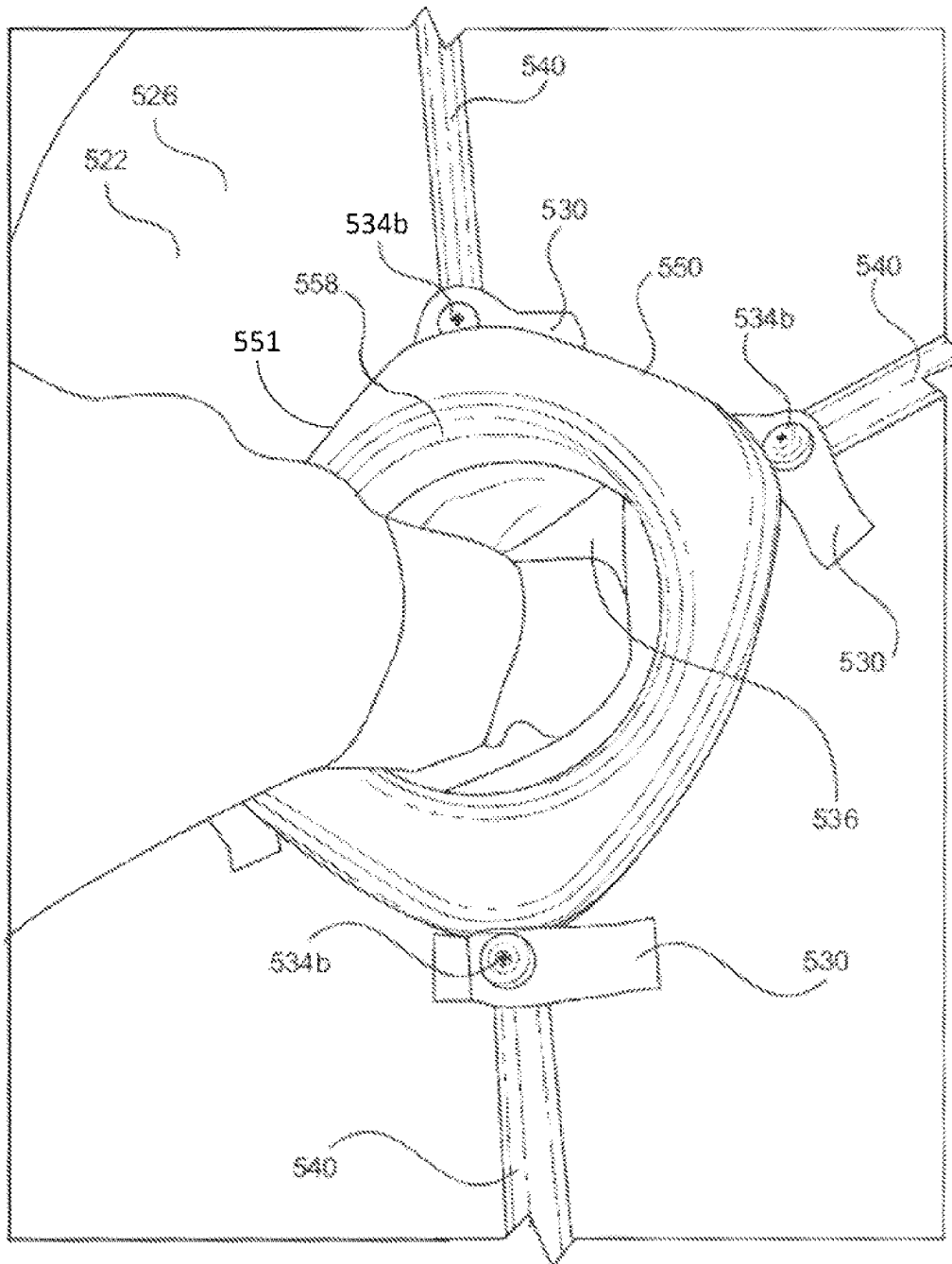


FIG. 49A

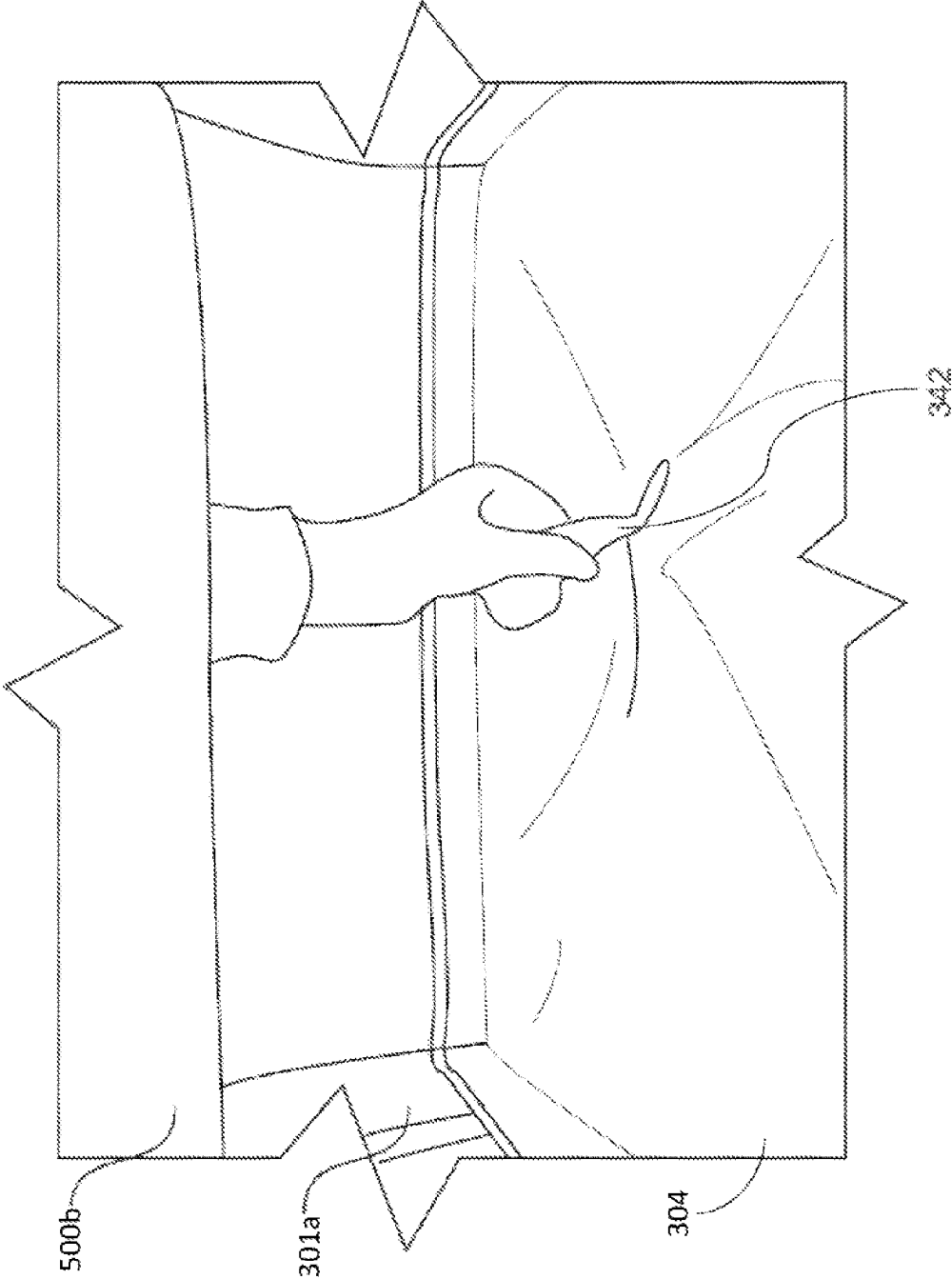


FIG. 49B

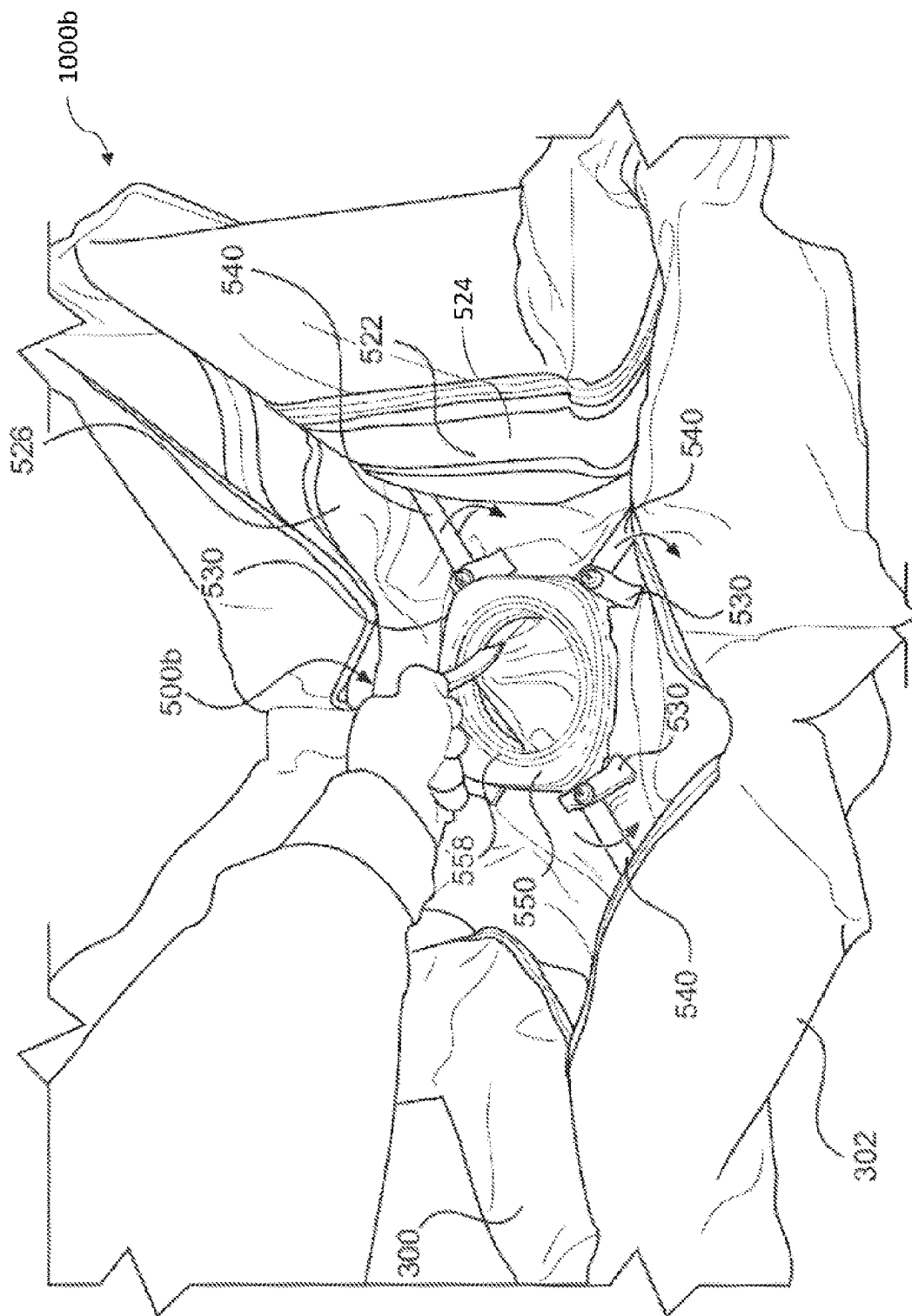


FIG. 49C

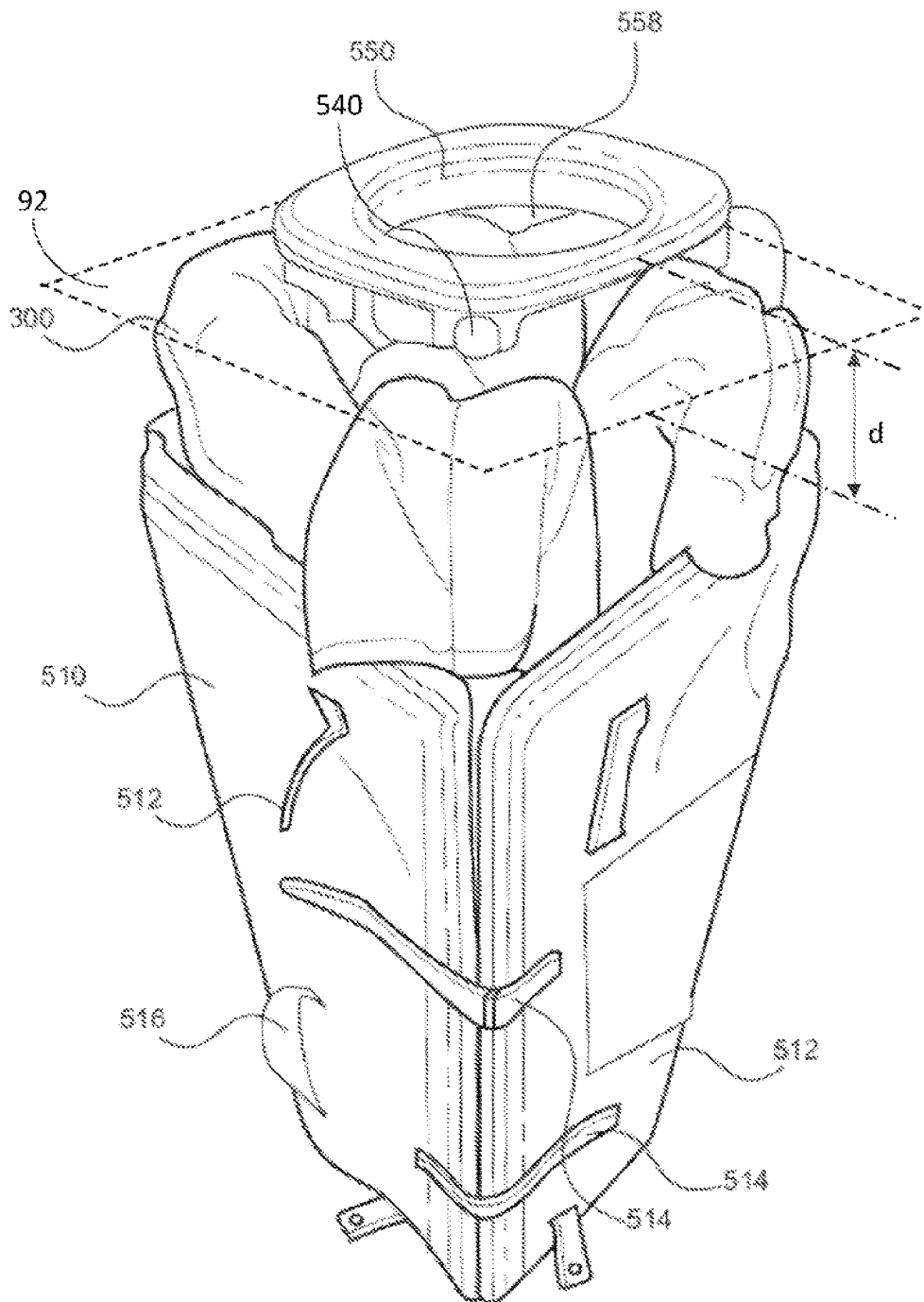


FIG. 49D

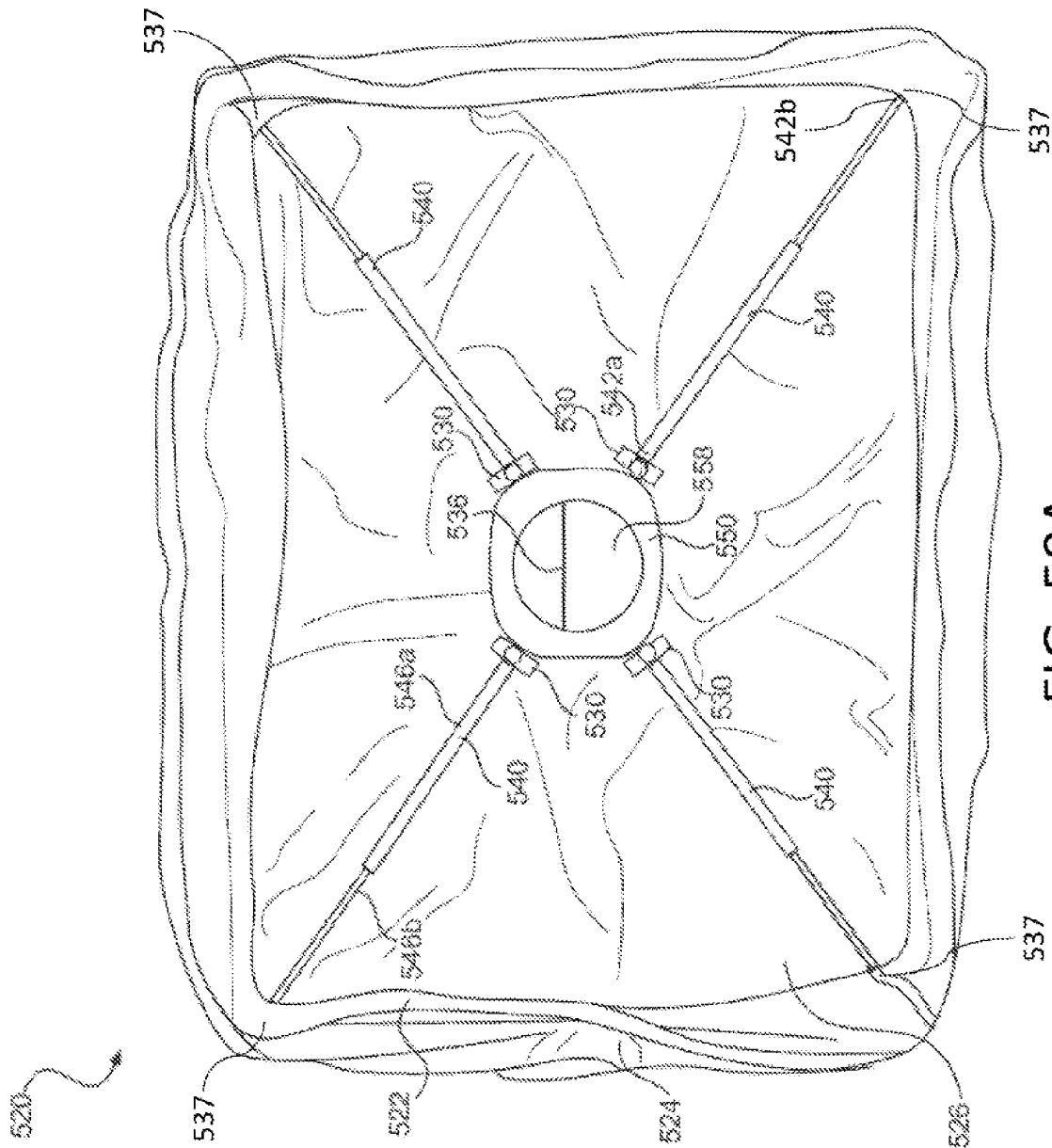


FIG. 50A

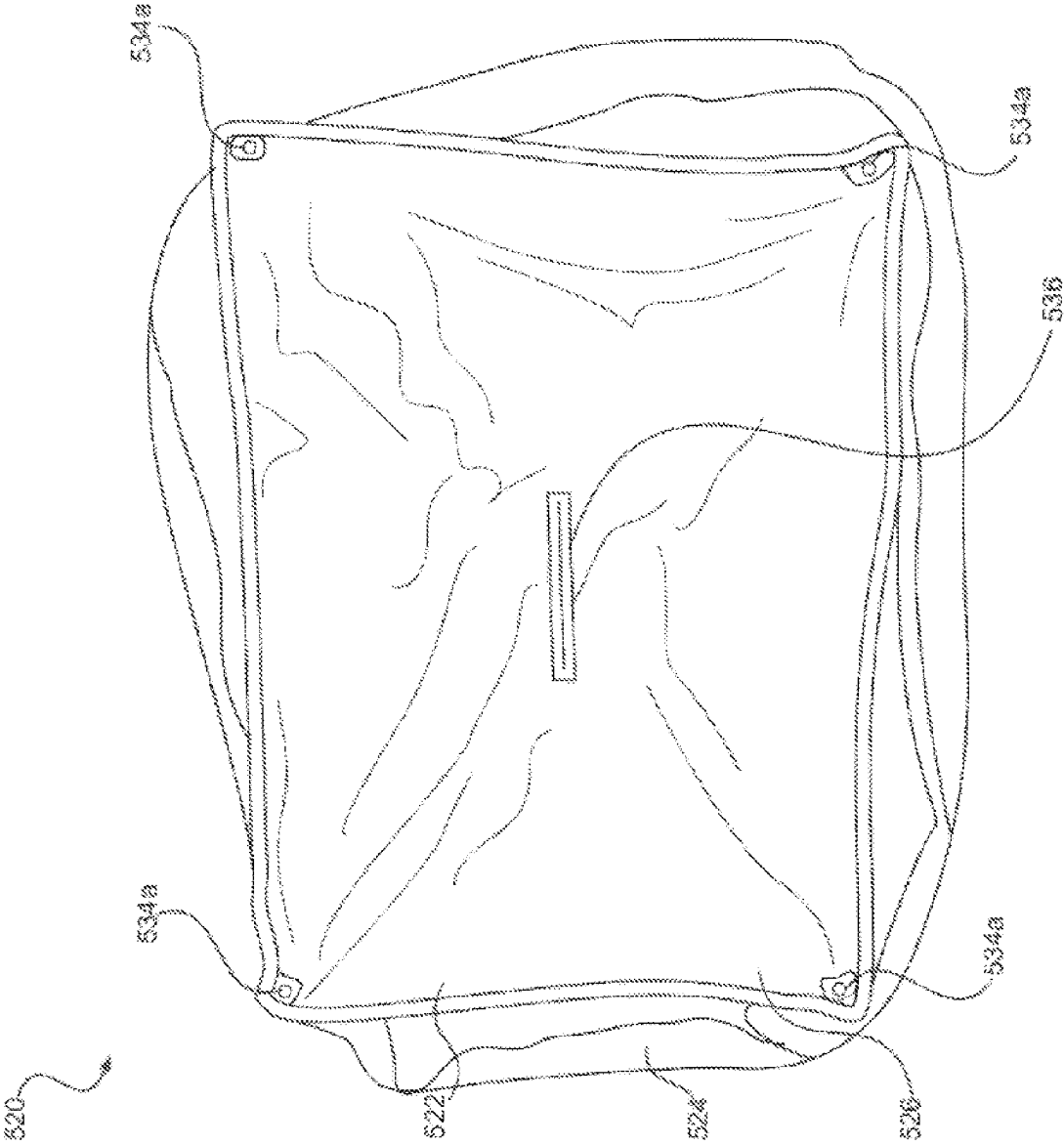


FIG. 50B

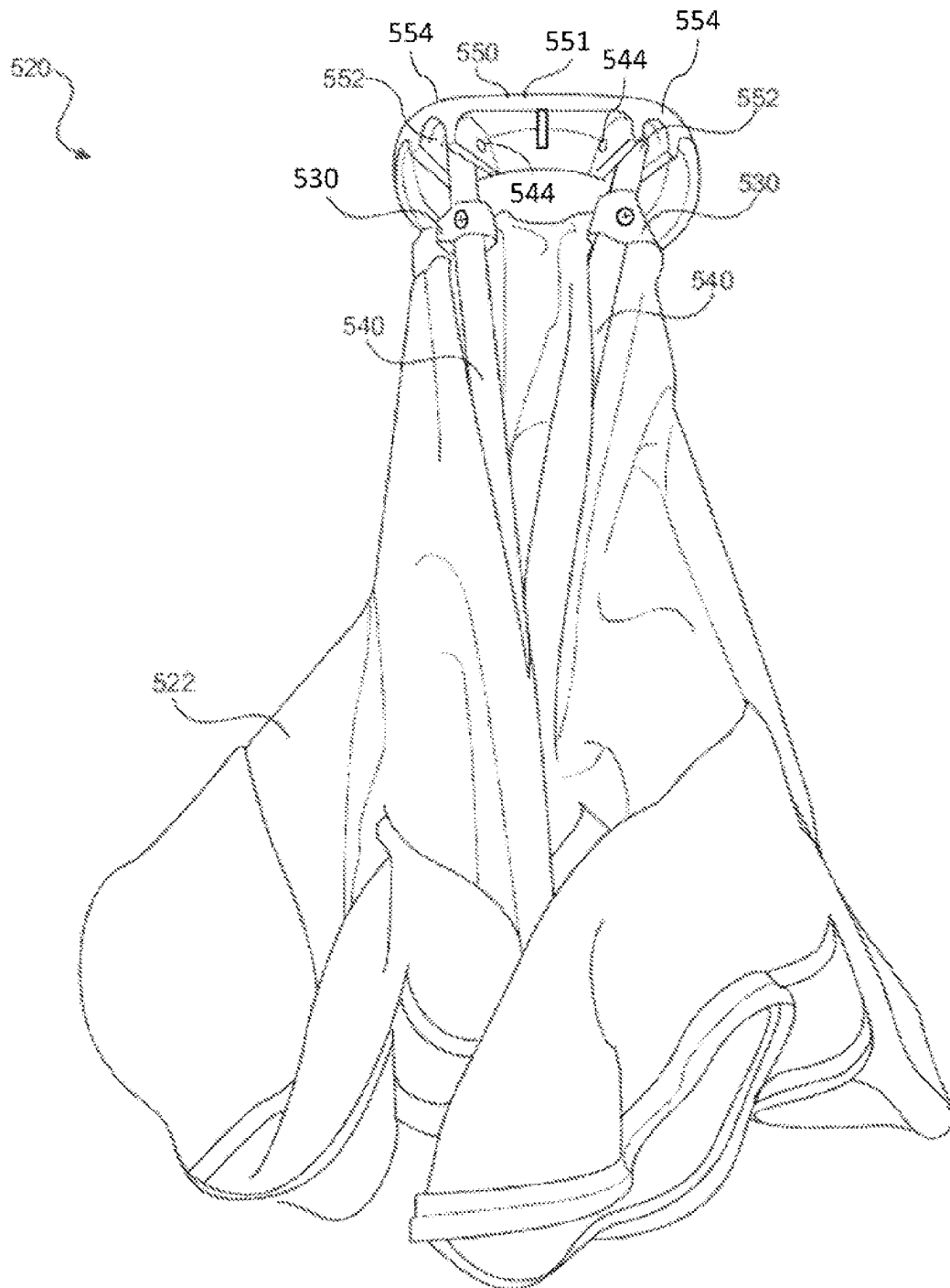


FIG. 50C

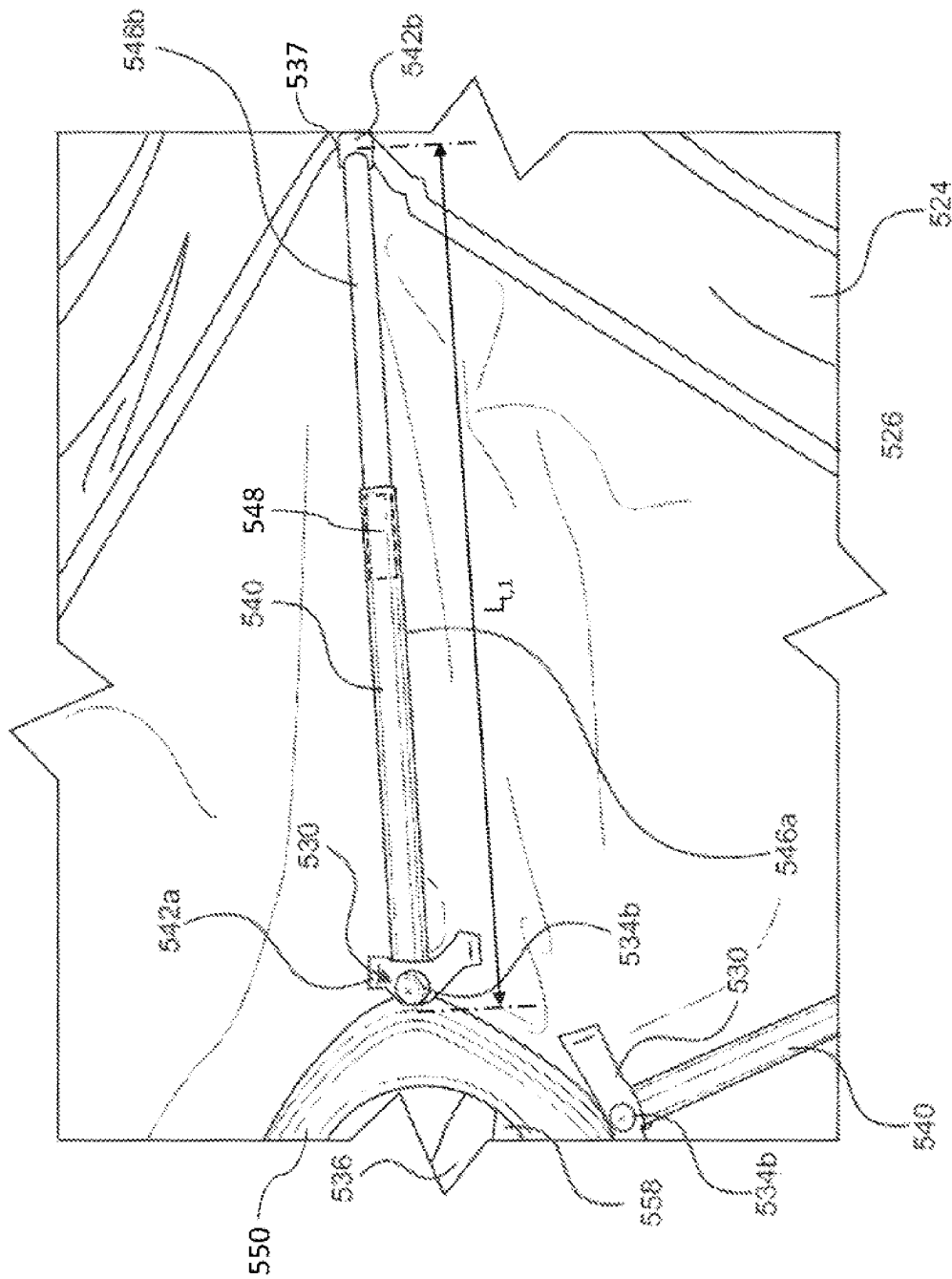


FIG. 51A

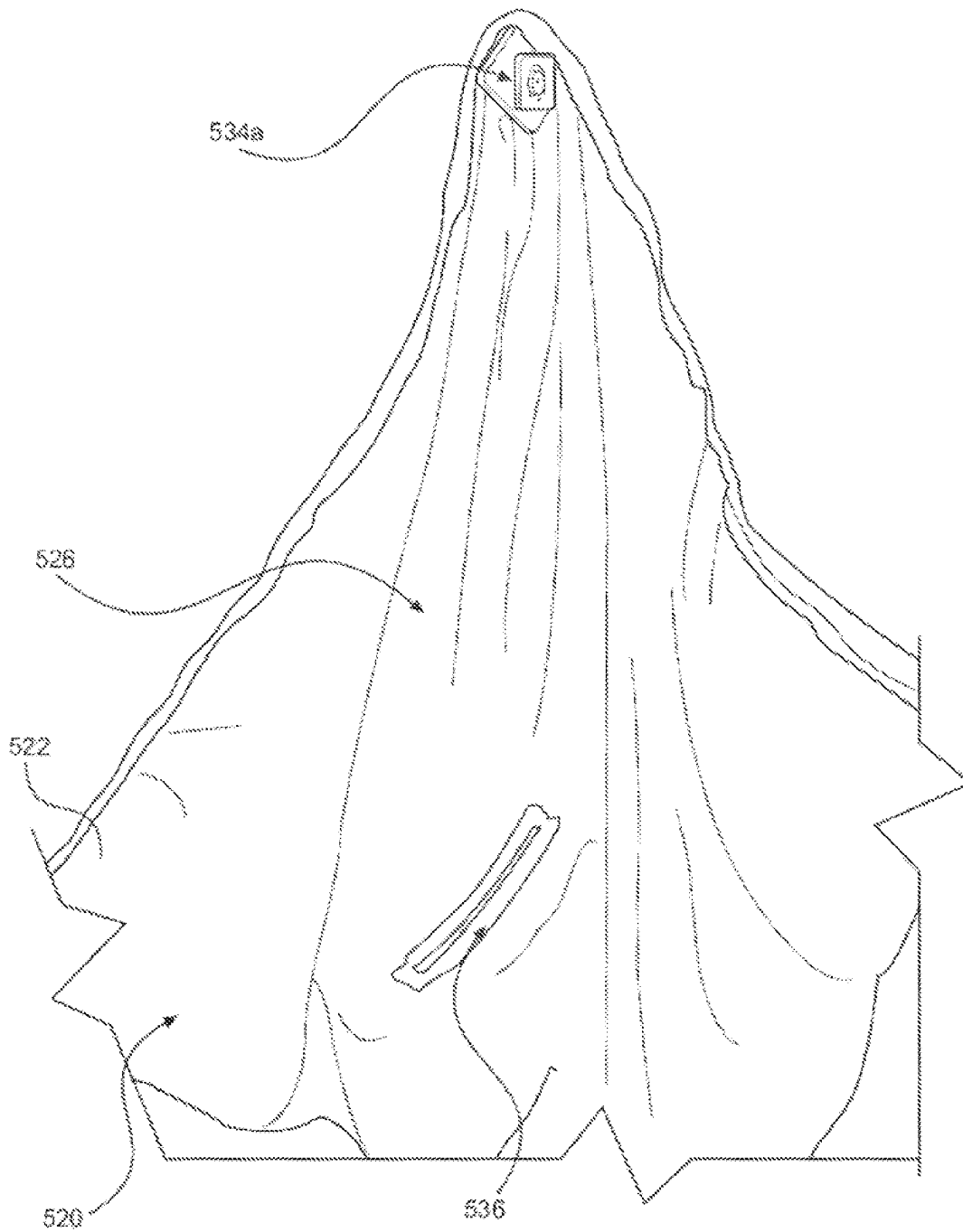


FIG. 51B

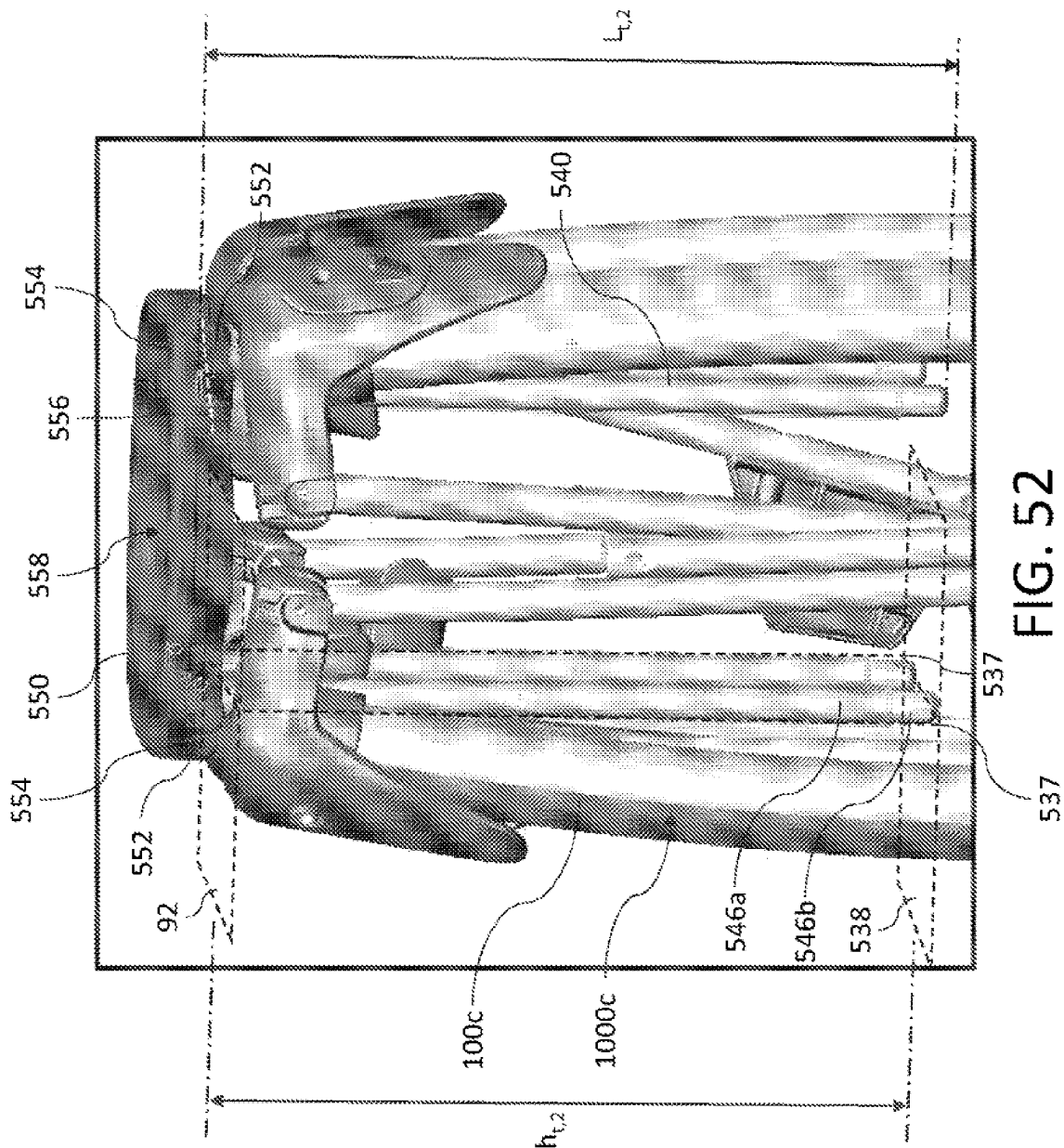


FIG. 52

BASSINET ACCESSORY FOR A PLAYARD

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a National Stage 371 Application of PCT Application No. PCT/US2021/013831, filed on Jan. 18, 2021, which claims a priority benefit to U.S. Provisional Application No. 63/021,950, filed on May 8, 2020, entitled, “X-Frame Playard with Toppers,” and U.S. Provisional Application No. 62/962,435, filed on Jan. 17, 2020, entitled, “X-Frame Playard.” Each of the aforementioned applications is incorporated by reference herein in its entirety.

BACKGROUND

A playard (also referred to herein as a “playpen” or a “game bed”) is a framed enclosure that provides a safe and comfortable space for a young child (e.g., an infant, a toddler) to sleep and play without significant supervision from a caregiver. The playard typically includes a support structure (e.g., a frame) that outlines an interior space of the playard; the playard also includes soft padding (also referred to herein as “soft goods”) placed within the interior space to provide a partially enclosed cushioned space to contain the child. Playards are generally foldable and/or collapsible to improve portability. For example, the caregiver may fold the playard for storage and/or transport and unfold the playard for use. Various types of playards have been manufactured and commercialized over the years with designs that have evolved, in part, depending on whether the playard is used primarily in outdoor settings or indoor settings.

FIG. 1A shows one example of a conventional outdoor playard 10a in an unfolded configuration. As shown, the playard 10a includes a frame 46 with multiple X-frame assemblies 20a that outline an interior space 11. Each X-frame assembly 20a includes X-tubes 22a and 22b that form a crossing pattern. In this example, the X-frame assemblies 20a are pivot-only X-frame assemblies where the X-tubes 22a and 22b are only rotatably coupled to each other and to other X-tubes to such that the frame 46 is foldable. As shown in the exploded-view inset in FIG. 1A, the playard 10a is provided with a latch mechanism 16 that attaches to the X-tubes 22a and 22b to lock the X-frame assemblies 20a in place when unfolded. Soft goods 12 are attached to the X-frame assemblies 20a and disposed along the sides and the floor of the interior space 11 for providing a partially enclosed space 13 for the child that is shaped and/or dimensioned to be similar to or smaller than the interior space 11 of the frame 46. As shown, the soft goods 12 includes webbing 14 along a top edge of the partially enclosed space 13 that functions as a top rail to increase the mechanical rigidity and stability of the playard 10a when the playard 10a is deployed. FIG. 1A also shows the playard 10a includes a canopy cover 40 disposed above the partially enclosed space 13 and mounted to the X-frame assemblies 20a to provide shade for a child.

FIG. 1B shows another example of a conventional outdoor playard 10b. As shown, the playard 10b includes a frame 46 with multiple pivot and slidable X-frame assemblies 20b coupled to adjoining leg support assemblies 24. Each leg support assembly 24 includes a leg tube 25, a corner (hidden beneath the soft goods 12) at the top of the leg tube 25, and a slider 26a or 26b that slides along the leg tube 25. The X-tubes 22a and 22b of each X-frame assembly 20 are coupled to respective sliders 26a and/or 26b and corners of the leg support assemblies 24. Thus, when the

playard 10b is being folded or unfolded, the X-tubes 22a and 22b undergo both rotation and displacement along the leg tubes 25 via the sliders 26a and/or 26b. Compared to the pivot-only X-frame assemblies 20a of the playard 10a, the pivot and slidable X-frame assemblies 20b of the playard 10b enable the playard 10b to be folded more compactly thus occupying less space in a folded configuration; additionally, the pivot and slidable X-frame assemblies 20b allow the frame 46 to provide a larger interior space 11 and, hence, a larger partially enclosed space 13 for a child when the playard 10b is in an unfolded configuration.

As before, the soft goods 12 may be attached to the leg support assemblies 24 and/or the X-frame assemblies 20b. The playard 10b also includes a pair of latch mechanisms 16a and 16b respectively mounted to sliders 26b on leg support assemblies 24 disposed on opposing sides of the playard 10b. As shown in the inset of FIG. 1B, the sliders 26b are different from the sliders 26a due to including features to lock the latch mechanisms 16a and 16b. The playard 10b also includes a canopy cover 40 disposed above the partially enclosed space 13 and mounted to the corners of the leg support assemblies 24.

FIG. 1C shows an example of a conventional indoor playard 10c. As shown, the playard 10c includes a frame 46 formed from multiple legs 30 and rigid top rails 32 to provide a rigid frame supporting soft goods 12. The frame 46 also includes a bottom support structure 34 so that the floor of the partially enclosed space 13 defined by the soft goods 12 is suspended off the ground. Compared to the outdoor playards 10a and 10b, the indoor playard 10c does not include X-frame assemblies to facilitate folding and/or unfolding. Instead, the top rails 32 are coupled to a hinge 36, which allows the playard 10c to be collapsed into a smaller form as shown in FIG. 1D. Additionally, the bottom support structure 34 is also foldable. Thus, to fold the playard 10c, the caregiver needs to first remove some of the soft goods 12, pull up a bottom hub to fold the bottom support structure 34 (step ‘A’ in FIG. 1D), and then unlock and fold the top rails 32 (step ‘B’ in FIG. 1D). The caregiver needs to perform these steps in reverse to setup the playard 10c.

FIG. 1C also shows the playard 10c includes a bassinet accessory 60 disposed within the partially enclosed space 13 to provide an elevated surface above the ground to support the child. The elevated surface may reduce the physical strain experienced by a caregiver when placing their child into the playard 10c and/or when taking their child out of the playard 10c by providing a more accessible and easier to reach space compared to the bottom of the playard 10c. The elevated surface of the bassinet accessory 60 also allows the caregiver to more easily monitor their child as well. Conventional bassinet accessories are typically configured to support infants and/or children weighing less than 15 lbs.

SUMMARY

The Inventors have recognized and appreciated that a foldable playard provides a caregiver a convenient and safe space for their child to play and/or sleep once the playard is setup, which alleviates the caregiver from having to continuously monitor their child. However, the Inventors have also recognized that conventional playards in some instances may be cumbersome to setup and/or stow away due, in part, to complicated mechanisms for folding, unfolding, latching and/or unlatching the playard (and correspondingly protracted procedures that the caregiver needs to perform while generally caring for their child). The complexity of conven-

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tional playards also results in a bulkier product, which is more difficult to handle and more expensive to manufacture and purchase as a consumer.

First, the Inventors have observed that conventional playards typically include various support structures, in addition to their frames, to provide more rigid boundaries outlining the interior space, so as to better contain the child and/or to increase the mechanical rigidity and stability of the frame. In many instances, one or more additional support structures are added to the frame of a conventional playard to ensure the playard meets various consumer safety standards related to the mechanical properties of the frame (e.g., American Society for Testing and Materials (ASTM) F406-19 entitled, "Standard Consumer Safety Specification for Non-Full-Size Baby Cribs/Play Yards").

With reference again to FIG. 1A and FIG. 1B, as noted above the respective frames 46 of the playards 10a and 10b include X-frame assemblies 20a and 20b to facilitate folding and/or unfolding of the frames 46. When unfolded, the X-frame assemblies 20a and 20b are disposed along the respective sides of the frames 46, thus providing a mechanically rigid and stable structure.

However, FIG. 1A shows the X-tubes 22a and 22b of the X-frame assembly 20a in the playard 10a, when unfolded, span the sides of the frame 46; this results in a top portion 47 of the interior space 11 above the X-frame assembly 20a that is not mechanically supported by the frame 46. If flexible, compliant soft goods 12 are placed over the frame 46 as shown in FIG. 1A, a child could potentially climb out of the playard 10a through the top portion 47 by folding and/or collapsing the soft goods. In view of the foregoing, as an additional support structure, the soft goods 12 includes an integrated webbing 14 that is pulled taut when the frame 46 is unfolded such that the webbing 14 mechanically functions as a top rail. In this manner, the webbing 14 provides a more rigid boundary spanning the top portions 47 of the interior space 11 to support the soft goods and to better keep the child within the playard 10a.

FIG. 1B similarly shows that the X-tubes 22a and 22b of the X-frame assembly 20b in the playard 10b, when unfolded, do not mechanically support the top portions 47 of the interior space 11 above the X-tubes 22a and 22b. Thus, similar to the playard 10a, the playard 10b includes webbing 14 that is directly coupled to the leg support assemblies 24 as an additional support structure. When the frame 46 of the playard 10b is unfolded, the webbing 14 is once again pulled taut to form a top rail and thereby provide a more rigid boundary spanning the top portions 47 of the interior space. It should be appreciated that without the webbing 14, the playards 10a and 10b are unlikely to comply with various consumer safety standards, such as ASTM F406-19.

As noted above in connection with FIG. 1C, the playard 10c includes rigid top rails 32 that connect adjacent legs 30. In this manner, the frame 46 of the playard 10c provides mechanical support structures that span the top and side boundaries of the interior space 11. However, a frame that only has vertical or nearly vertical legs and top rails is often prone to mechanical instability. For example, the frame may tilt to one side due to the bottom portion of the legs being mechanically unconstrained and/or due to backlash or slop between the joints connecting the rails and the legs together. This mechanical instability may be further exacerbated if the legs and the rails are configured to move relative to one another, e.g., to facilitate folding of the playard. Given this mechanical instability, to reinforce the frame 46 the playard

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10c includes an additional bottom support structure 34 that connects the legs 30 located at opposing corners of the frame 46.

The various support structures added to conventional playards as discussed above, and the various modifications made to the playards to accommodate these support structures, increase the complexity, number of parts, and cost of these playards.

For instance, the webbing 14 for the playards 10a and 10b needs to be sewn directly into the soft goods 12 or the X-frame assemblies 20a and 20b, and/or the leg support assemblies 24 need to incorporate additional structural features to directly attach to the soft webbing 14—both of which increase design complexity resulting in higher manufacturing costs. For the playard 10c, the rigid top rails 36 and the bottom support structure 34 need to include additional mechanisms (e.g., the hinge 36, hinges connecting the various members of the bottom support structure 34) to facilitate tear down and folding of the playard 10c, which increase the number of parts for manufacture and assembly. As shown in FIG. 1D, these additional mechanisms also make it more difficult for the caregiver to setup and tear down the playard 10c by adding additional steps (e.g., steps 'A' and 'B'). In particular, the playard 10c is especially difficult to unfold since the playard 10c tends to tip over and/or partially collapse when partially unfolded.

The Inventors also have recognized that conventional playards often include frames that compromise between the ease of folding and/or unfolding the frame and other aspects related to the practical use of the playard. For example, the playards 10a and 10b include X-frame assemblies 20a and 20b, respectively, which makes folding and/or unfolding the respective frames 46 appreciably easier for the caregiver. However, the X-frame assemblies 20a and 20b both span an appreciable portion (if not all) of the sides of the respective frames 46 as described above, which may interfere with the visibility of a child in the partially enclosed space 13 and thereby impede or obstruct a caregiver's ability to easily see the child in the playard.

More specifically, with reference again to FIG. 1A, the soft goods 14 in the playard 10a includes see-through portions along the sides of the partially enclosed space 13, which are intended to allow the caregiver to see their child. However, the X-tubes 22a and 22b in the pivot-only X-frame assemblies 20a span the entire sides of the partially enclosed space 13, thus obstructing the see-through portions of the soft goods 14 and, hence, limiting a caregiver's ability to visually check on their child in the partially enclosed space 13. For the playard 10b, the pivot and slidable X-frame assemblies 20b do not span the entire sides of the partially enclosed space 13. However, FIG. 1B shows the combination of the X-frame assemblies 20b and the soft goods 14 instead covers nearly the top half of the partially enclosed space 13, thus limiting the areas in which the caregiver can see into the partially enclosed space 13.

In another example, the frame 46 of the playard 10c allows the caregiver to readily see into the partially enclosed space 13 at the expense of using a more complicated folding/unfolding mechanism as described above. Indoor playards are also typically designed to be aesthetically pleasing for indoor settings (e.g., the indoor playard should match other indoor furniture), which can often lead to compromises in other areas such as ease of use. For instance, X-frame assemblies are often only used for outdoor playards because the appearance of X-tubes clashes with most indoor furniture.

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The Inventors further have observed that conventional playards often include complex latch mechanisms that are expensive to manufacture and difficult for consumers to use. For example, conventional playard frames that utilize pivot and slidable X-frame assemblies, such as the playard **10b** shown in FIG. 1B, often include multiple latch mechanisms disposed on opposing sides of the playard to prevent any one side of the playard frame from sagging downwards when locked in the unfolded configuration. In particular, as noted above FIG. 1B shows the playard **10b** includes a pair of latch mechanisms **16a** and **16b** disposed on opposing sides of the playard **10b**. To lock or unlock the playard **10b**, the caregiver needs to manually actuate each latch mechanism **16** one at a time, on different sides of the playard, which is inconvenient and cumbersome. In another example, the conventional indoor playard **10c** shown in FIG. 1C includes separate latch mechanisms for each hinge **36**. As described above, the caregiver needs to first lock each latch mechanism for each hinge **36** before unfolding the bottom support structure **34**, during which the playard **10c** may tip over and/or partially collapse if not held up properly by the caregiver.

The inclusion of multiple latch mechanisms increases the number of parts and, hence, the cost for manufacture. This drawback may be further exacerbated based on the placement and complexity of a given latch mechanism. For example, the latch mechanisms **16a** and **16b** in the playard **10b** are mounted to the sliders of the leg support assemblies **24**; as a result, the playard **10b** needs to include different types of sliders, i.e., the sliders **26b** forming part of the latch mechanisms **16a** and **16b**, and the different sliders **26a** for the remainder of the leg support assemblies **24**. Accordingly, this playard design increases the number of unique parts that need to be manufactured, which in turn increases manufacturing cost.

The Inventors have also observed that conventional playards also include various accessories to augment the functionality and/or environment for the child.

For example, the playard **10c** shown in FIG. 1C includes a bassinet accessory **60** to provide an elevated surface above the ground to support the child for their first several months of life. The Inventors have recognized and appreciated that a bassinet accessory provides caregivers a more convenient and accessible platform to place their child into the playard and/or to take their child out of the playard compared to the interior space of the playard (i.e., when the playard **10c** does not include the bassinet accessory **60**). The Inventors have also recognized a removable bassinet accessory effectively extends the lifetime use of the foldable playard from birth up until the child is typically able to climb out of the playard or weighs more than 30 lbs. However, the Inventors have also recognized that conventional bassinet accessories for playards often have to compromise between the accessibility of the elevated surface (e.g., how far the caregiver should reach into the playard to place their child into the bassinet accessory), ease of use (e.g., the procedure to fold and/or unfold the bassinet accessory and the foldable playard), and the overall size of the foldable playard and the bassinet accessory particularly when folded.

Bassinet accessories typically include a support structure to provide a flat surface for the child to sleep upon in order to meet various compliance standards (e.g., ASTM F2194 entitled, "Standard Consumer Safety Specification for Bassinets and Cradles"). For many conventional bassinet accessories, the support structure is a rigid structure that is not foldable (or unfoldable) with the playard frame. Thus, the bassinet accessory should be removed before folding the

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playard and/or installed when unfolding the playard, which adds additional steps for the caregiver to setup and/or tear down the playard. Additionally, the removal of the bassinet accessory requires the caregiver to provide extra space to store and/or transport the foldable playard and the bassinet accessory as separate items and may also increase the likelihood of the caregiver forgetting or losing the bassinet accessory especially when transporting the playard from one location to another location.

Bassinet accessories that fold and unfold together with the playard frame have been previously demonstrated to address, in part, the limitations associated with the rigid bassinet accessories described above. However, the Inventors have recognized conventional foldable bassinet accessories often achieve foldability with the playard by compromising other aspects of the bassinet accessory.

For example, the bassinet accessory **60** provides the playard **10c** with a relatively shallower elevated space to support the child (e.g., the top surface of the mattress is offset from the top rail **32** of the playard **10c** by a distance less than or equal to about 10 inches). This is achieved, in part, by utilizing a more complex folding mechanism that requires the user to assemble and disassemble part of the bassinet accessory **60** to facilitate unfolding and folding. For instance, FIG. 1E shows the bassinet accessory **60** for the playard **10c** includes bassinet soft goods **62** and two support tube assemblies **64** forming a support structure to support a mattress. As shown, each support tube assembly **64** includes support tubes **64a**, **64b**, and **64c** mounted to a bottom portion of the bassinet soft goods **62**.

To setup the bassinet accessory **60**, the caregiver should manually connect the support tube **64a** to the support tube **64b**, and connect the support tube **64c** to the support tube **64b**, to form a rigid support tube assembly **64** spanning the length of the bassinet accessory **60**. To tear down the bassinet accessory **60**, the caregiver should manually disconnect the support tubes **64a-64c** from one another. These additional steps not only make the bassinet accessory **60** more difficult to fold and/or unfold, but may also increase the likelihood of lost parts (e.g., the caregiver misplaces one of the support tubes separately from the mattress) and/or an improper setup especially if the caregiver does not properly connect the support tubes **64a-64c** together.

In some conventional foldable bassinet accessories, simpler folding mechanisms (e.g., a mechanism that does not require assembly of two or more components for deployment or disassembly for storage) have been used to simplify setup and/or tear down. However, these simpler folding mechanisms often result in an increase to the overall size of the playard in the folded configuration (e.g., a portion of the bassinet accessory extends appreciably beyond the envelope of the playard when folded) or results in a relatively deeper bassinet accessory (e.g., the top surface of the mattress is offset from the top rail **32** of the playard **10c** by a distance appreciably greater than 10 inches) to ensure the folding mechanism remains within the envelope of the folded playard. For the latter case, a deeper bassinet accessory results in the caregiver having to bend over further to place their child into the bassinet accessory and/or to take their child out of the bassinet accessory resulting in greater physical strain.

In another example, the playards **10a** and **10b** shown in FIG. 1A and FIG. 1B both include a canopy cover **40** to provide shade for a child when the playard is deployed in outdoor settings. However, the Inventors have recognized and appreciated that in some instances various accessories, and in particular canopy covers, often are prone to misuse

and premature detachment from the playard, and/or may compromise the safety of the child.

Generally, conventional canopy covers are supported by a separate canopy cover frame that directly mounts onto a top portion of the playard (e.g., the corners), which is already covered with soft goods. The presence of the soft goods can make it difficult for a caregiver to determine the proper location(s) on the playard where the canopy cover should be mounted, which can often result in incorrect canopy cover installations. Additionally, conventional canopy covers often are not attached securely to the playard due, in part, to the stack of multiple fabric layers in the soft goods. As a result, conventional canopy covers for outdoor playards are often prone to premature detachment due, for example, to a gust of wind.

Conventional canopy covers are also prone to being detached by a child placed within the partially enclosed space of the playard. For example, FIG. 1F shows the playard 10a of FIG. 1A with the canopy cover 40 pulled off a corner 28 by a child in the playard. As shown, a canopy bow 44 supports the canopy cover 40 over the playard 10a. The canopy bow 44 is attached to a canopy clip 42, which in turn should attach to the corner 28 covered by the soft goods 12. However, the combination of the canopy clip 42 not being securely attached to the corner 28 and the child's accessibility to the canopy clip 42 can lead to the removal of the canopy cover 40 by the child as shown in FIG. 1F. FIG. 1G shows another example where the child can further pull the canopy cover 40 into the partially enclosed space 13 of the playard 10a by pulling on the canopy bow 44 and/or the canopy clip 42.

In view of the foregoing observations by the Inventors, the present disclosure is thus directed to various inventive implementations of a foldable playard that is easier to operate (e.g., fold, unfold, latch and/or unlatch) as compared to conventional playards, structurally simpler with fewer parts for manufacture, and nonetheless sufficiently stable and rigid in structure so as to readily comply with various consumer safety standards (e.g., ASTM F406-19, referenced above).

In various inventive implementations, a foldable playard may generally include a frame that defines an interior space when unfolded, and soft goods that are mounted to the frame and partially disposed within the interior space to define a partially enclosed space for a child. In some implementations, a foldable playard includes an improved canopy cover assembly to cover the partially enclosed space (e.g., when the playard is deployed in an outdoor setting).

In one example of a frame for a foldable playard according to the present disclosure, the frame may be a closed frame that includes multiple leg support assemblies and X-frame assemblies arranged such that each leg support assembly is disposed along a side edge of the interior space, with the X-frame assemblies disposed between adjacent leg support assemblies along a side face of the interior space. The leg support assemblies enable the foldable playard to stand on the ground and the X-frame assemblies provide the structural support for the leg support assemblies as well as the mechanism to facilitate folding and/or unfolding of the playard. In some implementations, the leg support assemblies and the X-frame assemblies may define an interior space having a cross-section in the plane parallel to the ground that is polygonal in shape (e.g., a square, a rectangle, a hexagon).

Each leg support assembly of the frame of a foldable playard may include a leg tube, a corner mounted to a top end of the leg tube, a foot mounted to a bottom end of the

leg tube, and a slider that slides between the corner and the foot. The top and bottom ends of the leg tube may align with top and bottom vertices of the interior space, respectively. Each X-frame assembly may include at least one pair of X-tubes where each X-tube is rotatably coupled to at least another X-tube, the corner, and/or the slider. By coupling at least one of the X-tubes to the slider, the X-frame assembly becomes a pivot and slidable X-frame assembly in which the X-tubes are rotationally and translationally displaced when folding and/or unfolding the playard. In this manner, the combination of the X-frame assemblies and the leg support assemblies allows for a playard that folds into a smaller form occupying less volume and/or unfolds to provide a larger interior space and, hence, a larger partially enclosed space for the child as compared to conventional playards.

In one aspect, the X-frame assemblies of the frame of the foldable playard may be positioned sufficiently near a top portion of the interior space when the playard is deployed in an unfolded configuration such that each X-frame assembly effectively functions as a rigid top rail that mechanically connects adjacent leg support assemblies in the frame. Said in another way, the respective X-tubes of each X-frame assembly form a top perimeter structure that spans the top of the playard frame, thus outlining a top opening of the interior space. For example, each pair of X-tubes in each X-frame assembly may form a sufficiently shallow X-frame structure such that the X-tubes are mechanically similar to the rigid top rails in previous playards (e.g., the top rail 32 in the playard 10c).

However, unlike previous playards, the frames of the foldable playards disclosed herein are sufficiently rigid and stable with only X-frame assemblies coupling the leg support assemblies together. In other words, in example implementations, the frames of the foldable playards disclosed herein do not include a separate top rail (e.g., the webbing 14 of the playards 10a and 10b shown in FIG. 1A and FIG. 1B, or the top rail 32 of the playard 10c shown in FIG. 1C) or a bottom support structure (e.g., the bottom structure 34 of the playard 10c shown in FIG. 1C). Accordingly, the innovative frames described herein result in a more refined playard with sound mechanical stability using fewer parts.

In one aspect, the foldable playard frames disclosed in various examples herein achieve mechanical stability using fewer parts by reducing the length of the leg tubes as compared to conventional playards so as to make the frames less prone to being tilted and/or rotated (e.g., the resultant torque applied to a frame for a given force is reduced due to a shorter moment arm). As explained in greater detail below, in some implementations the length of a leg tube may be dimensioned based only on the portions of the foot and the corner that overlap with the leg tube and the distance the slider travels to sufficiently fold and/or unfold the frame.

Additionally, the dimensions and/or materials of the X-tubes employed in foldable playard frames disclosed in various examples herein may be chosen to provide sufficient mechanical rigidity to the frame. For example, the X-tubes may be formed from steel tubing with an exterior diameter of about 0.625 inches and a total length of about 24.5 inches. However, it should be appreciated the X-tubes may be formed from other materials (e.g., aluminum, carbon fiber) having different dimensions depending, in part, on the mechanical properties of the material and the desired dimensions of the interior space provided by the frame. In some implementations, as noted above, a frame comprising only leg support assemblies and X-frame assemblies as disclosed herein, without additional support structures, may satisfy the various mechanical rigidity, stability, and/or strength

requirements set forth in various consumer safety standards (e.g., ASTM F406-19, 7.3.3, 7.11).

It should be appreciated that soft goods may be coupled at various points along the frame so that the partially enclosed space formed by the soft goods opens properly when the playard is unfolded. However, the soft goods may generally be a compliant, flexible component that remains loose instead of being pulled taut and, hence, does not appreciably improve the mechanical rigidity and/or stability of the frame.

Additionally, by placing the X-frame assemblies of the frame near the top portion of the interior space, the sides of the frame are more exposed to provide a larger window for the caregiver to see their child when the child is placed within the interior space. Furthermore, soft goods attached to the frame may more readily cover the X-frame assemblies using less material. In some implementations, the soft goods may partially cover the X-frame assemblies to provide access to a latch mechanism (described in more detail below), while in other implementations the soft goods may completely cover the X-frame assemblies such that no portion of the X-frame assemblies are observable when the playard is unfolded (which may improve, in part, the aesthetic appearance of the playard for both outdoor and indoor settings).

As discussed in greater detail below, the “top portion” of foldable playard frame in a given example implementation may generally refer to the portion of the frame proximate to the top ends of the leg tubes and/or the corners of each leg support assembly. The leg tubes of the respective leg support assemblies may generally have substantially identical lengths. In some implementations, the top portion of the frame may be defined as having: 1) a top horizontal plane that intersects the top ends of the leg tubes and/or the corners; and 2) a bottom horizontal plane that is offset vertically from the top horizontal plane such that the X-tubes are located entirely within the top and bottom horizontal planes when the X-frame assembly is unfolded. In some implementations, the bottom horizontal plane may be offset from the top horizontal plane by a distance less than or equal to 30% of the total length of the leg tubes and, more preferably, less than or equal to 20% of the total length of the leg tubes.

As noted above, in some implementations a foldable playard frame may include one or more X-frame assemblies forming a single X-frame structure with one pair of X-tubes. Each X-tube in the pair of X-tubes may be rotatably coupled to a corner of one leg support assembly, a slider of another leg support assembly, and the other X-tube in the pair of X-tubes. In other example implementations, a foldable playard frame may include one or more X-frame assemblies forming a double X-frame structure with two pairs of X-tubes. In examples employing this double X-frame structure, each X-tube is coupled to either a slider or a corner of one leg support, the X-tube within the same pair of X-tubes, and another X-tube from another pair of X-tubes. In this manner, the frame may provide an interior space having a horizontal cross section in which the sides have different dimensions (e.g., an interior space with a rectangular shape).

In another aspect, a foldable playard frame according to the present disclosure may include a latch mechanism to maintain the frame in an unfolded configuration. In some implementations, the frame may only include a single latch mechanism to maintain the frame in the unfolded configuration. In some implementations, the single latch mechanism is configured such that, as a caregiver unfolds the frame (e.g., by moving the slider in one leg support assembly

towards the corner), the single latch mechanism is automatically actuated to lock the frame in the unfolded configuration. In this manner, the process of unfolding and locking the playard may be readily accomplished with the caregiver positioned at one side and/or one corner of the playard (i.e., the caregiver does not have to move around the playard to actuate multiple latch mechanisms). Furthermore, the caregiver may unfold and lock the playard using a single hand. For example, the single latch mechanism may automatically lock when the slider is displaced a sufficient distance along the leg tube.

In some implementations, the latch mechanism may be preferably disposed in the top portion of the frame as defined above. For example, the latch mechanism may include a latch member having a first end coupled to the corner of one leg support assembly and a second end that couples to a X-tube of one X-frame assembly or the slider. In this manner, the latch mechanism may be partially covered or, in some instances, fully covered by the soft goods.

The latch mechanism may also be coupled to various components of the frame including, but not limited to, an X-tube, a leg tube, a slider, and a corner. In some implementations, the latch mechanism may be coupled to the components of the X-frame assembly and/or the leg support assembly without having to modify the respective components of the X-frame assembly and the leg support assembly. For example, the latch mechanism may include a latch member that is rotatably coupled to the corner of one leg support assembly via a pin joint that also serves to rotatably couple an X-tube to the corner. In this manner, the playard may include a smaller number of unique parts for manufacture. In some implementations, the playard may include identical corners and/or identical sliders for the multiple leg support assemblies.

In some implementations, the latch mechanism may be a tool-less mechanism that is actuated in one or two steps by the caregiver. In one example, the latch member may couple respective components of the X-frame assembly and/or the leg support assembly to maintain an unfolded configuration via various attachment mechanisms including, but not limited to, a snap-fit connection, a spring-loaded pin, and a spring-loaded rotational lock off mechanism.

In some implementations, the latch mechanism may be a double-action latch that includes a latch member (e.g., mounted to the corner of one leg support assembly) and a latch boss (e.g., mounted to a X-tube of one X-frame assembly). The latch boss may include an undercut portion and the latch member may include a latch opening to receive the latch boss with a tab disposed within the latch opening to engage the undercut portion. In some implementations, the tab may include a slot and the undercut portion may include a rib to align the latch member and the latch boss when locking the latch mechanism. The undercut portion and the tab may be shaped such that the caregiver is unable to unlock the latch mechanism by pulling the latch member without applying an excessive amount of force (e.g., greater than 10 lbs of force). Instead, the caregiver may first squeeze the respective X-tubes of the X-frame assembly to displace the latch boss within the latch opening of the latch member to disengage the tab from the undercut portion. While squeezing the X-tubes together, the caregiver may then pull the latch member off the latch boss, thus unlocking the latch mechanism.

In yet another aspect, the foldable playard may include soft goods to define the partially enclosed space in which the child may play and/or sleep. Generally, the soft goods may cover a portion of the frame (e.g., the corners of the leg

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support assembly, a portion of the X-frame assemblies). In some implementations, the soft goods may be coupled directly to the frame (e.g., a corner) via one or more snap-fit connections. The soft goods may further include a semi-rigid tab disposed near the top edge of the soft goods to support a snap-fit connector to ensure the soft goods remain flush against the frame when attached (i.e., the top edge of the soft goods does not flip upwards to expose an interior portion of the soft goods). The soft goods may further include a floor portion that directly rests on the ground, as well as side portions, where the floor and side portions define the bottom and sides of the partially enclosed space. In some implementations, the side portions may be transparent and/or see-through (e.g., a mesh) to allow the caregiver to readily see their child in the playard.

In yet another aspect, a foldable playard according to the present disclosure may also include a canopy cover assembly, disposed on top of the playard frame and soft goods, to provide shade for the child within the playard. The canopy cover assembly may generally include multiple canopy support assemblies that provide a canopy cover frame or support structure. Each canopy support assembly may generally include a canopy bow that supports the canopy cover and a canopy clip to mount the canopy support assembly to the frame. In some implementations, different types of canopies (e.g., a half canopy, a full canopy) may be mounted onto the playard depending on the coverage desired by the caregiver.

In some implementations, the canopy clip may include snap-in features to directly couple the canopy clip the leg tube of one leg support assembly. In this manner, the canopy clip may be more securely attached to the frame (i.e., the canopy clip does not attach to a portion of the frame covered by soft goods), thereby reducing the likelihood the canopy cover assembly is accidentally detached from the frame. Each canopy clip may be further disposed outside the interior space along an exterior portion of one leg support assembly (e.g., proximate to the corner and/or the slider when the playard is unfolded). Additionally, the canopy bow may couple to the canopy clip such that a portion of the canopy bow is also disposed outside the interior space near the corner and/or the slider of the leg support assembly. The particular placement of the canopy clip and the portion of the canopy bow that overlaps the exterior portion of the frame may further limit the child's access to the various components of the canopy cover assembly, thus reducing the likelihood the child can detach and pull the canopy cover into the playard.

In yet another aspect, a foldable playard according to the present disclosure may include a bassinet accessory, disposed within the interior space of the frame and the partially enclosed space of the playard soft goods, to provide an elevated surface to support the child. The bassinet accessory may generally include a support structure that defines a relatively smaller partially enclosed space affiliated with the bassinet accessory to contain the child when the bassinet accessory is unfolded (e.g., the relatively smaller partially enclosed space of the bassinet accessory may be disposed within the partially enclosed space of the playard soft goods).

The support structure of the bassinet accessory may include bassinet soft goods with side surfaces and a bottom surface that at least partially define and surround the relatively smaller partially enclosed space of the bassinet accessory. The support structure may further include a hub and multiple support tubes that together form a rigid structure in the deployed unfolded configuration. Each support tube may

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be rotatably (e.g., pivotably) coupled to the hub to facilitate folding and unfolding of the bassinet accessory. The bassinet accessory may also include a mattress disposed on the hub and the support tubes in the deployed unfolded configuration to provide a cushioned surface for the child to rest upon. The mattress may be removable and foldable.

The bassinet accessory and, in particular, the support structure may fold and unfold together with the frame and the soft goods when installed on the foldable playard. The bassinet accessory may provide a relatively shallow partially enclosed space to improve accessibility for the caregiver. For example, the distance from the top surface of the mattress to the top side of the foldable playard may range between 7.5 inches and about 10 inches. More generally, the height of the bassinet accessory when installed on the playard, $h_{t,1}$, which is defined as the distance between respective bottom corner portions of the bassinet soft goods and the top of the foldable playard (e.g., a top horizontal plane defined by the playard), may range between 7.5 inches and about 12 inches.

The bassinet accessory may also include a folding mechanism that does not require assembly and/or disassembly when folding and unfolding the bassinet accessory together with the foldable playard. Instead, the hub and the support tubes may form a foldable structure with integrated mechanical stops and/or a locking mechanism (e.g., a hub latch) to maintain the hub and the support tubes in the desired unfolded configuration. In this manner, the procedure for folding and unfolding the foldable playard together with the installed bassinet accessory may be simplified compared to conventional bassinet accessories (e.g., the bassinet accessory 60 for the playard 10c).

In one example, the hub may be disposed at the center of the bottom surface of the bassinet soft goods and the support tubes may be disposed and oriented along the diagonal segments of the bottom surface. Said in another way, the support tubes may extend radially from the hub to the respective corner portions of the bottom surface of the bassinet soft goods. The support tubes may be further attached to the bassinet soft goods via one or more attachment mechanisms (e.g., a screw fastener, a strap) such that the bassinet soft goods and the support tubes move together. When folding or unfolding the bassinet accessory, the caregiver may pull up or push down on the hub, thus causing the support tubes and the bassinet soft goods to fold or unfold.

Additionally, the bassinet accessory may be disposed substantially within the interior space of the playard frame in both the unfolded and folded configurations such that the overall size of the foldable playard with the bassinet accessory remains substantially similar or the same as the foldable playard without the bassinet accessory. In this manner, the compact shape of the playard in the folded configuration is maintained for ease of storage and/or transport.

The bassinet accessory exhibiting the features described above (e.g., a relatively shallow height, a simple folding mechanism, and a compact size) may be accomplished in multiple ways. In one example, the support tubes may change in length between the folded and unfolded configurations. For instance, the hub may move in an upwards direction when folding the bassinet accessory and, conversely, in a downwards direction when unfolding the bassinet accessory. To ensure the hub does not extend appreciably outside the interior space of the playard as the playard is folded up, particularly given the relatively shallow height of the bassinet accessory, each support tube may be tele-

scoping (e.g., each support tube may include a first support tube and a second support tube telescopically coupled to the first support tube).

When the bassinet accessory is unfolded, the extended support tube may have an overall length, $L_{t,1}$, greater than the height, $h_{t,1}$, of the bassinet accessory. However, when the bassinet accessory is folded, the first support tube may telescopically move towards the second support tube. Thus, the overall length of the support tube changes from $L_{t,1}$ to a length, $L_{t,2}$ in the folded configuration, where $L_{t,2}$ is less than $L_{t,1}$. In various examples discussed in greater detail below, the length $L_{t,2}$ may be approximately less than or equal to the height $h_{t,1}$ of the bassinet accessory. It should be appreciated that the height of the bassinet accessory, $h_{t,1}$, may in some circumstances change somewhat when folding and unfolding the bassinet accessory (e.g., the bottom of the bassinet soft goods may fold and bunch up). However, in other circumstances, respective bottom corners of the bassinet accessory soft goods do not undergo significant vertical displacement between the folded and unfolded configurations. In any event, the above constraints imposed on the length of the support tube and height of the bassinet accessory in the respective folded and unfolded configurations may still be satisfied so as to mitigate substantial protrusion of the hub above a top of the playard in the folded configuration.

For this example, the bassinet accessory may not include a separate locking mechanism (e.g., a hub latch) to maintain the unfolded configuration. Instead, a combination of the integrated mechanical stops and the weight of the hub, the support tubes, the mattress, and/or the child may ensure the bassinet accessory remains in the deployed unfolded configuration. In this manner, the number of parts and the cost for manufacture of the bassinet accessory may be reduced.

In another example, an interior space of the playard below the bassinet accessory in the unfolded configuration may be utilized to contain the hub and/or the support tubes of the bassinet accessory when folding up the playard for storage and/or transport in the folded configuration. This may be accomplished, in part, given the shallow height of the bassinet accessory, which results in a larger portion of the interior space of the playard frame being disposed directly below the bassinet accessory. For instance, the height, h_b , corresponding to the distance from the ground to the bottom surface of the bassinet soft goods may be greater than or equal to about 18 inches. For this example, the hub may move in a downwards direction when folding the bassinet accessory and, conversely, in an upwards direction when unfolding the bassinet accessory. To ensure the hub and/or the support tubes remain contained within the interior space of the playard, the length of the support tube, L_s , may be approximately equal to or less than the height, h_b .

For this example, the integrated mechanical stops may limit further upward movement of the hub once the hub and the support tubes are in the desired unfolded configured (e.g., the hub and the support tubes form a substantially flat platform supporting the mattress). The hub may further include a hub latch that, when actuated, prevents the hub from moving downwards. Thus, the combination of the mechanical stops and the hub latch may maintain the bassinet accessory in the deployed unfolded configuration.

In one example, a frame for a foldable playard has a compact folded configuration for storage of the frame and a deployed unfolded configuration to support the foldable playard in an upright position on a ground surface to contain a child in an interior space of the foldable playard. The frame includes a plurality of leg support assemblies extending

upward from the ground surface when the frame is in the deployed unfolded configuration where each leg support assembly of the plurality of leg support assemblies includes a bottom end supported by the ground surface and a top portion opposite to the bottom end. The frame further includes a plurality of X-frame assemblies coupled to the plurality of leg support assemblies where each X-frame assembly of the plurality of X-frame assemblies is coupled to respective top portions of adjacent leg support assemblies of the plurality of leg support assemblies when the frame is in the deployed unfolded configuration such that, in the deployed unfolded configuration of the frame, the plurality of X-frame assemblies forms a top perimeter structure of the frame outlining the interior space of the foldable playard and the plurality of X-frame assemblies does not significantly impede visibility of the child when the child is in the interior space of the foldable playard. The plurality of X-frame assemblies constitutes the only interconnection in the frame between respective pairs of leg support assemblies of the plurality of leg support assemblies. Each leg support assembly may include a leg tube with an oval-shaped cross-section.

In another example, a foldable playard defining an interior space when in an unfolded configuration includes a plurality of leg support assemblies where each leg support assembly includes a leg tube disposed along a side edge of the interior space having a top end disposed at a top vertex of the interior space, a corner coupled to the top end of the leg tube, and a slider slidably coupled to the leg tube such that the slider is disposed proximate to the corner when the foldable playard is in the unfolded configuration. The foldable playard further includes a plurality of X-frame assemblies positioned at respective side faces of the interior space between adjacent leg support assemblies where each X-frame assembly of the plurality of X-frame assemblies forms a top rail between adjacent leg support assemblies. Additionally, the sliders in the plurality of leg support assemblies are identical, the corners in the plurality of leg support assemblies are identical, and respective pairs of leg support assemblies are only coupled together via at least one X-frame assembly of the plurality of X-frame assemblies. The leg tube may also have an oval-shaped cross-section.

In another example, a foldable playard defining an interior space when in an unfolded configuration includes a plurality of leg support assemblies where each leg support assembly includes a leg tube disposed along a side edge of the interior space having a top end disposed at a top vertex of the interior space, a corner coupled to the top end of the leg tube, and a slider slidably coupled to the leg tube such that the slider is disposed proximate to the corner when the foldable playard is in the unfolded configuration. The foldable playard further includes a plurality of X-frame assemblies positioned at respective side faces of the interior space between adjacent leg support assemblies of the plurality of leg support assemblies where each X-frame assembly of the plurality of X-frame assemblies forms a top rail between the adjacent leg support assemblies. The foldable playard further includes a single latch mechanism coupled to one leg support assembly of the plurality of leg support assemblies to maintain the foldable playard in the unfolded configuration when the latch mechanism is in a locked configuration. Additionally, respective pairs of adjacent leg support assemblies are only coupled together via one X-frame assembly of the plurality of X-frame assemblies. The leg tube may also have an oval-shaped cross-section.

In another example, a foldable playard defining an interior space when in an unfolded configuration includes a plurality

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of leg support assemblies where each leg support assembly includes a leg tube disposed along a side edge of the interior space having a top end disposed at a top vertex of the interior space, a corner disposed on the top end of the leg tube, and a slider slidably coupled to the leg tube such that the slider is disposed proximate to the corner when the foldable playard is in the unfolded configuration. The foldable playard further includes a plurality of X-frame assemblies positioned at respective side faces of the interior space where each X-frame assembly of the plurality of X-frame assemblies is coupled to adjacent leg support assemblies of the plurality of leg support assemblies. The foldable playard further includes a latch mechanism that directly couples together the corner of one leg support assembly of the plurality of leg support assemblies and a X-tube of one X-frame assembly of the plurality of X-frame assemblies when the latch mechanism is in a locked configuration where the latch mechanism provides the only mechanism to maintain the foldable playard in the unfolded configuration. Additionally, respective pairs of leg support assemblies are only coupled together via at least one X-frame assembly of the plurality of X-frame assemblies. The leg tube may also have an oval-shaped cross-section.

In another example, a foldable playard defining an interior space when in an unfolded configuration includes a plurality of leg support assemblies where each leg support assembly includes a leg tube disposed along a side edge of the interior space having a top end disposed at a top vertex of the interior space, a corner coupled to the top end of the leg tube, and a slider slidably coupled to the leg tube such that the slider is disposed proximate to the corner when the foldable playard is in the unfolded configuration. The foldable playard further includes a plurality of X-frame assemblies positioned at respective side faces of the interior space where each X-frame assembly of the plurality of X-frame assemblies is coupled to adjacent leg support assemblies. The foldable playard further includes a plurality of canopy support assemblies disposed, in part, above the interior space where each canopy support assembly includes a canopy bow disposed, in part, above the interior space and a canopy clip disposed outside the interior space proximate to a first leg support assembly of the plurality of leg support assemblies. The canopy clip includes one or more snap features directly coupled to the leg tube of the first leg support assembly and a canopy bow opening to receive a portion of the canopy bow to couple the canopy bow to the canopy clip. The foldable playard also includes a canopy cover supported by respective canopy bows of the plurality of canopy support assemblies to cover at least a portion of the interior space.

In another example, a foldable playard includes a leg support assembly. The leg support includes a leg tube having a top end, a corner disposed on the top end of the leg tube, and a slider slidably coupled to the leg tube. The foldable playard further includes a X-frame assembly coupled to the leg support assembly where the X-frame assembly includes a first X-tube rotatably coupled to the corner of the leg support assembly and a second X-tube rotatably coupled to the slider of the leg support assembly and the first X-tube. The foldable playard further includes a latch mechanism coupled to the leg support assembly and the X-frame assembly to maintain the foldable playard in an unfolded configuration when in a locked configuration where the latch mechanism includes a latch boss coupled to the second X-tube and disposed proximate to the slider of the leg support assembly having an undercut portion and a latch member coupled to the corner of the leg support assembly

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having a latch opening and a tab disposed within the latch opening. The undercut portion of the latch boss retains the tab of the latch member when the latch mechanism is engaged thereby maintaining the foldable playard in the unfolded configuration.

In yet another example, a foldable playard defining an interior space with a cross-sectional shape, in a plane parallel to a ground, forming a regular hexagon when in an unfolded configuration, includes six leg support assemblies. Each leg support assembly includes a leg tube arranged such that a longitudinal axis associated with the leg tube intersects a respective corner of the regular hexagon and further has a top end and a bottom end, a foot coupled to the bottom end of the leg tube to contact a ground to support the foldable playard, a corner coupled to the top end of the leg tube, and a slider slidably coupled to the leg tube and positioned between the foot and the corner where the slider is disposed proximate to the corner when the foldable playard is in the unfolded configuration and disposed proximate to the foot when the foldable playard is in a folded configuration. The foldable playard further includes six X-frame assemblies arranged such that each X-frame assembly is positioned along a side of the regular hexagon. Each X-frame assembly of the six X-frame assemblies forms a top rail between adjacent leg support assemblies. The six X-frame assemblies includes a first X-frame assembly disposed between and coupled to a first leg support assembly and a second leg support assembly of the six leg support assemblies where the first X-frame assembly includes a first X-tube having a first end rotatably coupled to the corner of the first leg support assembly and a second end rotatably coupled to the slider of the second leg support assembly and a second X-tube having a first end rotatably coupled to the corner of the second leg support assembly and a second end rotatably coupled to the slider of the first leg support assembly. The second X-tube is rotatably coupled to the first X-tube. The foldable playard further includes a latch mechanism coupled to only the first leg support assembly and only the first X-frame assembly to maintain the foldable playard in the unfolded configuration when in a locked configuration where the latch mechanism includes a latch boss coupled to one of the second X-tube and disposed proximate to the slider of the first leg support assembly having an undercut portion and a latch member having a first end coupled to the corner of the first leg support assembly, a latch opening disposed proximate to a pulling tab, and a tab disposed within the latch opening. The latch mechanism is changed to the locked configuration by moving the slider of the first leg support assembly towards the corner of the first leg support assembly until the latch member snaps onto the latch boss such that the tab of the latch member contacts the undercut portion of the latch boss and the central rib is disposed within the central slot. The latch mechanism is changed to an unlocked configuration by squeezing the first and second X-tubes together to release the tab of the latch member from the undercut portion of the latch boss and, while squeezing the first and second X-tubes together, pulling the latch member away from the latch boss. Additionally, respective pairs of leg support assemblies are only coupled together via at least one X-frame assembly of the plurality of X-frame assemblies, the sliders in the six leg support assemblies are identical, and the corners in the six leg support assemblies are identical.

It should be appreciated that all combinations of the foregoing concepts and additional concepts discussed in greater detail below (provided such concepts are not mutually inconsistent) are contemplated as being part of the

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inventive subject matter disclosed herein. In particular, all combinations of claimed subject matter appearing at the end of this disclosure are contemplated as being part of the inventive subject matter disclosed herein. It should also be appreciated that terminology explicitly employed herein that also may appear in any disclosure incorporated by reference should be accorded a meaning most consistent with the particular concepts disclosed herein.

BRIEF DESCRIPTION OF THE DRAWINGS

The skilled artisan will understand that the drawings primarily are for illustrative purposes and are not intended to limit the scope of the inventive subject matter described herein. The drawings are not necessarily to scale; in some instances, various aspects of the inventive subject matter disclosed herein may be shown exaggerated or enlarged in the drawings to facilitate an understanding of different features. In the drawings, like reference characters generally refer to like features (e.g., functionally similar and/or structurally similar elements).

FIG. 1A shows a conventional outdoor playard with a pivot-only X-frame assembly and a canopy cover.

FIG. 1B shows another conventional outdoor playard with a pivot and slidable X-frame assembly.

FIG. 1C shows a conventional indoor playard.

FIG. 1D shows the indoor playard of FIG. 1C being folded for storage or transport.

FIG. 1E shows the assembly of a bassinet accessory for the indoor playard of FIG. 1C.

FIG. 1F shows a conventional outdoor playard with a canopy cover assembly where the canopy cover is pulled off the corner of the X-frame assembly by a child located within the playard.

FIG. 1G shows another conventional outdoor playard with a canopy cover assembly where the canopy cover is pulled inside the interior space of the playard by a child located within the playard.

FIG. 2A shows a top perspective view of an exemplary playard forming a hexagonal-shaped interior space. The playard is in an unfolded configuration.

FIG. 2B shows a front view of the playard of FIG. 2A.

FIG. 2C shows a top view of the playard of FIG. 2A.

FIG. 2D shows a top perspective view of the playard of FIG. 2A in a folded configuration.

FIG. 2E shows a front view of the playard of FIG. 2D.

FIG. 2F shows a top view of the playard of FIG. 2D.

FIG. 3A shows a top perspective view of a X-frame assembly in the playard of FIG. 2A.

FIG. 3B shows a top view of the X-frame assembly of FIG. 3A.

FIG. 3C shows a top perspective view of a corner and a slider of a leg support assembly in the playard of FIG. 2A.

FIG. 3D shows a bottom perspective view of the corner and the slider of FIG. 3C.

FIG. 3E shows a top perspective view of a leg tube and a foot in the leg support assembly of FIG. 3C.

FIG. 4A shows an exploded top perspective view of the X-frame assembly of FIG. 3A and the leg support assembly of FIG. 3C.

FIG. 4B shows a magnified top perspective view of the corner and the slider in the leg support assembly and the X-tubes in the X-frame assembly of FIG. 4A.

FIG. 4C shows a magnified top perspective view of the leg tube and the foot in the leg support assembly of FIG. 4A.

FIG. 5A shows a top perspective view of the playard of FIG. 2A with soft goods.

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FIG. 5B shows a magnified view of top portion of the soft goods of FIG. 5A disposed over the corner of the leg support assembly in the playard of FIG. 2A.

FIG. 5C shows a magnified view of the top portion of FIG. 5B flipped upwards to show a tab and a snap-fit connector.

FIG. 6A shows a top perspective of a double-action latch mechanism in the playard of FIG. 2A.

FIG. 6B shows a top perspective of the double-action latch mechanism of FIG. 6A with the latch member removed.

FIG. 6C shows a magnified view of the latch member in the double-action latch mechanism of FIG. 6A.

FIG. 6D shows a magnified view of the latch boss in the double-action latch mechanism of FIG. 6A.

FIG. 6E shows an illustration for unlocking the double-action latch mechanism of FIG. 6A.

FIG. 7A shows a test being performed on the playard of FIG. 2A to evaluate the restraining force of the latch mechanism of FIG. 6A.

FIG. 7B shows a stability test being performed on the playard of FIG. 2A.

FIG. 8A shows a top perspective of the playard of FIG. 2A with soft goods and a flex lock latch mechanism with a latch opening. The playard is in an unfolded configuration.

FIG. 8B shows a magnified view of the flex lock latch mechanism of FIG. 8A.

FIG. 8C shows a perspective view of the flex lock latch mechanism of FIG. 8A with the soft goods removed and the flex lock latch mechanism in a locked configuration.

FIG. 8D shows a perspective view of the flex lock latch mechanism of FIG. 8C in an unlocked configuration.

FIG. 9A shows a top perspective view of the playard of FIG. 2A with soft goods and a flex lock latch mechanism with a latch member having a snap-fit connector. The playard is in an unfolded configuration.

FIG. 9B shows a magnified view of the flex lock latch mechanism of FIG. 9A.

FIG. 9C shows a top perspective view of the playard of FIG. 9A with the soft goods removed.

FIG. 9D shows a perspective view of the flex lock latch mechanism of FIG. 9C in a locked configuration.

FIG. 9E shows a perspective view of the flex lock latch mechanism of FIG. 9D in an unlocked configuration.

FIG. 9F shows a perspective view of the flex lock latch mechanism of FIG. 9E where the playard is partially folded after unlocking the flex lock latch mechanism.

FIG. 10 shows another flex lock latch mechanism with a latch member having a snap-fit connector where the latch member of the latch mechanism is coupled to a X-tube of an X-frame assembly.

FIG. 11A shows a top perspective view of the playard of FIG. 2A with soft goods and a flex lock latch mechanism with a hook structure. The playard is in an unfolded configuration.

FIG. 11B shows a magnified view of the flex lock latch mechanism of FIG. 11A.

FIG. 11C shows a perspective view of the flex lock latch mechanism of FIG. 11A with the soft goods removed and the flex lock latch mechanism in a locked configuration.

FIG. 11D shows a perspective view of the flex lock latch mechanism of FIG. 11C in an unlocked configuration.

FIG. 12A shows a top perspective view of the playard of FIG. 2A with a latch mechanism mounted to a slider and a corner of a leg support assembly. The playard is in an unfolded configuration.

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FIG. 12B shows a magnified view of the latch mechanism of FIG. 12A.

FIG. 13A shows a top perspective view of the playard of FIG. 2A with a latch mechanism mounted to a pair of X-tubes in the X-frame assembly. The playard is in an unfolded configuration.

FIG. 13B shows a perspective view of the playard of FIG. 13A in a folded configuration.

FIG. 13C shows a perspective of the X-frame assembly with the latch mechanism of FIG. 13A.

FIG. 13D shows an exploded view of the X-frame assembly with the latch mechanism of FIG. 13C.

FIG. 13E shows a perspective view of the latch mechanism of FIG. 13A in a locked configuration.

FIG. 13F shows a perspective view of the latch mechanism of FIG. 13E in an unlocked configuration.

FIG. 13G shows a top view of the latch mechanism of FIG. 13E.

FIG. 13H shows a top view of the latch mechanism of FIG. 13F.

FIG. 14A shows a top perspective view of the playard of FIG. 2A with a latch mechanism that includes a spring-loaded pin disposed at one end of a X-tube to engage with a leg tube. The playard is in an unfolded configuration.

FIG. 14B shows a side view of the latch mechanism of FIG. 14A in a locked configuration.

FIG. 14C shows a side view of the latch mechanism of FIG. 14B in an unlocked configuration.

FIG. 14D shows a side view of the latch mechanism of FIG. 14C after the playard is folded.

FIG. 15A shows a top perspective view of the playard of FIG. 2A with a latch mechanism that includes a snap-fit connector disposed at one end of a X-tube. The playard is in an unfolded configuration.

FIG. 15B shows a perspective view of the playard of FIG. 15A in a folded configuration.

FIG. 15C shows a side view of the latch mechanism of FIG. 15A in a locked configuration.

FIG. 15D shows a side view of the latch mechanism of FIG. 15A in an unlocked configuration and the playard in a folded configuration.

FIG. 16A shows a top perspective view of the playard of FIG. 2A with the latch mechanisms of FIGS. 13A and 14A installed. The playard is in an unfolded configuration.

FIG. 16B shows a perspective view of the playard of FIG. 16A in a folded configuration.

FIG. 17A shows a top perspective view of an exemplary playard forming a rectangular-shaped interior space with soft goods. The playard is in an unfolded configuration.

FIG. 17B shows another perspective view of the playard of FIG. 17A.

FIG. 17C shows a top perspective view of the playard of FIG. 17A in a folded configuration.

FIG. 17D shows a top perspective view of the playard of FIG. 17A in a partially unfolded configuration.

FIG. 18A shows a top perspective view of the playard of FIG. 17A with the soft goods removed.

FIG. 18B shows a magnified view of a corner and a slider of a leg support assembly in the playard of FIG. 18A.

FIG. 19A shows a top perspective view of the playard of FIG. 17C with the soft goods removed.

FIG. 19B shows a magnified view of the slider and a foot in the leg support assembly of FIG. 19A.

FIG. 20A shows a top perspective view of the playard of FIG. 17D with the soft goods removed.

FIG. 20B shows a top, side perspective view of the playard of FIG. 20A.

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FIG. 20C shows a top, front perspective view of the playard of FIG. 20A.

FIG. 20D shows a magnified view of the corner in the leg support assembly of FIG. 20A.

FIG. 20E shows a magnified view of the slider in the leg support assembly of FIG. 20A.

FIG. 21A shows a perspective view of the playard of FIG. 17D with the soft goods partially removed from the leg support assembly.

FIG. 21B shows a perspective view of the foot of the leg support assembly attached to the soft goods of FIG. 21A.

FIG. 22 shows a stability test being performed on the playard of FIG. 17A.

FIG. 23A shows a top, front perspective view of another exemplary playard forming a rectangular, convex-shaped interior space with soft goods. The rectangular playard is also shown with the bassinet accessory of FIG. 41A. The playard is in an unfolded configuration.

FIG. 23B shows a top perspective view of the playard of FIG. 23A.

FIG. 23C shows a front view of the playard of FIG. 23B.

FIG. 23D shows a top perspective view of the playard of FIG. 23B where the soft goods are shown as being transparent.

FIG. 23E shows a front view of the playard of FIG. 23D.

FIG. 24 shows a top perspective view of the playard of FIG. 23B with the soft goods removed.

FIG. 25A shows an exploded perspective view of a leg assembly having a wheel in the playard of FIG. 23A.

FIG. 25B shows an exploded perspective view of a leg assembly having a foot in the playard of FIG. 23A.

FIG. 26A shows a perspective view of the playard of FIG. 23A in a partially unfolded configuration.

FIG. 26B shows a cross-sectional view of a slider of a leg support assembly in the playard corresponding to the plane A-A of FIG. 26A.

FIG. 27A shows a magnified view of the slider and a corner of the leg support assembly in the playard of FIG. 23A.

FIG. 27B shows the soft goods attached to the corner of FIG. 27A.

FIG. 27C shows the soft goods removed from the corner of FIG. 27A.

FIG. 28A shows a top perspective view of the playard of FIG. 23A with a snap-fit latch mechanism disposed over the soft goods.

FIG. 28B shows a magnified view of the latch member of the latch mechanism of FIG. 28A.

FIG. 28C shows a perspective of a latch member in the latch mechanism of FIG. 28A.

FIG. 29A shows a top rail to corner post attachment test being performed on the playard of FIG. 23A.

FIG. 29B shows a testing apparatus mounted to the double X-frame assembly in the playard of FIG. 23A.

FIG. 29C shows the playard after conducting the test of FIG. 29A.

FIG. 29D shows the testing apparatus mounted to the double X-frame assembly in the playard of FIG. 23A.

FIG. 30A shows a strength test being applied to the double X-frame assembly in the playard of FIG. 23A.

FIG. 30B shows the playard of FIG. 30A after the strength test.

FIG. 30C shows the playard of FIG. 30B with the soft goods partially removed from the X-frame assembly.

FIG. 31 shows a stability test being performed on the playard of FIG. 23A.

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FIG. 32A shows a top perspective view of the playard of FIG. 2A with an exemplary canopy cover assembly that covers the entire interior space of the playard. The canopy cover is not shown.

FIG. 32B shows a front view of the playard and the canopy cover assembly of FIG. 32A.

FIG. 32C shows a top view of the playard and the canopy cover assembly of FIG. 32A.

FIG. 32D shows a magnified view of a canopy clip of a canopy support assembly in the canopy cover assembly of FIG. 32A coupled to the leg support assembly of the playard.

FIG. 32E shows a magnified view of the canopy clip of FIG. 32D.

FIG. 32F shows a perspective view of the canopy clip of FIG. 32D.

FIG. 33A shows a top view of the canopy clip of FIG. 32D being pressed onto the leg tube.

FIG. 33B shows a perspective view of the canopy clip of FIG. 32D where one lead-in feature is hooked onto the leg tube first and the canopy clip is rotated such that the other lead-in feature contacts the leg tube.

FIG. 34A shows a top perspective view of a hub in the canopy cover assembly of FIG. 32A.

FIG. 34B shows a bottom perspective view of the hub of FIG. 34A.

FIG. 35A shows a top, front perspective view of the playard of FIG. 2A with an exemplary canopy cover assembly that covers half the interior space of the playard and does not include a hub.

FIG. 35B shows a top, side perspective view of the playard and the canopy cover assembly of FIG. 35A.

FIG. 36A shows a top perspective view of the playard and the canopy cover assembly of FIG. 35A with the canopy cover removed.

FIG. 36B shows a front view of the playard and the canopy cover assembly of FIG. 36A.

FIG. 36C shows a top view of the playard and the canopy cover assembly of FIG. 36A.

FIG. 36D shows a perspective view of the canopy clip of the canopy support assembly in the canopy cover assembly of FIG. 36A.

FIG. 36E shows another perspective view of the canopy clip of FIG. 36D.

FIG. 37A shows a top, front perspective view of the playard of FIG. 2A with an exemplary canopy cover assembly that covers half interior space of the playard and includes a hub.

FIG. 37B shows a front view of the playard and the canopy cover assembly of FIG. 37A.

FIG. 37C shows a top view of the playard and the canopy cover assembly of FIG. 37A.

FIG. 38A shows a top perspective view of the hub of FIG. 37A.

FIG. 38B shows a bottom perspective view of the hub of FIG. 38A.

FIG. 39A shows a top perspective view of another hub that allows each canopy bow to pivot about a horizontal axis relative to the hub.

FIG. 39B shows a bottom perspective view of the hub of FIG. 39A.

FIG. 40A shows a top perspective view of another hub that allows each canopy bow to pivot about a vertical axis relative to the hub.

FIG. 40B shows a bottom perspective view of the hub of FIG. 40A.

FIG. 41A shows a top perspective view of the playard of FIG. 17A and an exemplary bassinet accessory installed on

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the playard with a hub that moves downwards when folding the playard and the bassinet accessory. The playard and the bassinet accessory are shown in an unfolded configuration.

FIG. 41B shows another top perspective view of the playard and the bassinet accessory of FIG. 41A in the unfolded configuration.

FIG. 41C shows a front side view of the playard of FIG. 23A with the bassinet accessory of FIG. 41A.

FIG. 42A shows a top perspective view of a mattress in the bassinet accessory of FIG. 41A that is partially folded and disposed in a partially enclosed space of the bassinet accessory.

FIG. 42B shows a top perspective view of the playard of FIG. 41A with the bassinet accessory removed and the mattress of FIG. 42A partially folded and disposed in a partially enclosed space defined by soft goods of the playard.

FIG. 43 shows a top perspective view of the playard and the bassinet accessory of FIG. 41A without the mattress revealing a hub and multiple support tubes of the bassinet accessory. The playard and the bassinet accessory are shown in the unfolded configuration.

FIG. 44A shows a magnified view of bassinet soft goods in the bassinet accessory corresponding to Inset A of FIG. 41B where the bassinet soft goods are coupled to soft goods in the playard via a zipper mechanism.

FIG. 44B shows a top perspective view of the bassinet accessory of FIG. 41A removed from the playard of FIG. 41A.

FIG. 45A shows a top perspective view of the playard and the bassinet accessory of FIG. 43 where the playard and the bassinet accessory are in the folded configuration.

FIG. 45B shows a top perspective view of the playard and the bassinet accessory of FIG. 43 where the playard and the bassinet accessory are partially unfolded and beginning to transition from the folded configuration to the unfolded configuration. The playard and the bassinet accessory may also be viewed as being partially folded and approaching the folded configuration.

FIG. 45C shows a top perspective view of the playard and the bassinet accessory of FIG. 43 where the playard and the bassinet accessory are partially unfolded and approaching the unfolded configuration. The playard and the bassinet accessory may also be viewed as being partially folded and beginning to transition to the folded configuration.

FIG. 46A shows a top perspective view of the hub with the hub latch and the support tubes of FIG. 43. The hub latch is shown in a locked state where rotational movement of the support tubes relative to the hub latch is constrained.

FIG. 46B shows a bottom perspective view of the hub, the hub latch, and the support tubes of FIG. 46A.

FIG. 47A shows a top perspective view of the hub with the hub latch and the support tubes of FIG. 43. The hub latch is shown in an unlocked state where rotational movement of the support tubes relative to the hub latch is permitted.

FIG. 47B shows a bottom perspective view of the hub, the hub latch, and the support tubes of FIG. 47A. Several support tubes are rotated to the folded configuration.

FIG. 48 shows a top perspective view of the playard of FIG. 17A and another exemplary bassinet accessory installed on the playard with a hub that moves upwards when folding the playard and the bassinet accessory. The playard and the bassinet accessory are shown in an unfolded configuration.

FIG. 49A shows a top perspective view of a user's hand reaching through respective openings of a hub and bassinet

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soft goods in the bassinet accessory of FIG. 48 to access a bottom portion of playard disposed below the bassinet accessory.

FIG. 49B shows a side view of the user's hand grasping a strap disposed on a bottom portion of soft goods in the playard of FIG. 48 to initiate folding of the playard and the bassinet accessory.

FIG. 49C shows a top perspective view of the user pulling the strap of FIG. 49B up and through the respective openings of the hub and the bassinet soft goods to fold the playard and the bassinet accessory.

FIG. 49D shows a top perspective view of the playard and the bassinet accessory of FIG. 49C where the playard and the bassinet accessory are in a folded configuration.

FIG. 50A shows a top view of the bassinet accessory of FIG. 48 in the unfolded configuration.

FIG. 50B shows a bottom view of the bassinet accessory of FIG. 50A in the unfolded configuration.

FIG. 50C shows a side view of the bassinet accessory of FIG. 50A in the folded configuration.

FIG. 51A shows a top view of a telescoping support tube in the bassinet accessory of FIG. 48 coupled to the hub and the bassinet soft goods where the support tube is in an extended state in the unfolded configuration.

FIG. 51B shows a bottom view of the bassinet soft goods of FIG. 51A with the support tubes attached to the bassinet soft goods.

FIG. 52 shows a perspective view of the hub and the support tubes of FIG. 48 installed on the playard of FIG. 23A. The playard is shown in the folded configuration and the support tubes are in a contracted state.

DETAILED DESCRIPTION

Following below are more detailed descriptions of various concepts related to, and implementations of, foldable playards that include; 1) a mechanically-sound rigid frame with a simpler construction compared to conventional playards that is easier to operate; 2) soft goods attached to the frame to provide a partially enclosed space for the child; optionally 3) a canopy cover assembly mounted to the frame to provide shade for the child; and optionally 4) a bassinet accessory coupled to the frame and/or the soft goods to provide an elevated surface to support the child. It should be appreciated that various concepts introduced above and discussed in greater detail below may be implemented in multiple ways. Examples of specific implementations and applications are provided primarily for illustrative purposes so as to enable those skilled in the art to practice the implementations and alternatives apparent to those skilled in the art.

The figures and example implementations described below are not meant to limit the scope of the present implementations to a single embodiment. Other implementations are possible by way of interchange of some or all of the described or illustrated elements. Moreover, where certain elements of the disclosed example implementations may be partially or fully implemented using known components, in some instances only those portions of such known components that are necessary for an understanding of the present implementations are described, and detailed descriptions of other portions of such known components are omitted so as not to obscure the present implementations.

In the discussion below, various examples of inventive foldable playards are provided, wherein a given example or set of examples showcases one or more particular features of a frame, a X-frame assembly, a leg support assembly, a latch mechanism, soft goods, a canopy cover assembly, and a

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bassinet accessory. It should be appreciated that one or more features discussed in connection with a given example of a foldable playard may be employed in other examples of foldable playards according to the present disclosure, such that the various features disclosed herein may be readily combined in a given foldable playard according to the present disclosure (provided that respective features are not mutually inconsistent).

Certain dimensions and features of the foldable playard are described herein using the terms "approximately," "about," "substantially," and/or "similar." As used herein, the terms "approximately," "about," "substantially," and/or "similar" indicates that each of the described dimensions or features is not a strict boundary or parameter and does not exclude functionally similar variations therefrom. Unless context or the description indicates otherwise, the use of the terms "approximately," "about," "substantially," and/or "similar" in connection with a numerical parameter indicates that the numerical parameter includes variations that, using mathematical and industrial principles accepted in the art (e.g., rounding, measurement or other systematic errors, manufacturing tolerances, etc.), would not vary the least significant digit.

An Exemplary Foldable Playard with X-Frame Assemblies

FIGS. 2A-2C show an exemplary frame 100a for a foldable playard in an unfolded configuration. As shown, the frame 100a may include multiple leg support assemblies 110a and multiple X-frame assemblies 140a that are arranged to outline and define an interior space 102. In particular, each leg support assembly 110a may be coupled to another adjacent leg support assembly 110a via a X-frame assembly 140a to form a closed frame structure (e.g., a frame that surrounds and separates the interior space 102 from the surrounding environment). As discussed further below in connection with FIG. 5A, in addition to the frame 100a, a foldable playard 1000a also includes soft goods 300 that are partially disposed within the interior space 102 to provide a padded, partially enclosed space 301 to contain a child 50. The soft goods 300 will be described in more detail below.

With reference again to FIG. 2A, the leg support assemblies 110a of the frame 100a may provide vertical or nearly vertical support stands that define the spatial extent of the interior space 102 in the unfolded configuration. In other words, the leg support assemblies 110a may define and/or otherwise be disposed along side edges 104 of the interior space 102. The X-frame assemblies 140a may provide the structural support to position and orient the leg support assemblies 110a as desired, as well as provide a mechanism to facilitate folding and/or unfolding of the frame 100a. As shown in FIG. 2A, each X-frame assembly 140a may define and/or otherwise be disposed on a side face 106 of the interior space 102 between adjacent side edges 104.

For the frame 100a shown in FIGS. 2A-2C, the interior space 102 has a horizontal cross-section (i.e., a cross-section in a plane parallel to a ground 90 supporting the frame 100a) shaped as a regular hexagon. However, it should be appreciated that in other implementations disclosed herein and discussed in further detail below, the number of leg support assemblies 110a and/or X-frame assemblies 140a may be adjusted to form interior spaces 102 with different horizontal cross-sectional shapes including, but not limited to a square,

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a rectangle, a pentagon, a hexagon, an octagon, a regular polygon, and an irregular polygon (i.e., the sides have different dimensions).

In some implementations, the interior space **102** may further form a three-dimensional volume shaped as a right prism. Said in another way, the leg support assemblies **110a** may be vertically oriented such that the horizontal cross-section of the interior space **102** is identical or substantially identical in shape and dimensions at any vertical position along the length of the leg support assemblies **110a**. In other implementations, the interior space **102** may form a three-dimensional volume shaped as a truncated pyramid where a bottom portion of the interior space **102** near the ground **90** is larger than a top portion of the interior space **102**. Said in another way, the leg support assemblies **110a** may be tilted when the frame **100a** is deployed such that the top portions of the leg support assemblies **110a** are positioned closer together than a bottom portion of the leg support assemblies **110a** so that the area of the horizontal cross-section of the interior space **102** decreases from the bottom portion to the top portion of the leg support assemblies **110a** if the leg support assemblies **110a** are substantially straight in shape. In one aspect, a frame **100a** forming a truncated pyramidal interior space **102** may be preferable for enhancing mechanical stability. The manner in which this geometry is achieved will be discussed in more detail below.

In the frame **100a** shown in FIG. 2A, each leg support assembly **110a** may include a leg tube **112** having a top end **113a** and a bottom end **113b** (see, for example, FIG. 4A), a foot **114** coupled to the bottom end **113b** to support the frame **100a** on the ground **90**, a corner **130** coupled to the top end **113a** of the leg tube **112**, and a slider **120** that is slidably coupled to the leg tube **112** and positioned between the foot **114** and the corner **130**. The top end **113a** of the leg tube **112** and/or the corner **130** may coincide with a top vertex **105** of the interior space **102** and the bottom end **113b** of the leg tube **112** and/or the foot **114** may coincide with a bottom vertex **107** of the interior space **102**.

In this implementation, each X-frame assembly **140a** may include a pair of X-tubes **142a** and **142b** that are arranged to cross one another to form a single X-shaped structure. The X-tubes **142a** and **142b** may be rotatably coupled to each other and to respective corners **130** and sliders **120** of adjacent leg support assemblies **110a**. Thus, the X-frame assemblies **140a** are pivot and slidable X-frame assemblies where the X-tubes **142a** and **142b** rotate relative to each other and the leg support assemblies **110a** and translate relative to the leg tubes **112** via movement of the sliders **120**. This enables the frame **100a** to be folded into a more compact structure that occupies less volume and/or allows for a larger interior space **102** compared to, for example, conventional playards with pivot-only X-frame assemblies.

In some implementations, the manner in which the multiple X-frame assemblies **140a** and the leg support assemblies **110a** are coupled to each other may enable a caregiver to fold and/or unfold the frame **100a** in a single step. For example, the caregiver may unfold the frame **100a** by moving the slider **120** in one leg support assembly **110a** towards the corner **130**. The motion of the slider **120**, in turn, causes the adjoining X-frame assemblies **140a** to rotate and translate. The motion of the adjoining X-frame assemblies **140a**, in turn, causes the sliders **120** in the adjacent leg support assemblies **110a** to move in a similar manner. This process may occur simultaneously for all X-frame assemblies **140a** and all sliders **120** resulting in the frame **100a** being unfolded as the caregiver moves the slider **120** for the one leg support assembly **110a**. Once the frame **100a** is

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unfolded, a latch mechanism **200a**, which will be described in more detail below, may be actuated to lock the frame **100a** in the unfolded configuration (e.g., the latch mechanism **200a** prevents the sliders **120** from sliding back down the respective leg tubes **112** towards the feet **114**).

In some implementations, the frame **100a** may be folded and/or unfolded with the feet **114** of the leg support assemblies **110a** remaining in contact with the ground **90**. The leg tubes **112** may also remain vertically upright or nearly vertically upright (e.g., leg tubes **112** may intentionally be tilted when the frame **100a** is unfolded to improve stability) as the frame **100a** is folded and/or unfolded. In this manner, the process of folding and/or unfolding the frame **100a** may be made easier for the caregiver. For example, the caregiver would not have to balance the frame **100a** from tipping over while setting up and/or tearing down the playard **1000a**.

With reference to FIG. 2B, in some implementations, the X-tubes **142a** and **142b** of each X-frame assembly **140a** may be positioned within a top portion **108** of the frame **100a** and/or the interior space **102** when the frame **100a** is unfolded. Said in another way, the X-frame assemblies **140a** may form a perimeter structure around the top portion **108** of the frame **100a** that outlines the horizontal cross section of the top opening of the interior space **102**. For example, FIG. 2C shows the X-frame assemblies **140a** form a top perimeter structure **109** that outlines a regular hexagon corresponding to the shape of the interior space **102**.

Positioning the X-tubes **142a** and **142b** in the top portion **108** of the frame when the frame is in the unfolded configuration provides several benefits to the frame **100a** and, in turn, to a foldable playard comprising the frame **100a**.

First, each X-frame assembly **140a** in the frame **100a** may function as a top rail that couples together two adjacent leg support assemblies **110a** and provides mechanical rigidity and stability to the frame **100a**. Said in another way, the X-frame assembly **140a** may be unfolded to such an extent that the X-tubes **142a** and **142b** form a shallow X-frame structure in the top portion **108** of the frame that effectively functions as a rigid top rail. For example, in the limit where the respective sliders **120** are positioned proximate to the respective corners **130** in adjacent leg support assemblies **110a**, the X-tubes **142a** and **142b** may be in near parallel alignment with one another when viewing the frame **100a** from the side or the front. Thus, each X-tube **142a** and **142b** may separately function as a top rail.

In some implementations, the leg support assemblies **110a** may only be coupled to one another via the X-frame assemblies **140a**. In other words, the frame **100a** may exclude other support structures, such as a separate compliant and/or rigid top rail (e.g., the webbing **14** of the playards **10a** and **10b** shown in FIG. 1A and FIG. 1B, the rigid top rails **32** of the playard **10c** shown in FIG. 1C) or a bottom support structure (e.g., the bottom support structure **34** of the playard **10c** shown in FIG. 1C), which may appreciably reduce the number of parts for manufacture and assembly. For example, as shown in FIGS. 2A-2C, the portion of the leg tubes **112** located between the bottom end **113b** and the slider **120** when the frame **100a** is unfolded may not be coupled to another portion of the frame **100a** (e.g., the bottom portions of the leg tubes **112** are mechanically unconstrained).

In some implementations, the frame **100a**, comprising only leg support assemblies **110a** and X-frame assemblies **140a** to couple the leg support assemblies **110a** together, may have sufficient mechanical rigidity, stability, and/or strength to meet the requirements set forth in various consumer safety standards (e.g., ASTM F406-19, 7.3.3, 7.11).

For example, FIG. 7B shows the playard **1000a** with the frame **100a** unfolded and with soft goods **300** installed undergoing a stability test (e.g., ASTM F406-19, 5.12, 8.17). For this test, the playard **1000a** was placed onto a flat piece of plywood and tilted at varying angles with a test weight disposed within the playard **1000a** leaning against one side of the frame **100a**. Based on this test, it was found the playard **1000a** did not tip even when tilted at an angle of 20 degrees with at least three feet **114** remaining in contact with the plywood base. This result exceeds the requirements set forth in ASTM F406-19, 8.17, which require the playard to maintain three contact points with the plywood base when tilted to an angle of 10 degrees.

This may be accomplished, in part, by tailoring the materials and/or dimensions of the X-tubes **142a** and **142b** to provide the mechanical properties that ensure the frame **100a** is mechanically rigid and stable when deployed. For example, the X-tubes **142a** and **142b** may be formed from steel tubing with an outer diameter of about 0.625 inches ($\frac{5}{8}$ inches) and a total length of about 24.5 inches. The term “about,” when used to describe the dimensions of the X-tubes **142a** and **142b**, are intended to cover manufacturing tolerances. For example, “about 0.625 inches” may correspond to the following dimensional ranges: 0.61875 to 0.63125 inches (+/-1% tolerance), 0.62 to 0.63 inches (+/-0.8% tolerance), 0.62125 to 0.62875 inches (+/-0.6% tolerance), 0.6225 to 0.6275 inches (+/-0.4% tolerance), 0.62375 to 0.62625 inches (+/-0.2% tolerance). Similar tolerances may be applied to describe the total length of the X-tubes **142a** and **142b**.

It should also be appreciated the X-tubes **142a** and **142b** may be formed from other materials including, but not limited to, aluminum and carbon fiber. The X-tubes **142a** and **142b** may also have different dimensions depending, in part, on the desired size of the frame **100a** and/or the interior space **102** and the mechanical properties of the materials used to form the X-tubes **142a** and **142b**. In some implementations, the X-frame assemblies **140a** may all have substantially identical or identical dimensions and/or shapes resulting in an interior space **102** with a horizontal cross-section shaped as a regular polygon. In some implementations, the frame **100a** may include X-frame assemblies **140a** having different dimensions and/or shapes resulting in an interior space **102** that is skewed in shape.

Additionally, with reference to FIG. 2B, the length L of the leg tubes **112**, defined as the distance between the top end **113a** and the bottom end **113b**, may generally be kept relatively small where possible in order to reduce the likelihood of the frame **100a** being tilted especially when a force is applied along the top portion **108** of the frame **100a**. For example, the length L may be chosen to ensure certain constraints on the frame **100a** are satisfied. These constraints include: (1) providing a desired height for the interior space **102**; (2) providing sufficient overlap with the foot **114** and the corner **130** to couple the foot **114** and corner **130** to leg tube **112**; and/or (3) providing sufficient room for the slider **120** to move between the foot **114** and the corner **130** to fold and/or unfold the frame **100a**. It should be appreciated that the lateral and vertical dimensions of the interior space **102** are coupled due, in part, to the rotational and translational motion of the X-frame assemblies **140a** (e.g., an increase in the lateral dimensions of the interior space **102** results in a corresponding increase in the vertical dimensions to ensure the X-frame assemblies **140a** have sufficient room to slide vertically along the leg tubes and hence fold).

In some implementations, the length L of the leg tubes **112** may be about 26 inches. Similar to the dimensions of the

X-tubes **142a** and **142b**, the term “about,” when used to describe the dimensions of the leg tube **112**, are intended to cover manufacturing tolerances. The tolerance values may be the same as the X-tubes **142a** and **142b**. In some implementations, the leg tubes **112** in the leg support assemblies **110a** may be substantially identical or identical. In some implementations, the leg tubes **112** may have different shapes and/or dimensions (e.g., some leg tubes **112** may be vertically oriented while other leg tubes **112** may be tilted when the frame **100a** is unfolded).

Second, another benefit provided by positioning the X-tubes **142a** and **142b** in the top portion **108** of the frame when the frame is in the unfolded configuration is that the X-frame assemblies **140a** occupy a smaller portion of the side faces **106** of the interior space **102** as compared to conventional playards with X-frame assemblies. When the soft goods **300** includes transparent and/or see-through side portions, the placement of the X-frame assemblies **140a** in the top portion **108** of the frame allows for greater visibility of the partially enclosed space **301** when the soft goods **300** are coupled to the frame **100a**. Said in another way, the X-frame assemblies **140a** do not appreciable visually obstruct and/or impede the caregiver from seeing their child when the child **50** is in the playard **1000a**.

Additionally, the soft goods **300** may use less material to cover the X-frame assemblies **140a**. In some implementations, the soft goods **300** may cover the corners **130** of the leg support assemblies **110a** and partially cover the X-frame assemblies **140a** such that the latch mechanism **200a**, when disposed in the top portion **108** of the frame **100a**, remains accessible to the caregiver. In some implementations, the soft goods **300** may fully cover the X-frame assemblies **140a** as well as the corners **130** and the sliders **120** of the leg support assemblies **110a** such that an observer may only see the leg tubes **112** and/or the feet **114** of the leg support assemblies **110a**. In this manner, the foldable playard **1000a** may be presented with a cleaner, more aesthetically desirable appearance to a consumer, in both indoor and outdoor settings.

As discussed above in connection with FIG. 2B, the top portion **108** may generally correspond to the portion of the frame **100a** proximate to the top ends **113a** of the leg tubes **112** and/or the corners **130** of each leg support assembly **110a**. More specifically, the top portion **108** may be defined as the portion of the frame **100a** located between a top horizontal plane **92** that intersects the top ends **113a** of the leg tubes **112** and/or the corners **130**, and a bottom horizontal plane **91** that is offset from the top horizontal plane **92** by an offset distance, x_1 , along the length of the respective leg tubes **112**. When the frame **100a** is unfolded, the X-tubes **142a** and **142b**, the sliders **120**, and the corners **130** are disposed within the top portion **108**. The offset distance, x_1 , may be defined as a fraction of the total length L of the leg tube **112** assuming the leg tubes **112** in the frame **100a** have identical lengths. In some implementations, the offset distance, x_1 , may be less than or equal to 30% of the total length, L , of the leg tubes **112** and, more preferably, less than or equal to 20% of the total length of the leg tubes **112**.

FIG. 2B also shows the frame **100a** may have an overall vertical height, H_1 , defined as the distance from the ground **90** to the top horizontal plane **92** along a vertical axis (i.e., normal to the ground) in the unfolded configuration. FIG. 2E similarly shows the frame **100a** may have an overall vertical height, H_2 , defined as the distance from the ground **90** to a top horizontal plane **92A** in the folded configuration. In some implementations, the height of the frame **100a** may remain substantially constant or constant between the folded

and unfolded configurations of the frame. In other words, the heights H_1 and H_2 may be equal or substantially similar and the planes **92** and **92A** are coplanar or substantially coplanar. In some implementations, however, the height of the frame **100a** may vary due, for example, to the leg support assemblies **110a** flaring outwards when the frame **100a** is unfolded as discussed in greater detail below. If the frame **100a** flares outwards in the unfolded configuration, the height H_2 may be somewhat greater than the height H_1 (i.e., the plane **92A** in the folded configuration may be disposed somewhat above the plane **92** in the unfolded configuration).

FIGS. 3A and 3B show additional views of the X-frame assembly **140a** in the frame **100a**. As shown, the X-tubes **142a** and **142b** may be rotatably coupled to each other via a pin joint **145**. The X-tube **142a** may have a first end **143a** rotatably coupled to the corner **130** of one leg support assembly **110a** via a pin joint **146a** and a second end **143b** rotatably coupled to the slider **120** of another leg support assembly **110a** via a pin joint **146b**. Similarly, the X-tube **142b** may be rotatably coupled to the corner **130** of one leg support assembly **110a** via a pin joint **146d** and rotatably coupled to the slider **120** of another leg support assembly **110a** via a pin joint **146c**.

The pin joints **145** and **146a-146d** may generally include a fastener (not shown) with a shaft inserted through openings **147** (see FIG. 4B) on the X-tubes **142a** and **142b** to allow rotational motion between the X-tubes **142a** and **142b**, the sliders **120**, and the corners **130**. The fastener may be various types of captive fasteners including, but not limited to, a rivet with a cap (e.g., a rolled rivet) and a bolt fastener with a nut.

Generally, the nominal dimensions and tolerances of the openings **147** and the shaft of the fastener affects the tightness or looseness of the pin joints **145** and **146a-146d**. If the opening **147** is dimensioned to interfere with the fastener (e.g., the size of the opening **147** is smaller than the size of the shaft of the fastener), the caregiver may have to apply a greater force to rotate the X-tubes **142a** and **142b**. In some instances, the pin joints **145** and **146a-146d** may be too tight such that the respective feet **114** of each leg support assembly **110a** do not contact the ground **90** when the frame **100a** is unfolded. For example, the caregiver may move the slider **120** of one leg support assembly **110a** towards the corresponding corner **130**, but the opposing sides of the frame **100a** may only be partially unfolded. In contrast, if the size of the opening **147** is appreciably larger than the fastener shaft, the pin joints **145** and **146a-146d** may allow the X-tubes **142a** and **142b** to rotate and/or translate along other unwanted axes of motion (e.g., the frame **100a** may wobble), which may compromise the mechanical stability of the frame **100a**. Thus, in some implementations, the nominal dimensions and tolerances of the opening **147** and the shaft of the fastener are particularly chosen to be sufficiently loose to ensure the feet **114** of the leg support assemblies **110a** contact the ground **90** while still being sufficiently tight to limit unwanted rotational and/or translation motion between the X-tubes **142a** and **142b** and/or the sliders **120** or corners **130**. For example, the tolerance (or clearance) between the shaft of the fastener and the edge of the opening **147** may be greater than or equal to about 0.010 inches and, more preferably, greater than or equal to about 0.015 inches.

As shown in FIG. 3A, the pin joint **145** may generally be located along the length of the respective X-tubes **142a** and **142b**. For example, the pin joint **145** may be positioned at an offset distance, z_1 , from the first end **143a** and an offset distance, z_2 , from the second end **143b**. In some implementations, the offset distances z_1 and z_2 may be equal, which

causes the respective first and second ends **143a** and **143b** of the X-tubes **142a** and **142b** to follow the same circular path when the X-tubes **142a** and **142b** are rotated. This, in turn, causes the orientation of the leg support assemblies **110a** to remain unchanged when the frame **100a** is being folded and/or unfolded. For example, the leg tubes **112** of each leg support assembly **110a** may remain vertically oriented for both folded and unfolded configurations.

In other implementations, however, the offset distances z_1 and z_2 may not be equal. For example, the offset distance z_2 may be larger than the offset distance z_1 , causing the first end **143a** of the X-tube **142a** to follow a smaller circular path and the second **143b** to follow larger circular path when the X-tube **142a** is rotated. The respective first and second ends **143a** and **143b** of the X-tube **142b** may similarly follow smaller and larger circular paths, respectively. This, in turn, may cause the leg support assemblies **110a** and, in particular, the leg tubes **112** to flare outwards when the frame **100a** is unfolded. In other words, the leg tubes **112** of the leg support assemblies **110a** may be tilted due to the rotational motion of the X-tubes **142a** and **142b** in the X-frame assemblies **140a** such that the top ends **113a** constitute the vertices of a smaller horizontal cross-section (parallel to the ground) than the bottom ends **113b** (i.e., the top ends **113a** are positioned closer to one another than the bottom ends **113b**). In this manner, the frame **100a** may define an interior space **102** with a truncated pyramidal interior shape as described above, which may be beneficial in improving the mechanical stability of the frame **100a** (e.g., the frame **100a** is less likely to be tilted over). With reference again to FIG. 2B, in some implementations the leg support assemblies **110a** may be flared outwards such that respective longitudinal axes **111a** associated with the leg tubes **110a** are tilted at an angle, Θ , relative to the ground **90**, wherein the angle ranges between 80 degrees and 88 degrees and, more preferably, between 83 degrees and 85 degrees.

Turning now to FIG. 3B, in some implementations the X-tubes **142a** and **142b** may also be bent in shape. For example, the first and second ends **143a** and **143b** of the X-tube **142a** may be aligned along a first axis **141a** while a central portion **144** of the X-tube **142a** is aligned along a second axis **141b** that is parallel to and offset from the axis **141a**. The X-tube **142b** may have a similar bent shape as the X-tube **142a**. In some implementations, the offset between the first and second axes **141a** and **141b** may be chosen to provide sufficient clearance between the X-tubes **142a** and **142b** such that the respective first and second ends **143a** and **143b** of the X-tubes **142a** and **142b** lie on the same plane (e.g., the side face **106** of the interior space **102**) as shown in FIG. 3B. This, in turn, allows the portions of the corners **130** and the sliders **120** to also lie on the same plane with the first and second ends **143a** and **143b** of the X-tubes **142a** and **142b**. In some implementations, aligning the corners **130** and sliders **120** in this manner may allow the frame **100a** to fold more compactly.

FIGS. 3C-3E show additional views of the leg support assemblies **110a** in the frame **100a**. As shown, the leg tube **112** may be a substantially elongated, hollow tube that defines that path along which the slider **120** travels when the frame **100a** is being folded and/or unfolded. In some implementations, the leg tube **112** may be substantially straight such that the slider **120** follows a straight path along the longitudinal axis **111a** (see FIGS. 2A-2C). However, it should be appreciated the leg tube **112** may also be curved in other implementations to define a correspondingly curved path for the slider **120** to follow. Examples of curved leg tubes **112** will be discussed in further detail below. In some

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implementations, the leg tube 112 may have a cross-section that remains constant along the length, L, of the leg tube 112. In some implementations, the leg tube 112 may have various cross-sectional shapes including, but not limited to a circle, an oval, and an oblong shape. The leg tube 112 may also be formed from various materials including, but not limited to steel, aluminum, and carbon fiber.

The slider 120 may include a base 121 that defines a through hole opening 122 shaped and/or dimensioned to surround the leg tube 112, thus enabling the slider 120 to slidably move along the leg tube 112. In some implementations, the shape of the opening 122 may conform with the cross-sectional shape of the leg tube 112. The slider 120 may further include an extended portion 124 coupled to one side of the base 121 to couple the X-tube 142a of one X-frame assembly 140a to the slider 120 via a fastener inserted through an opening on the extended portion 124 aligned to the opening 147 of the X-tube 142a (see, for example, the exploded views of FIGS. 4A and 4B). The extended portion 124 may also include a recessed opening to receive the end of the X-tube 142a that is coupled to the slider 120. The slider 120 may also include an extended portion 126 similar to the extended portion 124 that is disposed opposite from the extended portion 124 to couple the X-tube 142b of another X-frame assembly 140a to the slider 120 via another fastener inserted through an opening on the extended portion 126 aligned to the opening 147 of the X-tube 142b.

The extended portions 124 and 126 may generally be oriented at an angle relative to each other to align the respective X-tubes 142a and 142b from adjoining X-frame assemblies 140a along the desired geometry of the interior space 102. For example, the extended portions 124 and 126 may be rotated relative to one another by an obtuse angle of approximately 120 degrees corresponding to the angles between adjoining sides of a hexagon. In some implementations, the extended portions 124 and 126 may lie on the same horizontal plane. In some implementations, the extended portions 124 and 126 may be offset vertically from one another if the respective X-tubes 142a and 142b coupled to the slider 120 are not identical. In some implementations, the sliders 120 of the leg support assemblies 110a may be identical with one another, thus reducing the number of unique parts for manufacture.

The corner 130 may include a base 131 that defines an opening 132 to receive the top end 113a of the leg tube 112. In some implementations, the shape of the opening 132 may conform with the cross-sectional shape of the leg tube 112. Similar to the slider 120, the corner 130 may include extended portions 134 and 136 disposed on opposing sides of the base 131 to couple the X-tube 142b of one X-frame assembly 140a and the X-tube 142a of another X-frame assembly 140a to the corner 130 using a similar attachment mechanism as the slider 120, e.g., a fastener inserted through an opening aligned to the openings 147 of the X-tubes 142a and 142b (see, for example, the exploded views of FIGS. 4A and 4B). The extended portions 134 and 136 may each have recessed openings to receive respective ends of the X-tubes 142a and 142b.

The extended portions 134 and 136 may also be oriented at an angle relative to each other to align the respective X-tubes 142a and 142b from adjoining X-frame assemblies 140a along the desired geometry of the interior space 102. The extended portions 134 and 136 may also lie on the same horizontal plane and/or offset vertically from one another if the respective X-tubes 142a and 142b coupled to the corner 130 are not identical. In some implementations, the corners

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130 of the leg support assemblies 110a may be identical with one another, thus reducing the number of unique parts for manufacture.

FIG. 3C further shows the corner 130 may include a tab portion 138 that extends downwards along the leg tube 112 to support a snap-fit connector 139 to attach the soft goods 300 to the frame 100a. In some implementations, the tab portion 138 may be shaped and/or dimensioned to position the snap-fit connector 139 at a desired location along the leg tube 112. For example, the snap-fit connector 139 may be offset from the top end 113a to ensure the soft goods 300 overlaps and wraps around the top portion 108 of the frame 100a. In some implementations, an opening formed in the tab portion 138 to attach the snap-fit connector 139 to the corner 130 may also be used to securely couple the corner 130 to the leg tube 112 using the same fastener.

FIG. 3E shows a magnified view of the foot 114 of the leg support assembly 110a. As described above, the foot 114 supports the frame 100a and the foldable playard 1000a on the ground 90. As shown, the foot 114 may define an opening 115 to receive the bottom end 113b of the leg tube 112. In some implementations, the shape of the opening 115 may conform with the cross-sectional shape of the leg tube 112. The foot 114 may further include an opening 119 to securely couple the foot 114 to the leg tube 112 using, for example, a fastener (see, for example, FIG. 4C).

In some implementations, the foot 114 may also include a looped or ringed structure that extends from the base of the foot 114 to provide another attachment point to couple the soft goods 300 to the frame 100a. For example, FIG. 3C shows the foot 114 may include a D-ring 116 defining a D-shaped opening 117. The soft goods 300 may include a strap or a tether that is inserted through the D-shaped opening 117 and tied to the foot 114 to mechanically attach a bottom portion of the soft goods 300 to the frame 100a. As shown, the D-shaped opening 117 may be aligned such that a centerline axis 118 of the opening 117 is aligned substantially parallel with the longitudinal axis 111a of the leg tube 112. This orientation also allows the D-ring 116 to increase the area that the foot 114 contacts the ground 90, which may further improve the mechanical stability of the frame 100a. However, it should be appreciated the orientation and placement of the D-ring 116 may be varied in other implementations. For example, the D-ring 116 may be rotated 90 degrees relative to the ground such that the axis 118 of the opening 117 is perpendicular to the longitudinal axis 111a.

FIG. 5A shows the foldable playard 1000a with the soft goods 300 coupled to the frame 100a. As described above, the soft goods 300 defines a partially enclosed space 301 placed within the interior space 102 of the frame 100a to contain the child. In some implementations, the soft goods 300 may remain attached to the frame 100a as the frame 100a is folded and/or unfolded. As shown in FIG. 5A, the soft goods 300 may include a floor portion 304 that rests on the ground 90 when the playard 1000a is unfolded. The soft goods 300 may also include side portions 306 that define and surround the partially enclosed space 301. In some implementations, the side portions 306 may be transparent (e.g., a transparent plastic) or see-through (e.g., a mesh) so that a child in the playard is observable from outside the partially enclosed space 301. The soft goods 300 may also include one or more straps (e.g., a Velcro strap) and/or tethers to couple the soft goods 300 to each D-ring 116 of each foot 114 in the leg support assemblies 110a.

The soft goods 300 may also include a soft goods top portion 302 to wrap the soft goods 300 around the top portion 108 of the frame 100a. As shown in FIG. 5A, the soft

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goods top portion **302** may be formed from an opaque textile material with multiple layers of fabric to provide padding on the portions of the frame **100a** that are covered. The soft goods **300** also may include integrated snap-fit connectors **312** that couple to the snap-fit receivers **139** of the corners **130**. In some implementations, the soft goods **300** may include the same number of snap-fit connectors **312** such that the soft goods **300** attaches to each corner **130** of the frame **100a**. In some implementations, the snap-fit connector **312** may be disposed on a tab **310** that is attached to an interior piece of the soft goods **300** along the soft goods top portion **302** as shown in FIG. 5C. The tab **310** may stiffen the interior piece of the soft goods top portion **302** to ensure the soft goods top portion **302** remains flush against the frame **100a** (e.g., the soft goods top portion **302** does not curl upwards) when the snap-fit connector **312** is coupled to the snap-fit connector **139** on the corner **130** as shown in FIG. 5B. The tab **310** may be formed from a compliant material, such as polyethylene, and shaped to be stiffer than the surrounding textile material of the soft goods **300**.

FIGS. 6A-6D show multiple views of the latch mechanism **200a** disposed on the frame **100a**. As described above, the latch mechanism **200a** may lock the frame **100a** in the unfolded configuration. In particular, the latch mechanism **200a** may maintain the sliders **120** of the leg support assemblies **110a** proximate to the corresponding corners **130** such that the X-frame assemblies **140a** remain unfolded forming a shallow X-frame structure in the top portion **108** of the frame. Thus, the latch mechanism **200a** may provide sufficient mechanical restraints to support the various forces and/or torques applied to one or more of the sliders **120** (e.g., the weight of the X-tubes **142a** and **142b** acting on the slider **120**).

The latch mechanism **200a** may generally be coupled to and/or couple together various components of the frame **100a** including, but not limited to the slider **120**, the corner **130**, and the X-tubes **142a** or **142b**. Furthermore, the latch mechanism **200a** may be disposed, at least in part, within the top portion **108** of the frame **100a**. This may enable the latch mechanism **200a** to be at least partially covered by the soft goods **300**. For example, the latch mechanism **200a** may directly couple the corner **130** of one leg support assembly **110a** to a X-tube **142a** or **142b** of an adjoining X-frame assembly **140a** as shown in FIG. 6A.

The frame **100a** may generally include one or more latch mechanisms disposed on one or more leg support assemblies **110a** and/or the X-frame assemblies **140a**. For example, the frame **100a** may include latch mechanisms disposed on opposing sides of the frame **100a** to ensure the frame **100a**, when unfolded, maintains an even, unfolded shape (e.g., one side of the frame **100a** does not sag downwards relative to another side). However, in other implementations, a single latch mechanism is sufficient to lock the frame **100a** in the unfolded configuration while keeping the various leg support assemblies **110a** and X-frame assemblies **140a** unfolded evenly. For example, with reference again to FIGS. 2A-2C, these figures show that the frame **100a** includes a single latch mechanism **200a** disposed, in part, on one leg support assembly **110a** and one X-frame assembly **140a**. In some implementations, the latch mechanism **200a** may be configured to withstand a load greater than or equal to 10 lbs. before being disengaged or unlocked.

FIG. 6A shows the latch mechanism **200a** may include a latch member **210** (also referred to herein as a “flex lock”) with a top end **211a** coupled to the corner **130** of one leg support assembly **110a** and a latch boss **230** coupled to the X-tube **142a** of one X-frame assembly **140a**. The latch

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member **210** may include an opening **212** disposed at the first end **211a** that aligns with the opening on the corner **130** used to couple to the X-tube **142b**. In this manner, a single fastener may couple the latch member **210**, the corner **130**, and the X-tube **142b** together and the corner **130** may remain unmodified. In other words, the latch member **210** may be coupled to any one of the corners **130** in the leg support assemblies **110a** of the frame **100a** provided the latch boss **230** is coupled to one of the X-tubes **142a** and **142b** adjoining the leg support assembly **110a**. In some implementations, the latch member **210** may be coupled to the corner **130** via a pin joint connection or a rigid connection (e.g., in which the latch member **210** cannot be rotated relative to the corner **130**). The latch boss **230** may include an opening that is shaped and/or dimensioned to conform with the X-tube **142a**, thus enabling the latch boss **230** to be slid onto the X-tube **142a** for assembly. FIG. 6B shows the latch boss **230** may then be coupled to the X-tube **142a** using, for example, a fastener inserted through respective openings (not shown) on the latch boss **230** and the X-tube **142a**.

With reference again to FIG. 6A, the latch member **210** may include a latch opening **214** disposed at a second end **211b** of the latch member **210** located opposite from the first end **211a**. The latch opening **214** may be shaped and/or dimensioned to receive the latch boss **230**. In other words, the latch opening **214** may function as a latch catch. In this manner, the latch member **210** may directly couple the corner **130** to the X-tube **142b** by engaging with the latch boss **230**, thus holding the slider **120** in the top portion **108** of the frame **100a** near the corner **130**.

The latch member **210** may also include a tab **220** disposed at the second end **211b**. Generally, the latch member **210** may be a mechanically compliant component that bends when the caregiver pulls on the tab **220** to disengage the latch member **210** from the latch boss **230**. The latch member **210** may also have sufficient mechanical rigidity such that a restoring force is generated when bent by the caregiver. When the caregiver releases the tab **220**, the restoring force may return the latch member **210** back to its original shape. In some implementations, the latch member **210** may be formed from a plastic material. The latch member **210** may further have a sufficient thickness and/or be reinforced with integral rib structures to provide the desired mechanical rigidity.

In some implementations, the latch mechanism **200a** may be a double-action latch mechanism (e.g., the caregiver needs to perform two operations to unlock the latch mechanism). For example, FIG. 6C shows the latch opening **214** of the latch member **210** may include a tab **216** disposed within the latch opening **214**. FIG. 6D shows the latch boss **230** may include an undercut portion **232** that forms a notch or a slot between the X-tube **142a** and an end portion **236**. Thus, when the latch member **210** is coupled to the latch boss **230**, the tab **216** of the latch member **210** is disposed within the undercut portion **232** and retained by the end portion **236** of the latch boss **230**. In some implementations, the tab **216** may further define a slot **218** as shown in FIG. 6C, and the latch boss **230** may further include a rib **234** partially disposed within the undercut portion **232** as shown in FIG. 6D, that together facilitate alignment of the tab **216** to the undercut portion **232** to ensure the latch member **210** is properly engaged with the latch boss **230**.

To setup the frame **100a** and, by extension the playard **1000a**, the caregiver may initially move the slider **120** of one leg support assembly **110a** towards the corresponding corner **130** to partially unfold the frame **100a**. As the frame **100a** is being unfolded, the latch boss **230** disposed on the X-tube

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142a is displaced towards the latch member 210 coupled to the corner 130. Once the latch boss 230 reaches the latch member 210 and, in particular, the tab 216, further movement of the slider 120 along the leg tube 112 results in contact between the latch boss 230 and the tab 216, which causes the latch member 210 to be deflected outwards. In some implementations, the latch member 210 may include a lead-in feature on the tab 216 (not shown), such as a sloped or a ramped wall. The lead-in feature may allow the latch member 210 to be deflected more effectively as the latch boss 230 slides against the latch member 210 by orienting the contact force between the latch member 210 and the latch boss 230 along a direction that increases the magnitude of the torque applied to bend the latch member 210 (note that the pivot point of the latch member 210 is located at the mounting opening 212 as shown in FIG. 6A).

As the latch member 210 is deflected with further movement of the slider 120 along the leg tube 112, an internal restoring force is generated within the latch member 210, which is applied against the latch boss 230. As the caregiver continues to move the slider 120 towards the corner 130, the latch member 210 is deflected further outwards resulting in a higher magnitude restoring force being applied against the latch boss 230. When the slider 120 is moved sufficiently close to the corner 130, the latch boss 230 passes through the latch opening 214 and the restoring force causes the latch member 210 to snap back to its original position such that the latch boss 230 protrudes through the latch opening 214. Once the caregiver releases the slider 120, the slider 120 may move slightly downwards along the leg tube 112 due to gravity, causing the undercut portion 232 of the latch boss 230 to rest onto the tab 216 of the latch member 210.

FIG. 6E illustrates how a caregiver may transition the frame 100a and the playard 1000a to a folded configuration from the unfolded configuration by disengaging the double-action latch mechanism 200a. As shown in FIG. 6E, the caregiver may first squeeze the X-tubes 142a and 142b (as shown by the upward and downward arrows in FIG. 6E), which causes the slider 120 to move upwards along the leg tube 112, thus disengaging the tab 216 of the latch member 210 from the undercut portion 232 of the latch boss 230. While the caregiver is squeezing the X-tubes 142a and 142b together with one hand, the caregiver may then pull on the tab 220 of the latch member 210 with another hand to release the latch boss 230 from the latch opening 214 (as shown by the curved arrow in FIG. 6E). The “double-action” of the latch mechanism 200a is thus “squeeze-and-pull.” While holding the latch member 210, the caregiver may then release the X-tubes 142a and 142b and the slider 120 may then fall downwards along the leg tube 112 due, in part, to the weight of the X-frame assemblies 140a. The caregiver may then move the slider 120 downwards towards the foot 114 of the leg support assembly 110a, thus folding the playard 1000a.

With reference again to FIG. 6D, in some implementations of the double-action latch mechanism 200a, the undercut portion 232 and the end portion 236 of the latch boss 230 and the tab 216 of the latch member 210 may be shaped and/or dimensioned such that latch member 210 cannot be pulled off the latch boss 230 without applying an appreciably large force (e.g., a force greater than 20 lbs). For example, FIG. 7A shows a force test being applied to the double-action latch mechanism 200a, which shows the latch member 210 remains engaged to the latch boss 230 when a force greater than 24 lbs is applied to the tab 220.

It should be appreciated that, in other implementations, the playard 1000a and, in particular, the frame 100a may

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include other types of latching mechanisms. For example, FIGS. 8A-8D show a playard 1000a where the frame 100a includes a single-action latch mechanism 200b (e.g., the caregiver needs to perform only one operation to release the latch mechanism) instead of (or in addition to) the double-action latch mechanism 200a discussed immediately above.

Specifically, FIG. 8A shows the playard 1000a with the soft goods 300 installed onto the frame 100a, where the soft goods 300 covers the corners 130 of the leg support assemblies 110a and partially covers the X-frame assemblies 140a. In this manner, a portion of the single-action latch mechanism 200b is left exposed to provide access to the caregiver (see, for example, FIG. 8B). As shown in FIGS. 8C and 8D, the single-action latch mechanism 200b may also include a latch member 210 that is coupled at one end to the corner 130 via a fastener inserted through an opening 212 on the latch member 210. The latch member 210 may once again include a latch opening 214 to receive a latch boss 230. In this implementation, the latch boss 230 is shown coupled to the X-tube 142b of the X-frame assembly 140a.

The single-action latch mechanism 200b may be locked in a similar manner as the double-action latch mechanism 200a. Specifically, the slider 120 is moved towards the corner 130, which causes the latch boss 230 to initially deflect the latch member 210 until the latch boss 230 reaches the latch opening 214. At this point, the restoring force generated within the latch member 210 causes the latch member 210 to return to its original position with the latch boss 230 protruding through the latch opening 214. In this manner, the single-action latch mechanism 200b may hold the frame 100a in the unfolded configuration.

To unlock the single-action latch mechanism 200b and fold the frame 100a, the caregiver may pull on the tab 220 to deflect and/or bend the latching member 210 outwards, thus releasing the latch member 210 from the latch boss 230. As before, while the caregiver holds the latch member 210, the slider 120 may then move downwards along the leg tube 112 via a combination of gravity and the caregiver moving the slider 120 towards the foot 114 of the leg support assembly 110a as shown in FIG. 8D. In this manner, the playard 1000a may be folded.

FIGS. 9A-9F show another exemplary latch mechanism 200c installed on the frame 100a of the playard 1000a. FIG. 9A shows the frame 100a once again covered with soft goods 300. FIG. 9B shows the soft goods 300 only partially covers the X-frame assemblies 140a such that a bottom portion of the latch mechanism 200c is exposed. FIG. 9C shows the frame 100a without soft goods 300 attached. As shown, the latch mechanism 200c may be positioned on the frame 100a similar to the double-action latch mechanism 200a and the single-action latch mechanism 200b, i.e., the latch mechanism 200c is disposed in the top portion 108 of the frame 100a.

FIG. 9D shows the latch mechanism 200c may once again include a latch member 210 that is coupled to the corner 130 via a fastener inserted through an opening 212 at one end of the latch member 212. However, in this example, the latch member 210 may form a notch 240a that is shaped and/or dimensioned to form a snap-fit connection with the X-tube 142b. In this manner, the latch mechanism 200c may utilize fewer parts compared to the latch mechanisms 200a and 200b (e.g., the latch mechanism 200c only includes the latch member 210 and a fastener to couple the latch member 210 to the corner 130). As shown, the notch 240a may be shaped to conform with the cross-sectional shape of the X-tube 142b. As before, the latch member 210 may be a mechani-

cally compliant component that may be bent and/or deflected due to contact with the X-tube **142b** (e.g., when unfolding the frame **100a**) and/or by the caregiver pulling on the tab **220** disposed at the bottom end of the latch member **210** to release the latch member **210** from the X-tube **142b** (e.g., when folding the frame **100a**).

In the implementation shown in FIGS. 9A-9D, the frame **100a** and, by extension, the playard **1000a** may be setup once again by having the caregiver move the slider **120** of one leg support assembly **110a** towards the corresponding corner **130**. When the X-tube **142b** contacts the latch member **210** and, specifically, the tab **220**, the latch member **210** may be deflected outwards. The latch member **210** may further include a lead-in feature **222** (e.g., a sloped wall) to deflect the latch member **210** as the latch member **210** contacts the X-tube **142b**. The caregiver may then continue to move the slider **120** towards the corner **130** until the notch **240a** aligns with the X-tube **142b**.

In some implementations, the latch member **210** may be sufficiently compliant such that deflection of the latch member **210** does not produce an appreciable restoring force. Thus, the caregiver needs to press the latch member **210** to snap-fit the latch member **210** onto the X-tube **142b**. In other implementations, however, the latch member **210** may instead generate an internal restoring force when bent and/or deflected (e.g., the latch member **210** includes rib structures to increase the mechanical rigidity of the latch member **210**). The restoring force may be of sufficient magnitude to cause the notch **240a** to at least partially engage the X-tube **142b**. In some instances, the caregiver may still press the latch member **210** onto the X-tube **142b**, albeit with less force due to the restoring force generated in the latch member **210**, to ensure the latch member **210** is properly engaged to the X-tube **142b**. In yet other implementations, the restoring force may instead be sufficiently large to snap-fit connect the latch member **210** to the X-tube **142b** without any additional action by the caregiver.

Turning now to FIG. 9E, to unlock the latch mechanism **200c** the caregiver may pull on the tab **220** with sufficient force to disengage the notch **240a** from the X-tube **142b**. In implementations where the latch member **210** does not generate an appreciable restoring force, the caregiver may release the latch member **210**, and the slider **120** may then move downwards along the leg tube **112** via gravity and/or the caregiver actively moving the slider **120** as shown in FIG. 9F. In implementations where the latch member **210** generates an appreciable restoring force, the caregiver may hold the latch member **210** with one hand until the slider **120** moves a sufficient distance along the leg tube **112** such that the X-tube **142b** is no longer aligned with the notch **240a**.

Additionally, FIGS. 9D-9F show the corner **130**, in some implementations, may further include a hook **133** that protrudes outwards from the frame **100a**. The hook **133** may be used, in part, to pull the soft goods **300** taut around the frame **100a** and/or to function as a secondary restraining feature to prevent the soft goods **300** from prematurely detaching from the frame **100a**. In some implementations, the hook **133** may also be used as a locating feature to facilitate installation of the soft goods **300** onto the frame **100a**. FIGS. 9D-9F further show that, in some implementations, the corner **130** may not include the snap-fit connector **139** as before. Instead, a snap-fit connector **190** may be mounted directly onto the leg tube **112**.

FIG. 10 shows another exemplary latch mechanism **200d** coupled to the frame **100a**. The latch mechanism **200d** is a variant of the latch mechanism **200c** with the primary difference being the latch member **210** is coupled to the

X-tube **142a** instead of the corner **130** via a fastener inserted through the opening **212** and an opening on the X-tube **142a**. The latch mechanism **200d** may be locked and/or unlocked in the same manner as the latch mechanism **200c**. In some implementations, the latch member **210** of the latch mechanism **200d** may be dimensioned to be shorter in length due to the smaller separation distance between the X-tubes **142a** and **142b** compared to the latch member **210** of the latch mechanism **200c**.

FIGS. 11A-11D show another exemplary latch mechanism **200e** installed on the frame **100a** of the playard **1000a**. FIG. 11A shows the frame **100a** once again covered with soft goods **300**. FIG. 11B shows the soft goods **300** again partially covering the X-frame assemblies **140a** such that a bottom portion of the latch mechanism **200e** is exposed similar to the latch mechanisms **200a-200d**.

FIG. 11C shows the latch mechanism **200e** may again include a latch member **210** coupled to the corner **130** of one leg support assembly **110a** via a fastener inserted through the opening **212** at one end of the latch member **210**. In this example, the latch member **210** may include a hook structure **240b** near the tab **220**. As shown, the hook structure **240b** may provide a contoured surface upon which the X-tube **142b** may rest when the frame **100a** is unfolded. As before, the latch member **210** may be a mechanically compliant component that may be deflected and/or bent due to contact with the X-tube **142b** and/or the caregiver pulling on the tab **220** disposed at the bottom end of the latch member **210**.

The latch mechanism **200e** may lock the frame **100a** in the unfolded configuration in a similar manner to the latch mechanisms **200a-200d**. When the caregiver moves the slider **120** towards the corner **130**, the X-tube **142b** may contact the latch member **210** and deflect outwards. The latch member **210** may include a lead-in feature **222** formed between the hook structure **240b** and the bottom end of the latch member **210** to guide the X-tube **142b** moving against the latch member **210** and to deflect the latch member **210** outwards. Once the X-tube **142b** is disposed above the hook structure **240b**, the caregiver may release the slider **120** and the slider **120** may then move downwards along the leg tube **112** until the X-tube **142b** comes to rest on the hook structure **240b**.

In some implementations, the hook structure **240b** may be shaped such that the caregiver may release the latch mechanism **240b** by pulling on the tab **220** with sufficient force. In some implementations, the hook structure **240b** may be shaped to cradle the X-tube **142b** and/or the latch member **210** may be sufficiently rigid such that the latch member **210** acts as a double-action latch mechanism where the caregiver would have to apply an appreciably large force to disengage the latch member **210** from the X-tube **142b**. Instead, the caregiver may raise the slider **120** and/or squeeze the X-tubes **142a** and **142b** such that the X-tube **142b** is released from the hook structure **240b**. While the caregiver holds the X-tube **142b** above the hook structure **240b** with one hand, the caregiver may then pull the latch member **210** outwards to allow the X-tube **142b** to fall below the hook structure **240b** as shown in FIG. 11D.

FIGS. 12A and 12B show another exemplary latch mechanism **200f** that directly couples the slider **120** to the corner **130** in the frame **100a** of the foldable playard **1000a**. As shown in FIG. 12A, the frame **100a** may only include one latch mechanism **200f** coupled to one leg support assembly **110a** to support the multiple sliders **120** and/or X-frame assemblies **140a** when the frame **100a** is unfolded.

FIG. 12B shows the latch mechanism **200f** may include a latch member **243** disposed on the slider **120** of one leg

support assembly **110a** and a latch hook **242** disposed on the corresponding corner **130**. The latch member **243** may be integrally formed onto the slider **120** to form one single component or fabricated as a separate component that is then coupled to the slider **120** using, for example, a fastener or a snap-fit connection. In some implementations, the latch member **243**, when formed as a separate component, may be coupled to the openings of the slider **120** formed on the extended portions **124** and **126** to couple to the X-tubes **142a** and/or **142b** such that a single fastener couples the latch member **243**, the slider **120**, and one or more X-tubes **142a** and/or **142b** together. In this manner, the slider **120** may remain identical with the other sliders **120** in the frame **100a**.

The latch hook **242** may similarly be integrally formed onto the corner **130** to form one single component or fabricated as a separate component that is then coupled to the slider **120**. Similarly, the latch hook **242**, when formed as a separate component, may be coupled to the openings of the corner **130** formed on the extended portions **134** and **136** in a manner similar to the latch member **210** of the latch mechanism **200a** where the corner **130** remains unchanged and/or identical with the other corners **130** in the frame **100a**.

The latch member **243** may include a first end **241a** coupled to the slider **120** and a latch opening **244** disposed near a second end **241b** opposite from the first end **241a**. The latch opening **244** may be shaped to receive the latch hook **242** on the corner **130**. In some implementations, the latch hook **242** may have a contoured surface such that the portion of the latch member **243** forming the top side of the opening **244** rests upon the latch hook **242** when the latch mechanism **200f** is locked. In this manner, the latch mechanism **200f** may directly couple the slider **120** and the corner **130** together to hold the frame **100a** in the unfolded configuration. In some implementations, the latch opening **244** and the latch hook **242** may also be shaped to reduce or, in some instances, eliminate relative translational and/or rotational motion between the slider **120** and the corner **130** along axes of motion other than the longitudinal axis **111a**.

The latch member **243** may be a mechanically compliant component with a tab **220** disposed at the second end **241b** similar to the latch member **210** of the latch mechanism **200a**. Although the latch member **243** is disposed on the slider **120**, the latch member **243** may engage the latch hook **242** in a manner similar to the latch mechanisms **200a-200e**. As before, the caregiver may move the slider **120** towards the corner **130**. Once the tab **220** of the latch member **243** contacts the bottom surface of the latch hook **242**, the latch member **243** may be deflected outwards. As shown in FIG. **12B**, the bottom surface of the latch hook **242** may form a lead-in feature (e.g., a sloped surface) to guide the latch member **243** as it is deflected outwards. The latch member **243** may be sufficiently rigid to generate an internal restoring force when the latch member **243** is bent. Thus, once the slider **120** is moved sufficiently close to the corner **130** that the latch hook **242** is aligned with the latch opening **242**, the restoring force may cause the latch member **243** to snap back to its original form and the latch hook **242** may then protrude through the latch opening **242**.

Similar to the latch mechanism **200e**, the latch mechanism **200f** may be a single-action latch mechanism where the caregiver may release the latch member **243** from the latch hook **242** by pulling the tab **220** with sufficient force. In some implementations, the latch mechanism **200f** may be a double-action latch mechanism where the latch hook **242** may be sufficiently rigid and/or includes a sufficiently deep undercut portion such that the latch member **243** cannot be

released by pulling the tab **220** without applying excessive force (e.g., a force greater than 20 lbf). The caregiver should instead raise the slider **120** such that the portion of the latch member **243** forming the top side of the opening **244** is released from the latch hook **242**. While holding the slider **120** in the raised position, the caregiver may then pull the latch member **243** outwards so that the slider **120** may move downwards along the leg tube **112**.

FIGS. **13A-13H** show another exemplary latch mechanism **200g** that is mounted to the X-tubes **142a** and **142b** of one X-frame assembly **140a**. As shown in FIG. **13A**, the frame **100a** may include a single latch mechanism **200g** mounted to one X-frame assembly **140a** to support the frame **100a** in the unfolded configuration. In some implementations, the latch mechanism **200g** may be shaped and/or dimensioned to have the same or similar thickness as the X-frame assembly **140a** so that the latch mechanism **200g** does not protrude appreciably outwards from the frame **100a** particularly when the frame **100a** is in the folded configuration as shown in FIG. **13B**. In other words, the thickness of the latch mechanism **200g** may be the same or similar as the distance separating the outer exterior edge of the central portion **144** of the X-tube **142a** and the interior exterior edge of the central portion **144** of the X-tube **142b** in FIG. **3B**.

FIG. **13C** shows the latch mechanism **200g** may replace the pin joint **145** and, hence, may rotatably couple the X-tube **142a** to the X-tube **142b** such that the X-tubes **142a** and **142b** rotate about a rotation axis **252**. FIG. **13D** shows the latch mechanism **200g** may include a first housing **250a** disposed on an exterior portion of the frame **100a** and rigidly coupled to the X-tube **142b**. In particular, the first housing **250a** may include a notch **251a** and the X-tube **142b** may be formed with a flat section **148** within the central portion **144** that fits into the notch **251a**. Thus, the first housing **250a** may rotate together with the X-tube **142b**.

The latch mechanism **200g** may further include a second housing **250b** disposed within the interior space **102** of the frame **100a** and rigidly coupled to the X-tube **142a**. The second housing **250b** may also include a notch **251b** and the X-tube **142a** may also have a flat section **148** that fits into the notch **251b** so that the second housing **250b** rotates together with the X-tube **142a**. The first housing **250a** may be rotatably coupled to the second housing **250b** via a shaft or pin (not shown) inserted through respective openings in the first housing **250a**, the second housing **250b**, and the X-tubes **142a** and **142b** along the rotation axis **252** as shown in FIG. **13D**.

The first and second housings **250a** and **250b** may form a cavity to contain a locking gear **254**, which may translate along the rotation axis **252** relative to the first and second housings **250a** and **250b** to lock and/or unlock the latch mechanism **200g**. The cavity may further contain a return spring **253** disposed between the locking gear **254** and the second housing **250b** to impart a spring-bias force onto the locking gear **254** to maintain the latch mechanism **200g** in a locked configuration by default. The locking gear **254** may include a pair of latch key sections **256** that have interior sidewalls **257a** that define a channel **257c** shaped to restrict and lock the X-tubes **142a** and **142b** when the frame **100a** is unfolded (e.g., the X-tubes **142a** and **142b** are arranged to form a shallow X-frame structure). Said in another way, when the latch mechanism **200g** is locked, the flat sections **148** of the X-tubes **142a** and **142b** may be disposed within the channel **257c** where the sidewalls **257a** abut opposing sides of each of flat sections **148** to prevent rotation of the X-tubes **142a** and **142b**.

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When the playard 1000a is in the folded configuration, the locking gear 254 may be primarily disposed within the second housing 250b and the return spring 253 may be compressed due to the respective flat sections 148 of the X-tubes 142a and/or 142b contacting and/or pressing against the front portions 257b of the locking gear 254. To deploy the playard 1000a, the caregiver may once again move the slider 120 of at least one leg support assembly 110a and/or squeeze the X-tubes 142a and 142b of one X-frame assembly 140a together to unfold the frame 100a. As the X-tubes 142a and 142b are rotated, the respective flat sections 148 of the X-tubes 142a and 142b may slide along the front portions 257b of the locking gear 254, thus maintaining compression of the return spring 253. Once the X-tubes 142a and 142b are sufficiently rotated such that the respective flat sections 148 of the X-tubes 142a and 142b are aligned to match the geometry of the channel 257c, the spring 253 may then push the locking gear 254 outwards towards the first housing 250a such that the flat sections 144 are disposed within the channel 257c and constrained by the latch key sections 256 (see FIGS. 13E and 13G).

FIG. 13D further shows the latch mechanism 200g may include a release button 260 disposed, in part, within a recessed opening 259 formed along the front of the first housing 250a. The recessed opening 259 of the first housing 250a may be separated from the cavity formed between the first and second housings 250a and 250b by a recessed front surface of the first housing 250a. The release button 260 may be slidably coupled to the first housing 250a via the slot guides 258 and may include one or more tabs 262 that protrude through the recessed surface of the first housing 250a to contact front portions 257b of the latch key sections 256 on the locking gear 254.

To unlock the latch mechanism 200g, the caregiver may push the release button 260 into the recessed opening 259 causing the tabs 262 to press against the latch key sections 256 of the locking gear 254. The locking gear 254, in turn, is then displaced towards the second housing 250b along the rotation axis 252 resulting in compression of the return spring 253. Once the locking gear 254 is sufficiently displaced where the respective flat sections 148 of the X-tubes 142a and 142b are no longer disposed within the channel 257c, the caregiver may then rotate the X-tubes 142a and 142b and/or move the slider 120 of at least one leg support assembly 110a to fold the frame 100a (see FIGS. 13F and 13H). In some implementations, the depth of the recessed opening 259 and/or the length of the tabs 262 of the release button 260 may be tailored to ensure sufficient travel distance for the release button 260 to disengage the locking gear 254 from the X-tubes 142a and 142b. In some implementations, the release button 260 may remain disposed within the recessed opening 259 until the playard 1000a is unfolded.

FIGS. 14A-14D show another exemplary latch mechanism 200h integrated into the X-tube 142b of one X-frame assembly 140a to engage the slider 120 of one leg support assembly 110a in the frame 100a of the playard 1000a. FIG. 14A once again shows the frame 100a may only include a single latch mechanism 200h to support the frame 100a in the unfolded configuration.

FIG. 14B shows the latch mechanism 200h may include a latch 270 that is slidably coupled to the X-tube 142b and rotatably coupled to the slider 120 of one leg support assembly 110a. A return spring 272 may be disposed, at least in part, within an interior cavity of the X-tube 142b to impart a spring-bias force that pushes the latch 270 towards the leg tube 112. The leg tube 112 may include a latch opening 273

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shaped and/or dimensioned to receive at least a portion of the latch 270 (e.g., the tip of the latch 270).

When the frame 100a is sufficiently unfolded such that the slider 120 is positioned along the leg tube 112 to overlap the latch opening 273, the return spring 272 may push the latch 270 into the latch opening 272, thus locking the slider 120 and, by extension, the X-tube 142b in place. Since the X-tube 142b is movably coupled to the X-tube 142a, the corners 130 and sliders 120 of other leg support assemblies 110a, and the other X-frame assemblies 140a (via the other leg support assemblies 110a) in the frame 100a, the constraints applied to the slider 120 and the X-tube 142b by the latch mechanism 200h may maintain the frame 100a in the unfolded configuration.

FIG. 14B further shows the latch mechanism 200h may include a collar 271 coupled to the latch 270 to provide an actuator for the caregiver to move when unlocking the latch mechanism 200h. In some implementations, the latch 270 may be directly coupled to the collar 271 using, for example, a fastener inserted through an opening 276 on the collar and an opening (not shown) on the latch 270. The collar 271, in turn, may be slidably coupled to the second end 143b of the X-tube 142b. For example, the collar 271 may include a recessed opening (not shown) shaped to receive the second end 143b with sufficient depth to enable the collar 271 and, by extension, the latch 270 to slide along the X-tube 142b. To compensate for the respective lengths of the latch 270 and the collar 271, the X-tube 142b supporting the latch 270 and the collar 271 may be shorter in length compared to other X-tubes 142b in other X-frame assemblies 140a.

The latch 270 may be rotatably coupled directly to the slider 120 via a pin 274 inserted through the opening on the slider 120 (previously used to couple to the X-tube 142b in other X-frame assemblies 140a) and an opening 275 formed along the latch 270. In some implementations, the opening 275 may be a slot that is shaped and/or dimensioned to allow the latch 270 to slidably move relative to the slider 120 to facilitate insertion of the latch 270 into the latch opening 273.

In some implementations, the latch 270 may instead be disposed within the interior cavity of the X-tube 142b such that the overall length of the X-tube 142b remains the same as other X-tubes 142b in other X-frame assemblies 140a. The second end 143b of the X-tube 142b, however, may have an opening through which the latch 270 may pass through when engaging and/or disengaging the latch opening 273 on the leg tube 112. The collar 271 may be disposed outside the X-tube 142b and configured to slide together with the latch 270 along the length of the X-tube 142b. As before, the latch 270 may be coupled to the collar 271 via a fastener inserted through the opening 276 on the collar 271 and another opening (not shown) on the latch 270. The fastener may pass through the X-tube 142b via a slotted opening (not shown) that is shaped and/or dimensioned to be similar to the opening 275 on the latch 270.

The latch 270 and the X-tube 142b may be rotatably coupled to the slider 120. For example, the pin 274 may pass through the openings on the slider 120, the opening 275 on the latch 270, and the opening 147 on the X-tube 142b. The latch 270 may still have a slotted opening 275 to allow the latch 270 to slidably move relative to the slider 120 to engage and/or disengage the latch opening 273.

To unlock the latch mechanism 200h, the caregiver may move the collar 271 along the X-tube 142b to release the latch 270 from the latch opening 273 as shown in FIG. 14C. This causes the return spring 272 to be compressed, thus generating and/or increasing a spring-bias force applied to

the latch 270. While holding the collar 271, the slider 120 may then move downwards along the leg tube 112, thus folding the X-frame assembly 140a. Once the latch 270 is no longer aligned to the latch opening 273, the caregiver may release the collar 271 and continue folding the frame 100a. The spring-bias force applied to the latch 270 may cause the latch 270 to press against the exterior surfaces of the leg tube 112 as the slider 120 is moved towards the foot 114 and/or the surfaces of the slider 120 once the X-tube 142b is sufficiently rotated as shown in FIG. 14D. In some implementations, the end of the latch 270 may be shaped (e.g., curved or contoured) to allow the X-tube 142b to rotate smoothly when pressing against the leg tube 112 and/or the slider 120 as the frame 100a is being folded and/or unfolded.

FIGS. 15A-15D show yet another exemplary latch mechanism 200i mounted onto the frame 100a of the playard 1000a. Specifically, the latch mechanism 200i may be mounted to one end of the X-tube 142b (or 142a) of one X-frame assembly 140a to engage the slider 120 of one leg support assembly 110a. FIG. 15A once again shows the frame 100a may only include a single latch mechanism 200i to support the frame 100a in the unfolded configuration. The latch mechanism 200i may be shaped and/or dimensioned such that the latch mechanism 200i fits within the recessed opening of the extended section 126 (or 124) of the slider 120 together with the second end 143b of the X-tube 142b. In this manner, the latch mechanism 200i may not protrude outwards from the frame 100a even when the frame 100a is folded (see FIG. 15B), thus preserving the compact shape of the folded frame 100a.

FIG. 15C shows the latch mechanism 200i may include a latch base 280 coupled to the second end 143b of the X-tube 142b and rotatably coupled to the slider 120. In some implementations, a single fastener may couple the slider 120, the latch base 280, and the X-tube 142b together. As shown, the latch base 280 may include a latch member 284 that extends from the latch base 280. The latch member 284 may be a mechanically compliant component that is deformable and may have sufficient mechanical rigidity to generate a restoring force when deformed.

In some implementations, the latch base 280 may have a cylindrical shape and the latch member 284 may extend from the periphery of the latch base 280. The latch member 284 may have a curved and/or contoured shape as shown in FIGS. 15C and 15D. The latch member 284 may include an integrally formed latch catch 281 that is shaped to engage a latch opening 283 formed on a bottom surface 127 of the slider 120. The latch member 284 may further include a tab 282 disposed at the end of the latch member 284, which may be pulled to bend the latch member 282, thus releasing the latch catch 281 from the latch opening 283.

FIG. 15D shows the latch member 284 may be disposed between the sliders 120 from adjacent leg support assemblies 110a when the frame 100a is unfolded. When unfolding the frame 100a, the latch body 280 together with the latch member 284 may rotate with the X-tube 142b about the pin joint 146c relative to the slider 120 as the slider 120 moves up along the leg tube 112 towards the corner 130. As the latch body 280 rotates, the latch member 284 and, in particular, the latch catch 281, may initially contact the exterior portions of the slider 120, thus bending and/or deflecting the latch member 284. In some implementations, the latch catch 281 may include a lead-in feature to facilitate the deflection of the latch member 284 as the frame 100a is unfolded.

Once the slider 120 is positioned sufficiently close to the corner 130 and/or the X-tube 142b is sufficiently rotated

such that the latch catch 281 aligns with the latch opening 283, the restoring force generated by the deflection of the latch member 284 may insert the latch catch 281 into the latch opening 283. The latch catch 281 and the latch opening 283 may thus prevent further rotation of the X-tube 142b relative to the slider 120 and, hence, further movement of the slider 120 along the leg tube 112 to hold the frame 100a in the unfolded configuration.

To unlock the latch mechanism 200i, the caregiver may pull on the tab 282 with sufficient force to release the latch catch 281 from the latch opening 283. While holding the tab 282, the slider 120 may then move downwards along the leg tube 112 towards the foot 114, which causes the X-tube 142b and the latch body 280 to rotate relative to the slider 120. Once the latch catch 281 is no longer aligned with the latch opening 283, the caregiver may release the tab 282 and proceed with folding the frame 100a.

As described above, the frame 100a may generally include at least one latch mechanism to maintain the frame 100a and, by extension, the playard 1000a in the unfolded configuration. In some implementations, the frame 100a may include a single latch mechanism (e.g., one of the latch mechanisms 200a-200i) to lock the unfolded frame 100a, which may simplify the frame 100a by reducing the number of parts for manufacture. However, in other implementations, the frame 100a may include multiple latching mechanisms to ensure the various components of the frame 100a are kept evenly unfolded. Thus, it should be appreciated that in other implementations, the frame 100a may include combinations of one or more of the latch mechanisms 200a-200i described above.

FIGS. 16A and 16B show one example of a frame 100a that includes the latch mechanism 200g coupled to one X-frame assembly 140a and the latch mechanism 200h coupled to the X-tube of another X-frame assembly 140a and the slider 120 of one leg support assembly 110a. FIG. 16A shows the latch mechanisms 200g and 200i being used to maintain the frame 100a in the unfolded configuration. FIG. 16B shows the latch mechanisms 200g and 200i do not appreciably extend outwards from the frame 100a when the frame 100a is in the folded configuration.

As described above, the foldable playard may generally include a frame that outlines an interior space. The frame may include multiple leg support assemblies and X-frame assemblies that together define and/or align with the outer boundaries of the interior space. For example, the playard 1000a includes a frame 100a defining an interior space 102 with a horizontal cross-section shaped as a hexagon. It should be appreciated that the various implementations of the foldable playard described herein may define interior spaces having other geometries based, in part, on the number of leg support assemblies and/or the X-frame assemblies used for construction.

For example, the playard may outline an interior space with a square horizontal cross-section. The frame of the playard may include four identical leg support assemblies, which may be connected together using four identical X-frame assemblies where each X-frame assembly forms a single (or double) X-frame structure. As before, each X-frame assembly may couple adjacent leg support assemblies together.

In another example, FIGS. 17A-17D show an exemplary playard 1000b with a frame 100b that outlines an interior space 102 with a horizontal cross-section shaped as a rectangle. The frame 100a may include multiple leg support assemblies 110b defining and/or aligning with the respective side edges 104 of the interior space 102 when the frame

100b is unfolded to support the playard 1000b on the ground 90 (see, for example, FIG. 18A). The frame 100a may include a pair of X-frame assemblies 140a disposed on the smaller side faces 106 of the interior space 102 to couple together adjacent leg support assemblies 110b located on the shorter sides of the rectangular horizontal cross-section of the interior space 102. The frame 100 may further include a pair of X-frame assemblies 140b disposed on the larger side faces 106 of the interior space 102 to couple together adjacent leg support assemblies 110b located on the longer sides of the rectangular horizontal cross-section of the interior space 102. Thus, each leg support assembly 110b may couple to one X-frame assembly 140a and one X-frame assembly 140b.

To form the rectangular-shaped interior space 102, each X-frame assembly 140a may form a single X-frame structure, as described above, and each X-frame assembly 140b may form a double X-frame structure (i.e., two pairs of crossing X-tubes where each pair of X-tubes couples to one leg support assembly). The combination of the single and double X-frame structures allows the frame 100b to define an interior space 102 where the sides of the horizontal cross-section have different dimensions while enabling the X-frame assemblies 140a and 140b to couple to the same components of the leg support assembly 110b (e.g., the same slider 120 and corner 130) so that the leg support assemblies 110b, the X-frame assemblies 140a, and the X-frame assemblies 140b may fold and/or unfold together (see FIG. 17C). Furthermore, the double X-frame structure of the X-frame assembly 140b may also enable the leg support assemblies 110b and, in particular, the length of the leg tube 112 to be shorter compared to a single X-frame structure that spans the same length as the X-frame assembly 140b when deployed. Thus, the frame 100b may be more compact, particularly, when folded.

Similar to the frame 100a, the frame 100b may be unfolded with the feet 114 of the leg support assemblies 110b remaining in contact with the ground 90. Additionally, the leg tubes 112 may remain vertically upright or nearly vertically upright (e.g., leg tubes 112 may intentionally be tilted when the frame 100b is unfolded to improve stability) while the frame 100b is being folded and/or unfolded to make the process of setting up and/or tearing down the playard 1000b easier for the caregiver (see FIG. 17D).

Additionally, the X-frame assemblies 140a and 140b in the frame 100b may be disposed in the top portion 108 of the frame 100a to form a top perimeter structure along the interior space 102 (see FIG. 18A). As before, this may enable the respective X-tubes of the X-frame assemblies 140a and 140b to function as top rails to provide mechanical stability and rigidity to the frame 100b. In some implementations, the frame 100b may not include a separate compliant or rigid top rail and/or a bottom support structure.

In some implementations, the frame 100b with only X-frame assemblies 140a and 140b coupling the leg support assemblies 110b together may provide sufficient mechanical rigidity, stability, and/or strength to satisfy various consumer safety standards (e.g., ASTM F406-19). For example, FIG. 22 shows the playard 1000b subjected to a stability test. Similar to the playard 1000a, the playard 1000b was demonstrated to remain sufficiently stable (i.e., at least three feet 114 remained in contact with the underlying platform) when the playard 1000b was tilted more than 10 degrees.

FIGS. 17A and 17B further show the playard 1000b may include soft goods 300 coupled to the frame 100b and forming a partially enclosed space 301 disposed within the interior space 102 to contain the child 50. As before, the soft

goods 300 may be readily folded together with the frame 100b as shown in FIG. 17C. The soft goods 300 may include a floor portion 306 that rests on the ground 90 supporting the playard 1000b and side portions 304 that together define and surround the partially enclosed space 301. The floor portion 306 may include a removable mat to provide padding on the ground 90. The side portions 304 may be formed from transparent and/or see-through materials to allow the caregiver to monitor their child 50 when the child 50 is placed into the partially enclosed space 301. The soft goods 300 may include tethers and/or straps to attach the floor portion 306 to the bottom portions of the leg support assemblies 110b.

The soft goods 300 may further include a top portion 302, formed from an opaque textile material, to attach the soft goods 300 to the top portions of the leg support assemblies 110b as well as cover the top portion of the frame 100b. In particular, the soft goods 300 in the playard 1000b may fully cover one or more of the X-frame assemblies 140a and 140b, the corners 130 of the leg support assemblies 110b, and/or the sliders 120 of the leg support assemblies 110b. In some implementations, the soft goods 300 may fully cover the X-frame assemblies 140a and 140b as well as the sliders 120 and the corners 130 of the leg support assemblies 110b such that only the leg tubes 112 and/or the feet 114 are observable as shown in FIGS. 17A and 17B. As before, positioning the X-frame assemblies 140a and 140b in the top portion 108 of the frame 100b when the frame 100b is unfolded may also increase the visibility of the child 50 due to the larger visually unobstructed portions of the side faces 106.

As described above, conventional playards and especially, indoor playards, typically have to compromise between ease of use, visibility of the child, and/or the appearance of the playard (see, for example, playard 10c). In comparison, the playard 1000b may simultaneously improve ease of use, visibility of the child, and the overall appearance. First, the playard 1000b includes X-frame assemblies 140a and 140b that allow the frame 100b to be folded and/or unfolded in one step. For instance, the caregiver may move one slider 120 of one leg support assembly 110b to fold and/or unfold the frame 100b. Second, the X-frame assemblies 140a and 140b are positioned in the top portion 108 of the frame 100b when the playard 1000b is deployed, which allows for greater visibility of the child in the partially enclosed space 301 through the sides of the frame 100b. Third, aesthetically undesirable components, such as the X-tubes, the sliders 120, the corners 130, may be readily hidden by the top portion 302 of the soft goods 300 to provide a cleaner, more aesthetically desirable appearance.

FIG. 18A shows the frame 100b without the soft goods 300 attached in the unfolded configuration. As shown, each leg support assembly 110b may be similar to leg support assemblies 110a used in the frame 100a. For instance, the leg support assembly 110b includes a leg tube 112 with a top end 113a and a bottom end 113b, a corner 130 coupled to the top end 113a, a foot 114 coupled to the bottom end 113b, and a slider 120 that is slidably coupled to be leg tube 112 and disposed between the foot 114 and the corner 130. The top end 113a of the leg tube 112 and/or the corner 130 may align with a top vertex 105 of the interior space 102 and generally define a top horizontal plane 92 of the frame and hence a height H_1 of the frame between the ground surface 90 and the top horizontal plane 92. The bottom end 113b of the leg tube 112 and/or the foot 114 may align with a bottom vertex 107 of the interior space 102.

FIG. 18B further shows the leg tubes 112 may have a circular cross-sectional shape. The leg tubes 112 may also remain vertical or nearly vertical for both the folded and unfolded configurations. Thus, the interior space 102 may be shaped as right prism with rectangular base. The slider 120 may once again include a base 121 that defines a through hole opening 122 that surrounds the leg tube 112. The slider 120 may include extended portions 124 and 126 disposed on opposing sides of the base 121 to couple respective X-tubes (e.g., X-tubes 142a and 142d in FIG. 18B) of the X-frame assemblies 140a and 140b to the slider 120. The corner 130 may include a base 131 with a recessed opening (not shown) to receive the top end 113a of the leg tube 112. The corner 130 may further include a snap-fit connector 139 coupled to the base 131 instead of a tab 138 extending from the base 131 as in the leg support assembly 110a. Once again, the corner 130 may include extended portions 134 and 136 disposed on opposing sides of the base 131 to couple respective X-tubes (e.g., X-tubes 142b and 142c in FIG. 18D) of the X-frame assemblies 140a and 140b to the corner 130.

FIG. 19A shows the frame 100b in the folded configuration. FIG. 19B shows the slider 120 may be disposed proximate to the foot 114 when the frame 100b is folded. As described above and shown in FIGS. 18B and 19B, the X-frame assemblies 140a and 140b may couple to the same corner 130 and slider 120 of one leg support assembly 110b. Furthermore, the pin joints that connect the respective X-tubes of the X-frame assemblies 140a and 140b to the slider 120 or the corner 130 may be located along the same horizontal plane. Thus, the respective ends of the X-tubes of the X-frame assemblies 140a and 140b that couple to the leg support assembly 110b may travel the same distance along the leg tube 112 to fold and/or unfold both the X-frame assemblies 140a and 140b. This enables the slider 120 and the corner 130 to be thinner in size and, in turn, reduces the overall length, L, of the leg tube 112 such that the leg tube 112 only provides sufficient overlap to couple the foot 114 and the corner 130 to the leg tube 112 and sufficient clearance for the slider 120 to move a sufficient distance to fold and/or unfold the X-frame assemblies 140a and 140b. As shown in FIGS. 18B and 19B, the slider 120 may be disposed proximate to the corner 130 when the frame 100b is in the unfolded configuration and proximate to the foot 114 when the frame 100b is in the folded configuration. FIG. 19A also shows that, in the folded configuration, the frame has a height H_2 between the ground surface 90 and a top horizontal plane 92A defined by the frame. As noted above in connection with FIGS. 2B and 2E, the height of the frame 100b may remain substantially constant or constant between the folded and unfolded configurations of the frame. In other words, the heights H_1 and H_2 may be equal or substantially similar and the planes 92 and 92A are coplanar or substantially coplanar. In some implementations, however, the height of the frame 100b may vary (e.g., the height H_2 may be somewhat greater than the height H_1 and the plane 92A in the folded configuration may be disposed somewhat above the plane 92 in the unfolded configuration).

FIGS. 20A-20E show several views of the frame 100b in a partially unfolded/folded state. In particular, FIG. 20B shows the X-frame assembly 140a may once again include X-tubes 142a and 142b that are rotatably coupled to one another via a pin joint (e.g., a rolled rivet joint). As shown, the X-tube 142a may be rotatably coupled to the corner 130 of one leg support assembly 112b via a pin joint 146a and the slider 120 of another leg support assembly 112b via a pin joint 146b. Similarly, the X-tube 142b may be rotatably

coupled to the slider 130 of the one leg support assembly 112b via a pin joint 146c and the corner 130 of the other leg support assembly 112b via a pin joint 146d. Thus, the X-frame assembly 140a may operate in a similar or same manner as the X-frame assemblies 140a in the frame 100a.

FIG. 20C shows the X-frame assembly 140b may include two pairs of X-tubes, i.e., the X-tubes 142c and 142d as well as the X-tubes 142e and 142f. The X-tubes 142c and 142d may be rotatably coupled to each other via a pin joint 145 similar to the X-tubes 142a and 142b in the X-frame assembly 140a. Similarly, the X-tubes 142e and 142f may be rotatably coupled to each via another pin joint 145. Each pair of X-tubes 142c and 142d (or 142e and 142f) may be coupled to one leg support assembly 110b and to the other remaining pair of X-tubes. As shown, the X-tube 142c may be rotatably coupled to the corner 130 of one leg support assembly 110a via a pin joint 146e and the X-tube 142e via a pin joint 146f. The X-tube 142d may be rotatably coupled to the slider 120 of the one leg support assembly 110a via a pin joint 146g and to the X-tube 142e via a pin joint 146h. The X-tube 142e may be further rotatably coupled to the corner 130 of another leg support assembly 110b via a pin joint 146i. The X-tube 142f may be further rotatably coupled to the slider 120 of the other leg support assembly 110b via a pin joint 146j.

In some implementations, the shape and/or dimensions of the X-tubes 142c-142f may be substantially identical or identical with each other. The shape and/or dimensions of the X-tubes 142a and 142b of the X-frame assembly 140a may be different from the X-tubes 142c-142f of the X-frame assembly 140b depending, in part, on the desired dimensions of the rectangular-shaped interior space 102. However, in some implementations, the shape and/or dimensions of the X-tubes 142c-142f may also be substantially identical or identical with the X-tubes 142a and 142b of the X-frame assembly 140a.

FIG. 20C further shows the pair of pin joints 145 may be offset from the respective center points of the X-tubes 142c-142f. In particular, the pin joint 145 coupling the X-tubes 142c and 142d together may be positioned closer to the pin joints 146h and 146f than the pin joints 146e and 146g. Similarly, the pin joint 145 coupling the X-tubes 142e and 142f together may also be positioned closer to the pin joints 146h and 146f than the pin joints 146i and 146j. The position of the pin joints 145 along the X-tubes 142c-142f may be tailored to ensure the respective ends of the X-tubes 142c-142f align with the ends of the X-tubes 142a and 142b when coupled to the same corner 130 or slider 120.

For example, FIG. 20D shows the pin joint 146d coupling the X-tube 142b to the corner 130 and the pin joint 146e coupling the X-tube 142c to the same corner 130 lie on the same horizontal plane 150a. FIG. 20E similarly shows the pin joint 146b coupling the X-tube 142a to the slider 120 and the pin joint 146g coupling the X-tube 142d to the same slider 120 may also lie on the same horizontal plane 150b. As described above, aligning the pin joints in this manner may allow for a thinner slider 120 and corner 130, which, in turn, may reduce the overall length of the leg tube 112. However, it should be appreciated that in some implementations, the pin joints may not be aligned to the same horizontal plane. For example, FIG. 20E shows the extended portion 126 of the slider 120 and the pin joint 146g may be vertically raised (i.e., see extended portion 126-1 and pin joint 146g) above the extended portion 124 and the pin joint 146b.

FIGS. 21A and 21B show the soft goods 300 may attach to the frame 100b in a similar manner as in the frame 100a. Specifically, FIG. 21A shows the soft goods 300 may

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include a snap-fit connector **312** disposed on an interior portion of the top portion **302** to couple with the snap-fit connector **139** on the corner **130**. FIG. 21B shows the foot **114** of each leg support assembly **110b** may include a D-ring **116** that provides an opening to tie a tether **320** of the soft goods **300** to the bottom portion of the leg support assemblies **110b**. As shown, the tether **320** may form a closed via the snap-fit connector **322** coupled to another snap-fit connector (not shown) disposed at the base of the strap **320**.

In yet another example, FIGS. 23A-23E show a playard **1000c** with a frame **100c** that also outlines an interior space **102** with a horizontal cross-section shaped as a rectangle. However, the frame **100c** may include curved leg support assemblies **110c** resulting in the interior space **102** having a convex shape. In other words, the leg support assemblies **110c** curve outwards from the interior space **102** such that the size of the horizontal cross-section is larger at the mid-point of the leg support assemblies **110c** than the top or bottom portions of the leg support assemblies **110c**. In some implementations, a convex-shaped interior space **102** may provide the child **50** a larger volume to play and/or sleep compared to an interior space with straight leg support assemblies and the same footprint. Additionally, a convex-shaped interior space **102** may also provide a more aesthetically pleasing design.

As shown in FIG. 23A, the playard **1000c** may also include soft goods **300** that define a partially enclosed space **301** disposed within the interior space **102** of the frame **100c** for the child **50** to play and/or sleep. Similar to the playard **1000b**, the soft goods **300** in the playard **1000c** may include a floor portion **304** and side portions **306** that define and surround the partially enclosed space **301** as well as a top portion **302** that covers the top portion **108** of the frame **100c**. The soft goods **300** may include a removable mat placed onto the floor portion **304** to provide padding on the ground **90** supporting the playard **1000c**. The side portions **306** may also be formed from a transparent or see-through material. As before, the soft goods **300** may further include a removable mat that is placed onto the floor portion **304** to provide padding.

As shown in FIGS. 23D and 23E, the frame **100c** may include multiple leg support assemblies **110c** that each include at least a leg tube **112**, a slider **120**, and a corner **130**. Compared to the leg support assemblies **110a** and **110b**, the leg tube **112** may be curved along an axis **111b** such that the slider **120** moves along a curved path when the frame **100c** is folded and/or unfolded. The leg support assemblies **110c** may define and/or align with respective side edges **104** of the interior space **102** (see FIG. 24).

The leg support assemblies **110c** may further include either a foot **114** to support the playard **1000c** on the ground **90** or a wheel assembly **151** to more easily move and/or reorient the playard **1000c** after being unfolded. For example, FIG. 23D shows the leg support assemblies **110c** at one end of the interior space **102** may both include wheel assemblies **151**. Thus, the caregiver may pick up the playard **1000c** from the opposing end and pull the playard **1000c** with the wheel assemblies **151** rolling along the ground **90** to reposition the playard **1000c** as desired. In a manner similar to that illustrated in FIG. 18A, FIG. 23E shows that the frame **100c** has a height H_1 between the ground surface **90** and a top horizontal plane **92**.

FIG. 25A shows an exploded view of the leg support assembly **110c** with the wheel assembly **151**. As shown, the leg tube **112** may once again have a first end **113a** and a second end **113b**. The corner **130** may be coupled to the top end **113a** of the leg tube **112**. The wheel assembly **151** may

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include a base **152** that couples to the bottom end **113b** of the leg tube **112**. The wheel assembly **151** may further include a wheel **153** that is rotatably coupled to the base **152** via a wheel cover **154**. The slider **120** may thus be slidably coupled to the leg tube **112** such that the slider **120** is located between the base **152** of the wheel assembly **151** and the corner **130**. FIG. 25A also shows the frame **100c** may include a latch mechanism **200j** that directly couples the slider **120** to the corner **130**, which will be described in more detail below.

FIG. 25B shows an exploded view of the leg support assembly **110c** with the foot **114**. As shown, the leg tube **112**, the slider **120**, the corner **130**, and the foot **114** may be assembled in a similar manner to the leg support assemblies **110a** and **110b** as described above.

The frame **100c** may further include X-frame assemblies **140a**, disposed on the smaller curved side faces **106** of the interior space **102**, to couple adjacent leg support assemblies **110c** along the shorter sides of the rectangular cross-section of the interior space **102** (see FIG. 24). The frame **100c** may also include X-frame assemblies **140b**, disposed on the larger curved side faces **106** of the interior space **102**, to couple adjacent leg support assemblies **110c** along the longer sides of the rectangular cross-section of the interior space **102** (see FIG. 24). As before, the X-frame assemblies **140a** may form a single X-frame structure with one pair of X-tubes and the X-frame assemblies **140b** may form a double X-frame structure with two pairs of X-tubes.

The shape and/or dimensions of the respective X-tubes in the X-frame assemblies **140a** and **140b** and/or the location of the pin joints that rotatably couple each X-tube to another X-tube, the slider **120**, and/or the corner **130** may be tailored based, in part, on the desired dimensions of the interior space **102** similar to the frame **100b**. Additionally, in some implementations, the X-tubes of the X-frame assemblies **140a** and **140b** may be arranged such that the pin joints that couple the X-tubes to the same slider **120** or corner **130** of the leg support assembly **110c** are aligned along the same horizontal plane.

The X-frame assemblies **140a** and **140b** may once again be disposed within a top portion **108** of the frame **100c** and/or the interior space **102**. This enables the X-frame assemblies **140a** and **140b** to function as top rails to mechanically reinforce the frame **100c** while also eliminating other support structures, such as a separate top rail and/or a bottom support structure. The placement of the X-frame assemblies **140a** and **140b** may also provide a larger window for the caregiver to view their child **50** through the sides of the frame **100c**.

In some implementations, the soft goods **300** in the playard **1000c** may be divided into separate components, in part, to better conform with the geometry of the interior space **102**. For example, the side portions **306** and the floor portion **304** may be installed separately from the top portion **302**. To better conform with the shape of the interior space **102**, the side portions **306** may be mounted along an interior side of the leg tube **112** to reduce or, in some instances, prevent gaps from forming between the side portions **306** and the leg support assemblies **110c** (see, for example, FIG. 26A) when the playard **1000c** is unfolded. Said in another way, the side portions **306** of the soft goods **300** may be attached to the leg support assemblies **110c** to provide a seamless appearance with the leg tubes **112**, the feet **114**, and/or the wheel assemblies **151** being exposed along the exterior portion of the playard **1000c** as shown in FIGS. 23A-23C. Once the side portions **306** and the floor portion **304** of the soft goods **300** are installed, the top portion **302**

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may then be attached to the side portions 306 using, for example, a zipper connection (not shown), and subsequently coupled to the frame 100c to complete assembly.

This may be accomplished, in part, by incorporating a stiffener 330 into the side portions 306 of the soft goods 300, which may then be routed through a channel 171 formed along the leg tube 112. The stiffener 330 may be a compliant component, such as an extruded plastic rod that is inserted through a pocket formed along the respective corners of the side portions 306 located near the side edges 104 of the interior space 102. FIG. 26B shows the leg tube 112 may have an oblong cross-sectional shape with a curved side 172 that forms a recess along the interior side of the leg tube 112 facing the interior space 102. The channel 171 may be formed on the curved side 172 and may span a portion of or, in some instances, the entire length of the leg tube 112. As shown in FIG. 26B, the stiffener 330 may be inserted through the channel 171, thus holding the side portions 306 of the soft goods 300 against the leg tube 112.

The slider 120 in the leg support assembly 110c may still be allowed to move along the leg tube 112 even with the side portions 306 of the soft goods 300 installed onto the leg tube 112. For example, FIG. 26B shows the slider 120 may include a base 121 that defines a through hole opening 122 that only partially surrounds the leg tube 112 to guide the movement of the slider 120 along the leg tube 112. As shown, a slotted opening 128 may be formed along an interior side of the base 121 to allow the side portions 306 attached to the leg tube 112 to pass through the base 121 of the slider 120. In this manner, the slider 120 may move along the leg tube 112 unimpeded by the side portions 306 when the playard 1000c is folded and/or unfolded.

FIG. 26B further shows the slider 120 may once again include extended portions 124 and 126 disposed on opposing sides of the base 121 to couple to respective X-tubes of the X-frame assemblies 140a and 140b (e.g., X-tubes 142f and 142b).

FIG. 27A shows the corner 130 may once again include a base 131 with extended portions 134 and 136 disposed on opposing sides of the base 131 to couple to respective X-tubes of the X-frame assemblies 140a and 140b (e.g., X-tubes 142e and 142a). The corner 130 may further include a tab 138 that extends downwards along the leg tube 112 and outwards from the frame 100c to form an overhang portion. As shown in FIG. 27A, the slider 120 may be positioned underneath the overhang portion formed by the tab 138 and, hence, disposed between the leg tube 112 and the tab 138 of the corner 130 when the frame 100c is unfolded.

The corner 130 may be shaped in this manner to provide a hook structure for the top portion 302 of the soft goods 300 to wrap around, thus ensuring the corners 130 and the X-frame assemblies 140a and 140b are covered. In some implementations, the top portion 302 of the soft goods 300 may further include a pocket 331 to aid the caregiver in wrapping the soft goods 300 around the corners 130. Additionally, the soft goods 130 may primarily contact only the exterior surfaces of the corner 130, which may allow the corners of the playard 1000c to have a softer, gentler appearance. For example, the base 131 and the tab 138 of the corner 130 may have a smooth rounded shape for the top portion 302 of the soft goods 300 to wrap around. The top portion 302 of the soft goods 300 may include a snap-fit connector 312 disposed along an interior portion of the top portion 302 that couples to a corresponding snap-fit connector 139 on the corner 139 as shown in FIGS. 27B and 27C.

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In some implementations, the slider 120 may also include a rounded bottom section 170 positioned underneath the overhang portion of the tab 138 when the frame 100c is unfolded. As shown in FIGS. 26B and 27A, the rounded bottom section 170 may extend further outwards from the frame 100c than the tab 138 of the corner 130 to provide a lead-off feature to reduce or, in some instances, prevent a string or another tethered object from becoming entangled with the overhang portion of the corner 130.

As described above, the frame 100c may include the latch mechanism 200j to lock the frame 100c in the unfolded configuration by engaging the slider 120 of one leg support assembly 110c to the corresponding corner 130. Generally, the frame 100c may include one or more of the latch mechanisms 200j. For example, FIG. 28A shows the playard 1000c may include a single latch mechanism 200j coupled to one leg support assembly 110c. However, in other implementations, the playard 1000c may include another latch mechanism 200j coupled to another leg support assembly 110c on an opposite corner of the playard 1000c to ensure the frame 100c is evenly unfolded.

FIG. 28B shows the latch mechanism 200j may include a latch member 210 with a mounting base 224 at one end that is rigidly coupled to the slider 120 and a latch opening 214 disposed at an opposing end (see FIG. 28C) to receive a latch catch 291 disposed on the corner 130. The latch member 210 may be a mechanically compliant component with sufficient mechanical rigidity such that a restoring force is generated when the latch member 210 is bent and/or deflected. The latch member 210 may further include a tab 220, which may be pulled to bend the latch member 210 outwards from the frame 100c to release the latch member 210 from the latch catch 291. Additionally, the latch member 210 may include a lead-in portion 222 to facilitate engagement of the latch member 210 to the latch catch 291 when unfolding the playard 1000c.

FIG. 28B further show the latch mechanism 200j may be locked and/or unlocked with the soft goods 300 and, in particular, the top portion 302 covering the top portion 108 of the frame 100c. As shown, the latch catch 291 may protrude through an opening formed on the top portion 302 of the soft goods 300. The latch member 210 may be disposed over the top portion 302 when engaging with the latch catch 291. Thus, the latch member 210 may be left exposed. Furthermore, the internal restoring force generated by the latch member 210 may also cause at least a portion of the latch member 210 (e.g., the tab 220, the lead-in feature 222) to press onto the top portion 302 of the soft goods 300, thus further restraining the soft goods 300 against the corner 130. In other words, the latch member 210 may function as an integral escutcheon when engaged with the latch catch 291.

Similar to the playards 1000a and 1000b, the frame 100c of the playard 1000c may only include the leg support assemblies 110c and the X-frame assemblies 140a and 140b. In some implementations, the frame 100c may exhibit sufficient mechanical rigidity, stability, and strength to satisfy various consumer safety standards (e.g., ASTM F406-19). For example, FIGS. 29A-29D show the playard 1000c being subjected to a Top Rail to Corner Post Attachment test as set defined under ASTM F406-19, 7.11 and 8.30. As shown in FIGS. 29A and 29B, a torque is applied to one of the X-frame assemblies 140b by clamping a 24 inch long rod to the X-tubes of the X-frame assembly 140b and hanging a 15-20 lb weight onto the end of the rod. FIGS. 29C and 29D show that after applying the torque load for at least 10 seconds, the X-tubes of the X-frame assembly 140b were

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deformed, but the sliders **120** and the corners **130** coupled to the X-tubes did not crack and/or otherwise break, thus satisfying the requirements under ASTM F406-19, 7.11.

FIGS. 30A-30C show the playard **1000c** being subjected to another test to evaluate the mechanical strength and robustness of the X-frame assembly **140b** under ASTM F406-19, 7.3.3 and 8.11.2.4. As shown in FIG. 30A, a 100 lbf force was applied to the center of the X-frame assembly **140b** at a 45 degree angle relative to the floor for at least 15 seconds. FIGS. 30B and 30C show the X-tubes of the X-frame assembly **140b** were deformed and the rolled rivet joints connecting the X-tubes together were bent. However, the X-tubes, the rolled rivet joints, and the corners and sliders of the leg support assemblies did not crack and/or otherwise break, thus satisfying the requirements under ASTM F406-19, 7.3.3.

FIG. 31 further shows the playard **1000c** being subjected to a stability test where the playard **1000c** was placed onto playform and a load was applied to one side of the playard **1000c** from within the partially enclosed space **301**. Similar to the playards **1000a** and **1000b**, it was found at least three of the feet **114** and/or the wheels **151** of the playard **1000c** maintained contact with the underlying platform when the playard **1000c** was rotated more than 10 degrees, thus satisfying the requirements under ASTM F406-19 for stability.

In some implementations, the foldable playard, when deployed, may also provide a platform to support various accessories (also referred to herein as a “topper”) to augment the functionality of the playard. For example, FIG. 23D shows the frame **100c** of the playard **1000c** may include one or more topper supports **161** disposed on the respective X-tubes of the X-frame assemblies **140b**. The combination of the topper supports **161** and the corners **130** of the leg support assemblies **110c** may support one or more toppers **160** placed onto the top portion **108** of the frame **100c** when the playard **1000c** is in the unfolded configuration. The toppers **160** may be various accessories including, but not limited to a changing table, a bassinet, and a bouncer.

In another example, FIGS. 32A-32F show the frame **100a** in the unfolded configuration with a canopy cover assembly **400a**. The canopy cover assembly **400a** may be coupled to the frame **100a** and disposed, in part, above the interior space **102** of the frame **100a** to support a canopy cover **440** (see, for example, FIG. 32E) that covers the interior space **102**. The canopy cover **440** may be a compliant and/or flexible component formed from, for example, a textile material. For example, the playard **1000a** may be deployed in an outdoor setting, thus the canopy cover **440** may provide shade for the child **50** when placed in the partially enclosed space **301** of the playard **1000a**.

As shown in FIGS. 32A-32C, the canopy cover assembly **400a** may include multiple canopy support assemblies **410** that couple to each leg support assembly **110a** of the frame **100a**. Thus, the canopy cover assembly **400a** may fully cover the interior space **102** (i.e., the canopy cover assembly **400a** is a full canopy cover). In some implementations, the canopy support assemblies **410** may be substantially identical or identical with the other canopy support assemblies **410**.

Each canopy support assembly **410** may include a canopy bow **412** partially disposed above the interior space **102** to support the canopy cover **440** and a canopy clip **420a** to couple the canopy bow **412** to the frame **100a**. For the canopy cover assembly **400a**, the canopy bows **412** from each canopy support assembly **410** may be coupled together via a hub **450a** disposed above the interior space **102** as

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shown in FIG. 32A. In some implementations, the hub **450a** may be approximately aligned or aligned to the center of the interior space **102** when the canopy cover assembly **400a** is mounted to the frame **100a**, as shown in FIG. 32C. FIG. 32B further shows the respective canopy bows **412** of the canopy support assemblies **410** may form a frame or support structure where each canopy bow **412** is bent, in part, to define the desired shape of the canopy cover **440** when the canopy cover **440** is installed onto the canopy support assemblies **410**.

FIG. 32D shows the canopy clip **420a** may be disposed along an exterior portion of the frame **100a** (i.e., outside the interior space **102** of the frame **100a**) proximate to the slider **120** and the top portion **108** of the frame **100a** when the canopy clip **420a** is coupled to the leg support assembly **110a**. FIGS. 32E and 32F show the canopy clip **420a** may include a base **422** with snap-fit features **424** that form a snap-fit connector to directly couple the canopy clip **420a** to the leg tube **112** of one leg support assembly **110a**. Thus, the canopy cover assembly **400a** may be mounted to the frame **100a** without the use of any tools. Furthermore, the canopy cover assembly **400a** may be coupled to the frame **100a** without making any alterations or modifications to the frame **100a**. In this manner, the canopy cover assembly **400a** may not be limited for installation with only the frame **100a**, but instead the canopy cover assembly **400a** may be mounted onto the frames of other playards (e.g., other frames with six leg support assemblies). Said in another way, the canopy cover assembly **400a** may be a universally compatible accessory that the caregiver may separately purchase and/or install onto their playard.

The snap-fit features **424** may generally be shaped to conform with the cross-sectional shape of the leg tube **112** to ensure the canopy clip **420a** is securely coupled to the leg tube **112**. For example, FIG. 33A shows the snap-fit features **424** may form an oval-shaped channel that matches the oval-shaped cross-section of the leg tube **112**. In some implementations, the asymmetric cross-section of the leg tube **112** (e.g., the oval-shaped cross section) may ensure the canopy clip **420a** only couples to the leg tube **112** with a desired orientation and/or prevents unwanted rotation of the canopy clip **420a** when coupled to the leg tube **112**. In this manner, the canopy bow **4112** may be repeatedly and/or reliably positioned and/or oriented with respect to the frame **100a** such that the canopy cover **440**, when placed onto the canopy support assembly **410**, provides the desired coverage and/or aesthetic appearance. However, it should be appreciated that in other implementations, the shape of the snap-fit features **424** may be tailored to match the shape of the leg tubes most commonly used in various playard products (e.g., a circular-shaped leg tube). The snap-fit features **424** may further include lead-in features **425** to align the canopy clip **420a** to the leg tube **112** and/or to deflect the snap-fit features **424** outwards to facilitate engagement with the leg tube **112**.

In some implementations, the caregiver may thus align and press the canopy clip **420a** along the arrow shown in FIG. 33A to engage the snap-fit features **424** to the leg tube **112**. In some implementations, the caregiver may instead hook one of the snap-fit features **424** (e.g., via the corresponding lead-in feature **425**) onto the leg tube **112** and then rotate the opposing side of the canopy clip **420a** such that the other snap-fit feature **424** engages the leg tube **112** (e.g., via the corresponding lead-in feature **425**) as shown in FIG. 33B. Compared to conventional canopy cover assemblies, the canopy cover assembly **400a** may be more securely and reliably coupled to the frame **100a** by directly coupling the canopy clip **420a** to the leg tube **112** instead of a portion of

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the frame covered by soft goods. Thus, the canopy cover assembly 400a may be less susceptible to being removed accidentally by, for example, wind or the child 50 when placed into the partially enclosed space 301.

FIGS. 32E and 32F further show the canopy clip 420a may include a canopy bow opening 426 (e.g., also referred to herein as a “canopy bow socket 426”) formed, in part, on the base 422 to receive a first end 413a of the canopy bow 412. Once the first end 413a of the canopy bow 412 is inserted into the canopy bow opening 426, a fastener may be inserted through the opening 432 disposed on the side of the base 422 to securely couple the canopy bow 412 to the canopy clip 420a. In some implementations, the canopy clip 420a may alternatively utilize an integral snap finger to couple the canopy bow 412 to the canopy clip 420a via a snap-fit connection.

In some implementations, a portion of the canopy bow 412 may be disposed outside the interior space 102 of the frame 100a and positioned proximate to the top portion 108 of the frame 100a when coupled to the canopy clip 420a. For example, FIG. 32D shows a portion of the canopy bow 412 may be in substantially parallel or parallel alignment with the leg tube 112 and positioned next to the corner 130. By positioning the canopy bow 412 to overlap with the top portion 108 of the frame 100a, the canopy bow 412 is less susceptible to being pulled into the partially enclosed space 301 of the playard 1000a by the child 50 compared to conventional playards with canopy cover assemblies. For example, the canopy bow 412 may be more difficult to reach since the child 50 has to extend their arms over the corners 130 of the frame 100a to grab the canopy bow 412. Additionally, even if the child 50 manages to grab onto the canopy bow 412, they have less leverage to pull the canopy cover assembly 400a into the playard 1000a due to the canopy bow 412 overlapping the top portion 108 of the frame 100a and the canopy clip 420a positioned on an exterior portion of the frame 100a.

The canopy clip 420a may further include an alignment rib 430 that protrudes outwards from the base 422 towards the frame 100a. The alignment rib 430 may be used, in part, as an alignment feature to position the canopy clip 420a onto the leg support assembly 110a. For example, FIGS. 42D-F show the alignment rib 430 may be disposed between the top surface of the slider 120 and the bottom surface of the corner 130 such that the snap-fit features 424 are disposed just below the slider 120 when the canopy clip 420a is coupled to the leg tube 112. In some implementations, the alignment rib 430 may also prevent the canopy clip 420a from sliding downwards along the leg tube 112. For example, FIG. 32E shows the alignment rib 430 may contact the top surface of the slider 120 if the canopy clip 420a moves down along the leg tube 112.

In some implementations, the canopy cover 440 may be laid directly over and onto the canopy support assemblies 410. The canopy cover 440 may include one or more tethers 442 to pull and/or hold the canopy cover 440 taut along the canopy bows 412 of the canopy support assemblies 410. For example, FIG. 32E shows each tether 442 may be looped around a hook 428 disposed at the bottom of the base 422 of the canopy clip 420a.

The canopy bow 412, the canopy clip 420a, and/or the hub 450a may be formed from various materials including, but not limited to plastic and fiberglass. In some implementations, the canopy bow 412 may be formed as a single, mechanically compliant component that may bent into the desired shape to couple the canopy bow 412 to the hub 450a and/or the canopy clip 420a. In some implementations, the

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canopy bow 412 may be an assembly of components (e.g., tubes) coupled together via one or more shock cords or bungee cords. The tubes may be fitted to one another to form an assembly of tubes that mechanically function as a single, continuous rod. For example, FIG. 32E shows the canopy bow 412 may include an elastic cord 414 that passes through the canopy bow 412 to hold the various sections of the canopy bow 412 together. As shown, the elastic cord 414 may be terminated with a knot, which may be accessed by the caregiver through an opening 434 on the base 422 of the canopy clip 420a.

As described above, the canopy cover assembly 400a may include a hub 450a that couples the second ends 413b of each canopy bow 412 together to form a structure that covers the interior space 102 of the frame 100a. In some implementations, the canopy bows 412 may be coupled to the hub 450a prior to purchase by a consumer (e.g., the canopy cover assembly 400a may be assembled at a factory) or by a caregiver when installing the canopy cover assembly 400a onto the playard 1000a for the first time. In other words, the canopy bows 412 may remain coupled to the hub 450a for subsequent installations of the canopy cover assembly 400a such that the caregiver only needs to couple the respective canopy clips 420a to corresponding leg tubes 112 for setup.

In some implementations, the canopy bow 412 may be rigidly coupled to the hub 450a (i.e., the second 413b of the canopy bow 412 may not translate and/or rotate relative to the hub 450a). Thus, the canopy bows 412 of the canopy support assemblies 410 may be bent to facilitate attachment of the respective canopy clips 420a to the frame 100a. In some implementations, the second end 413b of the canopy bow 412 may be rotatably coupled to the hub 450a so that the canopy support assemblies 410 may be folded into a more compact structure for storage while remaining coupled to the hub 450a. For example, FIGS. 34A and 34B show the hub 450a may include a base 451 with multiple openings 452 to receive the second ends 413b of each canopy bow 412. The openings 452 may be aligned, in part, according to the relative locations of the leg support assemblies 110a of the frame 100a in the unfolded configuration. For example, the hub 450a may have six openings 452 disposed evenly around the periphery of the base 451 to align with the six leg support assemblies 110a, which may be arranged to form a hexagonal-shaped interior space 102.

Once the second end 413b of the canopy bow 412 is inserted into the opening 452, a pin 454 coupled to the second end 413b may be held in a corresponding slot 453 formed in the base 451 via, for example, a snap-fit connector. This allows the second end 413b of the canopy bow 412 to rotate relative to the base 451 via rotation of the pin 454 within the slot 453 about a rotation axis 460 as shown in FIG. 34B. In some implementations, the pin 454 may be integrally formed into the canopy bow 412. In some implementations, the pin 454 may be a separate component that is inserted through openings along the sides of the canopy bow 412 near the second end 413b.

The base 451 may further include a lip 457 to constrain the range of rotational motion of the canopy bow 412 relative to the hub 450a. For example, FIG. 34B shows the lip 457 may be disposed along the bottom side of the base 451, which causes the canopy bow 412 to bend when the canopy clip 420a attached to the first end 413a of the canopy bow 412 is positioned below the hub 450a. However, the canopy support assemblies 410 may be allowed to rotate such that the second end 413b of each canopy bow 412 is inserted through the opening 452 from the top side of the base 451 (i.e., the canopy clip 420a is positioned above the

hub 450a). In this manner, the canopy cover assembly 400a may be folded for storage and/or transport separately or together with the playard 1000a.

FIGS. 35A and 35B show the playard 1000a with the frame 100a and soft goods 300 and another exemplary canopy cover assembly 400b with the canopy cover 440 installed onto the playard 1000a. In this example, the canopy cover assembly 400b may cover half the interior space 102 (i.e., the canopy cover assembly 400b is a half canopy cover).

FIGS. 36A-36E show the canopy cover assembly 400b may once again include multiple canopy support assemblies 410 coupled to the frame 100a to provide a support structure that defines the desired shape of the canopy cover 440 when mounted to the canopy support assemblies 410. Compared to the canopy cover assembly 400a, however, the canopy support assemblies 410 of the canopy cover assembly 400b may include a canopy bow 412 that is directly coupled to two canopy clips 420b mounted to different leg support assemblies 110a of the frame 100a instead of a central hub. For example, FIGS. 36A and 36C show the canopy cover assembly 400b may include two canopy support assemblies 410 where the canopy bow 412 of each canopy support assembly 410 is coupled to two non-adjacent leg support assemblies 110a. The canopy bows 412 may overlap and/or cross one another as shown in FIG. 36C.

In this example, the canopy bow 412 may include multiple bow sections 416 coupled together via connectors 415. The connector 415 may be a tubular-shaped component that receives respective ends of two bow sections 416. In some implementations, each bow section 416 may be coupled to the connector 415 via a fastener inserted through a corresponding opening on the connectors 415 and/or an integral snap finger.

The canopy clip 420b may incorporate several of the same features as the canopy clip 420a described above. For example, FIGS. 36D and 36E show the canopy clip 420b may include a base 422 with snap-fit features 424, a canopy bow opening 426 to receive one end of the canopy bow 412, a mounting hole 432 to securely couple the canopy bow 412 to the canopy clip 420b, an opening 434 to access the elastic cord in the canopy bow 412, and a hook 428 to secure the tether 442 of the canopy cover 440 to the canopy clip 420. Compared to the canopy clip 420a, the canopy bow opening 426 of the canopy clip 420b may be tilted such that the portion of the canopy bow 412 coupled to the canopy clip 420b is oriented an angle relative to the leg tube 112 of the leg support assembly 110a to ensure the canopy bow 412 extends over a center portion of the interior space 102 as shown in FIG. 36C.

FIGS. 37A-37C show another exemplary canopy cover assembly 400c without the canopy cover 440 coupled to the frame 100a of the playard 1000a. The canopy cover assembly 400c may also cover half the interior space 102 similar to the canopy cover assembly 400b. However, the canopy support assemblies 410 of the canopy cover assembly 400c may be joined together by a hub 450b in the canopy cover assembly 400a. As shown, the canopy support assemblies 410 may include the canopy bows 412 and canopy clips 420a described above. In this example, the canopy support assemblies 410 may couple to enough leg support assemblies 110a to cover half the interior space 102 as shown in FIG. 37C.

FIGS. 38A and 38B show the hub 450b may once again include a base 451 with openings 452 to receive the second ends 413b of each canopy bow 412. As shown, the openings 452 may be formed as sockets that rigidly couple the second

ends 413b to the hub 450a such that the second end 413b of each canopy bow 412 is translationally and rotationally constrained to the hub 450b. In some implementations, the second end 413b may be coupled to the hub 450b via a fastener and/or a snap-fit connection.

FIGS. 39A and 39B show another hub 450c for the canopy cover assembly 400c, which allows the second end 413b of the canopy bow 412 to be rotatable relative to the base 451 so that the canopy cover assembly 400c may be folded. The hub 450c may incorporate several of the same features as the hub 450a described above. For example, the base 451 may include a slot 453 to receive a pin 454 mounted to the second end 413b of the canopy bow 412. The slot 453 and the pin 454 may allow the canopy bow 412 to rotate about the axis 460. The base 451 may further include a lip 457 disposed on a bottom side of the base 451 to limit the rotational motion of the canopy bow 412.

FIGS. 40A and 40B show yet another hub 450d for the canopy cover assembly 400c. As shown, the hub 450d may include a base 451 with an opening 456 that extends along the curved side of the base 451. The opening 456 may be shaped to receive the second ends 413b of multiple canopy bows 412 as shown in FIG. 40A. The base 451 may further include holes 455 on the top and bottom sides of the base 451 to couple the second end 413b of each canopy bow 412 to the base 451. In some implementations, a pin (not shown) may be inserted through the opening 455 and corresponding openings (not shown) on the canopy bow 412 such that the second end 413b of each canopy bow 412 may rotate about an axis 461 as shown in FIG. 40A. This, in turn, may enable the canopy cover assembly 400c to be folded by rotating each of the canopy bows 412 about a corresponding axis 461 to one side of the hub 451 such that the canopy bows 412 are approximately parallel or parallel with one another. In some implementations, a fastener may instead be inserted through the openings 455 to rigidly couple each canopy bow 412 to the hub 450d (i.e., the second end 413b of the canopy bow 412 does not rotate relative to the base 451).

Bassinet Accessory

The foldable playard may also include a bassinet accessory to provide an elevated surface to support a child in their first several months of life (e.g., an infant, a child weighing less than 15 lbs). Once the child outgrows the bassinet accessory, the bassinet accessory may be removed and the interior space of the foldable playard may be used to contain the child as described above. In this manner, the foldable playard may be reconfigured by the caregiver to adapt to the physical development of the child, thus extending the lifetime of the playard. When the bassinet accessory is installed on the playard, the playard may be considered as being in a "bassinet mode." When the bassinet accessory is removed from the playard, the playard may be considered as being in a "playard mode."

FIGS. 41A and 41B show the playard 1000b with an exemplary bassinet accessory 500a in the deployed unfolded configuration. As shown, the bassinet accessory 500a may be disposed within a top portion of the partially enclosed space 301 defined by the soft goods 300. The bassinet accessory 500a may define a separate relatively smaller partially enclosed space 501 disposed within the partially enclosed space 301 to contain the child in the unfolded configuration. The bassinet accessory 500a may generally include a support structure 520 that physically defines the partially enclosed space 501. The support structure 520 may also facilitate folding and unfolding of the bassinet accessory 500a together with the frame 100b and the soft goods 300, thus simplifying setup and tear down of the playard

1000b (i.e., the caregiver is not required to remove the support structure of the bassinet to fold the playard or install the support structure each time the playard is deployed).

The support structure **520** may include bassinet soft goods **522** with side surfaces **524** and a bottom surface **526** that physically surround at least a portion of the partially enclosed space **501**. The support structure **520** may further include a hub **550** and multiple support tubes **540** that together form a foldable structure. The hub **550** may be formed from a plastic material (e.g., via injection molding). The support tube **540** may be formed from various rigid materials including, but not limited to, aluminum and steel. In the unfolded configuration, the hub **550** and the support tubes **540** provide a rigid platform to support a mattress **510** (see, for example, FIG. **43**). The mattress **510**, in turn, may provide a cushioned surface **511** located above the ground surface **90** to support the child.

It should be appreciated the bassinet accessories disclosed herein may also be installed onto different playards (e.g., playards having frame shapes that are different than that shown in FIGS. **41A** and **41B**). For example, with reference again to FIG. **23A**, the bassinet accessory **500a** may also be installed on the foldable playard **1000c** in the same manner as the playard **1000b**.

The bassinet accessory **500a** may be dimensioned and/or shaped such that the partially enclosed space **501** extends laterally to the boundaries of the partially enclosed space **301** of the soft goods **300** and, in some instances, the interior space **102** of the frame **100b** when the soft goods **300** are disposed along the boundaries of the interior space **102**. For example, FIGS. **41A** and **41B** show the bassinet soft goods **522** may extend to the side portions **306** of the soft goods **300**. However, it should be appreciated that in other implementations, the bassinet accessory **500a** may be shaped and/or dimensioned such that a gap is formed between the side portions **306** of the soft goods **300** and the bassinet soft goods **522**. For example, with reference again to FIG. **23A**, this figure shows a gap is formed between the side portions **306** and the bassinet soft goods **522** due to the curved shape of the leg support assemblies **110c**.

In some implementations, the bassinet accessory **500a** may define a partially enclosed space **501** with a cross-sectional shape that conforms with the cross-sectional shape of the partially enclosed space **301** and, in some implementations, the interior space **102**. For example, FIGS. **41A** and **41B** show the partially enclosed space **501** may have a rectangular cross-sectional shape that extends to the side portions **306** of the soft goods **300**. In some implementations, the lateral dimensions of the partially enclosed space **501** may remain constant (or substantially constant as the bassinet soft goods **522** may be deformed by the soft goods **300**) such that the three-dimensional volume of the partially enclosed space **501** is shaped as a right prism.

FIGS. **41A** and **41B** also show the bassinet soft goods **522** may be coupled to the top portion **302** of the soft goods **300** such that the bassinet soft goods **522** hang below the top portion **302**. As a result, the bassinet accessory **500a** may be positioned below the top side of the playard **1000b**. For simplicity, the partially enclosed space **501** may include the space between the bottom surface **526** of bassinet soft goods **522** and the top side of the playard **1000b** (e.g., the top horizontal plane **92**). The presence of the bassinet accessory **500a** may further divide the partially enclosed space **301** such that a bottom portion **301a** of the partially enclosed space **300** is formed below the bassinet accessory **500a**.

The bassinet accessory **500a** may provide a relatively shallow partially enclosed space **501** to improve accessibil-

ity. This may enable the bassinet accessory **500a** to reduce the physical strain experienced by the caregiver since the caregiver would not bend over as much when directly placing the child into the partially enclosed space **301** of the soft goods **300** (or taking the child out of the partially enclosed space **301**). Additionally, the shallow bassinet accessory **500a** may also provide greater visibility of the child particularly when the caregiver is viewing the playard **1000b** from an elevated position (e.g., viewing the top of the playard **1000b**).

The bassinet accessory **500a** may be characterized by a height, $h_{t,1}$, defined as the distance from the respective bottom corner portions **537** of the bassinet soft goods **522** to the top horizontal plane **92** of the playard **1000b** in the unfolded configuration as shown in FIG. **41B**. The height, $h_{t,1}$, also corresponds to the height of the partially enclosed space **501**. In some implementations, the height, $h_{t,1}$, may range between 7.5 inches and about 12 inches. In some implementations, the bassinet accessory **500a** may also be characterized by a height, h_m , defined as the distance from the top surface **511** of the mattress **510** to the top horizontal plane **92** of the playard **1000b**. When the mattress **510** is not compressed (e.g., the child is not resting on the mattress **510**), the height, h_m , may range between 7.5 inches and about 10 inches. The bottom portion **301a** may also be characterized by a height, h_b , defined as the distance from the ground surface **90** to the bottom surface **526**. In some implementations, the height, h_b , may be greater than or equal to about 18 inches.

The term “about,” when used to describe the height dimensions $h_{t,1}$, h_b , and h_m , is intended to cover manufacturing tolerances and/or variations due to the deformation of the soft goods **300** and/or the bassinet soft goods **522**. For example, “about 12 inches” may correspond to a height ranging between 11.75 inches and 12.25 inches or between 11.5 inches and 12.5 inches. In another example, “about 10 inches” may correspond to a height ranging between 9.75 inches and 10.25 inches or between 9.5 inches and 10.5 inches. In another example, “about 18 inches” may correspond to a height ranging between 17.75 inches and 18.25 inches or between 17.5 inches and 18.5 inches.

In some implementations, the height, $h_{t,1}$, of the bassinet accessory **500a** and/or the height, h_b , of the bottom portion **301a** may remain substantially unchanged between the folded and unfolded configurations. For example, the support tubes **540** and the hub **550** may only cause the bassinet accessory **500a** to collapse along a lateral direction when folding the bassinet accessory **500a** together with the playard **1000b** (i.e., the lateral dimensions of the partially enclosed space **501** decrease while the height, $h_{t,1}$, remains substantially unchanged). Furthermore, the leg support assemblies **110a** may remain upright between the folded and unfolded configurations as described above, thus the height, h_b , may also remain substantially unchanged.

In another example, the frame **100b** may flare outwards when unfolded to improve, for example, the mechanical stability of the playard **1000b**. Alternatively, as shown in the side view of FIG. **41C**, a playard frame **100c** with curved legs, similar to that shown in FIG. **23E**, may be equipped with a bassinet accessory. In FIG. **41C**, the bassinet soft goods are not shown so as to reveal a relative position of the hub **550** and the support tubes **540** as viewed from the side in an unfolded configuration. Although the bassinet soft goods are not explicitly shown in FIG. **41C**, the figure nonetheless indicates that respective bottom corners **537** of the bassinet soft goods would be located at respective distal ends of the support tubes **540**. FIG. **41C** also shows the

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overall height H_1 of the frame **100c**, and the respective heights $h_{r,1}$ and h_b as discussed above.

Additionally, it should be appreciated that the bassinet accessory **500a** and, in particular, the bassinet soft goods **522** may fold and/or crumple when folding the playard **1000b**. These factors may contribute to small changes in the height, $h_{r,1}$, of the bassinet accessory **500a** and/or the height, h_b , of the bottom portion **301a** between the folded and unfolded configurations. For example, the height, $h_{r,1}$, in the unfolded configuration may change to the height, $h_{r,2}$, in the folded configuration (see, for example, FIG. **52**). However, the variations in the heights $h_{r,1}$ and h_b may be sufficiently small such that the support tubes **540** and the hub **550** remain disposed near to or within the interior space **102** of the playard **1000b**. Said in another way, the dimensional relations between, for example, the length of the support tube **540** and the heights $h_{r,1}$ and h_b , as described below, may remain substantially unaffected due to variations in the heights $h_{r,1}$ and h_b . In some implementations, the heights $h_{r,1}$ and h_b may increase or decrease by less than or equal to 1 inch.

In some implementations, the bassinet accessory **500a** may satisfy various consumer safety standards (e.g., ASTM F2194). Thus, the combination of the playard **1000b** and the bassinet accessory **500a** may satisfy ASTM F406, as described above, and ASTM F2194 together. For example, the bassinet accessory **500a** and, in particular, the hub **550** and the support tubes **540** may provide a sufficiently flat platform for the mattress **510** to rest upon such that the angle between neighboring segments **512** of the mattress **510** are less than 7 degrees. Additionally, the bassinet accessory **500a** may have no openings with a diameter ranging between 0.210 inches and 0.375 inches to prevent finger entrapment. The bassinet accessory **500a** may further have no components that undergo a scissoring or shearing motion. The bassinet accessory **500a** and, by extension, the playard **1000b** may support a static load of 54 lb or 3 times the manufacturer's recommended weight (whichever is greater) for at least 60 seconds. The bassinet accessory **500a** may be dimensioned and/or shaped such that any gaps between the edges of the mattress **510** and the side surfaces **524** of the bassinet soft goods **522** is less than 0.5 inches. Additionally, the height, h_m , from the top surface **511** of the mattress **510** to the top side **92** of the playard **1000b** may be greater than or equal to 7.5 inches.

FIGS. **42A** and **42B** show the mattress **510** may be removed from the bassinet accessory **500a** and/or the playard **1000b** for use in both the bassinet mode and the playard mode of the foldable playard **1000b**. Specifically, FIG. **42A** shows the playard **1000b** in the bassinet mode (i.e., the bassinet accessory **500a** is installed on the playard **1000b**) where the mattress **510** is disposed on top of the hub **550** and the support tubes **540**. FIG. **42B** shows the playard **1000b** in the playard mode (i.e., the bassinet accessory **500a** is removed from the playard **1000b**) where the mattress **510** is disposed on the floor portion **304** of the soft goods **300** (e.g., the mattress **510** rests on the ground). The mattress **510** may be a foldable component that provides a flat cushioned surface **511** for the child to play and/or sleep when unfolded and a compact structure for storage with the other components of the playard **1000b** when folded.

In some implementations, the mattress **510** may be a segmented mattress with multiple panels **512** that fold relative to each another along corresponding creases formed between adjoining panels **512**. For example, FIGS. **42A** and **42B** show the mattress **510** may include four panels **512** with one panel **512** folded for demonstration. In addition to

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providing a flat cushioned surface **511** to support the child, the mattress **510** may also wrap around the frame **100b**, the soft goods **300**, and the support structure **520** when the playard **1000b** is folded for storage (see, for example, FIG. **45A**). In some implementations, the mattress **510** may include one or more straps **514** disposed on a bottom side opposite the cushioned surface **511** to securely couple opposing panels **512** of the mattress **510** together to restrain the frame **100b** and thus, maintain the playard **1000b** in the folded configuration. The mattress **510** may further include a handle **516** for the caregiver to carry the playard **1000b** with the bassinet accessory **500a**.

FIG. **43** shows the bassinet accessory **500a** with the mattress **510** removed, thus exposing the hub **550**, the support tubes **540**, and the remaining portions of the bassinet soft goods **522**. The side surfaces **524** and the bottom surface **526** of the bassinet soft goods **522** may be formed of a compliant material including, but not limited to, a fabric, a mesh, and plastic. In some implementations, at least a portion of the side surfaces **524** may be transparent and/or see-through. Furthermore, the transparent and/or see-through portions of the bassinet soft goods **522** may overlap with the transparent and/or see-through portions of the soft goods **300** to effectively provide the caregiver one or more windows to monitor their child in the partially enclosed space **501**.

In some implementations, a top portion of the side surfaces **524** may be formed of a fabric material to attach the bassinet accessory **500a** to the soft goods **300** while the bottom portion of the side surfaces **524** may be formed of a transparent and/or see-through material. For the bassinet accessory **500a**, the bottom surface **526** of the bassinet soft goods **522** may not include an opening sufficiently large for a user to insert their hand through and into the bottom portion **301a** of the playard **1000b**. Said in another way, the bassinet soft goods **522** in the bassinet accessory **500a** may prevent the user from accessing the bottom portion **301a** of the playard **1000b**. However, it should be appreciated that in other implementations, the bottom surface **526** may include an opening, in part, to facilitate folding of the bassinet accessory and the playard together (see, for example, the bassinet accessory **500b**).

The bassinet accessory **500a** may generally be coupled to the soft goods **300** (or directly to the frame **100b**) via a coupling mechanism that allows the bassinet accessory **500a** to be readily removable from the playard **1000b** when, for example, the child outgrows the bassinet accessory **500a**. The bassinet accessory **500a** may generally be coupled to the soft goods **300** and/or the frame **100b** in several ways including, but not limited to, a zipper mechanism and straps (e.g., one strap connected to the bassinet accessory **500a** may extend over a portion of the soft goods **300** covering a corner **130** and clip onto a corresponding strap coupled to the frame **100b** via a buckle).

For example, FIG. **44A** shows the bassinet accessory **500a** may be coupled to the soft goods **300** via a zipper mechanism **527**. As shown, the top edges of the side surfaces **524** may support one set of zipper teeth **529** and a zipper handle **528** that couples to another set of zipper teeth **340** disposed on an interior bottom edge of the top portion **302** of the soft goods **300**. Thus, the bassinet accessory **500a**, via the bassinet soft goods **522**, may hang from the interior side of the top portion **302** of the soft goods **300**. Said in another way, the zipper mechanism **527** may be disposed within the partially enclosed space **501** along the interior sides of the

bassinet soft goods **522** and the top portion **302** such that the zipper mechanism **527** is not observable from the exterior of the playard **1000b**.

FIG. **43** shows the bassinet soft goods **522** may have a height, h_{sg} , that is less than the height, $h_{t,1}$, of the bassinet accessory **500a**. However, it should be appreciated that in other implementations, the bassinet soft goods **522** may extend over the top portion **302** of the soft goods **300** and couple to the soft goods **300** and/or the frame **100b** along the exterior of the playard **1000b**. For these implementations, the height, h_{sg} , may be approximately equal or equal to the height, $h_{t,1}$, of the bassinet accessory **500a**.

The caregiver may align and attach the zipper teeth **340** and **529** via the zipper handle **528** to install the bassinet accessory **500a** onto the playard **1000b**. Additionally, the caregiver may readily remove the bassinet accessory **500a** from the playard **1000b** by pulling on the zipper handle **528** to disengage the zipper teeth **340** and **529**. Once the bassinet accessory **500a** is removed from the playard **1000b**, the bassinet accessory **500a** may be folded as shown in FIG. **44B** and stowed separately.

The zipper mechanism **527** may generally span at least a portion of the side surfaces **524** to securely couple the bassinet accessory **500a** to the soft goods **300**. In some implementations, the bassinet accessory **500a** and the soft goods **300** may include multiple zipper mechanisms **527** that each span different portions of the side surfaces **524** such that collectively, the multiple zipper mechanisms **527** span the entirety of the top edges of the side surfaces **524**. The zipper mechanism **527** may generally reduce or, in some instances, eliminate unwanted openings formed between the side surfaces **524** and the top portion **302**.

As described above, the support tubes **540** and the hub **550** may form a foldable structure generally disposed on the bottom surface **526** of the bassinet soft goods **522** to facilitate folding and/or unfolding of the bassinet accessory **500a** together with the frame **100b**. As shown in FIG. **43**, the hub **550** may be disposed at or near the center of the bottom surface **526** and the support tubes **540** may extend radially from the hub **550** to the respective corner portions **537** of the bottom surface **526** of bassinet soft goods **522**. Said in another way, the support tubes **540** may be disposed along the diagonal segments of the bottom surface **526** (i.e., the line segments connecting the corners of the bottom surface **526** that do not share the same edge).

To facilitate folding and/or unfolding of the bassinet accessory **500a**, each support tube **540** may be rotatably coupled to the hub **550**. In particular, each support tube **540** may have a first end **542a** rotatably coupled to the hub **550** and a second end **542b** opposite the first end **542a** disposed at one corner portion **537** of the bassinet soft goods **522**. Additionally, the support tubes **540** and/or the hub **550** may be directly coupled to the bassinet soft goods **522** via one or more attachment mechanisms so that the bassinet soft goods **522** move together with the support tubes **540** and/or the hub **550** when folding and/or unfolding the bassinet accessory **500a**. The attachment mechanisms may include, but are not limited to, a strap, a screw fastener, a webbing tab, and a fabric tunnel.

In some implementations, the attachment mechanism(s) may be disposed at or near opposing ends **542a** and **542b** of each support tube **540** to ensure the center portion and the side portions of the bottom surface **526** of the bassinet soft goods **522** fold together with the support tubes **540** and the hub **550**. For instance, FIG. **43** shows the bottom surface **526** of the bassinet soft goods **522** may include a strap **530** that forms a fabric tunnel through which the support tube **540** is

inserted. The strap **530** may be disposed near the first end **542a** of the support tube **540** and sewn directly onto the bottom surface **526** of the bassinet soft goods **522**. For example, the support tube **540** may have a length, L_r , and the strap **530** may be offset from the end **542a** of the support tube **540** by a distance less than 50% of the length L_r . In some implementations, the strap **530** may be positioned sufficiently close to the hub **550** such that at least a portion of the strap **530** physically contacts the hub **550**. FIG. **44B** further shows the second end **542b** of each support tube **540** may be fastened directly to the bassinet soft goods **522** via a screw fastener **534a** inserted from the bottom side of the bottom surface **526** through an opening **532** at the corner portion **537**.

In the unfolded configuration, the support tubes **540** and the hub **550** provide a flat platform to support the mattress **510** as shown in FIG. **43** where the support tubes **540** are oriented substantially horizontal or horizontal along the bottom surface **526** of the bassinet soft goods **522**. In the folded configuration, the support tubes **540** rotate with respect to the hub **550** such that the support tubes **540** are oriented substantially vertical or vertical. For the bassinet accessory **500a**, the hub **550** moves upwards when unfolding the bassinet accessory **500a** and, conversely, downwards when folding the bassinet accessory **500a**.

In some implementations, the ends **542b** of each support tube **540** may remain stationary or substantially stationary with respect to the ground **90** (e.g., the bassinet soft goods **522** may deform causing the ends **542b** and/or the corner portions **537** to vary slightly as described above). In other words, the ends **542b** of each support tube may remain at a height, h_b , from the ground **90** even as the ends **542b** displace laterally when the bassinet accessory **500a** is folded and unfolded. Thus, as the hub **550** is displaced vertically, the support tubes **540** may rotate with respect to the hub **550** where the ends **542b** of each support tube **540** function as a pivot point that is constrained to move only laterally (e.g., a pin joint disposed in a slider joint).

In some implementations, the bassinet accessory **500a** and the playard **1000b** may be shaped and/or dimensioned such that the hub **550** and the support tubes **540** remain substantially within or entirely within the interior space **102** in both the folded and unfolded configurations. In other words, the bassinet accessory **500a** does not increase the overall size of the foldable playard **1000b**. This may be accomplished by tailoring the length, L_r , of each support tube **102** to be approximately less than or equal to the height, h_b , of the bottom portion **301a** separating the bottom surface **526** from the ground **90** in the unfolded configuration. Since the ends **542b** of each support tube **540** remain at the same or similar height, h_b , from the ground **90**, the support tube **102** does not extend past the feet **114** of the frame **100b** when it rotates from a horizontal orientation corresponding to the unfolded configuration to a vertical orientation corresponding to the folded configuration. In some implementations, the height, h_b , may be sufficiently greater than the length, L_r , of the support tube **540** such that the hub **550** is also contained entirely within the interior space **102** in the folded configuration.

It should be appreciated the support tubes **540** and the hub **550** of the bassinet accessory **500a** may remain within the interior space **102** of the playard **1000b** due, in part, to the relatively shallower height, $h_{t,1}$, of the partially enclosed space **501**, which results in a larger height, h_b , for the bottom portion **301a** for a given height, H , of the playard **1000b**. As a result, the support tubes **540** may be formed from a single rigid component, simplifying manufacture and assembly of

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the bassinet accessory 500a. However, it should be appreciated that, in other implementations, the length of the support tube may be changed between the folded and unfolded configurations to ensure the bassinet accessory remains substantially confined within the interior space 102 of the playard 1000b (see, for example, the telescoping support tubes 540 in the bassinet accessory 500b).

FIGS. 45A-45C show a series of figures that illustrate the process of unfolding the foldable playard 1000b and the bassinet accessory 500a. Specifically, FIG. 45A shows the foldable playard 1000b in the folded configuration. As shown, the bassinet accessory 500a is contained entirely within the interior space 102 of the playard 102 and, hence, is not observable in FIG. 45A. Furthermore, FIG. 45A shows the mattress 510 may wrap around the frame 100b to maintain the playard 1000b in the folded configuration.

To unfold the playard 1000b with the bassinet accessory 500a, the mattress 510 is first removed from the frame 100b. The caregiver may then pull the slider 120 towards the corner 130 of one leg support assembly 110a to at least partially unfold the frame 100b. In some implementations, the caregiver may pull the slider 120 until the latch mechanism 200a is engaged, thus locking the frame 100b in the unfolded configuration. Since the bassinet soft goods 522 are coupled to the soft goods 300, the bassinet accessory 500a may also at least partially unfold in response to the frame 100b unfolding. However, the weight (i.e., the gravitational force) of the support tubes 540 and the hub 550 may cause the bassinet accessory 500a to sag downwards even when the frame 100b is locked in the deployed unfolded configuration.

To prevent the support tubes 540 and the hub 550 from sagging downwards, the hub 550 may include a hub latch 570 with a release handle 576 that, when in a locked state, prevents the support tubes 540 from rotating relative to the hub 550. While unfolding the bassinet accessory 500a, the hub latch 570 may instead be in an unlocked state to allow the caregiver to pull the hub latch 570 and, in turn, rotate the support tubes 540. As shown in FIGS. 45B and 45C, the support tubes 540 may rotate towards a horizontal orientation corresponding to the unfolded configuration as the hub latch 570 is pulled upwards (see A in FIG. 45C). Once the bassinet accessory 500a is unfolded, the hub latch 570 may be rotated (see B in FIG. 45C) to change the hub latch 570 from an unlocked state to a locked state thus maintaining the support tubes 540 and the hub 550 at the desired unfolded configuration. The hub 550 may further include integrated mechanical stops 554 to prevent the hub 550 from moving further upwards once the hub 550 and the support tubes 540 are at the deployed unfolded configuration. This ensures the caregiver is unable to move the hub 550 past the desired unfolded configuration.

Additionally, conventional playards typically include a bottom support structure that folds with the frame. When unfolding the playard, the caregiver should bend over and reach through an opening in the bassinet soft goods to press down upon the bottom support structure to ensure the bottom support structure is properly unfolded. In contrast, the playard 1000b may not include a separate bottom support structure as described above. This means the caregiver does not have to bend over and reach down towards the floor portion 304 of the soft goods 300 when unfolding the bassinet accessory 500a together with the playard 1000b. Rather, the caregiver may pull on the hub latch 570, which is already positioned above the ground 90 when the bassinet accessory 500a is partially unfolded in response to the

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unfolding of the frame 100b. In this manner, the caregiver may experience less physical strain when unfolding the bassinet accessory 500a.

To fold the playard 1000b and the bassinet accessory 500a, the caregiver may release the hub latch 570 (and the latch mechanism 200a) and press down on the hub 550 and/or move the slider 120 of one leg support assembly 110a downwards towards the corresponding foot 114. In this manner, the bassinet accessory 500a may be unfolded and folded without assembling and disassembling, respectively, a portion of the bassinet accessory 500a unlike conventional bassinet accessories (e.g., the support tube assemblies 64 in the bassinet accessory 60).

FIGS. 46A and 46B show several views of the hub 550 and the hub latch 570 in the locked state. FIGS. 47A and 47B show several views of the hub 550 and the hub latch 570 in the unlocked state. As shown, the hub 550 may include a base 551 with a channel 552 to receive each support tube 540. The hub 550 may further include a pair of snap-fit hooks 555 for each channel 552 where each pair of snap-fit hooks 555 are disposed on opposing sides of the corresponding channel 552 and on a bottom side of the hub 550. The snap-fit hooks 555 are shaped to receive a pin 544 coupled to the support tube 540 to facilitate rotation of the support tube 540. Thus, each pair of snap-fit hooks 555 defines a rotation axis 556 about which the support tube 540 rotates with respect to the hub 550.

The channel 552 may extend from the edge of the base 551 to an end 567 located near the center of the base 551. The channel 552 may have a length, L_c , corresponding to the distance between the edge of the base 551 and the end 567. As shown, the channel 552 may have a notched opening on the top side of the base 551 that extends from the edge of the base 551 and terminates before reaching the end 567. The bottom side of the channel 552 may have a mechanical stop 554 (e.g., a section of the hub 550 that extends around the support tube 540 and across the channel 552) disposed at the edge of the base 551 and an opening 553 that extends from the mechanical stop 554 to the end 567 of the channel 552.

The features of the channel 552 (e.g., the notched opening, the mechanical stop 554, the opening 553) may be shaped, dimensioned, and positioned to constrain the rotational motion of the support tube 540. In particular, the channel 552 may only allow the support tube 550 to rotate between a horizontal orientation and a vertical orientation when folding or unfolding the bassinet accessory 500a. For example, the notched opening allows the support tube 540 to rotate such that the end 542b may be disposed above the hub 550 when folding the bassinet accessory 500a. In another example, the mechanical stop 554 may be shaped to physically contact the support tubes 540 once the support tubes 540 are oriented horizontally. In this manner, the mechanical stops 554 may limit the rotation of the support tubes 540 such that the hub 550 is unable to move past the desired unfolded configuration when unfolding the bassinet accessory 500a.

As described above, the hub 550 may further include a hub latch 570. When the hub latch 570 is in the locked state, the combination of the hub 550 and the hub latch 570 prevents the support tubes 540 from moving relative to the hub 550 and, hence, prevents the hub 550 from moving relative to the playard 1000b. In this manner, the hub latch 570 locks the bassinet accessory 500a in the unfolded configuration.

The hub latch 570 may be rotatably coupled to the base 551 via a rolled rivet 566 disposed at the center of the base 551. As shown in FIG. 46A, the hub latch 570 may include

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a base **572** disposed within a center opening **558** of the base **551**. The hub latch **570** may include a release handle **576** for the caregiver to grab and pull when unfolding the bassinet accessory **500a**. The hub **550** may further include multiple hooks **560** disposed on the bottom side of the base **551** and around the periphery of the base **572** of the hub latch **570** to provide additional mechanical support to the hub latch **570**. In particular, the hooks **560** may impose mechanical constraints that limit the hub latch **570** only to rotational motion about the rolled rivet **566**.

To lock the support tubes **540**, the hub latch **570** may include arms **574** for each support tube **540** that extend radially from the base **572**. FIG. 46B shows each arm **574** may be disposed over the opening **553** of a corresponding channel **552** in the locked state. Thus, the combination of the arm **574** and the mechanical stop **554** may effectively for a clamp that constrains and prevents movement of the support tube **540** relative to the hub **550**. FIGS. 47A and 47B show when the hub latch **570** is rotated to the unlocked state, the arms **574** no longer cover the openings **553** of each channel **552**, which allows the support tubes **540** to rotate relative to the hub **550** towards the folded configuration.

In some implementations, the hub **550** may further include a spring element **565** (e.g., a torsion spring) that generates a spring bias force to rotate the hub latch **570** towards the locked state. To ensure the hub latch **570** does not move past the locked state (e.g., the arms **574** move past the openings **553**), the hub **550** may include mechanical stops **562** (e.g., a rib that projects downwards from the base **551**) for the arms **574** to rest against. The mechanical stops **562** are positioned on the base **551** such that the arms **574** are disposed over the corresponding openings **553**.

FIG. 48 shows another exemplary bassinet accessory **500b** coupled to the playard **1000b**. As shown, the bassinet accessory **500b** may include a support structure **520** that defines a partially enclosed space **501** to contain the child in the unfolded configuration. The support structure **520** may include bassinet soft goods **522** with sides surfaces **524** and a bottom surface **526** that surround at least a portion of the partially enclosed space **501**. The support structure **520** may further include a hub **550** and support tubes **540** that form a foldable structure to facilitate folding and unfolding of the bassinet accessory **500b**. In the unfolded configuration, the support tubes **540** and the hub **550** may form a flat platform to support a mattress (not shown).

It should be appreciated that the bassinet accessory **500b** may also be installed onto other playards. For examples, FIG. 52 shows the bassinet accessory **500b** may be installed on the playard **1000c** described above.

The bassinet soft goods **522**, the support tubes **540**, and the hub **550** of the bassinet accessory **500b** may incorporate similar features described above for the bassinet accessory **500a**. For brevity, these features are not repeated below. Additionally, the shape and dimensions of the bassinet accessory **500b**, including the heights, $h_{t,1}$, h_b , and h_m , may be similar to or the same as the dimensions described above for the bassinet accessory **500a**. The bassinet accessory **500b** may also meet various consumer safety standards (e.g., ASTM F2194) as described above in relation to the bassinet accessory **500a**.

FIG. 48 shows the hub **550** may be disposed at or near the center of the bottom surface **526** and the support tubes **540** may extend radially from the hub **550** to the respective corner portions **537** of the bottom surface **526** of the bassinet soft goods **522** similar to the bassinet accessory **500a**. The support tubes **540** may be rotatably (e.g., pivotably) coupled to the hub **550** to facilitate folding and unfolding of the

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bassinet accessory **500b**. The support tubes **540** may also be coupled directly to the bassinet soft goods **522** via one or more attachment mechanisms such that the bassinet soft goods **522** move together with the support tubes **540** and the hub **550** when folding and unfolding the bassinet accessory **500b**. It should be appreciated that, in other implementations, the bassinet soft goods **522** may be coupled to the hub **550**.

In this example, the hub **550** moves upwards when folding the bassinet accessory **500b** and, conversely, downwards when unfolding the bassinet accessory **500b**. The benefit of this approach is that the bassinet accessory **500b** may maintain the deployed unfolded configuration without a separate locking mechanism (e.g., the hub latch **570**), thus simplifying the hub **500**. In the unfolded configuration, the support tubes **540** and the hub **550** may once again provide a flat platform to support the mattress **510** where the support tubes **540** are oriented substantially horizontal or horizontal along the bottom surface **526** of the bassinet soft goods **522**. In the folded configuration, the support tubes **540** rotate (e.g., pivot) with respect to the hub **550** such that the support tubes **540** are oriented substantially vertical or vertical and such that the ends **542b** of the support tubes are disposed below the hub **550** in the folded configuration.

The hub **550** may once again include integrated mechanical stops **554** to prevent the hub **550** from moving past the unfolded configuration once the support tubes **540** are aligned horizontally. Compared to the bassinet accessory **500a**, however, the weight of the hub **550** and/or the support tubes **540** does not cause the bassinet accessory **500b** to unfold. Rather, the weight of the hub **550**, the support tubes **540**, the child, and/or the mattress **510** apply a force that unfolds the bassinet accessory **500b** and thereafter maintains the bassinet accessory **500b** in the unfolded configuration. In this manner, the process of unfolding the bassinet accessory **500b** may be simplified.

One challenge, however, is that the bassinet accessory **500b** may provide a relatively shallow partially enclosed space **501**. For example, in the folded configuration, the length, $L_{t,1}$, of the support tubes **540**, is longer than the height, $h_{t,1}$, of the bassinet accessory **500b**. Similar to the bassinet accessory **500a**, the distal ends **542b** of each support tube **540** in the bassinet accessory **500b** may remain stationary or substantially stationary with respect to the ground **90**. In other words, the distal ends **542b** of each support tube may remain at a height, $h_{t,1}$, from the top horizontal plane **92** of the playard **1000b** as the ends **542b** displace laterally when the bassinet accessory **500b** is folded and unfolded. If the length of the support tubes **540** remains constant (e.g., the support tube is formed of a single rigid component), the rotation of the support tubes **540** from the horizontal orientation in the unfolded configuration to the vertical orientation in the folded configuration would cause the hub **550** and a portion of the support tubes **540** to protrude above the top horizontal plane **92** of the playard **1000b** in the folded configuration, thus increasing the overall size of the foldable playard **1000b** in the folded configuration.

To reduce the extent the bassinet accessory **500b** protrudes above the top horizontal plane **92** of the playard **1000b** in the folded configuration, the support tubes **540** may be telescoping such that the length, $L_{t,1}$, of the support tubes **540** in the unfolded configuration changes to a shorter length $L_{t,2}$, in the folded configuration. Thus, in some implementations, the length, $L_{t,1}$, of the support tubes **540** in the unfolded configuration is greater than the height, $h_{t,1}$, of the bassinet accessory **500b** while the length, $L_{t,2}$, of the support

tubes **540** in the folded configuration is approximately equal to or less than the height, $h_{t,1}$. It should be appreciated that, in some implementations, the height of the bassinet accessory **500b** may change between the folded and unfolded configurations. For example, FIG. **52** shows the bassinet accessory **500b** may have a height, $h_{t,2}$, in the folded configuration that differs from the height, $h_{t,1}$, in the unfolded configuration due, for example, to the deformation of the bassinet soft goods **522**. For these implementations, the length, $L_{t,1}$, of the support tubes **540** in the unfolded configuration remains greater than the height, $h_{t,1}$, and the length, $L_{t,2}$, of the support tubes **540** in the folded configuration is approximately equal to or less than the height, $h_{t,2}$.

To unfold the playard **1000b** with the bassinet accessory **500b**, the caregiver may remove the mattress **510** wrapped around the frame **100b** as before. Then, the caregiver may move a slider **120** towards a corner **130** of one leg support assembly **110a** to unfold the frame **100b**. Once the slider **120** is moved sufficiently to engage the latch mechanism **200a**, the frame **100b** is locked in the unfolded configuration. As before, the unfolding of the frame **100b** may cause the bassinet accessory **500b** to at least partially unfold. In some implementations, the weight of the hub **550** and the support tubes **540** may be sufficient to ensure the bassinet accessory **500b** unfolds without any external force applied by the caregiver. In some implementations, the caregiver may simply push down upon the hub **550** to unfold the bassinet accessory **500b**. In some implementations, the caregiver may place the mattress **510** onto the hub **550** and the weight of the mattress **510** may ensure the bassinet accessory **500b** is in the unfolded configuration. Similar to the bassinet accessory **500a**, the bassinet accessory **500b** may be unfolded without the caregiver having to reach down towards the floor portion **304**, which may reduce the physical strain experienced by the caregiver when unfolding the bassinet accessory **500b**.

FIGS. **49A-49D** show a series of figures that illustrate the process of folding the playard **1000b** and the bassinet accessory **500b**. FIG. **49A** shows the hub **550** may include a center opening **558** and the bottom surface **526** of the bassinet soft goods **522** may include a center opening **536**. To fold the playard **1000b** and the bassinet accessory **500b**, the caregiver may first disengage the latch mechanism **200a** on the frame **100b**. Then, the caregiver may extend their hand/arm through the center openings **558** and **536** to access the bottom portion **301a** of the playard **1000b**. FIG. **49B** shows the floor portion **304** of the soft goods **300** may include a strap **342**. When the caregiver reaches into the bottom portion **301a**, they may pull the strap **342** together with the floor portion **304** of the soft goods **300** in an upwards direction. FIG. **49C** shows the caregiver may continue to pull the strap **342** through the center openings **536** and **558**, which causes the floor portion **304** to contact the bassinet soft goods **522** and/or a portion of the hub **550**. As the caregiver continues to pull the strap **342** further, the contact between the floor portion **304** and the bassinet soft goods **522** and/or the hub **550** causes the hub **550** to move upwards and the support tubes **540** to rotate such that the ends **542b** move downwards relative to the hub **550** (see arrows in FIG. **49C**). The caregiver may continue to pull on the strap **342** until the playard **1000b** and the bassinet accessory **500b** are folded as shown in FIG. **49D**.

In some implementations, the playard **1000b** and the bassinet accessory **500b** may be folded without the caregiver having to insert their hand/arm through the center openings **536** and **558**. Instead, the caregiver may pull up on the hub **550** and/or move the slider **120** down towards the foot **114**

to fold the playard **1000b** and the bassinet accessory **500b**. Once the playard **1000b** is folded, the caregiver may lay the playard **1000b** on its side and press floor portion **304** into the interior space **102** before wrapping the mattress **510** around the frame **100b**. In this manner, the caregiver does not have to bend over and reach down to the floor portion **304**.

In some implementations, the length, $L_{t,2}$, of the support tubes **540** in the folded configuration may be tailored such that the hub **550** is disposed entirely within the interior space **102** (i.e., the hub **550** does not extend significantly beyond the top horizontal plane **92**). In some implementations, the length, $L_{t,2}$, of the support tubes **540** may be tailored such that the hub **550** protrudes above the top horizontal plane **92** with a bottom side of the hub **550** flush against the top horizontal plane **92**. This configuration may be preferential when the exterior width of the hub **550** is greater than or equal to the width of the interior space **102** in the folded configuration. Under these conditions, the lateral dimensions of the playard **1000b** may increase if the hub **550** is disposed within the interior space **102**, which may be undesirable. Thus, by positioning the hub **550** just above the playard **1000b**, the lateral dimensions of the frame **100b** in the folded configuration may be kept small (i.e., the lateral dimensions would be the same when the playard **1000b** does not include the bassinet accessory **500b**) without appreciably increasing the height of the playard **1000b** in the folded configuration. In some implementations, the top side of the hub **550** may extend above the top horizontal plane **92** of the playard **1000b** by a distance less than or equal to 1 inch.

FIGS. **50A-50C** show several views of the bassinet accessory **500b** removed from the playard **1000b**. As shown, the center opening **536** of the bassinet soft goods **522** may be aligned with the center opening **558** of the hub **550**. In some implementations, the center opening **536** may have a width that is equal to or smaller than the exterior width of the hub **550**. In other words, the center opening **536** may only be accessible through the center opening **558** and not from the sides of the hub **550**. It should be appreciated that, in other implementations, the hub **550** and/or the bassinet soft goods **522** may not include the center openings **536** and **558**, respectively. Instead, the caregiver may fold the bassinet accessory **500b** by pulling on the hub **550** as described above.

FIG. **50A** further shows each support tube **540** may have a first support tube **546a** coupled to the hub **550** and a second support tube **546b** telescopically coupled to the first support tube **546a**. As shown, the first support tube **546a** may have a larger width (or diameter) such that a portion of the second support tube **546b** may be disposed within the first support tube **546a**. It should be appreciated, however, that in other implementations, the first support tube **546a** may have a smaller width than the second support tube **546b** such that a portion of the first support tube **546a** is disposed within the second support tube **546b**. The relative lengths of the first and second support tubes **546a** and **546b** may be chosen to provide a desired length, $L_{t,1}$, in the unfolded configuration and a desired length, $L_{t,2}$, in the folded configuration. For example, the length, $L_{t,1}$, may be chosen such that the end **542b** extends to the corner portion **537** and the length, $L_{t,2}$, may be approximately equal to or less than the height, $h_{t,1}$ (or the height, $h_{t,2}$) as described above.

In some implementations, the support tube **540** may include a spring element (not shown) disposed within the first support tube **546a** to impart a bias force that extends the length of the support tube **540** (e.g., the spring element may move the second support tube **546b** away from the first support tube **546a**). Additionally, one or both of the support

tubes **546a** and **546b** may include a mechanical stop (not shown) that limits the extent the second support tube **546b** extends from the first support tube **546a**. Furthermore, the first support tube **546a** and the second support tube **546b** may overlap in the unfolded configuration. For example, FIG. **51A** shows an overlap section **548**. In some implementations, the overlap section **548** may have a length of about 1.5 inches to ensure the support tube **540** has sufficient mechanical rigidity to support the bassinet accessory **500b** in the unfolded configuration.

FIGS. **50A** and **50B** further show each support tube **540** may be directly coupled to the bottom surface **526** of the bassinet soft goods **522** via a strap **530** with a fastener **534b** disposed near the end **542a** of the first support tube **546a**. As shown in FIG. **51A**, the strap **530** may include a fastener **534a** to couple the strap **530** to the first support tube **546a**. The strap **530** may further be sewn directly into the bottom surface **526** to form a fabric tunnel that physically contacts the hub **550**. FIG. **51B** further shows a fastener **534a** may couple the bassinet soft goods **522** to the end **542b** of the second support tube **546b**. As shown, the fastener **534a** may be inserted through an opening (not shown) at or near the corner portion **537** from the bottom side of the bottom surface **526**.

Similar to the bassinet accessory **500a**, the bassinet accessory **500b** may be coupled to the top portion **302** of the soft goods **300** via multiple zipper mechanisms **527**. In this manner, the caregiver may readily remove the bassinet accessory **500b** from the playard **1000b** for cleaning or storage. FIG. **50C** shows the bassinet accessory **500b** folded for storage. The hub **550** may once again include a base **551** with multiple channels **552** to receive the support tubes **540**. The channel **552** may provide support for a pin **544** mounted to each support tube **540** to facilitate rotation of the support tube **540** relative to the hub **550**. As shown in FIG. **50C**, the top side of the channel **552** may be covered by a section of the base **551** corresponding to the mechanical stop **554** while the bottom side of the channel **552** may be exposed. Thus, support tube **540** may rotate such that the end **542b** of the support tube **540** is disposed below the hub **550** when folding the bassinet accessory **500b**. When the support tubes **540** are horizontally oriented in the unfolded configuration, the mechanical stops **544** may physically contact the support tubes **540** thus preventing the hub **550** from moving past the unfolded configuration.

FIG. **52** shows the bassinet accessory **500b** may be installed onto the playard **1000c** in a similar manner as the playard **1000b**. For clarity, the bassinet soft goods **522** are not shown. Instead, FIG. **52** shows a plane **538** corresponding to the respective bottom corner portions **537** of the bassinet soft goods **522** for reference. As shown, the hub **550** may be disposed above the top horizontal plane **92** of the playard **1000c** such that the bottom side of the hub **550** is flush with the top horizontal plane **92**. As described above, this arrangement may ensure the frame **100c** folds to its smallest lateral dimensions without appreciably increasing the height of the playard **1000c** due to the addition of the bassinet accessory **500b**. FIG. **52** also shows the support tube **540** in its contracted state where the second support tube **546b** is disposed nearly entirely within the first support tube **546a**.

CONCLUSION

All parameters, dimensions, materials, and configurations described herein are meant to be exemplary and the actual parameters, dimensions, materials, and/or configurations

will depend upon the specific application or applications for which the inventive teachings is/are used. It is to be understood that the foregoing embodiments are presented primarily by way of example and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein.

In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of respective elements of the exemplary implementations without departing from the scope of the present disclosure. The use of a numerical range does not preclude equivalents that fall outside the range that fulfill the same function, in the same way, to produce the same result.

Also, various inventive concepts may be embodied as one or more methods, of which at least one example has been provided. The acts performed as part of the method may in some instances be ordered in different ways. Accordingly, in some inventive implementations, respective acts of a given method may be performed in an order different than specifically illustrated, which may include performing some acts simultaneously (even if such acts are shown as sequential acts in illustrative embodiments).

All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety.

All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B,” when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In

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general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03.

The invention claimed is:

1. A foldable playard having a compact folded configuration for storage and a deployed unfolded configuration for placement on a ground surface to contain a child, the foldable playard comprising:

a frame defining an interior space to contain the child when the foldable playard is in the deployed unfolded configuration, the frame comprising:

a plurality of leg support assemblies extending upward from the ground surface when the frame is in the unfolded configuration, each leg support assembly of the plurality of leg support assemblies being disposed at a corresponding side edge of the interior space and only coupled to another leg support assembly of the plurality of leg support assemblies that is separated by a single side face via a folding mechanism;

soft goods coupled to the plurality of leg support assemblies; and

a bassinet accessory, coupled to the soft goods and disposed within the interior space, to support the child at an elevated position above the ground surface, wherein the frame, the soft goods, and the bassinet accessory fold together when changing the foldable playard to the compact folded configuration and unfold together when changing the foldable playard to the deployed unfolded configuration.

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2. The foldable playard of claim 1, wherein the bassinet accessory comprises:

a support structure, comprising:

bassinet soft goods having a plurality of side surfaces and a bottom surface; and

a plurality of support tubes, disposed along the bottom surface of the bassinet soft goods in the deployed unfolded configuration, to support a mattress.

3. The foldable playard of claim 2, wherein the bassinet accessory further comprises:

the mattress, disposed on the plurality of support tubes in the deployed unfolded configuration, to support the child.

4. The foldable playard of claim 3, wherein the mattress has a top surface that is offset from a top side of the foldable playard by a distance ranging between 7.5 inches and about 10 inches.

5. The foldable playard of claim 2, wherein respective bottom corner portions of the bassinet soft goods are offset from a top horizontal plane of the frame by a height in the deployed unfolded configuration, the height ranging between 7.5 inches and about 12 inches.

6. The foldable playard of claim 2, wherein:

respective bottom corner portions of the bassinet soft goods are offset from a top horizontal plane of the frame by a height in the deployed unfolded configuration; and

each support tube of the plurality of support tubes has a first length greater than the height in the deployed unfolded configuration that changes to a second length that is approximately equal to or less than the height in the compact folded configuration.

7. The foldable playard of claim 6, wherein:

each support tube of the plurality of support tubes includes an end; and

when the bassinet is installed on the foldable playard, each support tube of the plurality of support tubes is oriented vertically in the compact folded configuration such that the end is located above the top horizontal plane of the foldable playard by a distance less than or equal to about 1 inch.

8. The foldable playard of claim 6, wherein each support tube of the plurality of support tubes comprises:

a first support tube; and

a second support tube telescopically coupled to the first support tube.

9. The foldable playard of claim 8, wherein:

the bassinet accessory further comprises a hub disposed on a center portion of the bottom surface of the bassinet soft goods in the unfolded configuration; and

each support tube of the plurality of support tubes is pivotably coupled to the hub.

10. The foldable playard of claim 9, wherein the hub moves upwards away from the ground surface and the plurality of support tubes rotate relative to the hub when transitioning from the deployed unfolded configuration to the compact folded configuration.

11. The foldable playard of claim 10, wherein the hub is flush with the top horizontal plane of the foldable playard in the compact folded configuration.

12. The foldable playard of claim 9, wherein the hub moves downwards towards the ground surface and the plurality of support tubes rotate relative to the hub when transitioning from the deployed unfolded configuration to the compact folded configuration.

13. The foldable playard of claim 12, wherein the hub comprises a hub latch to constrain vertical downwards

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movement of the hub when actuated thereby maintaining the support structure in the deployed unfolded configuration.

14. The foldable playard of claim 1, wherein:

each leg support assembly of the plurality of leg support assemblies comprises a top portion, the top portions of respective leg support assemblies of the plurality of leg support assemblies defining a top horizontal plane of the foldable playard; and

the foldable playard has a height defined as a distance between the ground surface and the top horizontal plane that remains substantially constant between the deployed unfolded configuration and the compact folded configuration.

15. A bassinet accessory for a foldable playard, the foldable playard having a compact folded configuration for storage and a deployed unfolded configuration for placement in an upright position on a ground surface to contain a child, the foldable playard having a frame that defines an interior space to contain the child when the playard is in the deployed unfolded configuration, the bassinet accessory comprising:

a support structure couplable to the frame and disposable within the interior space, to support the child at an elevated position above the ground surface; and

bassinet soft goods coupled to the support structure, wherein the support structure and the bassinet soft goods are configured to fold together with the foldable playard when changing the foldable playard to the compact folded configuration and unfold together when changing the foldable playard to the deployed unfolded configuration.

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16. The bassinet accessory of claim 15, wherein the bassinet soft goods include a plurality of side surfaces and a bottom surface, the bassinet accessory further comprising:

a plurality of support tubes, disposed along the bottom surface of the bassinet soft goods in the deployed unfolded configuration, to support a mattress.

17. The bassinet accessory of claim 15, further comprising:

the mattress disposed on the plurality of support tubes in the deployed unfolded configuration.

18. The bassinet accessory of claim 16, wherein each support tube of the plurality of support tubes comprises a first support tube and a second support tube telescopically coupled to the first support tube.

19. The bassinet accessory of claim 18, further comprising:

a hub disposed on a center portion of the bottom surface of the bassinet soft goods in the unfolded configuration of the foldable playard,

wherein each support tube of the plurality of support tubes is pivotably coupled to the hub.

20. The bassinet accessory of claim 19, wherein the hub moves upwards away from the ground surface and the plurality of tubes rotate relative to the hub when transitioning from the deployed unfolded configuration of the foldable playard to the compact folded configuration of the foldable playard.

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